

Algebra

Important Formulas:-

- $(a + b)^2 = a^2 + b^2 + 2ab$
- $(a - b)^2 = a^2 + b^2 - 2ab$
- $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
- $(a + b)^2 - (a - b)^2 = 4ab$
- $(a + b)^2 - 4ab = (a - b)^2$
- $a^2 - b^2 = (a + b)(a - b)$
- $a^4 - b^4 = (a^2 + b^2)(a + b)(a - b)$
- $a^2 + \frac{1}{a^2} = (a + \frac{1}{a})^2 - 2$
 $= (a - \frac{1}{a})^2 + 2$
- $a^2 + b^2 + c^2 - ab - bc - ca =$
 $\frac{1}{2}\{(a - b)^2 + (b - c)^2 + (c - a)^2\}$
 If $a = b = c = k$ then,
 $a^2 + b^2 + c^2 = ab + bc + ca,$
- $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
- $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
- $a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$
- $a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$
- $a^3 + \frac{1}{a^3} = (a + \frac{1}{a})^3 - 3(a + \frac{1}{a})$
- $a^3 - \frac{1}{a^3} = (a - \frac{1}{a})^3 + 3(a - \frac{1}{a})$
- $a^3 + b^3 + c^3 - 3abc = (a + b + c)$
 $(a^2 + b^2 + c^2 - ab - bc - ca)$
 Or, $\frac{1}{2}(a + b + c)\{(a - b)^2 + (b - c)^2 + (c - a)^2\}$ Or,
 $(a + b + c)[(a + b + c)^2 - 3(ab + bc + ca)]$
 Or, $\frac{1}{2}(a + b + c)$
 $[3(a^2 + b^2 + c^2) - (a + b + c)^2]$
- If $(a + b + c) = 0$, then
 $a^3 + b^3 + c^3 = 3abc$
- If $(a - b - c) = 0$, then
 $a^3 - b^3 - c^3 = 3abc$
- If $(a + b - c) = 0$, then
 $a^3 + b^3 - c^3 = -3abc$
- If $a^3 + b^3 + c^3 = 3abc$,
 then either $a = b = c$ or $a + b + c = 0$
- $(a + b + c)^2 =$
 $a^2 + b^2 + c^2 + 2(ab + bc + ac)$
- $(a + b + c)^3 = a^3 + b^3 + c^3 +$
 $3(a + b)(b + c)(a + c)$

- $a^4 + a^2b^2 + b^4 = (a^2 + ab + b^2)$
 $(a^2 - ab + b^2)$
- If $(x^2 + y^2 + z^2) = 0$,
 then $x = 0, y = 0, z = 0$
- If $x + \frac{1}{x} = a$, then $x^3 + \frac{1}{x^3} = a^3 - 3a$
- If $x - \frac{1}{x} = a$, then $x^3 - \frac{1}{x^3} = a^3 + 3a$
- If $x + \frac{1}{x} = a$, then $x^2 + \frac{1}{x^2} = a^2 - 2$
- If $x - \frac{1}{x} = a$, then $x^2 + \frac{1}{x^2} = a^2 + 2$
- If $x + \frac{1}{x} = a$, then $x - \frac{1}{x} = \sqrt{a^2 - 4}$
- If $x - \frac{1}{x} = a$, then $x + \frac{1}{x} = \sqrt{a^2 + 4}$
- If $x + \frac{1}{x} = a$, then
 $x^2 - \frac{1}{x^2} = a\sqrt{a^2 - 4}$
- If $x - \frac{1}{x} = a$, then
 $x^2 - \frac{1}{x^2} = a\sqrt{a^2 + 4}$
- If $x + \frac{1}{x} = a$, then
 $x^4 + \frac{1}{x^4} = (a^2 - 2)^2 - 2$
- If $x - \frac{1}{x} = a$, then
 $x^4 + \frac{1}{x^4} = (a^2 + 2)^2 - 2$
- If $x + \frac{1}{x} = a$, then
 $x^5 + \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 + \frac{1}{x^3}) - (x + \frac{1}{x})$
- If $x - \frac{1}{x} = a$, then
 $x^5 - \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 - \frac{1}{x^3}) - (x - \frac{1}{x})$
- If $x + \frac{1}{x} = \pm\sqrt{3}$, then $x^6 = -1$
- If $x + \frac{1}{x} = 1$, then $x^3 = -1$
- If $x + \frac{1}{x} = -1$, then $x^3 = 1$
- If $x + \frac{1}{x} = 2$, then $x = 1$, similarly if
 $x + \frac{1}{x} = -2$, then $x = -1$

Some Special Case :-

- If $a^2 - ab + b^2 = 0$ then $a^3 + b^3 = 0$
- If $\frac{1}{a} - \frac{1}{b} = \frac{1}{a - b}$ then $a^3 + b^3 = 0$
- If $\frac{1}{b} + \frac{1}{a} = \frac{1}{a + b}$ then $a^3 - b^3 = 0$
- If $\frac{a}{b} + \frac{b}{a} = -1$ then $a^3 - b^3 = 0$

- If $\frac{a}{b} + \frac{b}{a} = 1$ then $a^3 + b^3 = 0$
- If $a^2 + a + 1 = 0$ then $a^3 - 1 = 0$ or $a^3 = 1$
- If $ab(a + b) = 1$ then $\frac{1}{a^3b^3} - a^3 - b^3 = 3$

Dividendo - Componendo rule:

$$\text{If } \frac{a}{b} = \frac{c}{d} \text{ then, } \frac{a+b}{a-b} = \frac{c+d}{c-d}$$

$$\text{If } \frac{a+b}{a-b} = \frac{a}{b} \text{ then } \frac{c}{d} = \frac{c+d}{c-d}$$

• If $(x - a)^2 + (y - b)^2 + (z - c)^2 = 0$
 then by putting $x = a, y = b, z = c$ in
 an expression we can find the value of
 the expression.

- If $x + y = a$, then xy will be
 maximum when $x = y$.
- If $xy = a$, then $(x + y)$ will be
 minimum when $x = y = \sqrt{a}$

Law of Indices :

$$(1) a^m \times a^n = a^{m+n} \quad (2) \frac{a^m}{a^n} = a^{m-n}$$

$$(3) (a^m)^n = a^{mn} \quad (4) \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$(5) a^0 = 1 \quad (6) a^{\frac{1}{n}} = \sqrt[n]{a}$$

If a is negative number and n is an odd
 number then $(-a)^n = -a^n$

Laws of surds :

- (1). $\sqrt[n]{a} = a^{\frac{1}{n}}$
- (2). $\sqrt[n]{a} \times \sqrt[n]{b} = \sqrt[n]{ab}$
- (3). $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$
- (4). $\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$
- (5). $(\sqrt[n]{a})^m = \sqrt[n]{a^m}$

Ex.1. If $3\sqrt{5}$ is given, find out the
 rationalizing factor.

Sol : $3\sqrt{5} \times \sqrt{5} = 3 \times 5 = 15$, so the
 rationalizing factor of $3\sqrt{5}$ is $\sqrt{5}$.

Rules for making a perfect square :

1. Middle term =
 $\pm \sqrt{\text{First term} \times \text{Third term}}$
2. First term = $\frac{(\text{Middle term})^2}{4 \times \text{Third term}}$
3. Third term = $\frac{(\text{Middle term})^2}{4 \times \text{First term}}$

Quadratic and cubic equation :

- For quadratic equation :
 $ax^2 + bx + c = 0$, then
 Sum of its zeros $(\alpha + \beta) = -\frac{b}{a}$

Product of its zeros ($\alpha\beta$) = $\frac{c}{a}$

where, α and β are roots of a given quadratic equation.

• **For cubic equation :**

$ax^3 + bx^2 + cx + d = 0$, then

Sum of its zeros ($\alpha + \beta + \gamma$) = $-\frac{b}{a}$

Product of its zeros ($\alpha\beta\gamma$) = $-\frac{d}{a}$

Sum of its zeros taken two at a time ($\alpha\beta + \beta\gamma + \gamma\alpha$) = $\frac{c}{a}$

where α , β and γ are roots of a given cubic equation.

Variety Questions

Q.1. If $x^4 + y^4 = x^2y^2$, then the value of $x^6 + y^6$ is :

SSC CPO 04/10/2023 (1st Shift)
(a) 1 (b) 3 (c) 2 (d) 0

Q.2. If $ax + by = 1$ and $bx + ay = \frac{2ab}{a^2 + b^2}$, then the value of x (in terms of a and b) is:

SSC CHSL 11/08/2023 (4th Shift)

- (a) $\frac{2b}{a^2 + b^2}$ (b) $\frac{a}{a^2 + b^2}$
(c) $\frac{b}{a^2 + b^2}$ (d) $\frac{2a}{a^2 + b^2}$

Q.3. If $x = (7 + 3\sqrt{5})$ then find the value of $x^2 + \frac{1}{x^2}$

SSC CHSL 09/08/2023 (4th Shift)

- (a) $\frac{580 + 315\sqrt{5}}{8}$ (b) $\frac{799 + 328\sqrt{5}}{8}$
(c) $\frac{799 + 315\sqrt{5}}{12}$ (d) $\frac{799 + 315\sqrt{5}}{8}$

Q.4. If $x^2 - 11x + 1 = 0$ what is the value of $x^8 - 14159x^4 + 11$?

SSC CHSL 08/08/2023 (1st Shift)
(a) 9 (b) 10 (c) 12 (d) 11

Q.5. Simplify:

$$\frac{(1-m)^3 + (m-n)^3 + (n-1)^3}{12(1-m)(m-n)(n-1)} = ?$$

SSC CHSL 07/08/2023 (3rd Shift)

- (a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{3}{2}$ (d) $\frac{1}{4}$

Q.6. For what positive value of k , the equation $3x^2 + 2kx + 3 = 0$ has real and equal roots?

SSC CHSL 07/08/2023 (2nd Shift)
(a) 12 (b) 1 (c) 3 (d) 9

Q.7. Simplify the following expression.

$$\frac{(a^2 + b^2 - c^2)^2 - (a^2 - b^2 + c^2)^2}{b^2 - c^2}$$

SSC CHSL 04/08/2023 (4th Shift)

(a) $3a^2$ (b) $4a^2$ (c) $5a^2$ (d) $2a^2$

Q.8. Find the value of $(a^3 + b^3 + c^3 - 3abc)$, where $a = 335$, $b = 215$ and $c = 180$.

SSC CHSL 04/08/2023 (3rd Shift)

- (a) 14472250 (b) 15421320
(c) 14502230 (d) 15452630

Q.9. Simplify the expression :

$$\frac{(u-v)^3 + (v-w)^3 + (w-u)^3}{(u^2-v^2)^3 + (v^2-w^2)^3 + (w^2-u^2)^3}$$

SSC CHSL 04/08/2023 (1st Shift)

- (a) $\frac{1}{(u+v)(v+w)(w+u)}$ (b) 1
(c) $\frac{3}{(u+v)(v+w)(w+u)}$ (d) 0

Q.10. If $x^3 = 270 + y^3$ and $x = (6 + y)$, then what is the value of $(x + y)$?

(given that $x > 0$ and $y > 0$)

SSC CHSL 03/08/2023 (4th Shift)

- (a) $2\sqrt{3}$ (b) $\sqrt{3}$ (c) $4\sqrt{3}$ (d) $4\sqrt{2}$

Q.11. If $p + q + r = 0$, then what is the simplified value of the expression

$$\left(\frac{p^2}{p^2 - qr} + \frac{q^2}{q^2 - pr} + \frac{r^2}{r^2 - pq} \right)$$

SSC CHSL 03/08/2023 (2nd Shift)

- (a) 0 (b) 2 (c) 1 (d) -1

Q.12. If $(x + \frac{1}{x}) = 5\sqrt{2}$, and $x > 1$, what is the value of $(x^6 - \frac{1}{x^6})$?

SSC CHSL 02/08/2023 (2nd Shift)

- (a) $22970\sqrt{23}$ (b) $23030\sqrt{23}$
(c) $23060\sqrt{23}$ (d) $22960\sqrt{23}$

Q.13. The difference between the cubes of two given natural numbers is 6272, while the positive difference between the two given numbers is 8. What is the sum of the cubes of the two given numbers?

SSC CGL 27/07/2023 (1st shift)

- (a) 9728 (b) 9684 (c) 8000 (d) 9600

Q.14. If $a = 101$, $b = 102$ and $c = 103$, then $a^2 + b^2 + c^2 - ab - bc - ca =$ _____.

SSC CGL 24/07/2023 (1st shift)

- (a) 2 (b) 4 (c) 3 (d) 6

Q.15. If $(x + \frac{1}{x}) = 5$, and $x > 1$, what is the value of $(x^8 - \frac{1}{x^8})$?

SSC CGL 18/07/2023 (3rd shift)

- (a) $60605\sqrt{21}$ (b) $60615\sqrt{21}$
(c) $60705\sqrt{21}$ (d) $60725\sqrt{21}$

Q.16. If $X - \frac{1}{X} = -6$, What will be the value of $X^5 - \frac{1}{X^5}$?

SSC CGL 18/07/2023 (2nd shift)

- (a) -8898 (b) -8896
(c) -8886 (d) -8892

Q.17. If $(x^2 + \frac{1}{x^2}) = 7$, and $0 < x < 1$, find the value of $x^2 - \frac{1}{x^2}$.

SSC CGL 17/07/2023 (1st shift)

- (a) $3\sqrt{5}$ (b) $4\sqrt{3}$ (c) $-4\sqrt{3}$ (d) $-3\sqrt{5}$

Q.18. If $x > 0$, and $x^4 + \frac{1}{x^4} = 254$, what is

the value of $x^5 + \frac{1}{x^5}$?

SSC CGL 14/07/2023 (4th shift)

- (a) $717\sqrt{2}$ (b) $723\sqrt{2}$ (c) $720\sqrt{2}$ (d) $726\sqrt{2}$

Q.19. If $\frac{2x}{x+2} + \frac{1}{2x+4} - \frac{12-x}{2x^2+4x}$

= $a + \frac{b}{x}$, where a and b are integers, then

the value of $ab =$ _____.

Graduate Level 27/06/2023 (Shift - 4)

- (a) 3 (b) -6 (c) 6 (d) -3

Q.20. If $6^x = 3^y = 2^z$, then what is the value of $\frac{1}{y} + \frac{1}{z} - \frac{1}{x}$?

SSC CHSL 20/03/2023 (2nd Shift)

- (a) 1 (b) 0 (c) 3 (d) 6

Q.21. For what value of m will the system of equations $17x + my + 102 = 0$ and $23x + 299y + 138 = 0$ have an infinite number of solutions?

SSC CGL Tier II (02/03/2023)

- (a) 221 (b) 223 (c) 220 (d) 219

Q.22. If $a + b + c = 5$, $a^3 + b^3 + c^3 = 85$ and $abc = 25$, then find the value of $a^2 + b^2 + c^2 - ab - bc - ca$.

SSC CGL 13/12/2022 (4th Shift)

- (a) 2 (b) 4 (c) 6 (d) 8

Q.23. If α, β are the roots of $6x^2 + 13x + 7 = 0$, then the equation whose roots are α^2, β^2 is:

SSC CGL 12/12/2022 (4th Shift)

- (a) $36x^2 - 87x + 49 = 0$
(b) $36x^2 - 85x + 49 = 0$
(c) $36x^2 - 85x - 49 = 0$
(d) $36x^2 + 87x - 49 = 0$

Q.24. If $x = 222$, $y = 223$ and $z = 224$, then find the value of $x^3 + y^3 + z^3 - 3xyz$.

SSC CGL 12/12/2022 (4th Shift)

- (a) 2007 (b) 2004 (c) 2006 (d) 2005

Q.25. What is the value of

$$(K - \frac{1}{K})(K^2 + \frac{1}{K^2})(K^4 + \frac{1}{K^4})$$

$$(K^8 + \frac{1}{K^8})(K^{16} + \frac{1}{K^{16}})(K^{32} + \frac{1}{K^{32}}) ?$$

SSC CGL 06/12/2022 (3rd Shift)

$$(a) \frac{K^{64} - \frac{1}{K^{64}}}{K + \frac{1}{K}} \quad (b) \frac{K^{32} - \frac{1}{K^{32}}}{K - \frac{1}{K}}$$

$$(c) \frac{K^{32} - \frac{1}{K^{32}}}{K + \frac{1}{K}} \quad (d) \frac{K^{32} + \frac{1}{K^{32}}}{K + \frac{1}{K}}$$

Q.26. Which of the following statements is correct ?

I. The value of $100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + \dots + 2^2 - 1^2$ is 5050.

II. If $8x + \frac{8}{x} = -16$ and $x < 0$, then the value of $x^{197} + x^{-197}$ is 2.

SSC CGL 02/12/2022 (4th Shift)

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

Q.27. If $mx^m - nx^n = 0$ then what is the

value of $\frac{1}{x^m + x^n} + \frac{1}{x^m - x^n}$ in terms of x^n ?

SSC CHSL 24/05/2022 (Morning)

$$(a) \frac{2mn}{x^n(n^2 + m^2)} \quad (b) \frac{2m}{x^n(m^2 - n^2)}$$

$$(c) \frac{2mn}{x^n(n^2 - m^2)} \quad (d) \frac{2mn}{x^n(m^2 - n^2)}$$

Q.28. $a^2 + b^2 = 25$, $x^2 + y^2 = 17$ and $ax + by = 8$ then find the value of $(ay - bx)$

SSC CHSL 13/4/2021 (Afternoon)

(a) 33 (b) 25 (c) 21 (d) 19

Q.29. If $2 = x + \frac{1}{1 + \frac{1}{5 + \frac{1}{2}}}$, then the

value of x is equal to:

SSC CGL Tier II (18/11/2020)

(a) $\frac{14}{13}$ (b) 1 (c) $\frac{15}{13}$ (d) $\frac{13}{15}$

Q.30. If $x^3 + 2x^2 - ax - b$ is exactly divisible by $(x^2 - 1)$, then the values of a and b are:

SSC CHSL 21/10/2020 (Afternoon)

(a) $a = -1$, and $b = 2$ (b) $a = 1$ and $b = -2$
(c) $a = 1$ and $b = 2$ (d) $a = 2$ and $b = 2$

Q.31. If $x = \sqrt[3]{5} + 2$, then the value of $x^3 - 6x^2 + 12x - 12$ is equal to:

SSC CHSL 21/10/2020 (Morning)

(a) 0 (b) 2 (c) 1 (d) -1

Q.32. $(ax + by)$ is a factor of:

SSC CHSL 16/10/2020 (Afternoon)

(a) $a^2x^2 + 2ab - b^2y^2$
(b) $a^2x^2 + 2abxy + b^2y^2$
(c) $a^2x^3 + 2abx + b^2y^2x$
(d) $a^2x^2 + 2ab + b^2y^2$

Q.33. If 'a' is a natural number, then $(7a^2$

+ 7a) is always divisible by:

SSC CHSL 16/10/2020 (Morning shift)

(a) 7 and 14 both (b) 7 only
(c) 14 only (d) 21 only

Q.34. The value of $(a^{\frac{2}{3}} + 2a^{\frac{1}{2}} + 3a^{\frac{1}{3}} + 2a^{\frac{1}{6}} + 1)(a^{\frac{1}{3}} - 2a^{\frac{1}{6}} + 1) - a^{\frac{1}{2}}(a^{\frac{1}{2}} - 2)$,

when $a = 7$, is:

SSC CHSL 13/10/2020 (Morning)

(a) 7 (b) 0 (c) 1 (d) $\sqrt{7}$

Q.35. If $a + b + c + d = 2$, then the maximum value of -

$(1 + a)(1 + b)(1 + c)(1 + d)$

SSC CHSL 18/03/2020 (Afternoon)

(a) $\frac{91}{9}$ (b) $\frac{81}{16}$ (c) $\frac{63}{22}$ (d) $\frac{54}{13}$

Q.36. The coefficient of x^2 in $(2x + y)^3$ is:

SSC CGL 05/03/2020 (Evening)

(a) $12y^2$ (b) 12 (c) $12y$ (d) 8

Q.37. If $x^{2a} = y^{2b} = z^{2c} \neq 0$ and $x^2 = yz$,

then the value of $\frac{ab + bc + ca}{bc}$ is:

SSC CGL 03/03/2020 (Morning)

(a) 3 (b) 3ac (c) 3bc (d) 3ab

Q.38. If $x = 5.51$, $y = 5.52$ and $z = 5.57$, then what is the value of

$x^3 + y^3 + z^3 - 3xyz$?

SSC CPO 11/12/2019 (Evening)

(a) 5.146 (b) 51.46
(c) 0.05146 (d) 0.5146

Q.39. The value of $\frac{1}{\sqrt{7} - 4\sqrt{3}}$ is closest to:

SSC CPO 09/12/2019 (Evening)

(a) 4.1 (b) 4.2 (c) 1.2 (d) 3.7

Q.40. If $(2x + 3y + 4)(2x + 3y - 5)$ is equivalent to $(ax^2 + by^2 + 2hxy + 2gx + 2fy + c)$, then what is the value of $\{(g + f - c)/abh\}$

SSC CPO 09/12/2019 (Morning)

(a) $\frac{35}{432}$ (b) $\frac{19}{108}$ (c) $\frac{19}{316}$ (d) $\frac{37}{216}$

Q.41. $(4x^3y - 6x^2y^2 + 4xy^3 - y^4)$ can be expressed as:

SSC CPO 09/12/2019 (Morning)

(a) $(x - y)^4 - x^4$ (b) $(x + y)^4 - y^4$
(c) $(x + y)^4 - x^4$ (d) $x^4 - (x - y)^4$

Q.42. Let $x = \sqrt[6]{27} - \sqrt{\frac{3}{4}}$ and

$y = \frac{\sqrt{45} + \sqrt{605} + \sqrt{245}}{\sqrt{80} + \sqrt{125}}$, then the value

of $x^2 + y^2$ is:

SSC CGL TIER II (13/09/2019)

(a) $\frac{223}{36}$ (b) $\frac{221}{36}$ (c) $\frac{221}{9}$ (d) $\frac{227}{9}$

Q.43. If $\sqrt{86} - 60\sqrt{2} = a - b\sqrt{2}$, then what will be the value of $\sqrt{a^2 + b^2}$, correct to one decimal place ?

SSC CGL TIER II (11/09/2019)

(a) 8.4 (b) 8.2 (c) 7.8 (d) 7.2

Q.44. If $a = \sqrt{8} - \sqrt{7}$ and $a = \frac{1}{b}$, then

$\frac{a^2 + b^2 - 3ab}{a^2 + ab + b^2}$ is equal to :

SSC CHSL 11/07/2019 (Evening)

(a) $\frac{27}{31}$ (b) $\frac{27}{32}$ (c) $\frac{29}{33}$ (d) $\frac{29}{31}$

Q.45. If $(a + b + 4)(ab + 4(a + b)) - 4ab = 0$, $a \neq -4$, $b \neq -4$, then,

$\left\{ \frac{1}{(a + b + 4)^{117}} - 2^{-234} \right\}$ is equal to :

SSC CHSL 11/07/2019 (Afternoon)

(a) $\frac{1}{4^{117}}$ (b) $\frac{1}{2^{117}}$ (c) $-\frac{1}{2^{234}}$ (d) 0

Q.46. $(x + y)^{\frac{1}{3}} + (z + y)^{\frac{1}{3}} = -(x + z)^{\frac{1}{3}}$ then

$(x^3 + y^3 + z^3)$ can be expressed as :

SSC CHSL 10/07/2019 (Morning)

(a) $\frac{1}{8}xyz$
(b) $(x + y)(y + z)(z + x)$
(c) $\frac{3}{8}(x + y)(y + z)(z + x)$
(d) $3xyz$

Q.47. The value of the expression

$\frac{1}{4}\{(a + \frac{1}{a})^2 - (a - \frac{1}{a})^2\}$ is :

SSC CHSL 09/07/2019 (Evening)

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) 1 (d) 4

Q.48. If $3^{\sqrt[4]{x}} + 4^{\sqrt[4]{x}} = 5^{\sqrt[4]{x}}$, then the value of x is :

SSC CHSL 08/07/2019 (Morning)

(a) 4 (b) 2 (c) 8 (d) 16

Q.49. If $x = 2 - \sqrt{3}$ then the value of $x^3 - x^{-3}$ is :

SSC CHSL 05/07/2019 (Evening)

(a) $-30\sqrt{3}$ (b) $30\sqrt{3}$ (c) $-30\sqrt{2}$ (d) $30\sqrt{2}$

Q.50. If $x \neq -1, 2$ and 5, then the simplified value of $\left\{ \frac{2(x^3 - 8)}{x^2 - x - 2} \times \right.$

$\left. \frac{x^2 + 2x + 1}{x^2 - 4x - 5} \right\} \div \frac{x^2 + 2x + 4}{3x - 15}$ is equal to :

SSC CHSL 03/07/2019 (Afternoon)

(a) $\frac{1}{6}$ (b) 6 (c) $\frac{3}{2}$ (d) $\frac{2}{3}$

Q.51. If x is real, and $x^4 - 5x^2 - 1 = 0$,

then the value of

$$(x^6 - 3x^2 + \frac{3}{x^2} - \frac{1}{x^6} + 1) \text{ is :}$$

SSC CHSL 03/07/2019 (Morning)

- (a) 126 (b) 110 (c) 116 (d) 96

Q.52. If $x + y = 7$ and $xy = 10$, then the value of $\frac{1}{x^3} + \frac{1}{y^3}$ is:

SSC CHSL 01/07/2019 (Evening)

- (a) 0.543 (b) 0.131 (c) 0.133 (d) 0.453

Q.53. If $(3x - 1)^3 + (4x - 3)^3 + (2x + 1)^3 = 3(3x - 1)(4x - 3)(2x + 1)$ and $x \neq \frac{1}{3}$ then $x = ?$

SSC CGL 10/06/2019 (Evening)

- (a) $\frac{1}{2}$ (b) 2 (c) $\frac{1}{4}$ (d) 1

Q.54. If $x + y + z = 0$, then the value of $(x^2 + y^2 + z^2) \div (z^2 - xy)$ is:

SSC CGL 10/06/2019 (Morning)

- (a) 1 (b) 2 (c) -2 (d) -1

Q.55. If $x^2 - 3x - 1 = 0$, then the value of $(x^2 + 8x - 1)(x^3 + x^{-1})^{-1}$ is:

SSC CGL 07/06/2019 (Afternoon)

- (a) $\frac{3}{8}$ (b) 8 (c) 1 (d) 3

Q.56. If $x = 2 - p$, then $x^3 + 6xp + p^3$, is equal to:

SSC CGL 07/06/2019 (Morning)

- (a) 12 (b) 6 (c) 8 (d) 4

Q.57. If $2x^2 + y^2 + 6x - 2xy + 9 = 0$, then the value of $4x^3 - y^3 + x^2y^2$ is:

SSC CGL 06/06/2019 (Afternoon)

- (a) 0 (b) 9 (c) -3 (d) -9

Q.58. If $ab + bc + ca = 8$ and $a^2 + b^2 + c^2 = 20$, then a possible value of $\frac{1}{2}(a + b + c)$ is:

$$c) [(a - b)^2 + (b - c)^2 + (c - a)^2] \text{ is}$$

SSC CGL 06/06/2019 (Morning)

- (a) 72 (b) 56 (c) 84 (d) 80

Q.59. If $x + y = 1$ and $xy(xy - 2) = 12$, then the value of $x^4 + y^4$ is:

SSC CGL 04/06/2019 (Evening)

- (a) 19 (b) 25 (c) 20 (d) 23

Q.60. If $16x^2 + 9y^2 + 4z^2 = 24(x - y + z) - 61$, then the value of $(xy + 2z)$ is:

SSC CGL 04/06/2019 (Afternoon)

- (a) 1 (b) 2 (c) 3 (d) 5

Q.61. If $(5\sqrt{5}x^3 - 81\sqrt{3}y^3) \div \sqrt{5}x - 3\sqrt{3}y = Ax^2 + By^2 + Cxy$, then the value of $(6A + B - \sqrt{15}C)$ is:

SSC CGL 04/06/2019 (Morning)

- (a) 10 (b) 9 (c) 15 (d) 12

Q.62. If $x + y + z = 19$,

$x^2 + y^2 + z^2 = 133$ and $xz = y^2$, then the difference between z and x is:

SSC CGL 04/06/2019 (Morning)

- (a) 5 (b) 3 (c) 6 (d) 4

Q.63. If $x^4 + x^{-4} = 194$, $x > 0$ then the value of $(x - 2)^2$ is:

SSC CGL 04/06/2019 (Morning)

- (a) 1 (b) 6 (c) 2 (d) 3

Q.64. If roots of $x^2 - 4x + a = 0$ are equal, then $a = ?$

SSC CPO 16/03/2019 (Evening)

- (a) -4 (b) 4 (c) 8 (d) -8

Practice Questions

SSC CHSL 2023 Tier - 2

Q.65. If $(a - 2b) = 8$ and $2ab = 9$, then find the value of $(a + 2b)$.

SSC CHSL Tier II (02/11/2023)

- (a) 12 (b) 6 (c) 10 (d) 8

Q.66. A and B each have a certain number of oranges. A says to B, "If you give me 10 of your oranges, I will have twice the number of oranges left with you." B replies, "If you give me 10 of your oranges, I will have the same number of oranges left with you." Find the number of oranges that A has.

SSC CHSL Tier II (02/11/2023)

- (a) 65 (b) 50 (c) 55 (d) 70

SSC CPO 2023 Tier - 1

Q.67. If $\frac{x}{y} + \frac{y}{x} = 1$ and $x + y = 2$, then

the value of $x^3 + y^3$ is:

SSC CPO 03/10/2023 (1st Shift)

- (a) 0 (b) 1 (c) 3 (d) 2

Q.68. If $\frac{A}{L} + \frac{M}{B} = 1$ and $\frac{B}{M} + \frac{N}{C} = 1$,

then the value of $\frac{L}{A} + \frac{C}{N}$ is:

SSC CPO 03/10/2023 (2nd Shift)

- (a) 1 (b) $\frac{B}{M}$ (c) $\frac{M}{B}$ (d) 0

Q.69. If $a \propto b$, $b \propto \frac{1}{c}$ and $c \propto d$, then the relation between a and b is:

SSC CPO 03/10/2023 (2nd Shift)

- (a) $a \propto \sqrt{d}$ (b) $a \propto d$ (c) $a \propto \frac{1}{d}$ (d) $a \propto d^2$

Q.70. If $A + \frac{1}{1 + \frac{1}{2 + \frac{1}{3}}} = \frac{9}{10}$, then the

value of A is:

SSC CPO 03/10/2023 (3rd Shift)

- (a) $\frac{3}{10}$ (b) $\frac{2}{5}$ (c) $\frac{1}{10}$ (d) $\frac{1}{5}$

Q.71. If $x(5 - \frac{2}{x}) = \frac{5}{x}$, then the value of

$x^2 + \frac{1}{x^2}$ is:

SSC CPO 03/10/2023 (3rd Shift)

- (a) $\frac{54}{23}$ (b) $\frac{53}{28}$ (c) $\frac{53}{27}$ (d) $\frac{54}{25}$

Q.72. The value of $\frac{(157 \times 157) + (157 \times 133) + (133 \times 133)}{(157 \times 157 \times 157) - (133 \times 133 \times 133)}$

is:

SSC CPO 04/10/2023 (2nd Shift)

- (a) 24 (b) $\frac{1}{290}$ (c) 290 (d) $\frac{1}{24}$

Q.73. Simplify the given expression.

$$\frac{(3.254 \times 3.254 \times 3.254) - (0.746 \times 0.746 \times 0.746)}{(3.254 \times 3.254) + (0.746 \times 0.746) + (3.254 \times 0.746)}$$

SSC CPO 04/10/2023 (2nd Shift)

- (a) 2.508 (b) 4.508 (c) 2.146 (d) 3.156

Q.74. The value of $\frac{a^2 - (b - c)^2}{(a + c)^2 - b^2} +$

$$\frac{b^2 - (a - c)^2}{(a + b)^2 - c^2} + \frac{c^2 - (a - b)^2}{(b + c)^2 - a^2} \text{ is:}$$

SSC CPO 04/10/2023 (3rd Shift)

- (a) 3 (b) -1 (c) 1 (d) 2

Q.75. Simplify the given expression.

$$(1 + x)^3 + (1 - x)^3 + (-2)^3$$

SSC CPO 04/10/2023 (3rd Shift)

- (a) $3(1 - x^2)$ (b) $6(1 - x^2)$
(c) $-6(1 - x^2)$ (d) $-3(1 - x^2)$

Q.76. Simplify the given expression.

$$(1 - 2x)^2 - (1 + 2x)^2$$

SSC CPO 05/10/2023 (1st Shift)

- (a) $-8x$ (b) $8x$
(c) $2 + 8x^2$ (d) $-(2 + 8x^2)$

Q.77. If $a + b + c = 5$ and $ab + bc + ca = 7$, then the value of $a^3 + b^3 + c^3 - 3abc$ is:

SSC CPO 05/10/2023 (1st Shift)

- (a) 20 (b) 25 (c) 15 (d) 30

Q.78. Simplify the given expression.

$$(x - 2y)(y - 3x) + (x + y)(x - y) + (x - 3y)(2x + y)$$

SSC CPO 05/10/2023 (3rd Shift)

- (a) $2x(x - 3y)$ (b) $2y(x + 3y)$
(c) $2x(x + 3y)$ (d) $2y(x - 3y)$

SSC MTS 2023 Tier - 1

Q.79. If $(x - 2)$ is a factor of $(2x^2 + 12kx - 25k)$, then what is the value of k ?

SSC MTS 01/09/2023 (2nd Shift)

- (a) 2 (b) 8 (c) 4 (d) 6

SSC CHSL 2023 Tier - 1

Q.80. If $2x + \frac{2}{x} = 5$, then the value of $(x^3 + \frac{1}{x^3} + 2)$ will be :

SSC CHSL 02/08/2023 (1st Shift)

- (a) $\frac{81}{11}$ (b) $\frac{81}{7}$ (c) $\frac{71}{8}$ (d) $\frac{81}{8}$

Q.81. If $(a - b) = 9$ and $(a^3 - b^3) = 4401$, find the value of ab .

SSC CHSL 02/08/2023 (2nd Shift)

- (a) 190 (b) 112 (c) 162 (d) 136

Q.82. If $(a + \frac{1}{a}) = 6$, then what is the value of $\frac{3}{4}(a^2 + \frac{1}{a^2})$?

SSC CHSL 02/08/2023 (2nd Shift)

- (a) 22.5 (b) 34 (c) 25.5 (d) 36

Q.83. If $a = \frac{1}{a - \sqrt{6}}$ ($a > 0$), then the value of $(a + \frac{1}{a})$ is :

SSC CHSL 02/08/2023 (3rd Shift)

- (a) $\sqrt{6}$ (b) $\sqrt{10}$ (c) $\sqrt{15}$ (d) $\sqrt{7}$

Q.84. If $\sqrt{\frac{a}{b}} = \frac{8}{3} + \sqrt{\frac{b}{a}}$ and $(a + b) = 30$, then what is the value of ab ?

SSC CHSL 03/08/2023 (1st Shift)

- (a) 64 (b) 28 (c) 81 (d) 36

Q.85. If $2x + 3y = 9$, and $xy = 3$, what is $8x^3 + 27y^3$?

SSC CHSL 03/08/2023 (1st Shift)

- (a) 235 (b) 244 (c) 243 (d) 234

Q.86. If $a^2 + \frac{1}{a^2} = \frac{7}{3}$, then what is the value of $(a^3 - \frac{1}{a^3})$?

SSC CHSL 03/08/2023 (3rd Shift)

- (a) $\frac{5}{3\sqrt{3}}$ (b) $\frac{10}{3\sqrt{3}}$ (c) $\frac{7}{3\sqrt{3}}$ (d) $\frac{8}{3\sqrt{3}}$

Q.87. If $a + b = 10$ and $a^2 + b^2 = 58$, find the value of ab .

SSC CHSL 03/08/2023 (3rd Shift)

- (a) 24 (b) 27 (c) 30 (d) 21

Q.88. If $(a + b + c) \neq 0$, then $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ is equal to:

SSC CHSL 03/08/2023 (4th Shift)

- (a) $a^3 + b^3 + c^3 - 3abc$ (b) $a^3 - b^3 + c^3 - 3abc$
(c) $a^3 + b^3 + c^3 - 3abc$ (d) $a^3 + b^3 + c^3 + 3abc$

Q.89. If $2a + 3b = 14$ and $2a - 3b = 10$, then find the value of 'ab'.

SSC CHSL 03/08/2023 (4th Shift)

- (a) 5 (b) 3 (c) 4 (d) 2

Q.90. If $(w + \frac{1}{w}) = 6$, then what will be the value of $(w - \frac{1}{w})$?

SSC CHSL 04/08/2023 (1st Shift)

- (a) $4\sqrt{2}$ (b) $\sqrt{2}$ (c) $3\sqrt{2}$ (d) $2\sqrt{2}$

Q.91. If $\sqrt{\frac{a}{b}} = \frac{8}{3} - \sqrt{\frac{b}{a}}$ and $a - b = 10$ then the value of ab is :

SSC CHSL 04/08/2023 (2nd Shift)

- (a) $32\frac{1}{7}$ (b) $32\frac{3}{7}$ (c) $32\frac{4}{7}$ (d) $32\frac{2}{7}$

Q.92. Which of the following is an identity?

SSC CHSL 04/08/2023 (2nd Shift)

- (a) $(a - b)^2 = a^2 - b^2$
(b) $(a + b)^2 = a^2 + b^2 + 2ab$
(c) $(a + b)^2 = a^2 + b^2$
(d) $(a - b)^2 = a^2 - b^2 - 2ab$

Q.93. If $x + \frac{2}{x} = 1$ then the value of $\frac{x^2 + 7x + 2}{x^2 + 13x + 2}$ is:

SSC CHSL 04/08/2023 (2nd Shift)

- (a) $\frac{5}{7}$ (b) $\frac{3}{7}$ (c) $\frac{4}{7}$ (d) $\frac{2}{7}$

Q.94. If $(x + \frac{1}{x}) = 7$ and $x > 1$, what is the value of $(x^3 - \frac{1}{x^3})$?

SSC CHSL 04/08/2023 (3rd Shift)

- (a) $174\sqrt{5}$ (b) $164\sqrt{5}$ (c) $154\sqrt{5}$ (d) $144\sqrt{5}$

Q.95. If $(y - \frac{1}{y}) = 9$, find the value of $(y^3 - \frac{1}{y^3})$

SSC CHSL 04/08/2023 (3rd Shift)

- (a) 766 (b) 756 (c) 702 (d) 729

Q.96. If $2x + 3y = 4$ and $4x^2 + 9y^2 = 64$, then what is the value of $8x^3 + 27y^3$.

SSC CHSL 04/08/2023 (4th Shift)

- (a) 253 (b) 235 (c) 352 (d) 325

Q.97. If $a + b + c = 15$ and $ab + bc + ca = 22$, then find the value of $a^2 + b^2 + c^2$.

SSC CHSL 07/08/2023 (1st Shift)

- (a) 131 (b) 141 (c) 161 (d) 181

Q.98. Simplify the following expression.

$$\{[(x-5)(x-1)] - [(9x-5)(9x-1)]\} \div 16x$$

SSC CHSL 07/08/2023 (1st Shift)

- (a) $2x(5x - 3)$ (b) $-(5x - 3)$
(c) $x(5x - 3)$ (d) $-6x(5x - 3)$

Q.99. If $x + \frac{1}{x} = 2$, then find the value of $x^{1823} + \frac{1}{x^{1929}}$

SSC CHSL 07/08/2023 (1st Shift)

- (a) 2 (b) 1 (c) 0 (d) -1

Q.100. The algebraic expression $2x^2 - 32$ is equal to_____.

SSC CHSL 07/08/2023 (2nd Shift)

- (a) $2(x - 4)^2$ (b) $(x + 4)(x - 4)$
(c) $2(x + 4)^2$ (d) $2(x + 4)(x - 4)$

Q.101. If $(x - \frac{1}{x}) = 0.4$, and $x > 0$, what is the value of $(x^2 + \frac{1}{x^2})$?

SSC CHSL 07/08/2023 (2nd Shift)

- (a) 4.16 (b) 3.32 (c) 1.84 (d) 2.16

Q.102. If $(x + \frac{1}{x}) = 5$, then the value of $(x^5 + \frac{1}{x^5})$ is:

SSC CHSL 07/08/2023 (2nd Shift)

- (a) 2535 (b) 2540 (c) 2515 (d) 2525

Q.103. If $x^2 + y^2 + z^2 = 628$ and $xy + yz + zx = 24$, then find the value of $(x + y + z)$.

SSC CHSL 07/08/2023 (3rd Shift)

- (a) 22 (b) 28 (c) 26 (d) 24

Q.104. if $x + \frac{1}{x} = -1$, then the value of $x^{15} + \frac{1}{x^{15}}$ will be:

SSC CHSL 07/08/2023 (3rd Shift)

- (a) 3 (b) 2 (c) -1 (d) 0

Q.105. Simplify the following expression.

$$\frac{x^3 + y^3 + z^3 - 3xyz}{(x - y)^2 + (y - z)^2 + (z - x)^2}$$

SSC CHSL 07/08/2023 (4th Shift)

- (a) $\frac{1}{3}(x + y + z)$ (b) $(x + y + z)$
(c) $\frac{1}{4}(x + y + z)$ (d) $\frac{1}{2}(x + y + z)$

Q.106. If $(x - \frac{1}{x-2}) = 7$, then the value of $(x - 2)^3 - \frac{1}{(x-2)^3}$ is:

SSC CHSL 07/08/2023 (4th Shift)

- (a) 130 (b) 115 (c) 140 (d) 125

Q.107. If $a + b + c = 16$, $ab + bc + ac = 81$, what is the value of $a^2 + b^2 + c^2$

SSC CHSL 08/08/2023 (1st Shift)

- (a) 94 (b) 98 (c) 89 (d) 87

Q.108. Simplify the following expression $(a + b + c)^2 - (a - b + c)^2 + 4ac$

SSC CHSL 08/08/2023 (1st Shift)

- (a) $4(bc + ac)$ (b) $2(ab + bc + ac)$
(c) $4(ab + bc + ac)$ (d) $4(bc + ab)$

Q.109. If $x^2 - 5\sqrt{5}x + 1 = 0$, and $x > 0$, what is the value of $(x^3 - \frac{1}{x^3})$?

SSC CHSL 08/08/2023 (2nd Shift)

- (a) 1331 (b) 1364 (c) 1296 (d) 1244

Q.110. If $x^4 + \frac{1}{x^4} = 194$, $x > 0$, then find

the value of $x^3 + \frac{1}{x^3} + x + \frac{1}{x}$.

SSC CHSL 08/08/2023 (3rd Shift)

(a) 76 (b) 66 (c) 56 (d) 46

Q.111. If $a + b + c = 5$ and $a^2 + b^2 + c^2 = 15$, then find the value of $a^3 + b^3 + c^3 - 3abc - 27$.

SSC CHSL 08/08/2023 (3rd Shift)

(a) 23 (b) 27 (c) 25 (d) 21

Q.112. If $x^2 + y^2 = 427$ and $xy = 202$, then find the value of $\frac{x+y}{x-y}$.

SSC CHSL 08/08/2023 (3rd Shift)

(a) $\sqrt{\frac{835}{23}}$ (b) $\sqrt{\frac{830}{29}}$

(c) $\sqrt{\frac{831}{23}}$ (d) $\sqrt{\frac{830}{23}}$

Q.113. If $(y - \frac{1}{y}) = -9$, what will be the value of $(y^5 - \frac{1}{y^5})$?

SSC CHSL 08/08/2023 (4th Shift)

(a) -62757 (b) -62748

(c) -62739 (d) -59049

Q.114. The algebraic expression $4x^2 - y^2 + 6y - 9$ is equal to _____.

SSC CHSL 09/08/2023 (1st Shift)

(a) $(2x + y - 3)(2x - y + 3)$

(b) $(2x - y - 3)(2x - y + 3)$

(c) $(2x + y - 3)^2$

(d) $(2x + y + 3)^2$

Q.115. If $x = (\sqrt{6} - 1)^{\frac{1}{3}}$, then the value of

$(x - \frac{1}{x})^3 + 3(x - \frac{1}{x})$ is:

SSC CHSL 09/08/2023 (1st Shift)

(a) $\frac{2\sqrt{6}-6}{5}$ (b) $\frac{4\sqrt{6}-6}{5}$

(c) $\frac{4\sqrt{6}-6}{3}$ (d) $\frac{4\sqrt{3}-6}{5}$

Q.116. If $(a + b + c) = 17$, and $(a^2 + b^2 + c^2) = 101$, find the value of $(a - b)^2 + (b - c)^2 + (c - a)^2$.

SSC CHSL 09/08/2023 (2nd Shift)

(a) 12 (b) 14 (c) 10 (d) 16

Q.117. If $\frac{x^2+1}{x} = 5$, then find the value

of $x^4 + \frac{1}{x^4} - 36$.

SSC CHSL 09/08/2023 (2nd Shift)

(a) 491 (b) 149 (c) 419 (d) 194

Q.118. If $a^2 + b^2 + c^2 = ab + bc + ca$, then

the value of $\frac{11a^4 + 13b^4 + 15c^4}{16a^2b^2 + 19b^2c^2 + 17c^2a^2}$ is:

SSC CHSL 09/08/2023 (3rd Shift)

(a) $1\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) $1\frac{3}{4}$

Q.119. Simplify the expression:

$(c + d)^2 - (c - d)^2$

SSC CHSL 09/08/2023 (3rd Shift)

(a) $4cd$ (b) $(c^2 + d^2)$ (c) $2(c^2 + d^2)$ (d) $2cd$

Q.120. Which of the following expressions is equal to the expression

$\frac{x^2 - 3x + 2}{x^2 - 4}$?

SSC CHSL 09/08/2023 (4th Shift)

(a) $\frac{x+1}{x-2}$ (b) $\frac{x-1}{x+2}$

(c) $\frac{x+1}{x+2}$ (d) $\frac{x-1}{x-2}$

Q.121. If $(l + m + n) = 0$, then what will

be the value of $(\frac{l^2}{mn} + \frac{m^2}{nl} + \frac{n^2}{lm})^2$?

SSC CHSL 10/08/2023 (2nd Shift)

(a) 9 (b) 16 (c) 4 (d) 1

Q.122. If $2x + 5y = 15$ and $xy = 6$,

then the value of $4x^2 + 25y^2$ is _____.

SSC CHSL 10/08/2023 (3rd Shift)

(a) 105 (b) 95 (c) 100 (d) 90

Q.123. If $x^2 - \sqrt{9.76}x + 1 = 0$ and

$x > 1$, the value of $(x^3 - \frac{1}{x^3})$ is:

SSC CHSL 10/08/2023 (3rd Shift)

(a) 21.042 (b) 24.024

(c) 21.024 (d) 24.042

Q.124. If $(5a + \frac{4}{a} - 2) = 13$ and $a > 0$,

what is the value of $(25a^2 + \frac{16}{a^2})$?

SSC CHSL 10/08/2023 (4th Shift)

(a) 158 (b) 157 (c) 185 (d) 175

Q.125. Simplify the given expression.

$\frac{(x^3 - y^3)(x + y)}{x^2 + xy + y^2}$

SSC CHSL 10/08/2023 (4th Shift)

(a) $x - y$ (b) $x^2 - y^2$ (c) $x + y$ (d) $x^2 + y^2$

Q.126. Simplify :-

$\frac{36a^2 - 49b^2}{6a + 7b}$

SSC CHSL 11/08/2023 (1st Shift)

(a) $7b - 6a$ (b) $6a + 7b$

(c) $6a - 7b$ (d) $\frac{1}{6a - 7b}$

Q.127. If $(a + b + c) = 17$, and

$(a^2 + b^2 + c^2) = 115$, find the value of

$(a + b)^2 + (b + c)^2 + (c + a)^2$.

SSC CHSL 11/08/2023 (2nd Shift)

(a) 402 (b) 408 (c) 394 (d) 404

Q.128. Simplify :

$(2a - 3b - c)^2 - (a + 2b + c)^2$

SSC CHSL 11/08/2023 (2nd Shift)

(a) $3a^2 + 5b^2 - 16ab + 2bc - 8ac$

(b) $3a^2 + 5b^2 - 8ab + 2bc - 6ac$

(c) $3a^2 + 5b^2 - 16ab - 6ac$

(d) $3a^2 + 5b^2 - 16ab + 2bc - 6ac$

Q.129. If $x^2 - 3.2x + 1 = 0$ and $x > 1$, the value of $(x^2 - \frac{1}{x^2})$ is :

SSC CHSL 11/08/2023 (3rd Shift)

(a) $16.8\sqrt{0.39}$ (b) $12.8\sqrt{0.39}$

(c) $16.8\sqrt{0.32}$ (d) $12.8\sqrt{0.32}$

Q.130. If $x^2 + y^2 + z^2 = 2(x + z - 1)$, then the value of $x^3 + y^3 + z^3$ is equal to :

SSC CHSL 11/08/2023 (3rd Shift)

(a) 6 (b) 1 (c) 2 (d) 8

Q.131. If $x + 3y = 6$, what is the value of $x^3 + 27y^3 + 54xy$?

SSC CHSL 14/08/2023 (1st Shift)

(a) 264 (b) 258 (c) 216 (d) 220

Q.132. If $p + q = 12$ and $pq = 14$, then find the value of $p^2 - pq + q^2$.

SSC CHSL 14/08/2023 (3rd Shift)

(a) 192 (b) 181 (c) 102 (d) 144

Q.133. Simplify the given expression,

$\frac{(x+3)^3 + (x-3)^3}{x^2 + 27}$

SSC CHSL 14/08/2023 (3rd Shift)

(a) $3x$ (b) x (c) $4x$ (d) $2x$

Q.134. Simplify the expression:

$\frac{1}{8} [\frac{1}{b-1} - \frac{1}{b+1} - \frac{2}{b^2+1} - \frac{4}{b^4+1}]$

SSC CHSL 14/08/2023 (4th Shift)

(a) $\frac{1}{b^8 - 1}$ (b) $\frac{8}{b^8 + 1}$

(c) $\frac{8}{b^8 - 1}$ (d) $\frac{1}{b^8 + 1}$

Q.135. If $(d + e + f) = 14$, $(d^2 + e^2 + f^2) = 96$, then find the value of $(de + ef + fd)$.

SSC CHSL 17/08/2023 (1st Shift)

(a) 75 (b) 25 (c) 50 (d) 100

Q.136. If $g = (2 - \sqrt{3})$, what will be the value of $(g^3 - \frac{1}{g^3})$?

SSC CHSL 17/08/2023 (1st Shift)

(a) $-30\sqrt{3}$ (b) 52 (c) $30\sqrt{3}$ (d) -52

Q.137. Simplify the expression:

$(2x + 13 - y)(2x - 13 - y)$

SSC CHSL 17/08/2023 (1st Shift)

(a) $4x^2 - y^2 - 4xy - 169$

(b) $4x^2 + y^2 + 4xy - 169$

(c) $4x^2 + y^2 - 4xy - 169$

(d) $4x^2 + y^2 - 4xy + 169$

Q.138. Simplify:

$$(3x - 2y)(3x + 2y)(9x^2 + 4y^2)$$

SSC CHSL 17/08/2023 (2nd Shift)

- (a) $81x^4 + 16y^4$ (b) $81y^4 - 16x^4$
(c) $81y^4 + 16x^4$ (d) $81x^4 - 16y^4$

Q.139. Simply the given expression :

$$(5p + 3q)(5p - 3q).$$

SSC CHSL 17/08/2023 (2nd Shift)

- (a) $25p^2 - 9q^2 + 30pq$ (b) $25p^2 - 9q^2$
(c) $25p^2 + 9q^2$ (d) $25p^2 + 9q^2 - 30pq$

Q.140. If $p^2 - 4p - 1 = 0$, then the value of

$$p^2 + 3p + \frac{1}{p^2} - \frac{3}{p} \text{ is :}$$

SSC CHSL 17/08/2023 (2nd Shift)

- (a) 40 (b) 50 (c) 65 (d) 30

Q.141. Simplify the expression:

$$\frac{a+b}{a-b} \div \frac{(a+b)^2}{(a^2-b^2)}$$

SSC CHSL 17/08/2023 (3rd Shift)

- (a) -1 (b) $(a+b)$ (c) 0 (d) 1

Q.142. If $3x - 2y = 3$ and $xy = 9$, then find the value of $3x + 2y$.

SSC CHSL 17/08/2023 (4th Shift)

- (a) 16 (b) 13 (c) 14 (d) 15

Q.143. If $px + qy = 1$ and $qx + py =$

$$\frac{2pq}{p^2 + q^2}, \text{ then } (x^2 + y^2)(p^2 + q^2) \text{ is equal to:}$$

SSC CHSL 17/08/2023 (4th Shift)

- (a) 2 (b) 3 (c) 0 (d) 1

SSC CGL 2023 Tier - 1

Q.144. If $7b - \frac{1}{4b} = 7$, then what is the

$$\text{value of } 16b^2 + \frac{1}{49b^2}?$$

SSC CGL 14/07/2023 (1st shift)

- (a) $\frac{80}{49}$ (b) $\frac{104}{7}$ (c) $\frac{120}{7}$ (d) $\frac{7}{2}$

Q.145. What is the value of $(3x^3 + 5x^2y + 12xy^2 + 7y^3)$, when $x = -4$ and $y = -1$?

SSC CGL 14/07/2023 (1st shift)

- (a) -329 (b) -359 (c) -361 (d) -327

Q.146. If $(a + b + c) = 16$, and $(a^2 + b^2 + c^2) = 90$, find the value of $(ab + bc + ca)$.

SSC CGL 14/07/2023 (1st shift)

- (a) 84 (b) 83 (c) 82 (d) 81

Q.147. What is the value of $a^3 + b^3 + c^3$, if $(a + b + c) = 0$?

SSC CGL 14/07/2023 (2nd shift)

- (a) $a^2 + b^2 + c^2 - 3abc$
(b) 0
(c) $3abc$
(d) $a^2 + b^2 + c^2 - ab - bc - ca$

Q.148. If $x + \frac{1}{x} = 7$, then the value of

$$x^6 + \frac{1}{x^6} \text{ is:}$$

SSC CGL 14/07/2023 (3rd shift)

- (a) 113682 (b) 103682
(c) 103882 (d) 103862

Q.149. If $a = 17$, $b = 13$, then find the value of the expression $(a^3 - b^3 - 3a^2b + 3ab^2)$.

SSC CGL 14/07/2023 (4th shift)

- (a) -64 (b) -27000 (c) 27000 (d) 64

Q.150. Simplify, $\frac{x^4 - 2x^2 + 1}{x^2 - 2x + 1}$.

SSC CGL 17/07/2023 (2nd shift)

- (a) $x^2 - 2x + 1$ (b) $x^2 + 2x + 2$
(c) $x^2 + 2x + 1$ (d) $x^2 + x + 1$

Q.151. If $(x - \frac{1}{x}) = \sqrt{6}$, and $x > 1$, what is

$$\text{the value of } (x^8 - \frac{1}{x^8})?$$

SSC CGL 17/07/2023 (2nd shift)

- (a) $1024\sqrt{15}$ (b) $992\sqrt{15}$
(c) $998\sqrt{15}$ (d) $1012\sqrt{15}$

Q.152. What is the value of the given expression?

$$(a + b + c)^2 - a^2 - b^2 - c^2$$

SSC CGL 17/07/2023 (3rd shift)

- (a) $2abc$ (b) $2(ab + bc + ca)$
(c) $2(a + b + c)$ (d) $2ab + bc - 2ca$

Q.153. What is the value of $\frac{X}{Y}$ if

$$\frac{X - 5Y}{X + 5Y} = \frac{7}{13}.$$

SSC CGL 17/07/2023 (4th shift)

- (a) $\frac{23}{7}$ (b) $\frac{24}{9}$ (c) $\frac{50}{3}$ (d) $\frac{100}{7}$

Q.154. If $(x + \frac{1}{x}) = 10$, what is the value

$$\text{of } (x^4 + \frac{1}{x^4})?$$

SSC CGL 17/07/2023 (4th shift)

- (a) 9604 (b) 9602 (c) 9600 (d) 9606

Q.155. The cube of the difference between two given natural numbers is 1728, while the product of these two given numbers is 108. Find the positive difference between the cubes of these two given numbers.

SSC CGL 18/07/2023 (1st shift)

- (a) 4104 (b) 5616 (c) 2160 (d) 5626

Q.156. If $(a + b + c) = 19$, and $(a^2 + b^2 + c^2) = 155$, find the value of $(a - b)^2 +$

$$(b - c)^2 + (c - a)^2.$$

SSC CGL 18/07/2023 (2nd shift)

- (a) 104 (b) 108 (c) 100 (d) 98

Q.157. If $a^2 + b^2 + c^2 = ab + bc + ac$, then the value of

$$\frac{11a^4 + 13b^4 + 17c^4}{17a^2b^2 + 9b^2c^2 + 15c^2a^2} \text{ is:}$$

SSC CGL 18/07/2023 (3rd shift)

- (a) 1 (b) 2 (c) 11 (d) 4

Q.158. The square of the difference between two given natural numbers is 324, while the product of these two given numbers is 144. Find the positive difference between the squares of these two given numbers.

SSC CGL 18/07/2023 (3rd shift)

- (a) 630 (b) 540 (c) 450 (d) 360

Q.159. If $p = \frac{\sqrt{2} + 1}{\sqrt{2} - 1}$ and $q = \frac{\sqrt{2} - 1}{\sqrt{2} + 1}$

$$\text{then find the value of } \frac{p^2}{q} + \frac{q^2}{p}$$

SSC CGL 18/07/2023 (4th shift)

- (a) 200 (b) 196 (c) 198 (d) 188

Q.160. If $(x + \frac{1}{x}) = 2\sqrt{2}$, and $x > 1$, what

$$\text{is the value of } (x^6 - \frac{1}{x^6})?$$

SSC CGL 19/07/2023 (1st shift)

- (a) $140\sqrt{2}$ (b) $116\sqrt{2}$ (c) $144\sqrt{2}$ (d) $128\sqrt{2}$

Q.161. If $(a + b + c) = 12$, and

$$(a^2 + b^2 + c^2) = 50, \text{ find the value of}$$

$$(a^3 + b^3 + c^3 - 3abc).$$

SSC CGL 19/07/2023 (2nd shift)

- (a) 36 (b) 24 (c) 42 (d) 48

Q.162. If $(3y - \frac{3}{y}) = 5$, find the value of

$$(y^2 + \frac{1}{y^2}).$$

SSC CGL 19/07/2023 (2nd shift)

- (a) $\frac{47}{9}$ (b) $\frac{49}{9}$ (c) $\frac{41}{9}$ (d) $\frac{43}{9}$

Q.163. If $x^2 - 7x + 1 = 0$, and $0 < x < 1$,

$$\text{what is the value of } x^2 - \frac{1}{x^2}?$$

SSC CGL 19/07/2023 (3rd shift)

- (a) $21\sqrt{5}$ (b) $-21\sqrt{5}$ (c) $28\sqrt{5}$ (d) $-28\sqrt{5}$

Q.164. If $(x + \frac{1}{x}) = 2$, then $x^7 + \frac{1}{x^{117}} =$

$$\underline{\hspace{2cm}}.$$

SSC CGL 19/07/2023 (4th shift)

- (a) 1 (b) 2 (c) 4 (d) 3

Q.165. If $(a + b + c) = 20$, and $a^2 + b^2 + c^2 = 152$, find the value of $a^3 + b^3 + c^3 - 3abc$.

SSC CGL 19/07/2023 (4th shift)

- (a) 560 (b) 640 (c) 480 (d) 720

Q.166. Simplify the expression $\frac{l^2 - m^2}{(l + m)^2}$,

$$\text{provided } (l + m) \neq 0.$$

SSC CGL 20/07/2023 (1st shift)

(a) $\frac{(l+m)}{(l-m)}$ (b) $\frac{(l-m)}{(l+m)}$ (c) 0 (d) 1

Q.167. If $x > 0$ and $x^4 + \frac{1}{x^4} = 142$, what is

the value of $x^7 + \frac{1}{x^7}$?

SSC CGL 20/07/2023 (2nd shift)

(a) $1561\sqrt{14}$ (b) $1563\sqrt{14}$
(c) $1560\sqrt{14}$ (d) $1562\sqrt{14}$

Q.168. If $a + b + c = 7$, $ab + bc + ca = 11$ and $abc = -1$, then $a^3 + b^3 + c^3$ is equal to:

SSC CGL 20/07/2023 (3rd shift)

(a) 101 (b) 107 (c) 109 (d) 111

Q.169. If $(x - \frac{1}{x}) = 6$, and $x > 0$, find the

value of $(x^2 - \frac{1}{x^2})$.

SSC CGL 20/07/2023 (4th shift)

(a) $12\sqrt{10}$ (b) $18\sqrt{10}$ (c) $24\sqrt{2}$ (d) $24\sqrt{10}$

Q.170. If $(x + \frac{1}{x}) = 3\sqrt{2}$, and $x > 1$, what

is the value of $(x^8 - \frac{1}{x^8})$?

SSC CGL 20/07/2023 (4th shift)

(a) $24364\sqrt{7}$ (b) $24384\sqrt{11}$
(c) $24384\sqrt{7}$ (d) $12192\sqrt{7}$

Q.171. What is the value of $(\frac{1}{a} - \frac{1}{b} - \frac{1}{c})$

if, $\frac{2a-5}{a} - \frac{4b-5}{b} + \frac{6c+5}{c} = 0$?

SSC CGL 21/07/2023 (2nd shift)

(a) $\frac{4}{5}$ (b) $-\frac{8}{5}$ (c) $\frac{2}{5}$ (d) $-\frac{12}{5}$

Q.172. Simplify the following expression.

$[(1+P)(1+P^2)(1+P^4)(1+P^8)(1+P^{16})(1-P)-1]$

SSC CGL 21/07/2023 (3rd shift)

(a) $-P^{32}$ (b) P^{32} (c) $(1+P^{32})$ (d) $(1-P^{32})$

Q.173. If x and y are positive numbers such that $x - y = 5$ and $xy = 150$, then the value of $(x + y)$ is:

SSC CGL 21/07/2023 (3rd shift)

(a) 45 (b) 25 (c) 35 (d) 15

Q.174. If $(x^2 - \frac{1}{x^2}) = 4\sqrt{6}$, and $x > 1$ what

is the value of $(x^3 - \frac{1}{x^3})$

SSC CGL 21/07/2023 (3rd shift)

(a) $20\sqrt{2}$ (b) $24\sqrt{2}$ (c) $18\sqrt{2}$ (d) $22\sqrt{2}$

Q.175. If $(a - b) = 1$ then what is the value of $(a^3 - b^3)$?

SSC CGL 21/07/2023 (4th shift)

(a) $a^2 + ab + b^2$ (b) $(a + b)^2 + 3ab$
(c) $a^2 - ab + b^2$ (d) $a^2 + 2ab + b^2$

Q.176. If $(x^2 + \frac{1}{x^2}) = 18$, and $x > 0$, what

is the value of $(x^3 + \frac{1}{x^3})$?

SSC CGL 21/07/2023 (4th shift)

(a) $36\sqrt{5}$ (b) $40\sqrt{5}$ (c) $46\sqrt{5}$ (d) $34\sqrt{5}$

Q.177. The value of $\frac{p^2 - (q-r)^2}{(p+r)^2 - q^2} +$

$\frac{q^2 - (p-r)^2}{(p+q)^2 - r^2} + \frac{r^2 - (p-q)^2}{(q+r)^2 - p^2}$

SSC CGL 24/07/2023 (1st shift)

(a) 1 (b) 2 (c) 0 (d) 3

Q.178. The difference between the cubes of two given natural numbers is 6272, while the positive difference between the two given numbers is 8. What is the product of the two given numbers?

SSC CGL 24/07/2023 (3rd shift)

(a) 160 (b) 240 (c) 200 (d) 320

Q.179. If $\frac{a}{b} = \frac{7}{6}$, then find the value of

the expression $\frac{6a+13b}{6a-13b}$.

SSC CGL 24/07/2023 (3rd shift)

(a) $-\frac{3}{10}$ (b) $\frac{10}{3}$ (c) $-\frac{10}{3}$ (d) $\frac{3}{10}$

Q.180. Simplify $\frac{1}{2+2p} + \frac{1}{2+2q} +$

$\frac{1}{2+2r}$, where $p = \frac{x}{y+z}$, $q = \frac{y}{z+x}$ and

$r = \frac{z}{x+y}$.

SSC CGL 24/07/2023 (4th shift)

(a) 1 (b) $x + y + z$ (c) 2 (d) $\frac{1}{2}$

Q.181. If $(y - \frac{1}{y}) = 4$, find the value of

$(y^6 + \frac{1}{y^6})$.

SSC CGL 25/07/2023 (1st shift)

(a) 5774 (b) 4096 (c) 5776 (d) 5778

Q.182. If $x^2 - 15x + 1 = 0$, what is the value of $x^4 - 223x^2 + 6$?

SSC CGL 25/07/2023 (1st shift)

(a) 9 (b) 5 (c) 6 (d) 0

Q.183. If $p + q + r = pqr = \frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1$, then find $p^3 + q^3 + r^3$.

SSC CGL 25/07/2023 (2nd shift)

(a) 1 (b) -1 (c) 5 (d) -5

Q.184. Simplify $\frac{256x^4 - 16y^4}{(80x^2 - 20y^2)(16x^2 + 4y^2)}$

SSC CGL 25/07/2023 (2nd shift)

(a) 5 (b) $\frac{1}{20}$ (c) $\frac{1}{5}$ (d) $\frac{2}{5}$

Q.185. If $7a - \frac{7}{a} + 4 = 0$, then find

$a^3 - \frac{1}{a^3} - 1$.

SSC CGL 25/07/2023 (2nd shift)

(a) $\frac{-995}{343}$ (b) $\frac{-875}{248}$
(c) $\frac{-694}{315}$ (d) $\frac{-765}{262}$

Q.186. Simplify the following expression.

$\frac{x^2 - 2x - 63}{x^2 + 14x + 49}$

SSC CGL 25/07/2023 (3rd shift)

(a) $\frac{x+9}{x+7}$ (b) $\frac{x-7}{x+7}$
(c) $\frac{x+7}{x-7}$ (d) $\frac{x-9}{x+7}$

Q.187. Simplify the expression

$\frac{s^2 + t^2 + 2st - u^2}{s^2 - t^2 - 2tu - u^2}$, provided $(s + t + u) \neq 0$.

SSC CGL 25/07/2023 (4th shift)

(a) $\frac{s+t-u}{s-t-u}$ (b) $\frac{s+t+u}{s-t+u}$
(c) $\frac{s-t-u}{s+t-u}$ (d) $\frac{s-t+u}{s+t+u}$

Q.188. If $x^2 - 8x - 1 = 0$, what is the value of $x^2 + \frac{1}{x^2}$?

SSC CGL 26/07/2023 (1st shift)

(a) 68 (b) 62 (c) 64 (d) 66

Q.189. If $(x^2 + \frac{1}{x^2}) = 6$, and $0 < x < 1$,

what is the value of $x^4 - \frac{1}{x^4}$?

SSC CGL 26/07/2023 (1st shift)

(a) $24\sqrt{2}$ (b) $-24\sqrt{2}$
(c) $-12\sqrt{10}$ (d) $12\sqrt{10}$

Q.190. If $x = \frac{1}{x-3}$, ($x > 0$), then the value of $x + \frac{1}{x}$ is:

SSC CGL 26/07/2023 (1st shift)

(a) $\sqrt{11}$ (b) $\sqrt{17}$ (c) $\sqrt{15}$ (d) $\sqrt{13}$

Q.191. If $x + \frac{1}{x} = 1$, then the value of

$\frac{x^2 + 7x + 1}{x^2 + 11x + 1} = ?$

SSC CGL 26/07/2023 (2nd shift)

(a) $\frac{3}{4}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$

Q.192. If $(x + \frac{1}{x}) = \sqrt{6}$, and $x > 1$,

what is the value of $(x^8 - \frac{1}{x^8})$?

SSC CGL 26/07/2023 (3rd shift)

(a) $120\sqrt{3}$ (b) $128\sqrt{3}$
(c) $112\sqrt{3}$ (d) $108\sqrt{3}$

Q.193. If $b = 5$, determine the value of an expression $(\frac{5}{b} + 5b)(\frac{25}{b^2} - 25 + 25b^2)$ using an identity.

SSC CGL 26/07/2023 (3rd shift)
(a) 25726 (b) 15626 (c) 15438 (d) 25636

Q.194. If $(x - \frac{1}{x}) = 10$, what is the value of $(x^4 + \frac{1}{x^4})$?

SSC CGL 26/07/2023 (3rd shift)
(a) 10404 (b) 10402 (c) 10406 (d) 10400

Q.195. Expand and simplify the algebraic expression: $(x - 5)^2 + (x + 3)^2 + 4x$
SSC CGL 26/07/2023 (4th shift)
(a) $2(x^2 - 17)$ (b) $2(x^2 + 17)$
(c) $(x^2 + 17)$ (d) $2(x^2 - 5x + 17)$

Q.196. If $2x + 3y = 16$ and $xy = 9$, then find $8x^3 + 27y^3$.
SSC CGL 27/07/2023 (1st shift)
(a) 1980 (b) 2980 (c) 1504 (d) 2189

Q.197. If $x + y = 25$ and $xy = 20$, then find the value of $x^3 + y^3$.
SSC CGL 27/07/2023 (1st shift)
(a) 13152 (b) 13125 (c) 14125 (d) 14152

Q.198. If $x^2 - \frac{1}{x^2} = 4\sqrt{2}$ what is the value of $x^4 - \frac{1}{x^4}$?

SSC CGL 27/07/2023 (2nd shift)
(a) $16\sqrt{2}$ (b) $8\sqrt{2}$ (c) $24\sqrt{2}$ (d) $32\sqrt{2}$

SSC Selection Post (Phase - XI)

Q.199. If $(x^2 + \frac{1}{x^2}) = 18$, and $x > 1$, what is the value of $(x^3 - \frac{1}{x^3})$?

Graduate Level 27/06/2023 (Shift - 4)
(a) 52 (b) 76 (c) 80 (d) 64

Q.200. What is the solution to the following system of linear equations?
 $3x - 9y + 4z = 5$
 $2x + 7y + z = 12$
 $3x - z = 0$

Graduate Level 28/06/2023 (Shift - 3)

(a) $x = \frac{143}{150}$, $y = \frac{31}{37}$, $z = \frac{143}{50}$

(b) $x = \frac{143}{150}$, $y = \frac{31}{30}$, $z = \frac{143}{50}$

(c) $x = \frac{137}{150}$, $y = \frac{31}{30}$, $z = \frac{143}{50}$

(d) $x = \frac{143}{150}$, $y = \frac{31}{30}$, $z = \frac{143}{51}$

Q.201. If $2a + b = 10$ and $2ab = 9$, then one of the values of $2a - b$ is:
Graduate Level 28/06/2023 (Shift - 3)
(a) 8 (b) 4 (c) 6 (d) 10

Q.202. If $x > 1$, and $x^2 + \frac{1}{x^2} = 2\sqrt{5}$, what is the value of $x^4 - \frac{1}{x^4}$?

Graduate Level 28/06/2023 (Shift - 3)
(a) $4\sqrt{5}$ (b) $4\sqrt{30}$ (c) $8\sqrt{5}$ (d) $8\sqrt{6}$

Q.203. If $x > 1$, and $x^4 + \frac{1}{x^4} = 79$, what is the value of $x - \frac{1}{x}$?

Graduate Level 30/06/2023 (Shift - 1)
(a) $2\sqrt{2}$ (b) $\sqrt{7}$ (c) $\sqrt{11}$ (d) $\sqrt{10}$

Q.204. If $x + y = \sqrt{13}$ and $x - y = \sqrt{11}$, then the value of $xy(x^2 + y^2)$ is:
Graduate Level 30/06/2023 (Shift - 1)
(a) 10 (b) 6 (c) $\sqrt{6}$ (d) $\sqrt{10}$

Q.205. If $ab + bc + ca = 0$, then find the value of $\frac{1}{a^2 - bc} + \frac{1}{b^2 - ca} + \frac{1}{c^2 - ab}$.
Graduate Level 30/06/2023 (Shift - 1)
(a) 0 (b) $\frac{1}{abc}$ (c) 1 (d) ab

Q.206. A varies directly with B. When $A = 52$, $B = 91$. Find the value of A when $B = 126$.
Higher Secondary 27/06/2023 (Shift - 3)
(a) 76 (b) 84 (c) 72 (d) 70

Q.207. For what value of q does the system of equations $38x + qy + 171 = 0$ and $46x + 414y + 207 = 0$ have infinite number of solutions?
Higher Secondary 28/06/2023 (Shift - 2)
(a) 380 (b) 345 (c) 342 (d) 350

Q.208. If $x = 14$, $y = 15$ and $z = 17$, then the value of $x^3 + y^3 + z^3 - 3xyz$ is:
Higher Secondary 28/06/2023 (Shift - 2)
(a) 333 (b) 322 (c) 312 (d) 222

Q.209. For what value of p does the system of equations $18x + 36y + 45 = 0$ and $px - 54y + 67 = 0$ have no solution?
Higher Secondary 30/06/2023 (Shift - 2)
(a) 54 (b) -27 (c) -36 (d) 27

Q.210. Solve $(\frac{9}{8} + \frac{8}{9})^2 - (\frac{9}{8} - \frac{8}{9})^2$.
Higher Secondary 30/06/2023 (Shift - 2)
(a) 4 (b) 1 (c) 8 (d) 2

SSC CHSL 2022 Tier - 1

Q.211. What is the value of $\frac{1+x^2}{1-x} \div \frac{1-x^4}{x-1} \times \frac{x(1-x)}{(1+x)}$?

SSC CHSL 09/03/2023 (1st Shift)

(a) $\frac{-x}{1+2x^2}$ (b) $\frac{-x}{1+x}$

(c) $\frac{-x}{(1+x)^2}$ (d) $\frac{-x}{1+x^2}$

Q.212. If $K + \frac{1}{K} - 2 = 0$ and $K > 0$, then

what is the value of $K^{17} + \frac{1}{K^{12}}$?

SSC CHSL 09/03/2023 (2nd Shift)
(a) 2 (b) 1 (c) 0 (d) 12

Q.213. What is the value of $(a + b + 2c)$

$(a^2 + b^2 + 4c^2 - ab - 2bc - 2ca)$?
SSC CHSL 09/03/2023 (3rd Shift)

(a) $a^3 - b^3 - c^3 - 6abc$
(b) $a^3 + b^3 + 4c^3 - 6abc$
(c) $a^3 - b^3 - 8c^3 - 3abc$
(d) $a^3 + b^3 + 8c^3 - 6abc$

Q.214. If $a^4 + \frac{1}{a^4} = 194$, then what is the value of $\frac{a^6 + 1}{a^3}$?

SSC CHSL 09/03/2023 (4th Shift)
(a) 46 (b) 50 (c) 52 (d) 48

Q.215. What is the value of $(a - b)(b + a)(a^2 + b^2)(2b^4 + 2a^4)$?
SSC CHSL 10/03/2023 (1st Shift)
(a) $a^4 - b^4$ (b) $a^8 - b^8$
(c) $2(a^8 - b^8)$ (d) $\frac{(a^8 - b^8)}{2}$

Q.216. If $\frac{(15)^3 + (4)^3}{(15^2 + 4^2 - k)} = 19$,

then what is the value of K?
SSC CHSL 10/03/2023 (3rd Shift)
(a) -60 (b) 120 (c) 60 (d) -120

Q.217. If $x^2 + \frac{1}{x^2} = 38$, then what is the value of $x - \frac{1}{x}$?

SSC CHSL 10/03/2023 (3rd Shift)
(a) 6 (b) 7 (c) 8 (d) 10

Q.218. Find the value of $p^3 - r^3$.
if $p + r = 21$, and $pr = 38$.
SSC CHSL 10/03/2023 (4th Shift)
(a) 6538 (b) 6113 (c) 6102 (d) 6851

Q.219. What is the value of $(a + b + c) \{(a - b)^2 + (b - c)^2 + (c - a)^2\}$?
SSC CHSL 13/03/2023 (1st Shift)
(a) $2a^3 + 2b^3 + 2c^3$
(b) $2a^3 + 2b^3 + 2c^3 - 6abc$
(c) $3abc$
(d) $6abc$

Q.220. What is the value of $(a^2 + b^2 + 4c^2 - ab - 2bc - 2ca)(a + b + 2c)$?
SSC CHSL 13/03/2023 (1st Shift)
(a) $a^3 + b^3 + 8c^3 - 8abc$
(b) $a^3 + b^3 + 6c^3 - 4abc$
(c) $4a^3 + b^3 + 8c^3 - 9abc$
(d) $a^3 + b^3 + 8c^3 - 6abc$

Q.221. If $x + \frac{1}{x} = -2$, then what is the value of $x^7 + x^{-7} + x^2 + x^{-2}$ ($x < 0$)
SSC CHSL 13/03/2023 (1st Shift)
(a) 4 (b) 2 (c) 1 (d) 0

Q.222. What is the value of $\frac{4x^2 + 9y^2 + 12xy}{144}$?
SSC CHSL 13/03/2023 (2nd Shift)

- (a) $(\frac{x}{6} + \frac{y}{4})^2$ (b) $(\frac{x}{3} + y)^2$
(c) $(\frac{x}{3} + \frac{y}{4})^2$ (d) $(\frac{x}{4} + \frac{y}{6})^2$

Q.223. What is the expansion of $(a - 4)^3$?
SSC CHSL 13/03/2023 (3rd Shift)
(a) $a^3 - 12a^2 + 48a + 64$
(b) $a^3 + 12a^2 + 48a + 64$
(c) $a^3 - 12a^2 + 48a - 64$
(d) $a^3 + 12a^2 + 48a - 64$

Q.224. If $x - y = 8$ and $8xy = 8$, then what is the value of $2x^3 - 2y^3$?
SSC CHSL 13/03/2023 (4th Shift)
(a) 536 (b) 1092 (c) 1072 (d) 1608

Q.225. If $a^3 = b^3 + 988$ and $a - b = 4$, then what is the value of $2a^2 + 2ab + 2b^2$?
SSC CHSL 14/03/2023 (1st Shift)
(a) 516 (b) 494 (c) 500 (d) 247

Q.226. If $a^3 + 3a^2 + 3a = 63$, then the value of $a^2 + 2a$ is:
SSC CHSL 14/03/2023 (2nd Shift)
(a) 22 (b) 19 (c) 15 (d) 8

Q.227. If $a + b + c = 6$ and $a^2 + b^2 + c^2 = 40$. then what is the value of $a^3 + b^3 + c^3 - 3abc$?
SSC CHSL 14/03/2023 (3rd Shift)
(a) 212 (b) 252 (c) 232 (d) 206

Q.228. If $9a + 9b + 9c = 81$ and $4ab + 4bc + 4ca = 160$, then what is the value of $6a^2 + 6b^2 + 6c^2$?
SSC CHSL 14/03/2023 (4th Shift)
(a) 1 (b) 3 (c) 4 (d) 6

Q.229. If $N + \frac{1}{N} = \sqrt{3}$, then find the value of $N^6 + \frac{1}{N^6} + 11$.
SSC CHSL 15/03/2023 (1st Shift)

- (a) - 11 (b) 9 (c) - 2 (d) 11

Q.230. What is the value of $\frac{(a+b)^2 - (a-b)^2}{ab}$?
SSC CHSL 15/03/2023 (1st Shift)
(a) 1 (b) 4 (c) 8 (d) 2

Q.231. What is the value of $(a - b)^3 - a^3 + b^3$?

SSC CHSL 15/03/2023 (2nd Shift)

- (a) $3ab(a - b)$ (b) $-3ab(a + b)$
(c) $3ab(a + b)$ (d) $-3ab(a - b)$

Q.232. Compute the expression $(299)^2$ using algebraic identities.

SSC CHSL 15/03/2023 (2nd Shift)

- (a) 89301 (b) 89402 (c) 88401 (d) 89401

Q.233. What is the value of $(k - \frac{1}{k})(k^2 +$

$$\frac{1}{k^2})(k^4 + \frac{1}{k^4})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})$$

SSC CHSL 15/03/2023 (2nd Shift)

- (a) $k^{64} - \frac{1}{k^{64}}$ (b) $\frac{k^{32} - \frac{1}{k^{32}}}{k - \frac{1}{k}}$
(c) $(k^{32} - \frac{1}{k^{32}})$ (d) $\frac{k^{32} - \frac{1}{k^{32}}}{k + \frac{1}{k}}$

Q.234. If $a^4 + \frac{1}{a^4} = 194$, then what is the value of $a^3 + \frac{1}{a^3}$?
SSC CHSL 15/03/2023 (3rd Shift)

- (a) 50 (b) 52 (c) 48 (d) 44

Q.235. If $x + \frac{1}{x} = \frac{K}{2}$, then what is the value of $\frac{x^8 + 1}{x^4}$?
SSC CHSL 16/03/2023 (1st Shift)

- (a) $\frac{K^4 - 16K^2 + 32}{16}$ (b) $\frac{K^4 - 8K^2 + 36}{32}$
(c) $\frac{K^4 - 8K^2 - 32}{16}$ (d) $\frac{K^4 - 8K^2 + 32}{16}$

Q.236. If $(p - q) = 6$, $(r - q) = 5$ and $(r - p) = 3$, then find the value of $(p^3 + q^3 + r^3 - 3pqr)/(p + q + r)$.

SSC CHSL 16/03/2023 (2nd Shift)

- (a) 35 (b) 45 (c) 30 (d) 40

Q.237. If $A + \frac{1}{A} = -1$, then find the value of $\frac{A^6 + A^3 - 1}{A^9 + A^3 - 1}$?
SSC CHSL 16/03/2023 (3rd Shift)

- (a) 0 (b) - 1 (c) 2 (d) 1

Q.238. If $(x + 1)^2 + (x + 2)^2 = 16$, then what is the value of $4x^2 + 12x + 40$?
SSC CHSL 16/03/2023 (3rd Shift)

- (a) 52 (b) 62 (c) 56 (d) 74

Q.239. What is the value of $a^2 + 4b^2 + 9c^2 - 4ab + 12bc - 6ca$?
SSC CHSL 16/03/2023 (4th Shift)

- (a) $(a + 2b - 3c)^2$ (b) $(2a - b - 3c)^2$
(c) $(2b - a - 3c)^2$ (d) $(a - 2b - 3c)^2$

Q.240. If $a + b + c = 1$, $ab + bc + ca = -1$

and $abc = -1$, then what is the value

$$a^3 + b^3 + c^3 ?$$

SSC CHSL 17/03/2023 (3rd Shift)

- (a) 3 (b) 5 (c) 2 (d) 1

Q.241. What is the value of

$$(a + b)^3 - a^3 - b^3 ?$$

SSC CHSL 17/03/2023 (3rd Shift)

- (a) $3ab(a - b)$ (b) $-3ab(a - b)$
(c) $3ab(a + b)$ (d) $-3ab(a + b)$

Q.242. If $\frac{1}{(x - 2)} + x = 8$, then what is the value of $\frac{1}{(x - 2)^2} + (x - 2)^2$?
SSC CHSL 17/03/2023 (4th Shift)

- (a) 38 (b) 36 (c) 40 (d) 34

Q.243. What is the value of $\frac{1 + X}{1 - X^2} \div$

$$\frac{1 + X}{1 - X^4} - \frac{1 - X^4}{1 - X} \times \frac{1 + X}{1 - X^2} ?$$

SSC CHSL 20/03/2023 (3rd Shift)

- (a) $\frac{2X(1 + X^2)}{(1 - X)}$ (b) $\frac{2X(1 + X^2)}{(X - 1)}$
(c) $(1 - X^2)$ (d) $(1 + X^2)$

Q.244. If $x + y + z = 0$, then what is the

$$\text{value of } \frac{x}{(yz)^2} + \frac{y}{(xz)^2} + \frac{z}{(xy)^2} ?$$

SSC CHSL 21/03/2023 (1st Shift)

- (a) $\frac{xyz}{3}$ (b) $\frac{3}{xyz}$ (c) xyz (d) $\frac{1}{xyz}$

Q.245. If $\frac{K}{K^2 - 2K + 1} = \frac{1}{6}$, then what

is the value of $\frac{1}{K^3} + K^3$?
SSC CHSL 21/03/2023 (2nd Shift)

- (a) 472 (b) 492 (c) 512 (d) 488

Q.246. If $(x - \frac{1}{x} = 4)$, then what is the value of $(x^6 + \frac{1}{x^6})$?
SSC CHSL 21/03/2023 (4th Shift)

- (a) 4689 (b) 4786 (c) 5832 (d) 5778

SSC CGL 2022 Tier - 2

Q.247. For what value of m will the system of equations $18x - 72y + 13 = 0$ and $7x - my - 17 = 0$ have no solution?

SSC CGL Tier II (06/03/2023)

- (a) 28 (b) 24 (c) 9 (d) 12

Q.248. If $2^x + 3^y = 17$ and $2^{x+2} - 3^{y+1} = 5$, then:

SSC CGL Tier II (07/03/2023)

- (a) $x = 1, y = 3$ (b) $x = 3, y = 3$
(c) $x = 3, y = 2$ (d) $x = 1, y = 2$

SSC CGL 2022 Tier - 1

Q.249. If $x^2 - 5x + 1 = 0$, then the value of

$$\frac{x^6 + x^4 + x^2 + 1}{5x^3}.$$

SSC CGL 01/12/2022 (1st Shift)

(a) 30 (b) 25 (c) 23 (d) 28

Q.250. x is a negative number such that $k + k^{-1} = -2$, then what is the value of

$$\frac{k^2 + 4k - 2}{k^2 + k - 5}?$$

SSC CGL 01/12/2022 (2nd Shift)

(a) 7 (b) 1 (c) -7 (d) -1

Q.251. If $(x + \frac{1}{x}) = 5\sqrt{2}$, then what is the

value of $(x^4 + x^{-4})$?

SSC CGL 01/12/2022 (2nd Shift)

(a) 2542 (b) 2650 (c) 2452 (d) 2302

Q.252. If $a + b = 11$ and $ab = 35$, then what is the value of $(a^4 + b^4)$?

SSC CGL 01/12/2022 (3rd Shift)

(a) 151 (b) 261 (c) 124 (d) 102

Q.253. If $x + y = 36$, then find $(x - 27)^3 + (y - 9)^3$.

SSC CGL 01/12/2022 (3rd Shift)

(a) 1 (b) 81 (c) 2y (d) 0

Q.254. If $\frac{a^2 + b^2 + c^2 - 1024}{ab - bc - ca} = -2$ and

$a + b = 5c$, where $c > 0$, then the value of c is _____.

SSC CGL 01/12/2022 (4th Shift)

(a) 8 (b) 4 (c) 12 (d) 5

Q.255. If $x = \frac{\sqrt{5} - 2}{\sqrt{5} + 2}$, then the value of x^2

+ x^{-2} is:

SSC CGL 02/12/2022 (1st Shift)

(a) 350 (b) 345 (c) 284 (d) 322

Q.256. If $x > 0$, and $x^4 + \frac{1}{x^4} = 2207$, what

is the value of $x^7 + \frac{1}{x^7}$?

SSC CGL 02/12/2022 (1st Shift)

(a) 710649 (b) 710647

(c) 710654 (d) 710661

Q.257. Simplify $\frac{x^2 + 2x + y^2}{x^3 - 5x^2}$ if $x + \frac{y^2}{x} = 5$

SSC CGL 02/12/2022 (2nd Shift)

(a) $\frac{5}{y^2}$ (b) $\frac{7}{y^2}$ (c) $-\frac{5}{y^2}$ (d) $-\frac{7}{y^2}$

Q.258. If $x + \frac{1}{x} = 8$, then find the value of

$$\frac{5}{x^2 - 8x + 2}?$$

SSC CGL 02/12/2022 (4th Shift)

(a) 3 (b) 4 (c) 0 (d) 5

Q.259. If $(4a + \frac{5}{a} + 5) = 14$, what is the

value of $(16a^2 + \frac{25}{a^2})$?

SSC CGL 03/12/2022 (1st Shift)

(a) 25 (b) 36 (c) 41 (d) 40

Q.260. If $x - \frac{1}{x} = 13$ then, what will be

the value of $x^4 + \frac{1}{x^4}$?

SSC CGL 03/12/2022 (1st Shift)

(a) 28561 (b) 29243 (c) 27887 (d) 29239

Q.261. The factors of $x^4 + x^2 + 25$ are:

SSC CGL 03/12/2022 (1st Shift)

(a) $(x^2 + 3x - 5)(x^2 - 3x + 5)$

(b) $(x^2 + 3x + 5)(x^2 - 3x + 5)$

(c) $(x^2 - 3x + 5)(x^2 - 3x + 5)$

(d) $(x^2 + 3x + 5)(x^2 + 3x + 5)$

Q.262. If $x = 3 + 2\sqrt{2}$, $x > 0$, then the value

of $\sqrt{x} - \frac{1}{\sqrt{x}}$ is:

SSC CGL 03/12/2022 (2nd Shift)

(a) 1 (b) $\sqrt{2}$ (c) 2 (d) $2\sqrt{2}$

Q.263. If $x + \frac{1}{x} = -2$, then what is the

value of $x^{17} + x^{-17} + x^{12} + x^{-12}$? ($x < 0$)

SSC CGL 03/12/2022 (2nd Shift)

(a) -2 (b) -1 (c) 1 (d) 0

Q.264. $(mx + n)$ is a factor of:

SSC CGL 03/12/2022 (4th Shift)

(a) $m^2 x^2 + 2nx + n^2$ (b) $m^2 x^2 + 2mnx + n^2$

(c) $m^2 x^2 + 2mx + n^2$ (d) $m^2 x^2 + 2mn + n^2$

Q.265. If $x + y + z = 0$, then what is the

value of $\frac{x^2}{(yz)} + \frac{y^2}{(xz)} + \frac{z^2}{(xy)}$?

SSC CGL 03/12/2022 (4th Shift)

(a) 1 (b) 0 (c) 2 (d) 3

Q.266. What is the possible value of $(a + b + c) - 3$, if $a^2 + b^2 + c^2 = 9$ and $ab + bc + ca = 8$?

SSC CGL 03/12/2022 (4th Shift)

(a) 5 (b) 3 (c) 9 (d) 2

Q.267. If $a - b = 2$ and $a^3 - b^3 = 80$, then what will be the value of ab ?

SSC CGL 05/12/2022 (2nd Shift)

(a) 12 (b) -24 (c) 24 (d) -12

Q.268. If $\frac{1}{x^2 + a^2} = x^2 - a^2$, then the value

of x is:

SSC CGL 05/12/2022 (2nd Shift)

(a) $(1 - a^4)^{\frac{1}{4}}$ (b) a

(c) $(a^4 - 1)^{\frac{1}{4}}$ (d) $(a^4 + 1)^{\frac{1}{4}}$

Q.269. Which of the following statement is correct?

I. The value of $100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + \dots + 22^2 - 21^2$ is 4840.

II. The value of $(k^2 + \frac{1}{k^2})(k - \frac{1}{k})$

$$(k^4 + \frac{1}{k^4})(k + \frac{1}{k})(k^4 - \frac{1}{k^4})$$

is $k^{16} - \frac{1}{k^{16}}$.

SSC CGL 05/12/2022 (2nd Shift)

(a) Neither I nor II (b) Only II

(c) Only I (d) Both I and II

Q.270. Select the correct algebraic expression.

SSC CGL 05/12/2022 (3rd Shift)

(a) $ab - a - b + 1 = (a - 1)(b - 1)$

(b) $ab + a - b + 1 = (1 - a)(1 - b)(1 - a)(1 + b)$

(c) $ab - a - b + 1 = (1 - a)(b - 1)$

(d) $ab - a - b + 1 = (a - 1)(1 - b)$

Q.271. If $x^2 + y^2 + z^2 = xy + yz + zx$ and $x = 1$, then find the value of

$$\frac{10x^4 + 5y^4 + 7z^4}{13x^2y^2 + 6y^2z^2 + 3z^2x^2}.$$

SSC CGL 05/12/2022 (4th Shift)

(a) 2 (b) 0 (c) -1 (d) 1

Q.272. If $x + \frac{1}{2x} = 3$, then evaluate

$$8x^3 + \frac{1}{x^3}.$$

SSC CGL 05/12/2022 (4th Shift)

(a) 212 (b) 216 (c) 180 (d) 196

Q.273. If $x^2 - 2xy = 84$ and $x - y = -10$, then the value of y is:

SSC CGL 05/12/2022 (4th Shift)

(a) 2 (b) 1 (c) 4 (d) 3

Q.274. If $x + \frac{1}{x} = 1$, then the value of

$x^{12} + x^9 + x^6 + x^3 + 1$ is:

SSC CGL 06/12/2022 (1st Shift)

(a) 1 (b) -1 (c) 0 (d) -2

Q.275. If $\frac{a}{b} + \frac{b}{a} = 1$ and $a + b = 2$, then the value of $a^3 + b^3$ is:

SSC CGL 06/12/2022 (2nd Shift)

(a) 0 (b) 3 (c) 1 (d) 2

Q.276. If $4x^2 + y^2 = 40$ and $xy = 6$, find the positive value of $2x + y$.

SSC CGL 06/12/2022 (2nd Shift)

(a) 8 (b) 6 (c) 5 (d) 4

Q.277. If $2a + 3b = 10$ and $ab = 3$, then find the value of $4a^2 + 9b^2$.

SSC CGL 06/12/2022 (3rd Shift)

(a) 60 (b) 62 (c) 64 (d) 66

Q.278. If $(4x - 7y) = 11$ and $xy = 8$, what is the value of $16x^2 + 49y^2$, given that x and y are positive numbers ?

SSC CGL 06/12/2022 (3rd Shift)
(a) 596 (b) 484 (c) 569 (d) 448

Q.279. If $x + \frac{1}{x} = 2$, then the value of

$x^{57} + \frac{1}{x^{57}}$ is :

SSC CGL 06/12/2022 (4th Shift)
(a) 1 (b) -2 (c) 0 (d) 2

Q.280. If $mx^m - nx^n = 0$, then what is the value of $\frac{1}{x^m + x^n} + \frac{1}{x^m - x^n}$ terms of x^n is:

SSC CGL 06/12/2022 (4th Shift)
Where $x, m, n > 0$

(a) $\frac{2mn}{\{(n^2 + m^2)\}}$ (b) $\frac{2mn}{\{x^n(m^2 - n^2)\}}$
(c) $\frac{-2mn}{\{x^n(m^2 + n^2)\}}$ (d) $\frac{2mn}{\{x^n(n^2 - m^2)\}}$

Q.281. If, for a non-zero x , $5x^2 + 7x + 5 = 0$ then the value of $x^3 + \frac{1}{x^3}$ is:

SSC CGL 06/12/2022 (4th Shift)
(a) $\frac{496}{125}$ (b) $\frac{532}{343}$ (c) $\frac{125}{532}$ (d) $\frac{182}{125}$

Q.282. If $x^2 + y^2 + 2y + 4x + 5 = 0$, then

$\frac{(x+y)}{(x-y)} =$ _____.

SSC CGL 07/12/2022 (1st Shift)
(a) -3 (b) $\frac{1}{3}$ (c) -1 (d) 3

Q.283. The value of $\frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{6(x-y)(y-z)(z-x)}$ where $x \neq y \neq z$ is equal to:

SSC CGL 07/12/2022 (1st Shift)
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{1}{9}$

Q.284. If $k + \frac{1}{k} + 2 = 0$ and $K < 0$, then

what is the value of $k^{10} + \frac{1}{k^{11}}$?

SSC CGL 07/12/2022 (2nd Shift)
(a) 1 (b) 0 (c) -1 (d) 2

Q.285. $3[a - \frac{1}{a}] + [a - \frac{1}{a}]^3 = ?$

SSC CGL 07/12/2022 (2nd Shift)
(a) $a^2 - \frac{1}{a^3}$ (b) $a^3 - \frac{1}{a^3}$
(c) $a^3 + \frac{1}{a^3}$ (d) $a^2 - \frac{1}{a^2}$

Q.286. If $x^4 + \frac{1}{x^4} = 194$ and $x^3 + \frac{1}{x^3} = ?$

SSC CGL 07/12/2022 (3rd Shift)
(a) 52 (b) 58 (c) 76 (d) 67

Q.287. $(a+1)^2 + (a+2)^2 = 16$ then what is the value of $40 + 12a + 4a^2$?

SSC CGL 07/12/2022 (4th Shift)
(a) 56 (b) 62 (c) 74 (d) 52

Q.288. Which of the following statements is correct ?

I. If $K + \frac{1}{K} = 12$, then $K^2 + \frac{1}{K^2} = 142$

II. The value of $(K^2 + \frac{1}{K^2})(K - \frac{1}{K})$

$(K^4 + \frac{1}{K^4})(K + \frac{1}{K})$ is $K^{16} - \frac{1}{K^{16}}$

SSC CGL 07/12/2022 (4th Shift)
(a) Only I (b) Neither I nor II
(c) Both I and II (d) Only II

Q.289. Find the remainder when we divide $3x^4 - 2x^2 + 4x - 1$ by $2x - 1$.

SSC CGL 07/12/2022 (4th Shift)
(a) 2 (b) 3 (c) $\frac{11}{16}$ (d) $\frac{15}{16}$

Q.290. What is the value of $(a+b)^2 - (a-b)^2$?

SSC CGL 08/12/2022 (1st Shift)
(a) $2(a^2 + b^2)$ (b) $4(a^2 + b^2)$ (c) $8ab$ (d) $4ab$

Q.291. If m and n are two positive real numbers such that $9m^2 + n^2 = 40$ and $mn = 4$, then the value of $3m + n$ is:

SSC CGL 08/12/2022 (1st Shift)
(a) 160 (b) 64 (c) 10 (d) 8

Q.292. The factors of $x^2 + 4y^2 + 4y - 4xy - 2x - 8$ are:

SSC CGL 08/12/2022 (1st Shift)
(a) $(x - 2y - 4)(x - 2y + 2)$
(b) $(x^2 - 2y - 4)(x^2 - 2y + 2)$
(c) $(x + 2y - 4)(x + 2y + 2)$
(d) $(x^2 - 2y - 4)(x^2 + 2y + 2)$

Q.293. If $y = 1 + \sqrt{3} + \sqrt{4}$, then the value of $2y^4 - 8y^3 - 6y^2 + 28y - 84$ is:

SSC CGL 08/12/2022 (2nd Shift)
(a) $40\sqrt{3}$ (b) $80\sqrt{3}$ (c) $20\sqrt{3}$ (d) $60\sqrt{3}$

Q.294. If $x = \frac{\sqrt{5} - \sqrt{4}}{\sqrt{5} + \sqrt{4}}$ and $y = \frac{\sqrt{5} + \sqrt{4}}{\sqrt{5} - \sqrt{4}}$

then the value of $\frac{x^2 - xy + y^2}{x^2 + xy + y^2}$?

SSC CGL 08/12/2022 (2nd Shift)
(a) $\frac{361}{363}$ (b) $\frac{341}{343}$ (c) $\frac{384}{387}$ (d) $\frac{321}{323}$

Q.295. If $\frac{2p}{p^2 - 5p + 1} = \frac{1}{10}$, $p \neq 0$, then

the value of $(p + \frac{1}{p})$ is :

SSC CGL 08/12/2022 (2nd Shift)
(a) 10 (b) 25 (c) 1 (d) 15

Q.296. Which of the following will satisfy $a^2 = b^2 + (ab)^2$ for the values a and b ?

SSC CGL 08/12/2022 (3rd Shift)

(a) $a = \sin x$, $b = \cot x$

(b) $a = \cos x$, $b = \tan x$

(c) $a = \cot x$, $b = \cos x$

(d) $a = \sin x$, $b = \tan x$

Q.297. If $abc = 5$, what is the value of

$(\frac{1}{1+a+b^{-1}} + \frac{1}{1+b+5c^{-1}} + \frac{1}{1+\frac{c}{5}+a^{-1}})$?

SSC CGL 08/12/2022 (3rd Shift)

(a) 5 (b) 1 (c) $\frac{1}{5}$ (d) $(a+b+c)$

Q.298. If $4(z+7)(2z-1) = Az^2 + Bz + C$, then the value of $A+B+C$ is:

SSC CGL 08/12/2022 (3rd Shift)
(a) 28 (b) 32 (c) 74 (d) 81

Q.299. Factorize the following expression: $p^3 + 27$

SSC CGL 08/12/2022 (3rd Shift)

(a) $(p-3)(p^2 - 3p + 9)$

(b) $(p+3)(p^2 + 3p + 9)$

(c) $(p+3)(p^2 - 3p + 9)$

(d) $(p+3)(p^2 - 3p - 9)$

Q.300. What is the value of $a^2 + b^2 + c^2$, if $a+b+c=9$ and $ab+bc+ca=23$?

SSC CGL 08/12/2022 (4th Shift)
(a) 22 (b) 32 (c) 49 (d) 35

Q.301. If $x + \frac{1}{x} = 2\sqrt{5}$ where $x > 1$, then the value of $x^3 - \frac{1}{x^3}$ is :

SSC CGL 08/12/2022 (4th Shift)
(a) 82 (b) 76 (c) 86 (d) 78

Q.302. If $x+y+z=0$, then the value of

$\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy}$ is:

SSC CGL 09/12/2022 (1st Shift)
(a) 3 (b) 1 (c) 0 (d) 2

Q.303. The value of

$\frac{(p-q)^3 + (q-r)^3 + (r-p)^3}{12(p-q)(q-r)(r-p)}$

where $p \neq q \neq r$, is equal to:

SSC CGL 09/12/2022 (1st Shift)
(a) $\frac{1}{9}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $\frac{1}{2}$

Q.304. What is the value of

$(a+b)^3 - a^3 + b^3$?

SSC CGL 09/12/2022 (1st Shift)

(a) $-3ab(a-b) + 2b^3$ (b) $3ab(a+b) + 2b^3$

(c) $3ab(a-b)b^3$ (d) $-3ab(a+b) + b^3$

Q.305. If $x^2 + \frac{1}{x^2} = 66$, the value of

$x - \frac{1}{x}$ is .

SSC CGL 09/12/2022 (2nd Shift)

(a) 10 (b) 8 (c) 9 (d) 6

Q.306. If $P - \frac{1}{P} = 6$, then what is the value of $P^4 + \frac{1}{P^4}$?

SSC CGL 09/12/2022 (3rd Shift)

(a) 1562 (b) 1432 (c) 1442 (d) 1444

Q.307. Simplify $x^4 - 15x^3 + 15x^2 - 15x + 40$; given $x = 14$.

SSC CGL 09/12/2022 (3rd Shift)

(a) 0 (b) 40 (c) 14 (d) 26

Q.308. If $y + \frac{1}{y} = 11$, then the value of

 $y^3 - \frac{1}{y^3}$ is:

SSC CGL 09/12/2022 (4th Shift)

(a) $345\sqrt{13}$ (b) $360\sqrt{13}$ (c) $352\sqrt{13}$ (d) $368\sqrt{13}$

Q.309. If $(x - \frac{1}{x})^2 = 12$, what is the value

of $(x^2 - \frac{1}{x^2})$, given that $x > 0$?

SSC CGL 09/12/2022 (4th Shift)

(a) $6\sqrt{2}$ (b) $8\sqrt{3}$ (c) $6\sqrt{3}$ (d) $8\sqrt{2}$

Q.310. $a^3 + b^3 + c^3 = 3abc$ if _____.

SSC CGL 09/12/2022 (4th Shift)

(a) $a + b + c = -3$ (b) $a + b + c = 0$ (c) $a + b + c = 1$ (d) $a + b + c = -1$

Q.311. If $m + \frac{1}{m-2} = 4$, then find the

value of $(m-2)^2 + \frac{1}{(m-2)^2}$.

SSC CGL 12/12/2022 (1st Shift)

(a) -2 (b) 4 (c) 0 (d) 2

Q.312. If $5x - \frac{5}{x} + 6 = 0$, then $x^2 + \frac{1}{x^2}$ is

SSC CGL 12/12/2022 (1st Shift)

(a) $\frac{86}{25}$ (b) $\frac{43}{12}$ (c) $\frac{81}{10}$ (d) $\frac{86}{11}$

Q.313. If $a = 26$ and $b = 22$, then the

value of $\frac{a^3 - b^3}{a^2 - b^2} - \frac{3ab}{a+b}$ is _____.

SSC CGL 12/12/2022 (1st Shift)

(a) $\frac{5}{3}$ (b) $\frac{13}{11}$ (c) $\frac{1}{3}$ (d) $\frac{11}{13}$

Q.314. What is the value of $a^2 + b^2 + c^2 - 2ab - 2bc + 2ca$?

SSC CGL 12/12/2022 (2nd Shift)

(a) $(2a + b + c)^2$ (b) $(a - b + c)^2$ (c) $(a - b - 2c)^2$ (d) $(a + 2b - c)^2$

Q.315. If $\frac{x}{8} + \frac{8}{x} = 1$, then the value of x^3 is:

SSC CGL 12/12/2022 (2nd Shift)

(a) -256 (b) -512 (c) 256 (d) 512

Q.316. If $p = 8.15$, $q = 9.06$ and $r =$

-17.21, then the value of $p^3 + q^3 + r^3 - 3pqr$ is:

SSC CGL 12/12/2022 (3rd Shift)

(a) -3.81 (b) -5.62 (c) 4.75 (d) 0

Q.317. If $x^4 + \frac{16}{x^4} = 15617$, $x > 0$, then

find the value of $x + \frac{2}{x}$.

SSC CGL 12/12/2022 (3rd Shift)

(a) $\sqrt{121}$ (b) $\sqrt{129}$ (c) $\sqrt{123}$ (d) $\sqrt{127}$

Q.318. If $x + y + z = 0$ and $x^2 + y^2 + z^2 = 40$, then what is the value of $xy + yz + zx$?

SSC CGL 12/12/2022 (4th Shift)

(a) -20 (b) 5 (c) -5 (d) -10

Q.319. Which of the following statement is correct?

I. If $x = 12$, $y = -2$ and $z = -10$, then $x^3 + y^3 + z^3 = 360$.

II. If $x + y = 48$ and $4xy = 128$, then $4x^2 + 4y^2 = 4480$.

SSC CGL 13/12/2022 (1st Shift)

(a) Neither I nor II (b) Only I

(c) Both I and II (d) Only II

Q.320. if $K + \frac{1}{K} + 2 = 0$ and $K < 0$, then

what is the value of $K^{11} + \frac{1}{K^4}$?

SSC CGL 13/12/2022 (1st Shift)

(a) 0 (b) -2 (c) -1 (d) -17

Q.321. if $x + \frac{1}{x} = 6$, then find the value of

$$\frac{3x}{2x^2 - 5x + 2}$$

SSC CGL 13/12/2022 (1st Shift)

(a) 1 (b) $\frac{3}{7}$ (c) $\frac{2}{3}$ (d) 0

Q.322. If $2p + q = 19$ and $8p^3 + q^3 = 361$, then find the value of pq .

SSC CGL 13/12/2022 (2nd Shift)

(a) 56 (b) 59 (c) 58 (d) 57

Q.323. If $\frac{a}{b} + \frac{b}{a} = -1$ and $a - b = 2$, then

the value of $a^3 - b^3$ is:

SSC CGL 13/12/2022 (3rd Shift)

(a) 0 (b) $\frac{1}{2}$ (c) 1 (d) -1

Q.324. If $x = 2 - 2^{\frac{1}{3}} + 2^{\frac{2}{3}}$ then find the

value of $x^3 - 6x^2 + 18x$.

SSC CGL 13/12/2022 (4th Shift)

(a) 40 (b) 33 (c) 45 (d) 22

Q.325. If $a = -12$, $b = -6$ and $c = 18$,

then what is the value of $\frac{2abc}{9}$.

SSC CGL 13/12/2022 (4th Shift)

(a) 554 (b) 288 (c) -288 (d) 144

SSC CPO 2022 Tier - 1

Q.326. If $x^4 + x^{-4} = 47$, $x > 0$, then what is the value of $x + \frac{1}{x} - 2$?

SSC CPO 09/11/2022 (Morning)

(a) 1 (b) 0 (c) 5 (d) 3

Q.327. The value of

$$\frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{(x^2-y^2)^3 + (y^2-z^2)^3 + (z^2-x^2)^3}$$
 where

 $x \neq y \neq z$, is:

SSC CPO 09/11/2022 (Afternoon)

(a) 0 (b) $\frac{1}{(x+y+z)}$ (c) $\frac{1}{(x+y)(y+z)(z+x)}$ (d) 1

Q.328. If $a + b + c = 10$; $a^2 + b^2 + c^2 = 38$, what is the value of $(a-b)^2 + (b-c)^2 + (c-a)^2$?

SSC CPO 09/11/2022 (Afternoon)

(a) 15 (b) 12 (c) 14 (d) 13

Q.329. If $16y^2 - k = (4y + \frac{3}{2})(4y - \frac{3}{2})$,

then the value of k is:

SSC CPO 09/11/2022 (Evening)

(a) $\frac{9}{4}$ (b) $\frac{11}{4}$ (c) $\frac{6}{4}$ (d) $\frac{7}{4}$

Q.330. The value of

$$\frac{\{(m^2 + n^2)(m-n) - (m-n)^3\}}{(m^2n - mn^2)}$$
 is:

SSC CPO 09/11/2022 (Evening)

(a) $m+n$ (b) $m-n$ (c) 2 (d) $\frac{m}{n}$

Q.331. If $x + y + z = 8$, and $x^2 + y^2 + z^2 = 20$ then the value of $x^3 + y^3 + z^3 - 3xyz$ is _____.

SSC CPO 09/11/2022 (Evening)

(a) 16 (b) 10 (c) 15 (d) -16

Q.332. If $\{(1 \div x) \times (x^3 - 2x^2) + (4^3 \div 2^5) \times x\} \div 25 = 1$, then the possible values of x is/are:

SSC CPO 09/11/2022 (Evening)

(a) -5 (b) 5 (c) 0 (d) Both 5 and -5

Q.333. The simplified form of $(7x + 4y)^2 + (7x - 4y)^2$ is:

SSC CPO 10/11/2022 (Morning)

(a) $98x^2 + 32y^2$ (b) $98x^2 + 32y^2 - 5xy$ (c) $32x^2 + 98y^2$ (d) $98x^2 - 32y^2$

Q.334. What are the values of x and y , respectively, from the following equations?

 $6x + 7y = 5xy$ $10y - 4x = 4xy$

SSC CPO 10/11/2022 (Morning)

(a) 3 and 4 (b) 4 and 5

(c) 2 and 4 (d) 2 and 5

Q.335. If $x + y = 10$, $2xy = 48$ and $x > y$,

then find $2x - y$.

SSC CPO 10/11/2022 (Morning)

- (a) 6 (b) 8 (c) 4 (d) 3

Q.336. If $a^2 + b^2 + 1 = 2a$, then the value of $a^4 + b^7$ is:

SSC CPO 10/11/2022 (Afternoon)

- (a) 1 (b) 0 (c) 2 (d) 4

Q.337. If a, b, c are all non-zero and $a + b + c = 0$, find the value of $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$?

SSC CPO 10/11/2022 (Afternoon)

- (a) 3 (b) 4 (c) 1 (d) $\frac{1}{2}$

Q.338. Simplify: $(x + y)^3 - (x - y)^3 - 6y(x^2 - y^2)$

SSC CPO 10/11/2022 (Evening)

- (a) $8y^3$ (b) x^3 (c) $8x^3$ (d) y^3

Q.339. If $x^2 + 6x + 1 = 0$ then the value of $(x + 6)^3 + \frac{1}{(x + 6)^3} = ?$

SSC CPO 10/11/2022 (Evening)

- (a) 245 (b) 216 (c) 186 (d) 198

Q.340. If a, b and c are positive integers such that $a^2 + b^2 = 82$ and $b^2 + c^2 = 65$, then the value of $2a + 7b - 3c$ is:

SSC CPO 11/11/2022 (Morning)

- (a) 2 (b) 5 (c) 49 (d) 1

Q.341. If $x(x - 5) = -1$, then the value of $x^3 (x^3 - 110) = ?$

SSC CPO 11/11/2022 (Morning)

- (a) 0 (b) -1 (c) 1 (d) 2

Q.342. If $a + \frac{1}{a} = p^2$, then find $a^2 + \frac{1}{a^2}$.

SSC CPO 11/11/2022 (Afternoon)

- (a) $p^2 - 2$ (b) $p^2 + 2$ (c) $p^4 - 2$ (d) $p^4 + 2$

Q.343. If $a = -9.6$, $b = 4.44$ and $c = 5.16$, then the value of $a^3 + b^3 + c^3 - 3abc$ is:

SSC CPO 11/11/2022 (Afternoon)

- (a) 0 (b) -1 (c) 2 (d) 1

Q.344. Find the value of the following equation. (Given $a < 10$)

$$\frac{(187 \div 17 \times a - 3 \times 3)}{(8^2 - 9 \times 7 + a^2)} = 1$$

SSC CPO 11/11/2022 (Afternoon)

- (a) 2 (b) 1 (c) 4 (d) 3

Q.345. If $\sqrt{1 + \frac{x}{529}} = \frac{24}{23}$, then the value of x is:

SSC CPO 11/11/2022 (Afternoon)

- (a) 15 (b) 27 (c) 47 (d) 30

Q.346. If the equation $k(21x^2 + 24) + rx + (14x^2 - 9) = 0$ and $k(7x^2 + 8) + px + (2x^2 - 3) = 0$ have both roots common, then

the value of $\frac{p}{r}$ is:

SSC CPO 11/11/2022 (Evening)

- (a) $\frac{1}{3}$ (b) $\frac{2}{5}$ (c) $\frac{4}{3}$ (d) $\frac{7}{5}$

SSC CGL 2021 Tier - 2

Q.347. What is the value of $\frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}} \div$

$$\frac{\sqrt{14} + \sqrt{10}}{\sqrt{14} - \sqrt{10}} + \frac{\sqrt{10}}{\sqrt{5}}?$$

SSC CGL Tier II (08/08/2022)

- (a) $\sqrt{2} + 2$ (b) $2\sqrt{2} + 2$
(c) $\sqrt{2} + 1$ (d) $2\sqrt{2} + 1$

Q.348. If $(x + \frac{1}{x})^2 = 3$, then what is the value of $x^6 + x^{-6}$?

SSC CGL Tier II (08/08/2022)

- (a) 6 (b) 2 (c) -6 (d) -2

Q.349. What is the solution of the following equations?

$$2X + 3Y = 12 \text{ and } 3X - 2Y = 5$$

SSC CGL Tier II (08/08/2022)

- (a) $X = 3, Y = 2$ (b) $X = 2, Y = 3$
(c) $X = -2, Y = 3$ (d) $X = 3, Y = -2$

Q.350. If $xy = -6$ and $x^3 + y^3 = 19$ (x and y are integers), then what is the value of

$$\frac{1}{x^{-1}} + \frac{1}{y^{-1}}?$$

SSC CGL Tier II (08/08/2022)

- (a) -1 (b) -2 (c) 1 (d) 2

SSC Selection Post (Phase - X)

Q.351. If $2^x = 3^y = 6^z$, then $\frac{1}{x} + \frac{1}{y} - \frac{1}{z}$ is equal to:

Graduate Level 01/08/2022 (Shift - 4)

- (a) 1 (b) $-\frac{1}{2}$ (c) 0 (d) $\frac{3}{2}$

Q.352. If $(a + b) = \sqrt{11}$ and $(a - b) = \sqrt{7}$ then find the value of $a^2 b^2 (a^2 + b^2)$.

Graduate Level 01/08/2022 (Shift - 4)

- (a) 11 (b) 7 (c) 5 (d) 9

Q.353. If arithmetic and geometric mean of x and y is 8 and $3\sqrt{7}$, respectively, then the value of $x^3 + y^3$ is:

Graduate Level 01/08/2022 (Shift - 4)

- (a) 1072 (b) 945 (c) 559 (d) 855

Q.354. The algebraic expression $25x^2 + 16y^2 + 9z^2 - 40xy + 24yz - 30zx$ is equal to:

Graduate Level 02/08/2022 (Shift - 3)

- (a) $(4y + 3z - 5x)(4x + 5y - 3z)$
(b) $(4y - 3z - 5x)(4x + 5y + 3z)$
(c) $(4y + 3z - 5x)^2$

(d) $(4y - 3z - 5x)^2$

Q.355. If $xy = 6$ and $x^2y + xy^2 + x + y = 63$, then the value of $x^2 + y^2$ is:

Graduate Level 02/08/2022 (Shift - 3)

- (a) 69 (b) 59 (c) 70 (d) 79

Q.356. Factorize the given algebraic expression. $x^3 + 27y^3 + 64z^3 - 36xyz$

Graduate Level 02/08/2022 (Shift - 3)

- (a) $(x - 3y - 4z)(x^2 + 9y^2 + 16z^2 - 3xy - 12yz - 4xz)$
(b) $(x + 3y + 4z)(x^2 + 9y^2 + 16z^2 - 3xy - 12yz - 4xz)$
(c) $(x + 3y + 4z)(x^2 + 9y^2 + 16z^2 - 12xy - 3yz - 4xz)$
(d) $(x + 3y + 4z)(x^2 + 9y^2 + 16z^2 + 3xy + 12yz + 4xz)$

Q.357. Simplify $(5x + 4y)^2 + (5x - 4y)^2$

Graduate Level 03/08/2022 (Shift - 3)

- (a) $32x^2 + 50y^2$ (b) $50x^2 + 32y^2$

- (c) $-80xy$ (d) $80xy$

Q.358. Simplify the expression

$$\left(\frac{p}{q}\right)^{10} + \left(\frac{q}{p}\right)^{10} \text{ if } \frac{p}{q} + \frac{q}{p} = 2.$$

Graduate Level 03/08/2022 (Shift - 3)

- (a) 2 (b) 0 (c) 1 (d) -1

Q.359. What must be subtracted from $x^3 - 3x^2y + 3xy^2 - y^3$ to get $x^3 - y^3$?

Graduate Level 04/08/2022 (Shift - 2)

- (a) $-3x^2y + 3xy^2$ (b) $3x^2y - 3xy^2$
(c) $x^3 - y^3$ (d) $-x^3 + y^3$

Q.360. If $a - b = 10$ and $ab = 4$, then the value of $a^3 - b^3 + 4(a + b)^2$ is:

Graduate Level 04/08/2022 (Shift - 2)

- (a) 1500 (b) 1584 (c) 1280 (d) 1623

Q.361. Find the value of $a^2 + b^2 + c^2 - 2ab + 2ac - 2bc$, if $a = x + y$, $b = x - y$ and $c = 2x - 1$.

Graduate Level 04/08/2022 (Shift - 4)

- (a) $(x - y - 1)^2$ (b) 0
(c) $(2x + 2y - 1)^2$ (d) $(2x - 2y - 1)^2$

Q.362. If $y + \frac{1}{y} = \sqrt{5}$, then the value of $y^6 + \frac{1}{y^6}$ is:

Graduate Level 05/08/2022 (Shift - 3)

- (a) 18 (b) 15 (c) 10 (d) 12

Q.363. If $P = \frac{x^2 - 100}{x^2 - 16}$ and $Q = \frac{x + 10}{x - 4}$,

then the value of $\frac{P}{Q}$ is:

Graduate Level 05/08/2022 (Shift - 3)

- (a) $\frac{x + 10}{x + 4}$ (b) $\frac{x - 10}{x + 4}$
(c) $\frac{x + 10}{x - 4}$ (d) $\frac{x - 10}{x - 4}$

Q.364. If $\frac{a}{2} = \frac{b}{3} = \frac{c}{4} = \frac{2a-3b+5c}{k}$,

then find the value of K.

Higher Secondary 01/08/2022 (Shift - 3)

- (a) 15 (b) 18 (c) 10 (d) 12

Q.365. if $\frac{x-1}{x} = 5$, then the value of

$1 + \frac{1}{x^2}$ is:

Higher Secondary 03/08/2022 (Shift - 4)

- (a) 16 (b) 18 (c) 17 (d) 27

Q.366. If $2x + \frac{1}{3x} = 12$, find $9x^2 + \frac{1}{4x^2}$.

Higher Secondary 04/08/2022 (Shift - 3)

- (a) 142 (b) 324 (c) 321 (d) 327

Q.367. What is the value of x .

if $\frac{4}{x} - \frac{3}{2x} = 20$?

Higher Secondary 05/08/2022 (Shift - 2)

- (a) $\frac{1}{4}$ (b) 8 (c) $\frac{1}{8}$ (d) 4

Q.368. If $x - y = 5$ and $x^3 - y^3 = 520$, then the value of xy is :

Higher Secondary 05/08/2022 (Shift - 2)

- (a) $\frac{79}{3}$ (b) $\frac{36}{5}$ (c) $\frac{34}{3}$ (d) $\frac{76}{5}$

SSC CHSL 2021 Tier - 1

Q.369. If $x + y + z = 0$, then what will be

the value of $(\frac{x^2}{yz}) + (\frac{y^2}{zx}) + (\frac{z^2}{xy})$?

SSC CHSL 24/05/2022 (Morning)

- (a) $\frac{3(x^2+y^2+z^2)}{xyz}$ (b) $x^2 + y^2 + z^2$
(c) $\frac{x^2 y^2 z^2}{x}$ (d) 3

Q.370. If $2a + b = 10$, $2ab = 9$, and $2a > b$, then find the value of $2a - b$.

SSC CHSL 24/05/2022 (Afternoon)

- (a) 12 (b) 6 (c) 8 (d) 10

Q.371. If $a + b = 12$ and $ab = 35$, find the value of $a^4 + b^4 - 10^3$?

SSC CHSL 24/05/2022 (Afternoon)

- (a) 2226 (b) 2026 (c) 2126 (d) 1226

Q.372. If $a + 2b = 27$ and $a^3 + 8b^3 = 5427$, then find the value of $2ab$

SSC CHSL 24/05/2022 (Evening)

- (a) 176 (b) 156 (c) 172 (d) 149

Q.373. If $x^2 = b - ax$ and $x^3(x^3 + c) = d$, then which of the following can be the values of c and d ?

SSC CHSL 25/05/2022 (Morning)

- (a) $c = a^3 - 3ab$ and $d = b^3$
(b) $c = -a^3 - 3ab$ and $d = b^3$
(c) $c = a^3 + 3ab$ and $d = b^3$

(d) $c = a^3 + 3ab$ and $d = -b^3$

Q.374. If $x + y + z = 0$, then find the value of $x^3 + y^3 + z^3 - 3xyz$.

SSC CHSL 25/05/2022 (Afternoon)

- (a) 1 (b) $x^3 + y^3 + z^3$ (c) 0 (d) $3xyz$

Q.375. If $r + \frac{64}{r} = 16$, then the value of

$r^4 + \frac{1}{r^3}$ is _____.

SSC CHSL 25/05/2022 (Evening)

- (a) $4096\frac{1}{512}$ (b) 512
(c) $512\frac{1}{4096}$ (d) 4096

Q.376. If $p = 11$, then the value of

$p(p^2 + 3p + 3)$ is:

SSC CHSL 25/05/2022 (Evening)

- (a) 1629 (b) 1225 (c) 1727 (d) 1111

Q.377. If $\frac{X^2}{y^2} + \frac{y^2}{X^2} = 7$, A possible of

$\frac{X^3}{y^3} + \frac{y^3}{X^3}$ is:

SSC CHSL 26/05/2022 (Morning)

- (a) 18 (b) 15 (c) 16 (d) 17

Q.378. If $a^2 - 4a + 1 = 0$, then the value

of $a^2 + a + \frac{1}{a} + \frac{1}{a^2}$ is :

SSC CHSL 26/05/2022 (Afternoon)

- (a) 10 (b) 1 (c) 18 (d) 16

Q.379. If $(x - \frac{1}{x}) = \frac{7}{3}$, what is the value

of $(x^3 - \frac{1}{x^3})$?

SSC CHSL 26/05/2022 (Afternoon)

- (a) $19\frac{20}{27}$ (b) $19\frac{19}{27}$ (c) $19\frac{7}{9}$ (d) $19\frac{2}{3}$

Q.380. If $(x + \frac{1}{x}) = \frac{11}{5}$, what is the value

of $(x^3 + \frac{1}{x^3})$?

SSC CHSL 26/05/2022 (Evening)

- (a) $4\frac{6}{125}$ (b) $5\frac{101}{125}$ (c) $10\frac{81}{125}$ (d) $17\frac{31}{125}$

Q.381. If $1 + x^2 + x^4 = 585$ and $1 - x + x^2 = 65$, then find the value of $1 + x + x^2$.

SSC CHSL 26/05/2022 (Evening)

- (a) 8 (b) 7 (c) 6 (d) 9

Q.382. If $a = 2022$, $b = 2021$ and $c = 2020$, then value of $a^2 + b^2 + c^2 - ab - bc - ca$ is :

SSC CHSL 26/05/2022 (Evening)

- (a) 2 (b) 4 (c) 3 (d) 1

Q.383. If $x^2 + 4y^2 = 40$, $xy = 6$, and $x > 2y$, then the value of $x - 2y$ is:

SSC CHSL 27/05/2022 (Morning)

- (a) 1 (b) 3 (c) 2 (d) 4

Q.384. If $x^2 - 8x + 1 = 0$, what is the value of $x^8 - 3842x^4 + 1$?

SSC CHSL 27/05/2022 (Evening)

- (a) -1 (b) 0 (c) 2 (d) 1

Q.385. $144x^2 - 36x + \frac{9}{4}$ can be expressed as the square of _____.

SSC CHSL 27/05/2022 (Evening)

- (a) $14x - \frac{3}{2}$ (b) $12x - \frac{9}{4}$
(c) $12x - \frac{3}{2}$ (d) $12x - 9$

Q.386. The sum of two numbers is 17 and their product is 34. Find the sum of their squares.

SSC CHSL 30/05/2022 (Morning)

- (a) 224 (b) 221 (c) 225 (d) 222

Q.387. If $8x^3 + 27y^3 + 64z^3 = 72xyz$, then the relation between x, y and z can be :

SSC CHSL 30/05/2022 (Morning)

- (a) $2x + 3y = -4z$ (b) $2x + y + z = 0$
(c) $2x - 3y + 4z = 0$ (d) $2x + 3y = 4z$

Q.388. The value of

$\frac{[(5.4)^3 - 0.064]}{[(5.4)^2 + 2.16 + 0.16]}$ is _____.

SSC CHSL 30/05/2022 (Morning)

- (a) 5 (b) 7 (c) 3 (d) 1

Q.389. If $x = 9$, then the value of $x^5 - 10x^4 + 10x^3 - 10x^2 + 10x - 1$ is :

SSC CHSL 30/05/2022 (Afternoon)

- (a) 10 (b) 8 (c) 9 (d) 1

Q.390. If $p^3 + 3p^2 + 3p = 26$, then the value of $p^2 + 2p$ is ?

SSC CHSL 30/05/2022 (Evening)

- (a) 8 (b) 12 (c) 10 (d) 15

Q.391. If $\frac{a}{b} + \frac{b}{a} = -1$, $a \neq 0$, $b \neq 0$,

then the value of $6(a^3 - b^3)$ is:

SSC CHSL 31/05/2022 (Morning)

- (a) 6 (b) 3 (c) 0 (d) 1

Q.392. If $p^4 = 4354 - \frac{1}{p^4}$, then the

value of $p^3 - \frac{1}{p^3}$ can be :

SSC CHSL 31/05/2022 (Morning)

- (a) 536 (b) 436 (c) 416 (d) 516

Q.393. If $x + y = 14$, then the value of $x^3 + y^3 + 42xy$ is:

SSC CHSL 31/05/2022 (Morning)

- (a) 2744 (b) 2644 (c) 2742 (d) 2714

Q.394. If $x = 5t$ and $y = \frac{1}{3}(t + 1)$, then

the value of t for which $x = 3y$ is:

SSC CHSL 31/05/2022 (Afternoon)

- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{5}$

Q.395. If $x > 0$ and $x^4 + \frac{1}{x^4} = 2207$,

what is the value of $(x^5 + \frac{1}{x^5})$?

SSC CHSL 31/05/2022 (Afternoon)

- (a) 15127 (b) 15134 (c) 15141 (d) 15130

Q.396. $(2x + 3y + 4z)(4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8xz) = ?$

SSC CHSL 31/05/2022 (Afternoon)

- (a) $8x^3 + 27y^3 + 64z^3 + 72xyz$
 (b) $8x^3 + 27y^3 + 64z^3 - 72xyz$
 (c) $8x^3 + 27y^3 - 64z^3 + 72xyz$
 (d) $8x^3 - 27y^3 - 64z^3 - 72xyz$

Q.397. Simplify :

$$3ab \times (a + b)^{-1} \times (a^{-1} + b^{-1})$$

SSC CHSL 31/05/2022 (Evening)

- (a) $\frac{1}{(a+b)}$ (b) 1 (c) $(a+b)$ (d) 3

Q.398. If $x^4 + \frac{1}{x^4} = 14159$, then a

possible value of $x + \frac{1}{x}$ is:

SSC CHSL 31/05/2022 (Evening)

- (a) 69 (b) 121 (c) 81 (d) 11

Q.399. If $8x^2 + 9x + 8 = 0$, then the value

of $x^3 + \frac{1}{x^3}$ is:

SSC CHSL 01/06/2022 (Morning)

- (a) $\frac{999}{212}$ (b) $\frac{199}{212}$ (c) $\frac{999}{512}$ (d) $\frac{199}{512}$

Q.400. Expand $x^2 + 2x + 3$ about $x = -2$

SSC CHSL 01/06/2022 (Afternoon)

- (a) $(x - 2)^2 - 2(x + 2) + 3$
 (b) $(x + 2)^2 - 2(x + 2) + 3$
 (c) $(x + 2)^2 + 2(x + 2) + 3$
 (d) $(x - 2)^2 - 2(x - 2) - 3$

Q.401. If $a + b = 2c$ then the value of

$$\frac{a}{a-c} + \frac{c}{b-c} \text{ is}$$

SSC CHSL 01/06/2022 (Afternoon)

- (a) $\frac{1}{2}$ (b) 1 (c) 0 (d) -1

Q.402. If $x + y + z = 25$, $x^3 + y^3 + z^3 = 85$ and $xyz = 20$, then find the value of $x^2 + y^2 + z^2 - xy - yz - zx$.

SSC CHSL 01/06/2022 (Evening)

- (a) 2 (b) 1 (c) 4 (d) 3

Q.403. Which among the below options can be a square root of ?

$$(x^2 - 14x + 49)(x^2 + 6x + 9)$$

SSC CHSL 02/06/2022 (Afternoon)

- (a) $(x - 4)(x + 9)$ (b) $(x - 1)(x + 17)$
 (c) $(x - 7)(x + 3)$ (d) $(x - 3)(x + 8)$

Q.404. If $(x^3 + y^3)^{\frac{2}{3}} = (x^2 + y^2)$, then

the value of $\frac{x^2 + y^2}{xy}$ $x \neq 0, y \neq 0$, is:

SSC CHSL 02/06/2022 (Afternoon)

- (a) $\frac{2}{3}$ (b) $x^2 + y$ (c) 1 (d) $x + y^2$

Q.405. The factors of $a^2 - 1 - 2x - x^2$ are

SSC CHSL 02/06/2022 (Evening)

- (a) $(a - x - 1)(a - x - 1)$
 (b) $(a + 1 + x)(a - 1 - x)$
 (c) $(a - x + 1)(a - x + 1)$
 (d) $(a - x + 1)(a - x - 1)$

Q.406. If $x - 2y = 3$ and $2xy = 5$, then find

the value of $(x + 2y)^2$

SSC CHSL 03/06/2022 (Morning)

- (a) 29 (b) 27 (c) 30 (d) 28

Q.407. If $a^2 + b^2 + \frac{1}{a^2} + \frac{1}{b^2} = 4$, $a \neq 0$,

$b \neq 0$, then the value of $a^4 + b^4$ is:

SSC CHSL 03/06/2022 (Morning)

- (a) 32 (b) 256 (c) 64 (d) 2

Q.408. If $a + b + c = 0$, then find the value

$$\text{of } \frac{a^2 + b^2 + c^2}{a^2 - bc}$$

SSC CHSL 03/06/2022 (Afternoon)

- (a) -1 (b) 1 (c) 2 (d) -2

Q.409. If $a - b = 3$ and $a^3 - b^3 = 999$, then

find the value of $a^2 - b^2$

SSC CHSL 03/06/2022 (Afternoon)

- (a) 60 (b) 62 (c) 64 (d) 63

Q.410. If

$$(a - 1)^2 + (b + 2)^2 + (c - 1)^2 = 0,$$

then the value of $a^2 + b^2 + c^2$

SSC CHSL 06/06/2022 (Morning)

- (a) 6 (b) 0 (c) 2 (d) 1

Q.411. $16a^2 + 4b^2 + 9c^2 - 16ab + 12bc - 24ca = ?$

SSC CHSL 06/06/2022 (Afternoon)

- (a) $(4a - 2b + 3c)^2$ (b) $(4a - 2b - 3c)^2$
 (c) $(4a + 2b - 3c)^2$ (d) $(4a + 2b + 3c)^2$

Q.412. If $p = 38$, then the value of

$p(p^2 + 3p + 3)$ is _____.

SSC CHSL 06/06/2022 (Evening)

- (a) 39313 (b) 59319 (c) 39318 (d) 59318

Q.413. If $a + b = \sqrt{7}$ and $a - b = \sqrt{5}$, then

the value of $8ab(a^2 + b^2) - (a - b)^2$

SSC CHSL 07/06/2022 (Morning)

- (a) 19 (b) 23 (c) 27 (d) 21

Q.414. If $P(x) = (x^3 - 8)(x + 1)$ and

$$Q(x) = (x^3 + 1)(x - 2)$$

the LCM of $P(x)$ and $Q(x)$ is:

SSC CHSL 07/06/2022 (Afternoon)

- (a) $(x^2 + 2x + 4)(x^2 + 4x + 1)$
 (b) $(x + 1)(x - 2)(x^2 + 2x + 4)(x^2 - x + 1)$
 (c) $(x + 1)^2(x - 2)^2(x^2 + 2x + 4)(x^2 + 4x + 1)$
 (d) $(x - 2)(x + 1)$

Q.415. If $x + y + z = 0$, then find the value

$$\text{of } \frac{x^2}{x^2 - yz} + \frac{y^2}{y^2 - xz} + \frac{z^2}{z^2 - xy}$$

SSC CHSL 07/06/2022 (Evening)

- (a) 2 (b) 1 (c) 3 (d) 0

Q.416. If $x = \sqrt{10} + \sqrt{11}$,

$y = \sqrt{10} - \sqrt{11}$, then value of

$$7x^2 - 50xy + 7y^2 = \underline{\hspace{2cm}}$$

SSC CHSL 08/06/2022 (Morning)

- (a) 344 (b) 704 (c) 1360 (d) 386

Q.417. What is the value of the following expression?

$$2^2 \left(\frac{x^a}{x^b}\right)^{(a+b)} \times 3^2 \left(\frac{x^b}{x^c}\right)^{(b+c)} \times 6^{-2} \left(\frac{x^c}{x^a}\right)^{(a+c)}$$

SSC CHSL 08/06/2022 (Afternoon)

- (a) 1 (b) 0 (c) 4 (d) 9

Q.418. If $x - \frac{1}{x} = 11$ and $x > 0$, what is

the value of $(x^2 - \frac{1}{x^2})$?

SSC CHSL 08/06/2022 (Afternoon)

- (a) $11\sqrt{123}$ (b) $55\sqrt{5}$
 (c) $11\sqrt{123}$ (d) $-55\sqrt{5}$

Q.419. If $px^3 + x^2 + 3x + q$ is exactly divisible by $(x + 2)$ and $(x - 2)$, then the values of p and q are:

SSC CHSL 08/06/2022 (Evening)

$$(a) p = -\frac{3}{4} \text{ and } q = 4$$

$$(b) p = \frac{3}{4} \text{ and } q = 4$$

$$(c) p = \frac{3}{4} \text{ and } q = -4$$

$$(d) p = -\frac{3}{4} \text{ and } q = -4$$

Q.420. If $x^3 + y^3 = 416$ and $x + y = 8$, then find $x^4 + y^4$

SSC CHSL 08/06/2022 (Evening)

- (a) 3002 (b) 3204 (c) 3004 (d) 3104

Q.421. If $2z = x + y$, then the value of

$$\frac{x}{x-z} + \frac{y}{y-z}$$

SSC CHSL 09/06/2022 (Morning)

(a) 0 (b) 1 (c) 2 (d) 5

Q.422. If $(ab + bc + ca) = 0$, then what is the value of

$$\left(\frac{1}{a^2 - bc} + \frac{1}{b^2 - ac} + \frac{1}{c^2 - ab}\right)?$$

SSC CHSL 09/06/2022 (Morning)

(a) 2 (b) 0 (c) 1 (d) $a + b + c$ **Q.423.** If $(x + 2)$ and $(x - 3)$ are the factors of $x^2 + k_1x + k_2$, then :

SSC CHSL 09/06/2022 (Afternoon)

(a) $k_1 = 1$ and $k_2 = -6$ (b) $k_1 = -1$ and $k_2 = -6$
(c) $k_1 = -1$ and $k_2 = 6$ (d) $k_1 = 1$ and $k_2 = 6$ **Q.424.** If $k + \frac{1}{k} = 2$, find the value of $8k \times k \times k$

SSC CHSL 09/06/2022 (Afternoon)

(a) 8 (b) 1 (c) 4 (d) 2

Q.425. For what value(s) of k will the expression $p + \frac{1}{9}\sqrt{p} + k^2$ be a perfect square ?

SSC CHSL 10/06/2022 (Afternoon)

(a) $k = \pm \frac{1}{8}$ (b) $k = \pm \frac{1}{9}$
(c) $k = \pm \frac{1}{21}$ (d) $k = \pm \frac{1}{18}$ **SSC CGL 2021 Tier - 1****Q.426.** If $x + y + 3 = 0$, then find the value of $x^3 + y^3 - 9xy + 9$.

SSC CGL 11/04/2022 (Morning)

(a) -18 (b) -36 (c) 18 (d) 36

Q.427. If $(4x + 2y)^3 + (4x - 2y)^3 = 16(Ax^3 + Bxy^2)$, then what is the value of $\frac{1}{2}(\sqrt{A^2 + B^2})$?

SSC CGL 11/04/2022 (Afternoon)

(a) 8 (b) 3 (c) 5 (d) 7

Q.428. If $x = 4 + \sqrt{15}$, what is the value of $(x^2 + \frac{1}{x^2})$?

SSC CGL 11/04/2022 (Evening)

(a) 48 (b) 54 (c) 72 (d) 62

Q.429. If $x + y + z = 11$, $xy + yz + zx = -6$, and $x^3 + y^3 + z^3 = 1604$, then the value of xyz is:

SSC CGL 12/04/2022 (Morning)

(a) 25 (b) 4 (c) 1 (d) 5

Q.430. If $a^2 + b^2 + 49c^2 + 18 = 2(b - 28c - a)$ then the value of $(a + b - 7c)$ is

SSC CGL 12/04/2022 (Afternoon)

(a) 4 (b) 3 (c) 2 (d) 1

Q.431. If $8k^6 + 15k^3 - 2 = 0$, then the positive value of $(k + \frac{1}{k})$ is:

SSC CGL 12/04/2022 (Evening)

(a) $2\frac{1}{2}$ (b) $2\frac{1}{8}$ (c) $8\frac{1}{2}$ (d) $8\frac{1}{8}$ **Q.432.** If $x - y + z = 0$ then find the value of $\frac{y^2}{2xz} - \frac{x^2}{2yz} - \frac{z^2}{2xy}$

SSC CGL 12/04/2022 (Evening)

(a) $\frac{3}{2}$ (b) $\frac{1}{2}$ (c) -6 (d) $-\frac{3}{2}$ **Q.433.** If $2\sqrt{2}x^3 - 3\sqrt{3}y^3 = (\sqrt{2}x - \sqrt{3}y)(Ax^2 - Bxy + Cy^2)$, then the value of $(A^2 + B^2 + C^2)$ is:

SSC CGL 13/04/2022 (Afternoon)

(a) 16 (b) 11 (c) 19 (d) 18

Q.434. If $x + y + z = 18$, $xyz = 81$ and $xy + yz + zx = 90$, then find the valueof $\sqrt[4]{x^3 + y^3 + z^3 + xyz}$

SSC CGL 13/04/2022 (Afternoon)

(a) 6 (b) 12 (c) 9 (d) 10

Q.435. If $a + b - c = 5$ and $ab - bc - ac = 10$, then find the value of $a^2 + b^2 + c^2$

SSC CGL 18/04/2022 (Morning)

(a) 40 (b) 5 (c) 45 (d) 15

Q.436. If $\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{3}$, then what is the value of $x^4 + \frac{1}{x^4}$?

SSC CGL 18/04/2022 (Evening)

(a) 531 (b) 721 (c) 623 (d) 527

Q.437. If $a^3 + b^3 = 218$ and $a + b = 2$, then the value of $\sqrt{(1 - ab)}$ is:

SSC CGL 19/04/2022 (Morning)

(a) 5 (b) 3 (c) 4 (d) 6

Q.438. If $x^2 - 5x - 1 = 0$, what is the value of $\frac{x^6 - x^4 + x^2 - 1}{x^3}$?

SSC CGL 19/04/2022 (Afternoon)

(a) 145 (b) 140 (c) 130 (d) 135

Q.439. If $(x + y)^3 - (x - y)^3 - 3y(2x^2 - 3y^2) = ky^3$ then find the value of k

SSC CGL 19/04/2022 (Evening)

(a) 10.5 (b) 8 (c) 11 (d) 10

Q.440. If $\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{5}$, $x \neq 0$ then what is the value of $\frac{(x^4 + \frac{1}{x^2})}{(x^2 + 1)}$.

SSC CGL 20/04/2022 (Afternoon)

(a) 42 (b) 44 (c) 48 (d) 46

Q.441. If $a + b + c = 6$, $a^2 + b^2 + c^2 = 32$, and $a^3 + b^3 + c^3 = 189$, then the value of $4abc$ is:

SSC CGL 20/04/2022 (Evening)

(a) 12 (b) 16 (c) 9 (d) 8

Q.442. If $x^4 - 79x^2 + 1 = 0$, then a value of $x + x^{-1}$ can be:

SSC CGL 21/04/2022 (Afternoon)

(a) 9 (b) 5 (c) 7 (d) 8

Q.443. If $4x - 3y = 12$ and $xy = 5$, thefind the value of $\frac{16x^2 + 9y^2}{8}$

SSC CGL 21/04/2022 (Evening)

(a) 33 (b) 18 (c) 3 (d) 44

Q.444. If $(a + \frac{1}{a} + 3)^2 = 16$, where a is a non-zero real number, then findthe value of $a^2 + \frac{1}{a^2}$.

SSC CGL 21/04/2022 (Evening)

(a) 3 (b) 47 (c) 49 (d) 7

SSC CGL 2020 Tier - 2**Q.445.** If $a + b + c = 1$, $ab + bc + ca = -22$ and $abc = -40$, then what is the value of $a^3 + b^3 + c^3$?

SSC CGL Tier II (29/01/2022)

(a) -51 (b) 27 (c) 67 (d) -53

Q.446. If $2x^2 + 5x + 1 = 0$, then one of the values of $x - \frac{1}{2x}$ is:

SSC CGL Tier II (29/01/2022)

(a) $\frac{\sqrt{13}}{2}$ (b) $\frac{\sqrt{17}}{2}$ (c) $\frac{5}{2}$ (d) $\frac{13}{2}$ **Q.447.** $\frac{22\sqrt{2}}{4\sqrt{2} - \sqrt{3} + \sqrt{5}} = a + \sqrt{5}b$, with $a, b > 0$, then what is the value of $(ab) : (a + b)$?

SSC CGL Tier II (29/01/2022)

(a) 7 : 8 (b) 4 : 7 (c) 7 : 4 (d) 8 : 7

Q.448. If $2x - y = 2$ and $xy = \frac{3}{2}$, then what is the value of $x^3 - \frac{y^3}{8}$?

SSC CGL Tier II (29/01/2022)

(a) $\frac{13}{4}$ (b) $-\frac{5}{4}$ (c) $\frac{9}{2}$ (d) $\frac{5}{2}$ **Q.449.** If $x = \sqrt{1 + \frac{\sqrt{3}}{2}} - \sqrt{1 - \frac{\sqrt{3}}{2}}$ then the value of $\frac{\sqrt{3} - x}{\sqrt{3} + x}$ is:

(corrected to two decimal places)

SSC CGL Tier II (29/01/2022)

(a) 0.25 (b) 0.19 (c) 0.27 (d) 0.17

Q.450. If $x^2 - \sqrt{7}x + 1 = 0$, then what is the value of $x^5 + \frac{1}{x^5}$?

SSC CGL Tier II (29/01/2022)

(a) $25\sqrt{7}$ (b) $21\sqrt{7}$ (c) $27\sqrt{7}$ (d) $19\sqrt{7}$

Q.451. If $x = 32.5$, $y = 34.6$ and $z = 30.9$, then the value of $x^3 + y^3 + z^3 - 3xyz$ is $0.98k$, where k is equal to:

SSC CGL Tier II (03/02/2022)

(a) 1,033 (b) 933 (c) 1,026 (d) 921

Q.452. If $x + \frac{1}{x} = 3$, $x \neq 0$, then the value of $x^7 + \frac{1}{x^7}$ is:

SSC CGL Tier II (03/02/2022)

(a) 749 (b) 843 (c) 746 (d) 849

Q.453. If $x^4 + y^4 + x^2y^2 = 17\frac{1}{16}$ and $x^2 - xy + y^2 = 5\frac{1}{4}$, then one of the values of $(x - y)$ is:

SSC CGL Tier II (03/02/2022)

(a) $\frac{5}{2}$ (b) $\frac{3}{4}$ (c) $\frac{5}{4}$ (d) $\frac{3}{2}$

SSC MTS 2020 Tier - 1

Q.454. If x and y are two numbers having opposite signs, and $x^2 : y^2 = 16 : 81$, then the value of $\frac{3x + 4y}{2x - 3y}$?

SSC MTS 05/10/2021 (Morning)

(a) $\frac{12}{25}$ (b) $-\frac{27}{40}$ (c) $-\frac{24}{35}$ (d) $\frac{48}{35}$

Q.455. If $(11a^3 - 15b^3) : (10a^3 + 4b^3) = 5 : 7$, then $(9a + 5b) : (5a - 3b)$ is equal to:

SSC MTS 06/10/2021 (Afternoon)

(a) 10 : 7 (b) 13 : 5 (c) 15 : 4 (d) 14 : 9

SSC CGL 2020 Tier - 1

Q.456. If $x + y = 4$ and $\frac{1}{x} + \frac{1}{y} = \frac{16}{15}$, then what is the value of $(x^3 + y^3)$?

SSC CGL 13/08/2021 (Morning)

(a) 18 (b) 19 (c) 21 (d) 16

Q.457. If $x - y = 11$ and $\frac{1}{x} - \frac{1}{y} = \frac{11}{24}$, then what is the value of $x^3 - y^3 + x^2y^2$?

SSC CGL 13/08/2021 (Afternoon)

(a) 1331 (b) 1115 (c) 1105 (d) 1307

Q.458. If $16x^2 + y^2 = 48$ and $xy = 2$, $x, y > 0$, then the value of $(64x^3 + y^3)$ is:

SSC CGL 13/08/2021 (Afternoon)

(a) 320 (b) 340 (c) 300 (d) 240

Q.459. If $x - \frac{1}{x} = 5$, $x \neq 0$, then what is the value of $\frac{x^6 - 5x^3 - 1}{x^6 + 7x^3 - 1}$?

SSC CGL 13/08/2021 (Afternoon)

(a) $\frac{41}{45}$ (b) $\frac{45}{41}$ (c) $\frac{45}{49}$ (d) $\frac{49}{45}$

Q.460. If $x^4 + \frac{1}{x^4} = 727$, $x > 1$, then

what is the value of $(x - \frac{1}{x})$?

SSC CGL 13/08/2021 (Evening)

(a) 6 (b) -6 (c) -5 (d) 5

Q.461. If $x - \frac{1}{x} = 1$, then what is the value of $x^8 + \frac{1}{x^8}$?

SSC CGL 13/08/2021 (Evening)

(a) 3 (b) 119 (c) 47 (d) -1

Q.462. If $a^3 + b^3 = 405$ and $a + b = 9$, then the value of ab is:

SSC CGL 16/08/2021 (Morning)

(a) 15 (b) 10 (c) 12 (d) 8

Q.463. If $(2x - \frac{3}{x}) = 2$, then what is the value of $(16x^4 + \frac{81}{x^4})$?

SSC CGL 16/08/2021 (Morning)

(a) 184 (b) 328 (c) 180 (d) 220

Q.464. If $a - \frac{12}{a} = 1$, where $a > 0$, then the value of $a^2 + \frac{16}{a^2}$ is:

SSC CGL 16/08/2021 (Afternoon)

(a) 15 (b) 17 (c) 11 (d) 19

Q.465. If $4x^4 - 37x^2 + 9 = 0$, $x > \sqrt{\frac{3}{2}}$, then what is the value of $8x^3 - \frac{27}{x^3}$?

SSC CGL 16/08/2021 (Afternoon)

(a) 215 (b) -215 (c) 35 (d) -35

Q.466. If $x + y = 2$ and $\frac{1}{x} + \frac{1}{y} = \frac{18}{5}$, then the value of $(x^3 + y^3)$

SSC CGL 16/08/2021 (Evening)

(a) $4\frac{2}{3}$ (b) $4\frac{3}{5}$ (c) $3\frac{1}{3}$ (d) $3\frac{1}{5}$

Q.467. If $x^2 + 4y^2 = 53$ and $x - 2y = 5$, then what is the value of $x^3 - 8y^3$?

SSC CGL 16/08/2021 (Evening)

(a) -85 (b) 335 (c) 155 (d) 85

Q.468. If $a^2 + c^2 + 17 = 2(a - 8b - 2b^2)$, then what is the value of $(a^3 + b^3 + c^3)$?

SSC CGL 17/08/2021 (Morning)

(a) 9 (b) -7 (c) 10 (d) -4

Q.469. If $4x^4 = 5x^2 - 1$, $x > \frac{1}{\sqrt{2}}$ then what is the value of $(2x^2 - x - 1)$?

SSC CGL 17/08/2021 (Evening)

(a) 0 (b) 1 (c) -2 (d) 2

Q.470. If $x^2 + 9y^2 + 4z^2 = 12(x - 2y + 2z) - 88$, then the value of $(x - 3y + z)$ is:

SSC CGL 17/08/2021 (Evening)

(a) 10 (b) 13 (c) 11 (d) 5

Q.471. Simplify the following expression.

$$\frac{(375 + 125)^2 - (125 - 375)^2}{375 \times 375 - 125 \times 125}$$

SSC CGL 18/08/2021 (Morning)

(a) $\frac{15}{8}$ (b) $\frac{3}{4}$ (c) $\frac{3}{2}$ (d) $\frac{27}{28}$

Q.472. If $x^2 - 5\sqrt{2}x + 1 = 0$, then what is the value of $\frac{(x^3 + \frac{1}{x})}{x^2 + 1}$?

SSC CGL 18/08/2021 (Afternoon)

(a) $\frac{12\sqrt{2}}{5}$ (b) $\frac{24\sqrt{2}}{5}$ (c) $\frac{26\sqrt{2}}{5}$ (d) $\frac{18\sqrt{2}}{5}$

Q.473. What is the constant term in the expansion of $(5x^2 - \frac{1}{x})^3$?

SSC CGL 18/08/2021 (Afternoon)

(a) 5 (b) -15 (c) 15 (d) 75

Q.474. If $a^3 - b^3 = 2349$ and $(a - b) = 9$, then $(a + b)^2 - ab$ is equal to:

SSC CGL 18/08/2021 (Evening)

(a) 261 (b) 280 (c) 229 (d) 244

Q.475. If $a + b + c = 0$, then what is the value of $\frac{(b+c)^2}{bc} + \frac{(c+a)^2}{ca} + \frac{(a+b)^2}{ab}$.

SSC CGL 18/08/2021 (Evening)

(a) 1 (b) -3 (c) -1 (d) 3

Q.476. If $a^2 + b^2 + c^2 + 216 = 12(a + b - 2c)$, then $\sqrt{ab - bc - ca}$ is:

SSC CGL 20/08/2021 (Morning)

(a) $6\sqrt{5}$ (b) $4\sqrt{5}$ (c) $3\sqrt{5}$ (d) $8\sqrt{5}$

Q.477. If $(56\sqrt{7}x^3 - 2\sqrt{2}y^3) \div (2\sqrt{7}x - \sqrt{2}y) = Ax^2 + By^2 - Cxy$, then find the value of $A + B - \sqrt{14}C$.

SSC CGL 20/08/2021 (Afternoon)

(a) 19 (b) 10 (c) 58 (d) 38

Q.478. If $x - y = 4$ and $x^3 - y^3 = 316$, $y > 0$ then the value of $x^4 - y^4$ is:

SSC CGL 20/08/2021 (Evening)

(a) 2482 (b) 2320 (c) 2500 (d) 2401

Q.479. If $8 + 2px^2 - 36x - 27x^3 = (2 - 3x)^3$, then what is the value of p ?

SSC CGL 23/08/2021 (Morning)

(a) 27 (b) 54 (c) 9 (d) -27

Q.480. If $2x^2 - 7x + 5 = 0$, then what is the value of $x^3 + \frac{125}{8x^3}$?

SSC CGL 23/08/2021 (Morning)

(a) $12\frac{5}{8}$ (b) $16\frac{5}{8}$ (c) $10\frac{5}{8}$ (d) $18\frac{5}{8}$

Q.481. If $x + \frac{1}{x} = \frac{17}{4}$, $x > 1$, then what is

the value of $x - \frac{1}{x}$?

SSC CGL 23/08/2021 (Morning)

- (a) $\frac{9}{4}$ (b) $\frac{3}{2}$ (c) $\frac{8}{3}$ (d) $\frac{15}{4}$

Q.482. If $9(a^2 + b^2) + c^2 + 20 = 12(a + 2b)$, then the value of $\sqrt{6a + 9b + 2c}$ is:

SSC CGL 23/08/2021 (Evening)

- (a) 3 (b) 4 (c) 2 (d) 6

Q.483. If $2x + 3y + 1 = 0$, then what is the value of $(8x^3 + 8 + 27y^3 - 18xy)$?

SSC CGL 24/08/2021 (Morning)

- (a) 7 (b) - 9 (c) - 7 (d) 9

Q.484. Simplify the following expression.

$$\frac{5(a^6 - b^6)^3 + 5(b^6 - c^6)^3 + 5(c^6 - a^6)^3}{2(a^3 - b^3)^3 + 2(b^3 - c^3)^3 + 2(c^3 - a^3)^3}$$

SSC CGL 24/08/2021 (Afternoon)

- (a) $\frac{5}{2}(a^3 + b^3)(b^3 + c^3)(c^3 + a^3)$
 (b) $\frac{5}{2}(a^3 + b^3)(b^3 - c^3)(c^3 - a^3)$
 (c) $\frac{5}{2}(a^3 - b^3)(b^3 + c^3)(c^3 + a^3)$
 (d) $\frac{5}{2}(a^3 - b^3)(b^3 - c^3)(c^3 + a^3)$

Q.485. If $y = 2x + 1$, then what is the value of $(8x^3 - y^3 + 6xy)$?

SSC CGL 24/08/2021 (Afternoon)

- (a) - 15 (b) 1 (c) - 1 (d) 15

Q.486. Given that $x^8 - 34x^4 + 1 = 0$, $x > 0$.

What is the value of $(x^3 - x^{-3})$?

SSC CGL 24/08/2021 (Evening)

- (a) 12 (b) 14 (c) 18 (d) 16

SSC CHSL 2020 Tier - 1

Q.487. If $x^4 + y^4 + x^2y^2 = 117$ and $x^2 + y^2 - xy = 3(4 + \sqrt{3})$, then the value of $(x^2 + y^2)$ will be:

SSC CHSL 12/04/2021 (Morning)

- (a) 9 (b) $6\sqrt{3}$ (c) 12 (d) $13\sqrt{3}$

Q.488. If $x^2 - 3x + 1 = 0$, then the

Value of $2(x^8 + \frac{1}{x^8}) - 5(x^2 + \frac{1}{x^2})$ is:

SSC CHSL 12/04/2021 (Evening)

- (a) 3479 (b) 4379 (c) 4370 (d) 4279

Q.489. If $x - y - z = 0$, then the value of $(x^2 + y^2 + z^2) \div (y^2 + zx)$ is:

SSC CHSL 12/04/2021 (Evening)

- (a) - 1 (b) 2 (c) - 2 (d) 1

Q.490. Given that $(2x + y)^3 - (x + 2y)^3 = (x - y)[A(x^2 + y^2) + Bxy]$, then the value of $(2A - B)$ is:

SSC CHSL 13/04/2021 (Afternoon)

- (a) 7 (b) 6 (c) 0 (d) 1

Q.491. If $x^3 + y^3 = 468$ and $x + y = 12$, then the value of $x^4 + y^4$ will be:

SSC CHSL 13/04/2021 (Evening)

- (a) 3620 (b) 2036 (c) 3025 (d) 3026

Q.492. If $a^2 + b^2 + c^2 + 170 = 2(8a + 5b - 9c)$, then the value of $\sqrt{4a + 8b - c}$ will be.

SSC CHSL 13/04/2021 (Evening)

- (a) 15 (b) 12 (c) 9 (d) 8

Q.493. If $x + y + z = 5$, $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$, $xyz = 12$ and $x^3 + y^3 + z^3 = 151$, then the value of $(x^2 + y^2 + z^2)$ is:

SSC CHSL 15/04/2021 (Morning)

- (a) 24 (b) 23 (c) 22 (d) 21

Q.494. If $x + y = 5$ and $\frac{1}{x} + \frac{1}{y} = \frac{20}{9}$, then the value of $(x^3 + y^3)$ will be:

SSC CHSL 15/04/2021 (Morning)

- (a) $\frac{205}{4}$ (b) $\frac{635}{8}$ (c) $\frac{635}{4}$ (d) $\frac{365}{4}$

Q.495. If $(7x + 3)^3 + (x - 2)^3 + 27(2x - 5)^3 = 9(7x + 3)(x - 2)(2x - 5)$ then the value of $5x + 3$ is:

SSC CHSL 15/04/2021 (Morning)

- (a) 8 (b) 6 (c) 10 (d) 2

Q.496. If $(x + \frac{2}{x}) = 7$, then what is the value of $(2x^2 + \frac{8}{x^2})$?

SSC CHSL 16/04/2021 (Morning)

- (a) 44 (b) 94 (c) 50 (d) 90

Q.497. If $(\frac{x}{y} + 1) = 4$, then what is the value of $(\frac{x^2 + y^2}{y^2})$?

SSC CHSL 16/04/2021 (Morning)

- (a) 12 (b) 22 (c) 14 (d) 10

Q.498. If $a^2 + 4b^2 + 25c^2 + 18 = 2(a - 2b + 20c)$, then what is the value of $(a + 2b + 5c)$?

SSC CHSL 16/04/2021 (Afternoon)

- (a) 3 (b) 4 (c) 6 (d) 5

Q.499. Simplify the following expression.

$$\frac{0.8 \times 0.8 \times 0.8 + 0.6 \times 0.6 \times 0.6}{0.08 \times 0.08 + 0.06 \times 0.06 - 0.08 \times 0.06}$$

SSC CHSL 16/04/2021 (Evening)

- (a) 14 (b) 1.4 (c) 140 (d) 160

Q.500. If $x^2 + y^2 = 45$ and $x - y = 5$ then what is the value of $x^3 - y^3$?

SSC CHSL 16/04/2021 (Evening)

- (a) 150 (b) 250 (c) - 25 (d) 275

Q.501. If $a + b + c = 4$, $ab + bc + ca = -14$ and $abc = -18$, then the value of

$$\sqrt{4a^3 + 4b^3 + 4c^3 - 36}$$
 will be:

SSC CHSL 19/04/2021 (Evening)

- (a) 26 (b) 24 (c) 25 (d) 18

Q.502. Using algebraic identities, simplify the following expression.

$$\frac{(x^4 + x^2 + 1)}{(x^2 + x + 1)}$$

SSC CHSL 04/08/2021 (Morning)

- (a) $(x^2 - 2x + 1)$ (b) $(x^2 + x + 1)$
 (c) $(x^2 + 2x + 1)$ (d) $(x^2 - x + 1)$

Q.503. If $a + b = p$, $ab = q$ the $(a^4 + b^4)$ is equal to:

SSC CHSL 04/08/2021 (Morning)

- (a) $p^4 - 2p^2q^2 + q^2$ (b) $p^4 - 4p^2q^2 + 2q^2$
 (c) $p^4 - 4p^2q + q^2$ (d) $p^4 - 4p^2q + 2q^2$

Q.504. If $\sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{6}$, then the value of $x^6 + \frac{1}{x^6}$ will be:

SSC CHSL 04/08/2021 (Afternoon)

- (a) 2270 (b) 2502 (c) 2702 (d) 2712

Q.505. If $x^2 + 1 - 2x = 0$, $x > 0$, then $x^2(x^2 - 2) =$ _____.

SSC CHSL 04/08/2021 (Evening)

- (a) 1 (b) $\sqrt{2}$ (c) - 1 (d) 0

Q.506. If $x^2 + 4y^2 + 3z^2 + \frac{19}{4} = 2\sqrt{3}(x + y + z)$, then the value of $(x - 4y + 3z) =$?

SSC CHSL 05/08/2021 (Morning)

- (a) $\frac{\sqrt{3}}{3}$ (b) $2\sqrt{3}$ (c) $\sqrt{3}$ (d) $\frac{\sqrt{3}}{2}$

Q.507. If $9x^2 - 6x + 1 = 0$, then the value of $27x^3 + (27x^3)^{-1}$ will be:

SSC CHSL 05/08/2021 (Afternoon)

- (a) 8 (b) 4 (c) 2 (d) 1

Q.508. If $(4x - 5)^3 + (x - 2)^3 + 27(2x - 5)^3 = 9(4x - 5)(x - 2)(2x - 5)$ then the value of $(x + \frac{3}{2})$ will be:

SSC CHSL 05/08/2021 (Afternoon)

- (a) $\frac{3}{2}$ (b) $\frac{1}{2}$ (c) $\frac{7}{2}$ (d) $\frac{5}{2}$

Q.509. If $a + b = 24$ and $a^2 + b^2 = 306$, where $a > b$, then the value of $4a - 5b$ is:

SSC CHSL 05/08/2021 (Evening)

- (a) 18 (b) 20 (c) 12 (d) 15

Q.510. If $a + b - c = 0$, then what is the value of $\frac{(b-c)^2}{4bc} + \frac{(c-a)^2}{4ca} - \frac{(a+b)^2}{4ab}$?

SSC CHSL 06/08/2021 (Morning)

- (a) $\frac{1}{2}$ (b) $-\frac{3}{4}$ (c) $-\frac{1}{2}$ (d) $\frac{3}{4}$

Q.511. Simplify the following expression: $(2a - b - 3c)(4a^2 + b^2 + 9c^2 + 2ab + 6ac -$

3bc)

SSC CHSL 06/08/2021 (Morning)

(a) $-8a^3 + b^3 + 27c^3$

(b) $8a^3 + b^3 + 27c^3$

(c) $8a^3 - b^3 - 27c^3 - 18abc$

(d) $8a^3 - b^3 - 27c^3 + 18abc$

Q.512. If $(x^2 + \frac{1}{49x^2}) = 15\frac{5}{7}$ then what

is the value of $(x + \frac{1}{7x})$?

SSC CHSL 06/08/2021 (Afternoon)

(a) 7 (b) ± 7 (c) ± 4 (d) 4

Q.513. If $(3x + 2y)^3 + (3x - 2y)^3 = 3kx(3x^2 + 4y^2)$, then the value of k will be:

SSC CHSL 09/08/2021 (Morning)

(a) 18 (b) 9 (c) 3 (d) 6

Q.514. If $x + y = 27$ and $x^2 + y^2 = 425$

then the value of $(x - y)^2$ will be:

SSC CHSL 09/08/2021 (Afternoon)

(a) 225 (b) 169 (c) 121 (d) 144

Q.515. If $x + \frac{1}{15x} = 3$, then the value of

$9x^3 + \frac{1}{375x^3}$ will be:

SSC CHSL 10/08/2021 (Morning)

(a) 237.6 (b) 376.2 (c) 273.6 (d) 367.2

Q.516. If $2x + 3y + 4z = 11$, $8x^3 + 27y^3 + 64z^3 = 105$, and $xyz = 1$, then the value of $4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8xz$ is:

SSC CHSL 10/08/2021 (Afternoon)

(a) 3 (b) 4 (c) 5 (d) 6

Q.517. If $(x - 1.5)^3 + (x - 4)^3 + (x - 3.5)^3 = 3(x - 1.5)(x - 4)(x - 3.5)$, then what is the value of x?

SSC CHSL 10/08/2021 (Evening)

(a) 1 (b) 3 (c) 6 (d) 9

Q.518. If $x^4 + x^{-4} = 47$, $x > 0$, then the value of $(2x - 3)^2$ is:

SSC CHSL 11/08/2021 (Morning)

(a) 9 (b) 3 (c) 5 (d) 7

Q.519. If $a^2 + b^2 + c^2 + 48 = 8(a + b + c)$, then the value of $\sqrt[3]{a^3 - b^3 + c^3}$?

SSC CHSL 11/08/2021 (Morning)

(a) 6 (b) 4 (c) 3 (d) 2

Q.520. If $a - \frac{24}{a} = 5$, where $a > 0$, then the

value of $a^2 + \frac{64}{a^2}$ is:

SSC CHSL 11/08/2021 (Afternoon)

(a) 45 (b) 56 (c) 60 (d) 65

Q.521. If $3x - 2y + 3 = 0$, then what will

be the value of $27x^3 + 54xy + 30 - 8y^3$?

SSC CHSL 12/08/2021 (Morning)

(a) -27 (b) 57 (c) -57 (d) 3

Q.522. If $a = \frac{\sqrt{5} + 2}{\sqrt{5} - 2}$ and $b = \frac{\sqrt{5} - 2}{\sqrt{5} + 2}$,

then the value of $2a^2 + 2b^2 - 5ab$ is equal to:

SSC CHSL 12/08/2021 (Afternoon)

(a) 693 (b) 649 (c) 635 (d) 639

Q.523. If $x^2 + (4 - \sqrt{3})x - 1 = 0$, then what is the value of $x^2 + \frac{1}{x^2}$?

SSC CHSL 12/08/2021 (Evening)

(a) $9 - 8\sqrt{3}$ (b) $21 - 12\sqrt{3}$

(c) $21 - 8\sqrt{3}$ (d) $17 - 8\sqrt{3}$

SSC CPO 2020 Tier - 1

Q.524. If $a^2 + b^2 + c^2 + 216 = 12 \times$

 $(a + b - 2c)$ then $\sqrt{ab - bc + ca}$ is:

SSC CPO 23/11/2020 (Morning)

(a) 6 (b) 4 (c) 3 (d) 8

Q.525. The value of

$\frac{0.325 \times 0.325 + 0.175 \times 0.175 + 25 \times 0.00455}{5 \times 0.0065 \times 3.25 - 7 \times 0.175 \times 0.025}$

$\frac{0.5}{1.5}$ is:

SSC CPO 23/11/2020 (Evening)

(a) -3 (b) 3 (c) 0 (d) -1

Q.526. If $x^2 + 8y^2 - 12y - 4xy + 9 = 0$, then the value of $(7x - 8y)$ is:

SSC CPO 23/11/2020 (Evening)

(a) 12 (b) 5 (c) 9 (d) 21

Q.527. If $x + y + z = 19$, $x^2 + y^2 +$

$z^2 = 133$ and $xz = y^2$, $x > z > 0$, what

is the value of $(x - z)$?

SSC CPO 25/11/2020 (Morning)

(a) -2 (b) 0 (c) -5 (d) 5

SSC CGL 2019 Tier - 2

Q.528. Given that $x^8 - 34x^4 + 1 = 0$

 $x > 0$, What is the value of $(x^3 + x^{-3})$?

SSC CGL Tier II (15/11/2020)

(a) $5\sqrt{8}$ (b) $6\sqrt{6}$ (c) $5\sqrt{6}$ (d) $6\sqrt{8}$

Q.529. If $9x^2 + y^2 = 37$ and

 $xy = 2$, $x, y > 0$, then the value of $(27x^3 + y^3)$ is:

SSC CGL Tier II (15/11/2020)

(a) 217 (b) 207 (c) 301 (d) 259

Q.530. The value of $\frac{0.0203 \times 2.92}{0.7 \times 0.0365 \times 2.9}$

$\div \frac{(12.12)^2 - (8.12)^2}{(0.25)^2 + (0.25)(19.99)}$ is:

SSC CGL Tier II (15/11/2020)

(a) 0.05 (b) 0.5 (c) 0.1 (d) 0.01

Q.531. If $x - \frac{1}{x} = 5$, $x \neq 0$, then what is

the value of $\frac{x^6 + 3x^3 - 1}{x^6 - 8x^3 - 1}$?

SSC CGL Tier II (16/11/2020)

(a) $\frac{13}{12}$ (b) $\frac{11}{13}$ (c) $\frac{3}{8}$ (d) $\frac{4}{9}$

Q.532. The value of $\frac{27 \times (0.25)^3 + 125(0.05)^3}{(0.75)^2 - 0.25 \times 0.5}$ is:

SSC CGL Tier II (16/11/2020)

(a) 1 (b) 0.25 (c) 0.75 (d) 0.5

Q.533. If $\frac{8 + 2\sqrt{3}}{3\sqrt{3} + 5} = a\sqrt{3} - b$, then the

value of $a + b$ is equal to:

SSC CGL Tier II (18/11/2020)

(a) 18 (b) 15 (c) 16 (d) 24

Q.534. If $x(3 - \frac{2}{x}) = \frac{3}{x}$ then the value

of $x^3 - \frac{1}{x^3}$ is equal to:

SSC CGL Tier II (18/11/2020)

(a) $\frac{61}{27}$ (b) $\frac{52}{27}$ (c) $\frac{8}{27}$ (d) $\frac{62}{27}$

Q.535. The value of $5 - \frac{8 + 2\sqrt{15}}{4}$

 $\frac{1}{8 + 2\sqrt{15}}$ is equal to:

SSC CGL Tier II (18/11/2020)

(a) $\frac{2}{3}$ (b) 1 (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

SSC CHSL 2019 Tier - 1

Q.536. If $a + b + c = 2$, $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$,

 $ac = \frac{4}{b}$ and $a^3 + b^3 + c^3 = 28$, find thevalue of $a^2 + b^2 + c^2$.

SSC CHSL 12/10/2020 (Morning)

(a) 12 (b) 8 (c) 6 (d) 10

Q.537. If $x + y + z = 10$, $x^3 + y^3 + z^3 = 75$ and $xyz = 15$, then find the value of

$x^2 + y^2 + z^2 - xy - yz - zx$

SSC CHSL 12/10/2020 (Afternoon)

(a) 6 (b) 3 (c) 5 (d) 4

Q.538. If $a - b = 4$ and $a^3 - b^3 = 88$, then find the value of $a^2 - b^2$.

SSC CHSL 12/10/2020 (Afternoon)

(a) $6\sqrt{6}$ (b) $9\sqrt{6}$ (c) $7\sqrt{6}$ (d) $8\sqrt{6}$

Q.539. If the sum of two numbers is 11 and the sum of their squares is 65, then the sum of their cubes will be:

SSC CHSL 12/10/2020 (Evening)

(a) 355 (b) 576 (c) 407 (d) 615

Q.540. If $2^{x+y-2z} = 8^{8z-5-y}$;

$$5^{4y-6z} = 25^{y+z}; 3^{4x-3z} = 9^{x+z} \text{ then}$$

the value of $2x + 3y + 5z$ is:

SSC CHSL 13/10/2020 (Morning)

- (a) 56 (b) 44 (c) 32 (d) 28

Q.541. Find the value

$$\frac{1}{(9-4\sqrt{5})^2} + \frac{1}{(9+4\sqrt{5})^2}$$

SSC CHSL 13/10/2020 (Morning)

- (a) 322 (b) 424 (c) 246 (d) 286

Q.542. If $x^4 + \frac{1}{x^4} = \frac{257}{16}$, then find

$$\frac{8}{13} \left(x^3 + \frac{1}{x^3} \right), \text{ where } x > 0.$$

SSC CHSL 13/10/2020 (Afternoon)

- (a) 5 (b) 4 (c) 6 (d) 8

Q.543. If $a + b = 8$ and $a + a^2b + b + ab^2 = 128$, then the positive value of $a^3 + b^3$ is:

SSC CHSL 13/10/2020 (Afternoon)

- (a) 96 (b) 224 (c) 344 (d) 152

Q.544. Find the factors of the expression $3x^2 - 5x - 8$.

SSC CHSL 13/10/2020 (Evening)

- (a) $(x-1)$ and $(3x-8)$
 (b) $(x+1)$ and $(3x-8)$
 (c) $(x+1)$ and $(3x+8)$
 (d) $(x-1)$ and $(3x+8)$

Q.545. $(a+2b)^2 - (a-2b)^2$ is equal to:

SSC CHSL 14/10/2020 (Morning)

- (a) $10ab$ (b) $6ab$ (c) $8ab$ (d) $4ab$

Q.546. The value of

$$5\sqrt{3} + 7\sqrt{2} - \sqrt{6} - \frac{23}{\sqrt{2} + \sqrt{3} + \sqrt{6}} \text{ is:}$$

SSC CHSL 14/10/2020 (Afternoon)

- (a) 0 (b) 16 (c) 12 (d) 10

Q.547. If $a + b = 27$ and $a^3 + b^3 = 5427$, then find ab .

SSC CHSL 14/10/2020 (Evening)

- (a) 143 (b) 135 (c) 176 (d) 149

Q.548. Find the value of x .

$$\sqrt[3]{15625} - \sqrt{x} = 4$$

SSC CHSL 14/10/2020 (Evening)

- (a) 625 (b) 343 (c) 441 (d) 81

Q.549. The sum of two numbers is 47 and their product is 550. Find the sum of their squares.

SSC CHSL 15/10/2020 (Afternoon)

- (a) 1109 (b) 986 (c) 876 (d) 1209

Q.550. If $x = 2 + \sqrt{3}$, then find the value of $x^4 - 8x^3 + 16x^2$.

SSC CHSL 15/10/2020 (Afternoon)

- (a) -1 (b) 1 (c) 0 (d) 2

Q.551. $2x - 3y$ is a factor of:

SSC CHSL 16/10/2020 (Afternoon)

- (a) $4x^2 + 2x - 3y + 9y^2 - 12xy$

(b) $8x^3 + 27y^3$

(c) $4x^2 + 9y^2 + 12xy$

(d) $4x^2 + 2x - 3y + 36y^2 + 12xy$

Q.552. If $a : b = 3 : \sqrt{5}$, then the value of $(2a + b) : (3a - 2b)$ is:

SSC CHSL 16/10/2020 (Evening)

(a) $\frac{1}{64}(64 + 21\sqrt{5})$ (b) $\frac{1}{62}(64 + 21\sqrt{5})$

(c) $\frac{1}{63}(64 + 21\sqrt{5})$ (d) $\frac{1}{61}(64 + 21\sqrt{5})$

Q.553. If $x + y = 4$ and $x^3 + y^3 = 12$, then the value of $x^4 + y^4 = ?$

SSC CHSL 16/10/2020 (Evening)

(a) $\frac{146}{9}$ (b) $\frac{146}{3}$ (c) $\frac{146}{7}$ (d) $\frac{146}{5}$

Q.554. If $x = 3 + 2\sqrt{2}$, then the value of

$$x^2 + \frac{1}{x^2} \text{ is:}$$

SSC CHSL 20/10/2020 (Afternoon)

- (a) 30 (b) 36 (c) 32 (d) 34

Q.555. If $x + \frac{4}{x} - 4 = 0$, then the value of $x^2 - 4$ is equal to:

SSC CHSL 20/10/2020 (Afternoon)

- (a) 2 (b) 4 (c) 0 (d) 1

Q.556. If $x^3 + 27y^3 + 64z^3 = 36xyz$, then the relationship between x , y and z is:

SSC CHSL 20/10/2020 (Evening)

(a) $x + y + z = 0$ (b) $x - 3y + 4z = 0$

(c) $x + 3y = 4z$ (d) $x + 3y + 4z = 0$

Q.557. If $x(5 - \frac{2}{x}) = \frac{5}{x}$, then the value of $x^2 + \frac{1}{x^2}$ is equal to:

SSC CHSL 21/10/2020 (Morning)

(a) $2\frac{4}{25}$ (b) $2\frac{1}{25}$ (c) $\frac{4}{25}$ (d) $2\frac{3}{25}$

Q.558. If $\sqrt{x} = \sqrt{3} - \sqrt{5}$, then the value of $x^2 - 16x + 6$ is:

SSC CHSL 21/10/2020 (Afternoon)

- (a) 0 (b) 4 (c) 2 (d) -2

Q.559. If $x = \frac{\sqrt{3}}{2}$, then the value of

$$\frac{\sqrt{1+x} + \sqrt{1-x}}{\sqrt{1+x} - \sqrt{1-x}} \text{ is equal to:}$$

SSC CHSL 21/10/2020 (Afternoon)

- (a) 3 (b) $\sqrt{3}$ (c) 2 (d) $\sqrt{2}$

Q.560. If $x + \frac{1}{x} = \sqrt{3}$, then the value of

$$x^{18} + x^{12} + x^6 + 1 \text{ is:}$$

SSC CHSL 21/10/2020 (Evening)

- (a) 0 (b) 2 (c) 3 (d) 1

Q.561. If $a^2 + b^2 + 2b + 4a + 5 = 0$, then

the value of $\frac{2a-3b}{2a+3b}$ is equal to:

SSC CHSL 26/10/2020 (Morning)

- (a) $\frac{1}{7}$ (b) $\frac{2}{7}$ (c) $\frac{3}{7}$ (d) $\frac{2}{5}$

Q.562. If $2a + \frac{1}{a} = 4$, then the value of

$$a^2 + \frac{1}{4a^2} \text{ is:}$$

SSC CHSL 26/10/2020 (Morning)

- (a) 3 (b) 4 (c) 5 (d) 12

Q.563. If $A = \frac{1+2x}{1-2x}$ and $B = \frac{1-2x}{1+2x}$,

then the value of $\frac{A+B}{A-B}$ is:

SSC CHSL 26/10/2020 (Afternoon)

(a) $x + \frac{1}{4x}$ (b) $x - \frac{1}{4x}$

(c) $\frac{1}{4x} - x$ (d) $x^2 + \frac{1}{4x^2}$

Q.564. If $x = 3 + 2\sqrt{2}$, then the value of $\sqrt{x} - \frac{1}{\sqrt{x}}$ is:

SSC CHSL 26/10/2020 (Evening)

- (a) 2 (b) 1 (c) 0 (d) 3

Q.565. If $a + b + c = 0$, then

$$\left(\frac{2a^2}{3bc} + \frac{2b^2}{3ca} + \frac{2c^2}{3ab} \right) \text{ is equal to:}$$

SSC CHSL 26/10/2020 (Evening)

- (a) 3 (b) 4 (c) 1 (d) 2

Q.566. If the value of

$$\frac{3x\sqrt{y} + 2y\sqrt{x}}{3x\sqrt{y} - 2y\sqrt{x}} - \frac{3x\sqrt{y} - 2y\sqrt{x}}{3x\sqrt{y} + 2y\sqrt{x}} \text{ is same}$$

as that of $\sqrt{x}\sqrt{y}$, then which of the following relations between x and y is correct?

SSC CHSL 18/03/2020 (Morning)

(a) $9x - 4y = 36$ (b) $9x + 4y = 24$

(c) $9x + 4y = 36$ (d) $9x - 4y = 24$

Q.567. The value of $1 + 3 + 5 + 7 + \dots (2n-1)$ is:

SSC CHSL 18/03/2020 (Evening)

(a) $(2n-1) \times (2n-1)$ (b) $\frac{n}{2}$

(c) $n \times n$ (d) $\frac{n(n+1)}{2}$

Q.568. If $2(a^2 + b^2) = (a + b)^2$ then,

SSC CHSL 18/03/2020 (Evening)

- (a) $b = 2a$ (b) $a = -b$ (c) $a = b$ (d) $a = 2b$

Q.569. Find the value of x , if

$$21\sqrt{x} + 20\sqrt{x} = 29\sqrt{x},$$

SSC CHSL 18/03/2020 (Evening)

- (a) 4 (b) 3 (c) 0 (d) 2

Q.570. The value of

$$[(a^2 - b^2)^3 + (b^2 - c^2)^3 + (c^2 - a^2)^3] \div$$

$$[(a-b)^3 + (b-c)^3 + (c-a)^3] \text{ is equal}$$

to: (Given $a \neq b \neq c$)

SSC CHSL 19/03/2020 (Morning)

(a) $(a+b)(b+c)(c+a)$

(b) $(a^2 - b^2) + (b^2 - c^2) + (c^2 - a^2)$

(c) $(a^2 + b^2) + (b^2 + c^2) + (c^2 + a^2)$

(d) $(a-b)(b-c)(c-a)$

Q.571. If $A = \frac{x-1}{x+1}$ then, the value of

$A - \frac{1}{A}$ is:

SSC CHSL 19/03/2020 (Evening)

(a) $\frac{x^2-1}{-4(2x+1)}$ (b) $\frac{-4x}{x^2-1}$

(c) $\frac{x^2-1}{-4(2x-1)}$ (d) $\frac{-4(2x-1)}{x^2-1}$

SSC CGL 2019 Tier - 1

Q.572. If x, y and z are three integers such that $x + y = 8, y + z = 13$ and $z + x =$

17, then the value of $\frac{x^2}{yz}$ is:

SSC CGL 03/03/2020 (Morning)

(a) 1 (b) 0 (c) $\frac{18}{11}$ (d) $\frac{7}{5}$

Q.573. The expression $(a + b - c)^3 + (a - b + c)^3 - 8a^3$ is equal to:

SSC CGL 03/03/2020 (Afternoon)

(a) $6a(a-b+c)(c-a-b)$

(b) $3a(a+b-c)(a-b+c)$

(c) $6a(a+b-c)(a-b+c)$

(d) $3a(a-b+c)(c-a-b)$

Q.574. $(a+b-c+d)^2 - (a-b+c-d)^2 = ?$

SSC CGL 03/03/2020 (Evening)

(a) $4a(b+d-c)$

(b) $4a(b-d+c)$

(c) $2a(b+c-d)$

(d) $2a(a+b-c)$

Q.575. The value of $27a^3 - 2\sqrt{2}b^3$ is equal to:

SSC CGL 03/03/2020 (Evening)

(a) $(3a - \sqrt{2}b)(9a^2 - 2b^2 - 3\sqrt{2}ab)$

(b) $(3a - \sqrt{2}b)(9a^2 - 2b^2 + 6\sqrt{2}ab)$

(c) $(3a - \sqrt{2}b)(9a^2 + 2b^2 + 3\sqrt{2}ab)$

(d) $(3a + \sqrt{2}b)(9a^2 + 2b^2 + 3\sqrt{2}ab)$

Q.576. If $5x + \frac{1}{3x} = 4$, then what is the

value of $9x^2 + \frac{1}{25x^2}$?

SSC CGL 04/03/2020 (Morning)

(a) $\frac{174}{125}$ (b) $\frac{144}{125}$ (c) $\frac{114}{25}$ (d) $\frac{119}{25}$

Q.577. On simplification,

$$\frac{x^3 - y^3}{x[(x+y)^2 - 3xy]} \div \frac{y[(x-y)^2 + 3xy]}{x^3 + y^3} \times \frac{(x+y)^2 - (x-y)^2}{x^2 - y^2}$$

is equal to:

SSC CGL 04/03/2020 (Morning)

(a) 4 (b) 1 (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

Q.578. If $12x^2 - 21x + 1 = 0$, then what is the value of $9x^2 + (16x^2)^{-1}$?

SSC CGL 04/03/2020 (Evening)

(a) $\frac{429}{8}$ (b) $\frac{465}{16}$ (c) $\frac{417}{16}$ (d) $\frac{453}{8}$

Q.579. If $P = \frac{x^3 + y^3}{(x-y)^2 + 3xy}$, $Q =$

$$\frac{(x+y)^2 - 3xy}{x^3 - y^3} \text{ and } R = \frac{(x+y)^2 + (x-y)^2}{x^2 - y^2}$$

, then what is the value of $(P \div Q) \times R$?

SSC CGL 05/03/2020 (Morning)

(a) $2(x^2 + y^2)$

(b) $4xy$

(c) $x^2 + y^2$

(d) $2xy$

Q.580. Find the value of .

$$\frac{x^2(x-4)^2}{(x+4)^2 - 4x} \div \frac{(x^2-4x)^3}{(x+4)^2} \times \frac{64-x^3}{16-x^2}$$

SSC CGL 05/03/2020 (Afternoon)

(a) $\frac{x+4}{x-4}$

(b) $\frac{x+4}{x(x-4)}$

(c) $\frac{x+4}{x(4-x)}$

(d) $\frac{x-4}{x+4}$

Q.581. If $3^a = 27^b = 81^c$ and $abc = 144$,

then the value of $12\left(\frac{1}{a} + \frac{1}{2b} + \frac{1}{5c}\right)$ is:

SSC CGL 06/03/2020 (Morning)

(a) $\frac{17}{120}$

(b) $\frac{18}{10}$

(c) $\frac{18}{120}$

(d) $\frac{33}{10}$

Q.582. If the value of $(a+b-2)^2 + (b+c-5)^2 + (c+a-5)^2 = 0$, then the value

of $\sqrt{(b+c)^a + (c+a)^b - 1}$ is:

SSC CGL 06/03/2020 (Morning)

(a) 2

(b) 1

(c) 3

(d) 0

Q.583. If $a^3 + \frac{1}{a^3} = 52$, then the value of

$2(a + \frac{1}{a})$ is:

SSC CGL 06/03/2020 (Evening)

(a) 6

(b) 2

(c) 8

(d) 4

Q.584. If $b+c = ax, c+a = by$ and $a+b = cz$, then the value of

$$\frac{1}{9} \left[\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} \right]$$

is:

SSC CGL 06/03/2020 (Evening)

(a) $\frac{1}{3}$

(b) 0

(c) $\frac{1}{9}$

(d) 1

Q.585. If $a^4 + \frac{1}{a^4} = 50, a > 0$, then find

the value of $a^3 + \frac{1}{a^3}$

SSC CGL 07/03/2020 (Evening)

(a) $\sqrt{2(1 + \sqrt{13})}(-1 + 2\sqrt{13})$

(b) $\sqrt{2(1 - \sqrt{13})}(-1 + 2\sqrt{13})$

(c) $\sqrt{2(1 + \sqrt{13})}(-1 - 2\sqrt{13})$

(d) $\sqrt{2(1 + \sqrt{13})} + (-1 + 2\sqrt{13})$

Q.586. Find the product of $(a+b+2c)(a^2 + b^2 + 4c^2 - ab - 2bc - 2ca)$.

SSC CGL 07/03/2020 (Evening)

(a) $a^3+b^3+8c^3-6abc$ (b) $a^3+b^3+6c^3-6abc$

(c) $a^3+b^3+8c^3-2abc$ (d) $a^3+b^3+8c^3-abc$

Q.587. If $a^2 + b^2 - c^2 = 0$, then the value of

$$\frac{2(a^6+b^6-c^6)}{3a^2b^2c^2}$$

is:

SSC CGL 09/03/2020 (Afternoon)

(a) 2

(b) 0

(c) 1

(d) 3

Q.588. If $a = 2b = 8c$ and $a + b + c = 13$

then the value of $\frac{\sqrt{a^2+b^2+c^2}}{2c}$ is:

SSC CGL 09/03/2020 (Afternoon)

(a) $\frac{9}{2}$

(b) $\frac{5}{6}$

(c) $-\frac{9}{2}$

(d) $-\frac{5}{6}$

Q.589. If x, y and z are three numbers such that $x + y = 13, y + z = 15$ and

$z + x = 16$, then the value of $\frac{xy+xz}{xyz}$ is:

SSC CGL 09/03/2020 (Afternoon)

(a) $\frac{5}{18}$

(b) $\frac{5}{36}$

(c) $\frac{18}{5}$

(d) $\frac{36}{5}$

Q.590. If $x + 3y + 2 = 0$ then the value of $x^3 + 27y^3 + 8 - 18xy$ is:

SSC CGL 09/03/2020 (Evening)

(a) -2

(b) 2

(c) 0

(d) 1

SSC CPO 2019 Tier - 1

Q.591. What is the simplified value of :

$$\frac{1}{8} \left\{ \left(x + \frac{1}{y} \right)^2 - \left(x - \frac{1}{y} \right)^2 \right\}$$

SSC CPO 09/12/2019 (Morning)

(a) $\frac{x}{2y}$

(b) $\frac{x}{y}$

(c) $\frac{4x}{y}$

(d) $\frac{2x}{y}$

Q.592. The value of

$$\frac{0.325 \times 0.325 + 0.175 \times 0.175 - 25 \times 0.00455}{5 \times 0.0065 \times 3.25 - 7 \times 0.175 \times 0.025}$$

lies between:

SSC CPO 09/12/2019 (Evening)

(a) 0.25 and 0.35 (b) 0.05 and 0.15

(c) 0.15 and 0.25 (d) 0.35 and 0.45

Q.593. What is the value of

$$\frac{0.74 \times 1.23 \times 0.13}{(0.37)^3 + (0.41)^3 - 8(0.39)^3} ?$$

SSC CPO 11/12/2019 (Morning)

(a) $-\frac{1}{3}$

(b) 1

(c) -1

(d) $\frac{1}{3}$

Q.594. If $a + b + c = 6$ and $a^2 + b^2 + c^2 = 38$, then what is the value of -

$$a(b^2 + c^2) + b(c^2 + a^2) + c(a^2 + b^2) + 3abc$$

SSC CPO 11/12/2019 (Morning)

(a) 3

(b) -3

(c) 6

(d) -6

Q.595. If $\sqrt{x} + \frac{1}{\sqrt{x}} = 3, x > 0$, then

$$x^2(x^2 - 47) = ?$$

SSC CPO 11/12/2019 (Evening)

- (a) 0 (b) 2 (c) - 2 (d) - 1

Q.596. If x and y are real numbers, then the least possible value of

$$4(x-2)^2 + (y-3)^2 - 2(x-3)^2 \text{ is:}$$

SSC CPO 11/12/2019 (Evening)

- (a) 3 (b) - 4 (c) 1 (d) - 8

Q.597. The value of

$$\frac{4.669 \times 4.669 - 9 \times (0.777)^2}{(4.669)^2 + (2.331)^2 + 14(0.667)(2.331)}$$
 is
(1 - k), where k = ?

SSC CPO 11/12/2019 (Evening)

- (a) 0.666 (b) 0.334 (c) 1 (d) 2.338

Q.598. The value of $\sqrt{11 + 2\sqrt{18}}$ is closest to ?

SSC CPO 11/12/2019 (Evening)

- (a) 4.8 (b) 4.4 (c) 3.8 (d) 4.1

Q.599. The value of $\sqrt{6 - \sqrt{17 - 2\sqrt{72}}}$ is closest to :

SSC CPO 12/12/2019 (Morning)

- (a) 2.1 (b) 2.4 (c) 2.7 (d) 1.7

Q.600. If $\frac{8x}{2x^2 + 7x - 2} = 1$, $x > 0$, then

$$\text{what is the value of } x^3 + \frac{1}{x^3}$$

SSC CPO 12/12/2019 (Morning)

- (a) $\frac{3}{8}\sqrt{17}$ (b) $\frac{3}{4}\sqrt{17}$
(c) $\frac{5}{8}\sqrt{17}$ (d) $\frac{5}{4}\sqrt{17}$

Q.601. If $(x^3 + \frac{1}{x^3} - k)^2 + (x + \frac{1}{x} - p)^2 = 0$, where k and p are real numbers and $x \neq 0$, the $\frac{k}{p}$ is equal to:

SSC CPO 13/12/2019 (Morning)

- (a)
- $p^2 + 1$
- (b)
- $p^2 + 3$
- (c)
- $p^2 - 1$
- (d)
- $p^2 - 3$

Q.602. If $x^6 - 512y^6 = (x^2 + Ay^2)(x^4 - Bx^2y^2 + Cy^4)$, then what is the value of (A + B - C) ?

SSC CPO 13/12/2019 (Evening)

- (a) - 72 (b) 72 (c) - 80 (d) 48

Q.603. Solve the following $(a + b + c)(ab + bc + ca) - abc = ?$

SSC CPO 13/12/2019 (Evening)

- (a) $(a + b)(b + c)(c - a)$
(b) $(a - b)(b - c)(c - a)$
(c) $(a + b)(b - c)(c + a)$
(d) $(a + b)(b + c)(c + a)$

Q.604. The value of $\frac{1}{\sqrt{17 + 12\sqrt{2}}}$ is closest to _____.

SSC CPO 13/12/2019 (Evening)

- (a) 1.2 (b) 0.04 (c) 1.4 (d) 0.18

Q.605. If $a^3 - b^3 = 216$ and $a - b = 6$, then $(a + b)^2 - ab$ is equal to:

SSC CPO 15/03/2019 (Evening)

- (a) 38 (b) 42 (c) 52 (d) 36

Q.606. If $a^3 + b^3 = 432$ and $a + b = 12$, then $(a + b)^2 - 3ab$ is equal to:

SSC CPO 16/03/2019 (Afternoon)

- (a) 42 (b) 52 (c) 36 (d) 38

SSC CGL 2018 Tier - 2

Q.607. If $a^2 + b^2 + c^2 + 96 = 8(a + b - 2c)$, then $\sqrt{ab - bc + ca}$ is equal to :

SSC CGL Tier II (11/09/2019)

- (a) 6 (b)
- $2\sqrt{2}$
- (c) 4 (d)
- $2\sqrt{3}$

Q.608. If $x = \sqrt{1 + \frac{\sqrt{3}}{2}} - \sqrt{1 - \frac{\sqrt{3}}{2}}$,

then the value of $\frac{\sqrt{2-x}}{\sqrt{2+x}}$ will be closest

to :

SSC CGL Tier II (11/09/2019)

- (a) 0.17 (b) 0.12 (c) 1.4 (d) 1.2

Q.609. If $x + \frac{1}{16x} = 3$, then the value of $16x^3 + \frac{1}{256x^3}$ is :

SSC CGL Tier II (12/09/2019)

- (a) 423 (b) 441 (c) 432 (d) 414

Q.610. If $x = \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} + \sqrt{3}}$ and y is the reciprocal of x, then what is the value of $(x^3 + y^3)$?

SSC CGL Tier II (12/09/2019)

- (a) 488 (b) 504 (c) 472 (d) 476

Q.611. If $\sqrt{10 - 2\sqrt{21}} + \sqrt{8 + 2\sqrt{15}} = \sqrt{a} + \sqrt{b}$, where a and b are positive integers, then the value of $\sqrt{ab}$ is closest to :

SSC CGL Tier II (12/09/2019)

- (a) 4.6 (b) 5.9 (c) 6.8 (d) 7.2

Q.612. $ab(a - b) + bc(b - c) + ca(c - a)$ is equal to :

SSC CGL Tier II (13/09/2019)

- (a) $(a+b)(b-c)(c-a)$ (b) $(a-b)(b+c)(c-a)$
(c) $(a-b)(b-c)(c-a)$ (d) $(b-a)(b-c)(c-a)$

Q.613. Given That

$$(5x - 3)^3 + (2x + 5)^3 + 27(4 - 3x)^3 = 9(3 - 5x)(2x + 5)(3x - 4), \text{ then the value of } (2x + 1) \text{ is :}$$

SSC CGL Tier II (13/09/2019)

- (a) - 13 (b) 15 (c) - 15 (d) 13

Q.614. If $\frac{3(x^2+1)-7x}{3x} = 6$, $x \neq 0$ then the

value of $\sqrt{x} + \frac{1}{\sqrt{x}}$ is :

SSC CGL Tier II (13/09/2019)

- (a) $\sqrt{\frac{25}{3}}$ (b) $\frac{\sqrt{11}}{3}$ (c) $\sqrt{\frac{35}{3}}$ (d) $\sqrt{\frac{31}{3}}$

SSC CGL 2018 Tier - 1

Q.615. If $x + y + z = 19$, $xy + yz + zx = 114$,

then the value of $\sqrt{x^3 + y^3 + z^3 - 3xyz}$ is

SSC CGL 04/06/2019 (Afternoon)

- (a) 21 (b) 17 (c) 19 (d) 13

Q.616. If $x = a + \frac{1}{a}$ and $y = a - \frac{1}{a}$ then

$\sqrt{x^4 + y^4 - 2x^2y^2}$ is equal to:

SSC CGL 06/06/2019 (Morning)

- (a) $16a^2$ (b) 8 (c) $\frac{8}{a}$ (d) 4

Q.617. If $a + \frac{1}{a} = 3$, then $(a^4 + \frac{1}{a^4})$ is

equal to :

SSC CGL 06/06/2019 (Evening)

- (a) 77 (b) 47 (c) 81 (d) 27

Q.618. If $x^4 - 6x^2 - 1 = 0$, then the value of $x^6 - 5x^2 + \frac{5}{x^2} - \frac{1}{x^6} + 5 = ?$

SSC CGL 07/06/2019 (Morning)

- (a) 219 (b) 209 (c) 204 (d) 239

Q.619. If $\sqrt{x} - \frac{1}{\sqrt{x}} = 4$, then $x^2 + \frac{1}{x^2}$ is

equal to :

SSC CGL 10/06/2019 (Afternoon)

- (a) 192 (b) 326 (c) 322 (d) 256

Q.620. If $\sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{7}$, then $x^3 + \frac{1}{x^3}$ is

equal to:

SSC CGL 11/06/2019 (Evening)

- (a) 140 (b) 130 (c) 120 (d) 110

Answer Key :-

1.(d)	2.(b)	3.(d)	4.(b)
5.(d)	6.(c)	7.(b)	8.(a)
9.(a)	10.(c)	11.(b)	12.(b)
13.(a)	14.(c)	15.(a)	16.(c)
17.(d)	18.(a)	19.(b)	20.(b)
21.(a)	22.(a)	23.(b)	24.(a)
25.(a)	26.(a)	27.(c)	28.(d)
29.(c)	30.(c)	31.(c)	32.(b)
33.(a)	34.(c)	35.(b)	36.(c)
37.(a)	38.(c)	39.(d)	40.(a)
41.(d)	42.(a)	43.(c)	44.(a)

Pinnacle				Day: 39th - 48th				Algebra			
45.(d)	46.(c)	47.(c)	48.(d)	229.(b)	230.(b)	231.(d)	232.(d)	413.(a)	414.(b)	415.(a)	416.(a)
49.(a)	50.(b)	51.(a)	52.(c)	233.(d)	234.(b)	235.(a)	236.(a)	417.(a)	418.(b)	419.(d)	420.(d)
53.(b)	54.(b)	55.(c)	56.(c)	237.(d)	238.(b)	239.(d)	240.(d)	421.(c)	422.(b)	423.(b)	424.(a)
57.(a)	58.(a)	59.(b)	60.(d)	241.(c)	242.(d)	243.(b)	244.(b)	425.(d)	426.(a)	427.(c)	428.(d)
61.(d)	62.(a)	63.(d)	64.(b)	245.(d)	246.(d)	247.(a)	248.(c)	429.(a)	430.(a)	431.(a)	432.(a)
65.(c)	66.(d)	67.(a)	68.(a)	249.(c)	250.(b)	251.(d)	252.(a)	433.(c)	434.(a)	435.(b)	436.(d)
69.(c)	70.(d)	71.(d)	72.(d)	253.(d)	254.(a)	255.(d)	256.(b)	437.(d)	438.(d)	439.(c)	440.(d)
73.(a)	74.(c)	75.(c)	76.(a)	257.(d)	258.(d)	259.(c)	260.(d)	441.(a)	442.(a)	443.(a)	444.(b)
77.(a)	78.(d)	79.(b)	80.(d)	261.(b)	262.(c)	263.(d)	264.(b)	445.(d)	446.(b)	447.(a)	448.(a)
81.(d)	82.(c)	83.(b)	84.(c)	265.(d)	266.(d)	267.(a)	268.(d)	449.(c)	450.(d)	451.(a)	452.(b)
85.(c)	86.(b)	87.(d)	88.(c)	269.(c)	270.(a)	271.(d)	272.(c)	453.(a)	454.(c)	455.(c)	456.(b)
89.(c)	90.(a)	91.(a)	92.(b)	273.(c)	274.(a)	275.(a)	276.(a)	457.(b)	458.(a)	459.(c)	460.(d)
93.(c)	94.(d)	95.(b)	96.(c)	277.(c)	278.(c)	279.(d)	280.(d)	461.(c)	462.(c)	463.(a)	464.(b)
97.(d)	98.(b)	99.(a)	100.(d)	281.(d)	282.(d)	283.(b)	284.(b)	465.(a)	466.(a)	467.(b)	468.(b)
101.(d)	102.(d)	103.(c)	104.(b)	285.(b)	286.(a)	287.(b)	288.(a)	469.(a)	470.(b)	471.(c)	472.(b)
105.(d)	106.(c)	107.(a)	108.(c)	289.(c)	290.(d)	291.(d)	292.(a)	473.(c)	474.(a)	475.(d)	476.(a)
109.(b)	110.(c)	111.(a)	112.(c)	293.(a)	294.(d)	295.(b)	296.(c)	477.(c)	478.(b)	479.(a)	480.(b)
113.(c)	114.(a)	115.(b)	116.(b)	297.(b)	298.(b)	299.(c)	300.(d)	481.(d)	482.(b)	483.(a)	484.(a)
117.(a)	118.(c)	119.(a)	120.(b)	301.(b)	302.(a)	303.(c)	304.(b)	485.(c)	486.(b)	487.(c)	488.(b)
121.(a)	122.(a)	123.(c)	124.(c)	305.(b)	306.(c)	307.(d)	308.(b)	489.(b)	490.(d)	491.(d)	492.(c)
125.(b)	126.(c)	127.(d)	128.(d)	309.(b)	310.(b)	311.(d)	312.(a)	493.(b)	494.(d)	495.(a)	496.(d)
129.(b)	130.(c)	131.(c)	132.(c)	313.(c)	314.(b)	315.(b)	316.(d)	497.(d)	498.(b)	499.(c)	500.(d)
133.(d)	134.(a)	135.(c)	136.(a)	317.(b)	318.(a)	319.(a)	320.(a)	501.(a)	502.(d)	503.(d)	504.(c)
137.(c)	138.(d)	139.(b)	140.(d)	321.(b)	322.(d)	323.(a)	324.(d)	505.(c)	506.(c)	507.(c)	508.(c)
141.(d)	142.(d)	143.(d)	144.(c)	325.(b)	326.(a)	327.(c)	328.(c)	509.(d)	510.(b)	511.(c)	512.(c)
145.(d)	146.(b)	147.(c)	148.(b)	329.(a)	330.(c)	331.(d)	332.(d)	513.(d)	514.(c)	515.(a)	516.(a)
149.(d)	150.(c)	151.(b)	152.(b)	333.(a)	334.(c)	335.(b)	336.(a)	517.(b)	518.(c)	519.(b)	520.(d)
153.(c)	154.(b)	155.(b)	156.(a)	337.(a)	338.(a)	339.(d)	340.(d)	521.(d)	522.(d)	523.(c)	524.(a)
157.(a)	158.(b)	159.(c)	160.(a)	341.(b)	342.(c)	343.(a)	344.(b)	525.(b)	526.(c)	527.(d)	528.(a)
161.(a)	162.(d)	163.(b)	164.(b)	345.(c)	346.(a)	347.(c)	348.(d)	529.(a)	530.(a)	531.(a)	532.(a)
165.(a)	166.(b)	167.(a)	168.(c)	349.(a)	350.(c)	351.(c)	352.(d)	533.(a)	534.(d)	535.(b)	536.(b)
169.(a)	170.(c)	171.(a)	172.(a)	353.(a)	354.(c)	355.(a)	356.(b)	537.(b)	538.(d)	539.(c)	540.(b)
173.(b)	174.(d)	175.(a)	176.(d)	357.(b)	358.(a)	359.(a)	360.(b)	541.(a)	542.(a)	543.(d)	544.(b)
177.(a)	178.(b)	179.(c)	180.(a)	361.(c)	362.(a)	363.(b)	364.(a)	545.(c)	546.(c)	547.(c)	548.(c)
181.(d)	182.(b)	183.(a)	184.(c)	365.(c)	366.(c)	367.(c)	368.(a)	549.(a)	550.(b)	551.(a)	552.(d)
185.(a)	186.(d)	187.(a)	188.(d)	369.(d)	370.(c)	371.(b)	372.(a)	553.(a)	554.(d)	555.(c)	556.(d)
189.(b)	190.(d)	191.(b)	192.(c)	373.(c)	374.(c)	375.(a)	376.(c)	557.(a)	558.(c)	559.(b)	560.(a)
193.(b)	194.(b)	195.(b)	196.(c)	377.(a)	378.(c)	379.(b)	380.(a)	561.(a)	562.(a)	563.(a)	564.(a)
197.(c)	198.(c)	199.(b)	200.(b)	381.(d)	382.(c)	383.(d)	384.(b)	565.(d)	566.(d)	567.(c)	568.(c)
201.(a)	202.(c)	203.(b)	204.(b)	385.(c)	386.(b)	387.(a)	388.(a)	569.(a)	570.(a)	571.(b)	572.(c)
205.(a)	206.(c)	207.(c)	208.(b)	389.(b)	390.(a)	391.(c)	392.(a)	573.(a)	574.(a)	575.(c)	576.(c)
209.(b)	210.(a)	211.(c)	212.(a)	393.(a)	394.(c)	395.(a)	396.(b)	577.(a)	578.(c)	579.(a)	580.(b)
213.(d)	214.(c)	215.(c)	216.(c)	397.(d)	398.(d)	399.(c)	400.(b)	581.(d)	582.(c)	583.(c)	584.(c)
217.(a)	218.(d)	219.(b)	220.(d)	401.(b)	402.(b)	403.(c)	404.(a)	585.(a)	586.(a)	587.(a)	588.(a)
221.(d)	222.(a)	223.(c)	224.(c)	405.(b)	406.(a)	407.(d)	408.(c)	589.(a)	590.(c)	591.(a)	592.(a)
225.(b)	226.(c)	227.(b)	228.(d)	409.(d)	410.(a)	411.(b)	412.(d)	593.(a)	594.(d)	595.(d)	596.(b)

597.(a)	598.(b)	599.(b)	600.(c)
601.(d)	602.(c)	603.(d)	604.(d)
605.(d)	606.(c)	607.(c)	608.(a)
609.(a)	610.(a)	611.(b)	612.(d)
613.(b)	614.(d)	615.(c)	616.(d)
617.(b)	618.(b)	619.(c)	620.(d)

Solutions :-**Sol.1.(d) Given,**

$$x^4 + y^4 = x^2 y^2 \Rightarrow x^4 + y^4 - x^2 y^2 = 0$$

$$x^6 + y^6 = x^{(2)^3} + y^{(2)^3}$$

$$= (x^2 + y^2)(x^4 + y^4 - x^2 y^2)$$

$$x^6 + y^6 = (x^2 + y^2) \times 0 = 0$$

$$\text{Sol.2.(b)} \quad ax + by = 1 \text{ --- (1)}$$

$$\text{And, } bx + ay = \frac{2ab}{a^2 + b^2} \text{ --- (2)}$$

Put $b = 0$ in equation (1) we get;

$$ax + 0 \times y = 1 \Rightarrow a = x = 1$$

Now, put $b = 0$ in equation (2) we get;

$$0 + ay = \frac{2a \times 0}{a^2 + 0^2} \Rightarrow ay = 0 \Rightarrow y = 0$$

So, put $(a = 1 \text{ and } b = 0)$ in the options

Value of $x = 1$ obtained from option b

$$\text{Sol.3.(d)} \quad x = (7 + 3\sqrt{5}) \Rightarrow x^2 = 94 + 42\sqrt{5}$$

$$\frac{1}{x} = \frac{1}{(7 + 3\sqrt{5})} \times \frac{(7 - 3\sqrt{5})}{(7 - 3\sqrt{5})}$$

$$= \frac{(7 - 3\sqrt{5})}{4} \Rightarrow \frac{1}{x^2} = \frac{94 - 42\sqrt{5}}{16}$$

$$\text{Now, } x^2 + \frac{1}{x^2}$$

$$= (94 + 42\sqrt{5}) + \frac{94 - 42\sqrt{5}}{16}$$

$$= (94 + 42\sqrt{5}) + \frac{47 - 21\sqrt{5}}{8}$$

$$= \frac{94 \times 8 + 42\sqrt{5} \times 8 + 47 - 21\sqrt{5}}{8}$$

$$= \frac{799 + 315\sqrt{5}}{8}$$

$$\text{Sol.4.(b)} \quad x + \frac{1}{x} = 11$$

$$x^2 + \frac{1}{x^2} = 11^2 - 2 = 119$$

$$x^4 + \frac{1}{x^4} = 119^2 - 2 = 14159$$

$$\text{Now, } x^8 - 14159x^4 + 11$$

$$x^8 - (x^4 + \frac{1}{x^4})x^4 + 11$$

$$x^8 - x^8 - \frac{x^4}{x^4} + 11 = 10$$

$$\text{Sol.5.(d)} \quad \text{If } (a + b + c) = 0, \text{ Then } a^3 +$$

$$b^3 + c^3 = 3abc$$

$$\frac{(1-m)^3 + (m-n)^3 + (n-1)^3}{12(1-m)(m-n)(n-1)}$$

$$= \frac{3(1-m)(m-n)(n-1)}{12(1-m)(m-n)(n-1)} = \frac{1}{4}$$

$$\text{Sol.6.(c)} \quad 3x^2 + 2kx + 3 = 0$$

$$D = b^2 - 4ac$$

For real roots, D must be equal to zero.

$$0 = (2k)^2 - 4 \times 3 \times 3$$

$$k^2 = \frac{36}{4} = 9 \Rightarrow k = 3$$

$$\text{Sol.7.(b)} \quad \frac{(a^2 + b^2 - c^2)^2 - (a^2 - b^2 + c^2)^2}{b^2 - c^2} =$$

$$\frac{(a^2 + b^2 - c^2 + a^2 - b^2 + c^2) \times (a^2 + b^2 - c^2 - a^2 + b^2 - c^2)}{b^2 - c^2}$$

$$= \frac{2a^2 \times 2(b^2 - c^2)}{b^2 - c^2} = 4a^2$$

$$\text{Sol.8.(a)} \quad a = 335, b = 215 \text{ and } c = 180$$

$$a^3 + b^3 + c^3 - 3abc = \left(\frac{a+b+c}{2}\right)$$

$$\{(a-b)^2 + (b-c)^2 + (c-a)^2\}$$

$$= \left(\frac{335 + 215 + 180}{2}\right)$$

$$\{(120)^2 + (35)^2 + (155)^2\}$$

Using digital sum :-

$$= \frac{10}{2} \{(3)^2 + (8)^2 + (11)^2\}$$

$$= 5\{9 + 64 + 121\}$$

$$= 5\{9 + 1 + 4\} = 5\{5\} = 25 = 7$$

digital sum

Only option (a) has 7 as a digital sum.

$$\text{Sol.9.(a)} \quad \text{If } a + b + c = 0, \text{ then } a^3 + b^3 + c^3 = 3abc$$

$$\frac{(u-v)^3 + (v-w)^3 + (w-u)^3}{(u^2-v^2)^3 + (v^2-w^2)^3 + (w^2-u^2)^3}$$

$$= \frac{3(u-v)(v-w)(w-u)}{3(u^2-v^2)(v^2-w^2)(w^2-u^2)}$$

$$= \frac{1}{(u+v)(v+w)(w+u)}$$

$$\text{Sol.10.(c)} \quad x^3 - y^3 = 270, \text{ and } x - y = 6$$

$$(x^3 - y^3) = (x - y)\{x^2 + xy + y^2\}$$

$$270 = 6(36 + 3xy) \Rightarrow xy = 3$$

$$\text{Now, } (x + y) = \sqrt{(6)^2 + 4xy}$$

$$= \sqrt{36 + 4 \times 3} = 4\sqrt{3}$$

$$\text{Sol.11.(b)} \quad p + q + r = 0 \text{(given)}$$

$$\text{Let, } p = 0 \text{ then } q = -r$$

Now,

$$\left(\frac{p^2}{p^2 - qr} + \frac{q^2}{q^2 - pr} + \frac{r^2}{r^2 - pq}\right)$$

$$\left(0 + \frac{q^2}{q^2 - 0} + \frac{r^2}{r^2 - 0}\right) = 2$$

$$\text{Sol.12.(b)} \quad x + \frac{1}{x} = 5\sqrt{2}$$

$$x^3 + \frac{1}{x^3} = (5\sqrt{2})^3 - 3(5\sqrt{2})$$

$$= 250\sqrt{2} - 15\sqrt{2} = 235\sqrt{2}$$

$$\Rightarrow x^3 - \frac{1}{x^3} = \sqrt{\left(x^3 + \frac{1}{x^3}\right)^2 - 4}$$

$$= \sqrt{(235\sqrt{2})^2 - 4}$$

$$= \sqrt{110450 - 4} = \sqrt{110446} = 49\sqrt{46}$$

$$\text{Now, } x^6 - \frac{1}{x^6} = (x^3 + \frac{1}{x^3})(x^3 - \frac{1}{x^3})$$

$$= 235\sqrt{2} \times 49\sqrt{46}$$

$$= 235 \times 49 \times 2 \times \sqrt{23} = 23030\sqrt{23}$$

$$\text{Sol.13.(a)} \quad a^3 - b^3 = 6272 \text{ and } a - b = 8$$

$$a^3 - b^3 = (a - b)[(a - b)^2 + 3ab]$$

$$6272 = 8[8^2 + 3ab]$$

$$784 = 64 + 3ab$$

$$720 = 3ab \Rightarrow ab = 240$$

$$(a + b)^2 = (a - b)^2 + 4ab$$

$$(a + b)^2 = 8^2 + 4 \times 240 = 1024$$

$$\Rightarrow a + b = 32$$

$$\text{Now, } a^3 + b^3 = (a + b)[(a + b)^2 - 3ab]$$

$$a^3 + b^3 = 32[32^2 - 3 \times 240]$$

$$= 32[1024 - 720] \Rightarrow 32 \times 304 = 9728$$

Short trick :-

$$a^3 - b^3 = 6272, \text{ and } a - b = 8,$$

By hit and trial,

$a = 20$ and $b = 12$ satisfy the given equation.

Now,

$$a^3 + b^3 = 8000 + 1728 = 9728$$

$$\text{Sol.14.(c)} \quad a^2 + b^2 + c^2 - ab - bc - ca$$

$$= \frac{(a-b)^2 + (b-c)^2 + (c-a)^2}{2}$$

$$= \frac{(101-102)^2 + (102-103)^2 + (103-101)^2}{2}$$

$$= \frac{1 + 1 + 4}{2} = 3$$

$$\text{Sol.15.(a)} \quad \left(x + \frac{1}{x}\right) = 5$$

$$x^2 + \frac{1}{x^2} = (5)^2 - 2 = 23$$

$$\text{And, } x^4 + \frac{1}{x^4} = (23)^2 - 2 = 527 \text{ --- (1)}$$

$$\text{Now, } x - \frac{1}{x} = \sqrt{5^2 - 4} = \sqrt{21}$$

$$x^2 - \frac{1}{x^2} = (x + \frac{1}{x})(x - \frac{1}{x}) = 5\sqrt{21}$$

$$\text{And, } x^4 - \frac{1}{x^4} = (x^2 + \frac{1}{x^2})(x^2 - \frac{1}{x^2})$$

$$= 23 \times 5\sqrt{21} = 115\sqrt{21} \text{ --- (2)}$$

$$\text{Now, } x^8 - \frac{1}{x^8} = (x^4 + \frac{1}{x^4})(x^4 - \frac{1}{x^4})$$

$$= 527 \times 115\sqrt{21} = 60605\sqrt{21}$$

$$\text{Sol.16.(c)} \quad x - \frac{1}{x} = -6$$

$$x^2 + \frac{1}{x^2} = (-6)^2 + 2 = 38$$

$$x^3 - \frac{1}{x^3} = (-6)^3 + 3 \times (-6)$$

$$= -216 - 18 = -234$$

$$x^5 - \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 - \frac{1}{x^3}) - (x - \frac{1}{x})$$

$$= 38 \times (-234) - (-6)$$

$$= -8892 + 6 = -8886$$

Sol.17.(d) $x^2 + \frac{1}{x^2} = 7$

$$x + \frac{1}{x} = \sqrt{7+2} = 3$$

As $0 < x < 1$, so value of $(x - \frac{1}{x})$ will always be -ve.

$$x - \frac{1}{x} = -\sqrt{9-4} = -\sqrt{5}$$

$$x^2 - \frac{1}{x^2} = (x + \frac{1}{x})(x - \frac{1}{x})$$

$$= (3)(-\sqrt{5}) = -3\sqrt{5}$$

Sol.18.(a) $x^4 + \frac{1}{x^4} = 254$

$$x^2 + \frac{1}{x^2} = \sqrt{254+2} = 16$$

$$x + \frac{1}{x} = \sqrt{16+2} = 3\sqrt{2} \text{ And,}$$

$$x^3 + \frac{1}{x^3} = (3\sqrt{2})^3 - 3 \times 3\sqrt{2}$$

$$= 54\sqrt{2} - 9\sqrt{2} = 45\sqrt{2}$$

Now,

$$x^5 + \frac{1}{x^5} = (x^3 + \frac{1}{x^3}) \times (x^2 + \frac{1}{x^2}) - (x + \frac{1}{x})$$

$$= 45\sqrt{2} \times 16 - 3\sqrt{2} = 717\sqrt{2}$$

Sol.19.(b)

$$\frac{2x}{x+2} + \frac{1}{2x+4} - \frac{12-x}{2x^2+4x} = a + \frac{b}{x}$$

$$\Rightarrow \frac{4x^2+x-(12-x)}{2x^2+4x} = \frac{4x^2+2x-12}{2x^2+4x}$$

$$= \frac{2x^2+x-6}{x^2+2x} = \frac{(x+2)(2x-3)}{x(x+2)} = 2 + \frac{-3}{x}$$

On comparing we get,

$$a = 2 \text{ and } b = -3$$

$$\text{So, } ab = -6$$

Sol.20.(b) $6^x = 3^y = 2^z = k$ (let)

$$\text{Then, } k^{\frac{1}{z}} = 2, k^{\frac{1}{y}} = 3, k^{\frac{1}{x}} = 6$$

$$\Rightarrow k^{\frac{1}{x}} \rightarrow 2 \times 3 \Rightarrow k^{\frac{1}{x}} = k^{\frac{1}{y}} \times k^{\frac{1}{z}}$$

$$\text{therefore, } k^{\frac{1}{x}} = k^{\frac{1}{y}} \times k^{\frac{1}{z}}$$

$$\Rightarrow \frac{1}{x} = \frac{1}{y} + \frac{1}{z} \Rightarrow \frac{1}{y} + \frac{1}{z} - \frac{1}{x} = 0$$

Sol.21.(a) $17x + my + 102 = 0$

$$23x + 299y + 138 = 0$$

For infinite number of solutions,

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow \frac{17}{23} = \frac{m}{299}$$

$$m = \frac{17 \times 299}{23} = 221$$

Sol.22.(a) Formula : $a^3 + b^3 + c^3 - 3abc$

$$= (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$$

$$\Rightarrow 85 - 3 \times 25 = 5(a^2+b^2+c^2-ab-bc-ca)$$

$$\Rightarrow 85 - 75 = 5(a^2+b^2+c^2-ab-bc-ca)$$

$$\Rightarrow 10 = 5(a^2+b^2+c^2-ab-bc-ca)$$

$$\Rightarrow a^2+b^2+c^2-ab-bc-ca = \frac{10}{5} = 2$$

Sol.23.(b) Since, α, β are the roots of

$$6x^2 + 13x + 7 = 0. \text{ then,}$$

$$\alpha + \beta = \frac{-13}{6} \text{ and } \alpha\beta = \frac{7}{6}$$

$$\alpha^2 + \beta^2 = \left(\frac{-13}{6}\right)^2 - 2 \times \frac{7}{6} = \frac{169}{36} - \frac{7}{3} = \frac{85}{36}$$

So, the equation whose roots are α^2, β^2 is given by :

$$x^2 - \left(\frac{85}{36}\right)x + \frac{49}{36} = 0 \Rightarrow 36x^2 - 85x + 49 = 0$$

Sol.24.(a) Formula used :

$$x^3 + y^3 + z^3 - 3xyz = \frac{1}{2}(x+y+z)$$

$$\left[(x-y)^2 + (y-z)^2 + (z-x)^2\right]$$

$$= \frac{(222+223+224)}{2} [(222-223)^2$$

$$+ (223-224)^2 + (224-222)^2]$$

$$= \frac{669}{2} [(-1)^2 + (-1)^2 + (2)^2]$$

$$= \frac{669}{2} \times 6 = 2007$$

Sol.25.(a) Dividing and multiplying by

$$\left(k + \frac{1}{k}\right), \text{ we get}$$

$$\frac{\left(k + \frac{1}{k}\right)\left(k - \frac{1}{k}\right)\left(k^2 + \frac{1}{k^2}\right)\left(k^4 + \frac{1}{k^4}\right)\left(k^8 + \frac{1}{k^8}\right)\left(k^{16} + \frac{1}{k^{16}}\right)\left(k^{32} + \frac{1}{k^{32}}\right)}{\left(k + \frac{1}{k}\right)}$$

$$= \frac{\left(k^2 - \frac{1}{k^2}\right)\left(k^2 + \frac{1}{k^2}\right)\left(k^4 + \frac{1}{k^4}\right)\left(k^8 + \frac{1}{k^8}\right)\left(k^{16} + \frac{1}{k^{16}}\right)\left(k^{32} + \frac{1}{k^{32}}\right)}{\left(k + \frac{1}{k}\right)}$$

$$= \frac{\left(k^4 - \frac{1}{k^4}\right)\left(k^4 + \frac{1}{k^4}\right)\left(k^8 + \frac{1}{k^8}\right)\left(k^{16} + \frac{1}{k^{16}}\right)\left(k^{32} + \frac{1}{k^{32}}\right)}{\left(k + \frac{1}{k}\right)}$$

$$= \frac{\left(k^8 - \frac{1}{k^8}\right)\left(k^8 + \frac{1}{k^8}\right)\left(k^{16} + \frac{1}{k^{16}}\right)\left(k^{32} + \frac{1}{k^{32}}\right)}{\left(k + \frac{1}{k}\right)}$$

$$= \frac{\left(k^{16} - \frac{1}{k^{16}}\right)\left(k^{16} + \frac{1}{k^{16}}\right)\left(k^{32} + \frac{1}{k^{32}}\right)}{\left(k + \frac{1}{k}\right)}$$

$$= \frac{\left(k^{32} - \frac{1}{k^{32}}\right)\left(k^{32} + \frac{1}{k^{32}}\right)}{\left(k + \frac{1}{k}\right)} = \frac{\left(k^{64} - \frac{1}{k^{64}}\right)}{\left(k + \frac{1}{k}\right)}$$

Sol.26.(a) Statement 1.

Sum of squares of n even natural no.

$$= \frac{2}{3} \times (n)(n+1)(2n+1)$$

Sum of squares of n odd natural no.

$$= \frac{1}{3} \times n \times (2n+1) \times (2n-1)$$

$$= (100^2 + 98^2 + 96^2 + \dots + 2^2) -$$

$$(99^2 + 97^2 + 95^2 + \dots + 1^2)$$

$$= \frac{2}{3} \times (50)(50+1)(2 \times 50+1) - \frac{1}{3}$$

$$\times 50 \times (2 \times 50+1) \times (2 \times 50-1)$$

$$= \frac{1}{3} \times 50 \times 101 \times (102-99) = 5050$$

Statement 1 is correct.

Statement 2 :-

$$8x + \frac{8}{x} = -16 \Rightarrow x + \frac{1}{x} = -2 \Rightarrow x = -1$$

$$\text{Now, } x^{197} + x^{-197} = -2$$

Statement 2 is incorrect.

Sol.27.(c) $mx^m - nx^n = 0 \Rightarrow mx^m = nx^n$

$$\Rightarrow \frac{m}{n} = \frac{x^n}{x^m}$$

$$\frac{1}{x^m + x^n} + \frac{1}{x^m - x^n} = \frac{1}{n+m} + \frac{1}{n-m}$$

$$= \frac{2n}{n^2 - m^2} = \frac{2mn}{m(n^2 - m^2)} = \frac{2mn}{x^n(n^2 - m^2)}$$

Sol.28.(d) $(ax + by)^2 + (ay - bx)^2 = 8^2 + p^2$

$$\Rightarrow a^2x^2 + b^2y^2 + 2abxy + a^2y^2 + b^2x^2 - 2abxy$$

$$= 64 + p^2$$

$$\Rightarrow a^2x^2 + b^2y^2 + a^2y^2 + b^2x^2 = 64 + p^2$$

$$\Rightarrow a^2(x^2 + y^2) + b^2(x^2 + y^2) = 64 + p^2$$

$$\Rightarrow (x^2 + y^2)(a^2 + b^2) = 64 + p^2$$

$$\Rightarrow 25 \times 17 = 425 = 64 + p^2$$

$$\Rightarrow p^2 = 425 - 64 = 361 \Rightarrow p = 19$$

Short-Trick: Value putting $a = 4, b = 3,$
 $x = -1$ and $y = 4$

Sol.29.(c) $2 = x + \frac{1}{1 + \frac{1}{5 + \frac{1}{2}}}$

$$2 = x + \frac{1}{1 + \frac{2}{11}} \Rightarrow 2 = x + \frac{11}{13} \Rightarrow x = \frac{15}{13}$$

Sol.30.(c) Given:- $x^3 + 2x^2 - ax - b$ is divisible by $x^2 - 1$.

Therefore, $x^2 = 1$ satisfies the equation

$$\text{and } x^3 + 2x^2 - ax - b = 0 \text{ at } x^2 = 1$$

Or $x = +1$ or -1

$$\Rightarrow x^3 + 2x^2 - ax - b = 0 \quad (\because x = 1)$$

$$\Rightarrow 1 + 2 - a - b = 0 \Rightarrow 3 - a - b = 0$$

$$\Rightarrow a + b = 3 \dots (i)$$

$$\Rightarrow x^3 + 2x^2 - ax - b = 0 \quad (\because x = -1)$$

$$\Rightarrow -1 + 2 + a - b = 0 \Rightarrow 1 + a - b = 0$$

$$\Rightarrow a - b = -1 \dots (ii)$$

From eq. (i) and (ii):- $a = 1$ and $b = 2$

Sol.31.(c) $x = \sqrt[3]{5} + 2 \Rightarrow x - 2 = \sqrt[3]{5}$

Cubing both sides, we get

$$(x - 2)^3 = 5 \Rightarrow x^3 - 6x^2 + 12x - 8 = 5$$

$$\Rightarrow x^3 - 6x^2 + 12x - 12 = 1$$

Sol.32.(b) $(ax + by)$, on squaring it,

$$(ax + by)^2 = a^2x^2 + 2abxy + b^2y^2$$

So,

$(ax + by)$ is a factor of $a^2x^2 + 2abxy + b^2y^2$

Sol.33.(a) $7a^2 + 7a = 7a(a + 1)$

a and $a + 1$ are two consecutive number and out of two consecutive number one is odd and one is even and even is always divisible by 2

So, the complete number is divisible by 14 and 7 both.

Short-Trick:

Put the value of $a = 1$

$$7a^2 + 7a = 7(1)^2 + 7(1) = 14$$

So, number is divisible by both 7 and 14

Sol.34.(c) $(a^{\frac{2}{3}} + 2a^{\frac{1}{2}} + 3a^{\frac{1}{3}} + 2a^{\frac{1}{6}} + 1)$

$$(a^{\frac{1}{3}} - 2a^{\frac{1}{6}} + 1) - a^{\frac{1}{2}} (a^{\frac{1}{2}} - 2)$$

Let $a^{\frac{1}{6}} = p$ then,

$$\begin{aligned} &= (p^4 + 2p^3 + 3p^2 + 2p + 1)(p^2 - 2p + 1) - p^3(p^3 - 2) \\ &= p^6 + 2p^5 + 3p^4 + 2p^3 + p^2 - 2p^5 - 4p^4 - 6p^3 - 4p^2 - 2p + p^4 + 2p^3 + 3p^2 + 2p + 1 - p^6 + 2p^3 = 1 \end{aligned}$$

Short-Trick:

As the equation is independent of a , so you can put $a = 0$ for the solution Which gives 1 as answer

Sol.35.(b) $a + b + c + d = 2$ (given)

$$\text{Let } a = b = c = d = \frac{1}{2}$$

Therefore, $(1 + a)(1 + b)(1 + c)(1 + d)$

$$= (1 + \frac{1}{2})(1 + \frac{1}{2})(1 + \frac{1}{2})(1 + \frac{1}{2})$$

$$= (\frac{3}{2})(\frac{3}{2})(\frac{3}{2})(\frac{3}{2}) = \frac{81}{16}$$

Sol.36.(c)

$$(2x + y)^3 = 8x^3 + y^3 + 12x^2y + 6xy^2$$

Coefficient of $x^2 = 12y$

Sol.37.(a) Given, $x^{2a} = y^{2b} = z^{2c} \neq 0$ (1) and $x^2 = yz$ (2)

Let $x = y = z = m$...(where m is any integer value) put these value in eq (2)

$$x^2 = yz \Rightarrow m^2 = m \times m$$

.....(condition satisfied)

put these value in eq (1)

$$x^{2a} = y^{2b} = z^{2c} \Rightarrow m^{2a} = m^{2b} = m^{2c} \neq 0$$

.....(3)....(m is an integer value)

in equation (3) since base is same then power must be same

$$\Rightarrow 2a = 2b = 2c \Rightarrow a = b = c$$

$$\text{let } a = b = c = k \Rightarrow \frac{ab + bc + ca}{bc}$$

$$\Rightarrow \frac{(k \times k) + (k \times k) + (k \times k)}{k \times k} = \frac{3k^2}{k^2} = 3$$

Short trick : On putting the values as, $a = b = c = 1$ and $x = y = z = 2$, all equations are satisfied.

Sol.38.(c)

$$x^3 + y^3 + z^3 - 3xyz = \frac{1}{2}(x + y + z)$$

$$\begin{aligned} &[(x - y)^2 + (y - z)^2 + (z - x)^2] \\ &= \frac{1}{2}(5.51 + 5.52 + 5.57)[(5.51 - 5.52)^2 \\ &\quad + (5.52 - 5.57)^2 + (5.57 - 5.51)^2] \\ &= \frac{16.6}{2} \times [(-0.01)^2 + (-0.05)^2 + (0.06)^2] \\ &= 8.3[\frac{1+25+36}{10000}] = \frac{83 \times 62}{100000} = 0.05146 \end{aligned}$$

$$\text{Sol.39.(d)} \frac{1}{\sqrt{7-4\sqrt{3}}} \dots\dots\dots(i)$$

We know that, $7 - 4\sqrt{3} = (2 - \sqrt{3})^2$

Putting the values in eq (i), We get

$$= \frac{1}{\sqrt{(2 - \sqrt{3})^2}} = \frac{1}{2 - \sqrt{3}}$$

On solving we get,

$$= \frac{1 \times (2 + \sqrt{3})}{(2 - \sqrt{3})(2 + \sqrt{3})} = \frac{(2 + \sqrt{3})}{4 - 3}$$

$$= 2 + \sqrt{3} = 2 + 1.73 = 3.73$$

Sol.40.(a) On solving the equation

$$(2x + 3y + 4)(2x + 3y - 5)$$

$$= 4x^2 + 6xy - 10x + 6xy + 9y^2 - 15y + 8x + 12y - 20$$

$$= 4x^2 + 9y^2 + 12xy - 2x - 3y - 20 \dots\dots\dots \text{eq(1)}$$

On comparing equation (1) with

$$(ax^2 + by^2 + 2hxy + 2gx + 2fy + c)$$

We get,

$$a = 4, b = 9, h = 6, g = -1, f = -\frac{3}{2}, c = -20$$

Now, putting these values in equation

$$\left\{ \frac{(g + f - c)}{abh} \right\} = \left(\frac{(-1 - \frac{3}{2}) + 20}{216} \right) = \frac{35}{432}$$

Sol.41.(d)

Let us assume the value of $x = 3$ and $y = 2$

Now, putting the value in equation

$$(4x^3y - 6x^2y^2 + 4xy^3 - y^4)$$

$$= 4 \times 27 \times 2 - 6 \times 9 \times 4 + 4 \times 3 \times 8 - 16$$

$$= 216 - 216 + 96 - 16 = 80$$

Now, check for these values in the four options.

Only option (d) is satisfied.

$$= x^4 - (x - y)^4 = 3^4 - (3 - 2)^4 = 80$$

$$\text{Sol.42.(a)} \text{ Given, } x = \sqrt[6]{27} - \sqrt[6]{6\frac{3}{4}}$$

$$\text{and } y = \frac{\sqrt{45} + \sqrt{605} + \sqrt{245}}{\sqrt{80} + \sqrt{125}}$$

$$\Rightarrow x^2 = (\sqrt[6]{27} - \sqrt[6]{6\frac{3}{4}})^2$$

$$= \sqrt[3]{27} + \frac{27}{4} - 2 \times \sqrt[6]{27} \times \sqrt[6]{6\frac{3}{4}}$$

$$= 3 + \frac{27}{4} - 9 = \frac{3}{4}$$

$$y = \frac{\sqrt{45} + \sqrt{605} + \sqrt{245}}{\sqrt{80} + \sqrt{125}}$$

$$y = \frac{3\sqrt{5} + 11\sqrt{5} + 7\sqrt{5}}{4\sqrt{5} + 5\sqrt{5}} = \frac{7}{3} \Rightarrow y^2 = \frac{49}{9}$$

$$x^2 + y^2 = \frac{3}{4} + \frac{49}{9} = \frac{223}{36}$$

Sol.43.(c) Given,

$$\sqrt{86 - 60\sqrt{2}} = a - b\sqrt{2} \dots(1)$$

$$\text{But } \sqrt{86 - 60\sqrt{2}}$$

$$\Rightarrow \sqrt{36 + 50 - 2 \times 6 \times 5\sqrt{2}}$$

$$\Rightarrow \sqrt{(6 - 5\sqrt{2})^2} \Rightarrow (6 - 5\sqrt{2})$$

Comparing this with equation (1)

$$a = 6, b = 5$$

Now,

$$\sqrt{a^2 + b^2} = \sqrt{6^2 + 5^2} = \sqrt{61} \approx 7.8$$

Sol.44.(a) Given, $a = \sqrt{8} - \sqrt{7}$ and

$$a = \frac{1}{b} \Rightarrow b = \frac{1}{\sqrt{8} - \sqrt{7}}$$

Apply rationalization

$$\Rightarrow b = \frac{1}{\sqrt{8} - \sqrt{7}} \times \frac{\sqrt{8} + \sqrt{7}}{\sqrt{8} + \sqrt{7}} = \sqrt{8} + \sqrt{7}$$

$$\Rightarrow a + b = a + \frac{1}{a} \dots(a = \frac{1}{b})$$

$$= \sqrt{8} - \sqrt{7} + \sqrt{8} + \sqrt{7} = 2\sqrt{8}$$

$$\Rightarrow a^2 + \frac{1}{a^2} = a^2 + b^2 = (2\sqrt{8})^2 - 2 = 30$$

Now,

$$ab = (\sqrt{8} - \sqrt{7})(\sqrt{8} + \sqrt{7}) = 1$$

$$\Rightarrow \frac{a^2 + b^2 - 3ab}{a^2 + ab + b^2} = \frac{30 - 3(1)}{30 + 1} = \frac{27}{31}$$

Sol.45.(d) Short-Trick: Put $a = 1, b = -1$

$$(a + b + 4)\{ab + 4(a + b)\} - 4ab = 0$$

$$\Rightarrow (1 - 1 + 4)\{1(-1) + 4(1 - 1)\} - 4$$

$$\times 1 \times -1 = 0$$

$$\Rightarrow (4)(-1) + 4 = 0 \Rightarrow -4 + 4 = 0$$

Condition satisfied.

$$\left\{ \frac{1}{(a + b + 4)^{117}} - 2^{-234} \right\}$$

$$\Rightarrow \left\{ \frac{1}{(1 - 1 + 4)^{117}} - 2^{-234} \right\}$$

$$\Rightarrow \frac{1}{(2^2)^{117}} - 2^{-234} \Rightarrow \frac{1}{(2)^{234}} - 2^{-234}$$

$$\Rightarrow 2^{-234} - 2^{-234} = 0$$

$$\text{Sol.46.(c)} (x + y)^{\frac{1}{3}} + (z + y)^{\frac{1}{3}} = -(x + z)^{\frac{1}{3}}$$

$$\Rightarrow (x + y)^{\frac{1}{3}} + (z + y)^{\frac{1}{3}} + (x + z)^{\frac{1}{3}} = 0$$

$$\Rightarrow [(x + y)^{\frac{1}{3}}]^3 + [(z + y)^{\frac{1}{3}}]^3 + [(x + z)^{\frac{1}{3}}]^3$$

$$= 3(x + y)^{\frac{1}{3}}(z + y)^{\frac{1}{3}}(x + z)^{\frac{1}{3}}$$

$$\Rightarrow 2(x + y + z) = 3[(x + y)(z + y)(x + z)]^{\frac{1}{3}}$$

Cubing both sides,

$$\Rightarrow 8(x + y + z)^3 = 27[(x + y)(z + y)(x + z)]$$

$$\Rightarrow x^3 + y^3 + z^3 + 3(x + y)(y + z)(z + x)$$

$$= \frac{27}{8}[(x + y)(z + y)(x + z)]$$

$$\Rightarrow x^3 + y^3 + z^3 = (\frac{27}{8} - 3)[(x + y)(z + y)]$$

$$(x + z)]$$

$$\Rightarrow x^3 + y^3 + z^3 = \frac{3}{8}[(x+y)(z+y)(x+z)]$$

Sol.47.(c) $\frac{1}{4}\{(a + \frac{1}{a})^2 - (a - \frac{1}{a})^2\}$

$$\Rightarrow \frac{1}{4}\{((a)^2 + (\frac{1}{a})^2 + 2a \times \frac{1}{a}) - \{(a)^2 + (\frac{1}{a})^2 - 2a \times \frac{1}{a}\}\}$$

$$\Rightarrow \frac{1}{4}(a^2 + \frac{1}{a^2} + 2 - a^2 - \frac{1}{a^2} + 2)$$

$$\Rightarrow \frac{1}{4} \times 4 = 1$$

Short-Trick:

We know that ,

$$(a + b)^2 - (a - b)^2 = 4ab$$

$$= \frac{1}{4}\{(a + \frac{1}{a})^2 - (a - \frac{1}{a})^2\}$$

$$= \frac{1}{4} \times 4a \times \frac{1}{a} = 1$$

Sol.48.(d) Given, $3\sqrt[4]{x} + 4\sqrt[4]{x} = 5\sqrt[4]{x}$

We know that 3, 4 and 5 are triplet pairs

$$\Rightarrow 3^2 + 4^2 = 5^2$$

Comparing it with the question term

$$\sqrt[4]{x} = 2 \Rightarrow (\sqrt[4]{x})^4 = 2^4 \Rightarrow x = 16$$

Sol.49.(a) Given,

$$x = 2 - \sqrt{3} \Rightarrow \frac{1}{x} = \frac{1}{2 - \sqrt{3}}$$

Applying rationalization

$$\Rightarrow \frac{1}{2 - \sqrt{3}} \times \frac{2 + \sqrt{3}}{2 + \sqrt{3}} = 2 + \sqrt{3}$$

$$\Rightarrow x - \frac{1}{x} = (2 - \sqrt{3}) - (2 + \sqrt{3}) = -2\sqrt{3}$$

$$\Rightarrow x^3 - \frac{1}{x^3} = (-2\sqrt{3})^3 + 3(-2\sqrt{3})$$

$$(\text{If } x - \frac{1}{x} = k, \text{ then } x^3 - \frac{1}{x^3} = k^3 + 3k)$$

$$= -24\sqrt{3} - 6\sqrt{3} = -30\sqrt{3}$$

Sol.50.(b) Short-Trick: Put $x = 0$

$$\left\{ \frac{2(x^3 - 8)}{x^2 - x - 2} \times \frac{x^2 + 2x + 1}{x^2 - 4x - 5} \right\} \div \frac{x^2 + 2x + 4}{3x - 15}$$

$$\Rightarrow \left\{ \frac{2(0 - 8)}{0 - 2} \times \frac{0 + 1}{0 - 5} \right\} \div \frac{0 + 4}{0 - 15}$$

$$\Rightarrow (8 \times \frac{1}{-5}) \div (\frac{4}{-15}) = 6$$

Sol.51.(a) Given,

$$x^4 - 5x^2 - 1 = 0 \Rightarrow x^2 - \frac{1}{x^2} = 5$$

$$\Rightarrow (x^6 - 3x^2 + \frac{3}{x^2} - \frac{1}{x^6} + 1)$$

$$= \{x^6 - \frac{1}{x^6} - 3(x^2 - \frac{1}{x^2}) + 1\}$$

$$\Rightarrow \{(x^2)^3 - \frac{1}{(x^2)^3} - 3(x^2 - \frac{1}{x^2}) + 1\}$$

$$\Rightarrow (x^2 - \frac{1}{x^2})^3 + 1 \Rightarrow 5^3 + 1 = 126$$

Sol.52.(c) Given, $x + y = 7$ and $xy = 10$

$$(x + y)^2 = x^2 + y^2 + 2xy$$

$$\Rightarrow (7)^2 = x^2 + y^2 + 2(10)$$

$$\Rightarrow x^2 + y^2 = 49 - 20 = 29$$

Also,

$$x^3 + y^3 = (x + y)(x^2 + y^2 - xy)$$

Put the values of $(x + y)$,

$$(x^2 + y^2) \text{ and } (xy)$$

$$x^3 + y^3 = (7)(29 - 10) = 133$$

Now,

$$\frac{1}{x^3} + \frac{1}{y^3} \Rightarrow \frac{x^3 + y^3}{(xy)^3} = \frac{133}{10^3} = 0.133$$

Short-Trick:

Using hit and trial method :-

$$\text{put } x = 5 \text{ and } y = 2$$

$$x + y = 7 \Rightarrow 5 + 2 = 7$$

$$xy = 10 \Rightarrow 5 \times 2 = 10$$

Both conditions satisfied clearly

5 and 2 are the values of x and y respectively.

$$\frac{1}{x^3} + \frac{1}{y^3} \Rightarrow \frac{1}{5^3} + \frac{1}{2^3}$$

$$\Rightarrow \frac{1}{125} + \frac{1}{8} = \frac{8 + 125}{125 \times 8} = 0.133$$

Sol.53.(b) Given,

$$(3x - 1)^3 + (4x - 3)^3 + (2x + 1)^3$$

$$= 3(3x - 1)(4x - 3)(2x + 1)$$

$$\Rightarrow (3x - 1)^3 + (4x - 3)^3 + (2x + 1)^3 -$$

$$3(3x - 1)(4x - 3)(2x + 1) = 0$$

We know that

$$a^3 + b^3 + c^3 - 3abc = 0, \text{ Only if either } a + b + c = 0 \text{ or } a = b = c$$

Case 1:

$$(3x - 1) + (4x - 3) + (2x + 1) = 0$$

$$3x + 4x + 2x - 1 - 3 + 1 = 0 \Rightarrow 9x - 3 = 0$$

$$\Rightarrow x = \frac{1}{3} \text{ (But } x \neq \frac{1}{3})$$

Case 2 :

$$(3x - 1) = (4x - 3) = (2x + 1)$$

$$(3x - 1) = (4x - 3) \text{ or } (4x - 3)$$

$$= (2x + 1) \text{ or } (3x - 1) = (2x + 1)$$

$$4x - 3x = 3 - 1 \text{ or } 4x - 2x = 1 + 3 \text{ or } 3x - 2x$$

$$= 1 + 1$$

$$\Rightarrow x = 2 \text{ or } x = \frac{4}{2} = 2 \text{ or } x = 2$$

Sol.54.(b) Put $x = 1, y = 1$ and $z = -2$

$$x + y + z = 0 \Rightarrow 1 + 1 + (-2) = 0$$

$$0 = 0 \text{(condition satisfied)}$$

$$(x^2 + y^2 + z^2) \div (z^2 - xy)$$

$$\Rightarrow [(1)^2 + (1)^2 + (-2)^2] \div [(-2)^2 - 1 \times 1]$$

$$\Rightarrow 6 \div 3 = 2$$

Sol.55.(c) Given,

$$x^2 - 3x - 1 = 0 \Rightarrow x^2 - 1 = 3x \Rightarrow x - \frac{1}{x} = 3$$

$$\text{Now, } (x^2 + 8x - 1)(x^3 + x^{-1})^{-1}$$

$$\Rightarrow \frac{(x^2 + 8x - 1)}{(x^3 + \frac{1}{x})} \Rightarrow \frac{(x^3 + 8x^2 - x)}{(x^4 + 1)}$$

$$\Rightarrow \frac{x^2(x - \frac{1}{x} + 8)}{x^2(x^2 + \frac{1}{x^2})} \Rightarrow \frac{x - \frac{1}{x} + 8}{x^2 + \frac{1}{x^2}}$$

$$\Rightarrow \frac{3 + 8}{3^2 + 2} = 1 \dots (\text{If } x - \frac{1}{x} = k, \text{ then } x^2 + \frac{1}{x^2} = k^2 + 2)$$

Sol.56.(c) $x = 2 - p \Rightarrow x + p = 2$

Cubing both sides

$$(x + p)^3 = 2^3$$

$$\Rightarrow x^3 + p^3 + 3xp(x + p) = 8$$

$$\Rightarrow x^3 + p^3 + 3xp(2) = 8$$

$$\Rightarrow x^3 + 6xp + p^3 = 8$$

Sol.57.(a) $2x^2 + y^2 + 6x - 2xy + 9 = 0$

$$\Rightarrow x^2 + 6x + 9 + x^2 + y^2 - 2xy = 0$$

$$\Rightarrow (x + 3)^2 + (x - y)^2 = 0$$

Sum of the squares of two numbers is equal to zero only if the numbers themselves are zero.

$$\Rightarrow (x + 3) = 0 \text{ and } (x - y) = 0 \Rightarrow x = y = -3$$

Now,

$$4x^3 - y^3 + x^2y^2 \Rightarrow 3x^3 + x^4 \dots (x = y)$$

$$\Rightarrow x^3(3 + x) \Rightarrow (-3)^3(3 - 3) = 0$$

Sol. 58.(a) Given, $ab + bc + ca = 8$ and

$$a^2 + b^2 + c^2 = 20$$

We know that,

$$\Rightarrow (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) = 20 + 2(8) = 36$$

$$\Rightarrow (a + b + c) = 6$$

$$\Rightarrow \frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$$

$$= a^3 + b^3 + c^3 - 3abc = (a + b + c)\{(a + b + c)^2 - 3(ab + bc + ca)\}$$

$$\Rightarrow 6\{(6)^2 - 3 \times 8\} = 6(36 - 24) = 6 \times 12 = 72$$

Sol.59.(b) $x + y = 1$

Squaring both sides

$$x^2 + y^2 + 2xy = 1 \Rightarrow x^2 + y^2 = 1 - 2xy$$

Again squaring both sides

$$x^4 + y^4 + 2x^2y^2 = 1 + 4x^2y^2 - 4xy$$

$$x^4 + y^4 = 1 + 2x^2y^2 - 4xy$$

$$x^4 + y^4 = 1 + 2(x^2y^2 - 2xy)$$

$$\text{Put } (x^2y^2 - 2xy = 12) = 1 + 2(12) = 25$$

Sol.60.(d)

$$16x^2 + 9y^2 + 4z^2 = 24(x - y + z) - 61$$

$$\Rightarrow (16x^2 - 24x + 9) + (9y^2 + 24y + 16)$$

$$+ (4z^2 - 24z + 36) = 0$$

$$\Rightarrow (4x - 3)^2 + (3y + 4)^2 + (2z - 6)^2 = 0$$

$$\Rightarrow x = \frac{3}{4}, y = -\frac{4}{3}, z = 3$$

Therefore,

$$xy + 2z = \frac{3}{4} \times (-\frac{4}{3}) + 2 \times 3$$

$$= (-1) + 6 = 5$$

Sol.61.(d) We know that,

$$(a^3 - b^3) = (a - b)(a^2 + b^2 + ab)$$

$$\text{So, } 5\sqrt{5}x^3 - 81\sqrt{3}y^3$$

$$= (\sqrt{5}x - 3\sqrt{3}y)[(\sqrt{5}x)^2 + (3\sqrt{3}y)^2 + \sqrt{5}x \times 3\sqrt{3}y]$$

$$= (\sqrt{5}x - 3\sqrt{3}y)(5x^2 + 27y^2 + 3\sqrt{15}xy)$$

Put this value in question term,

$$\frac{5\sqrt{5}x^3 - 81\sqrt{3}y^3}{(\sqrt{5}x - 3\sqrt{3}y)} = Ax^2 + By^2 + Cxy$$

$$\Rightarrow \frac{(\sqrt{5}x - 3\sqrt{3}y)(5x^2 + 27y^2 + 3\sqrt{15}xy)}{(\sqrt{5}x - 3\sqrt{3}y)}$$

$$= Ax^2 + By^2 + Cxy$$

$$\Rightarrow (5x^2 + 27y^2 + 3\sqrt{15}xy)$$

$$= Ax^2 + By^2 + Cxy$$

Comparing the coefficients of x^2 , y^2 and xy

$$A = 5, B = 27 \text{ and } C = 3\sqrt{15}$$

$$(6A + B - \sqrt{15}C) = 6(5) + 27 - (\sqrt{15} \times 3\sqrt{15}) = 12$$

Sol.62.(a) We know that

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

Put the given values

$$19^2 = 133 + 2(xy + yz + y^2)$$

$$114 = y(x + y + z) \Rightarrow y = 6$$

$$x + z = 19 - 6 = 13 \Rightarrow xz = 6^2 = 36$$

Only 9 and 4 are such pairs whose sum is 13 and multiplication is 36

$$\text{Required difference} = 9 - 4 = 5$$

$$\text{Sol.63.(d)} x^4 + \frac{1}{x^4} = 194$$

$$\Rightarrow x^2 + \frac{1}{x^2} = \sqrt{194 + 2} = 14$$

$$\Rightarrow x + \frac{1}{x} = \sqrt{14 + 2} = 4$$

$$= x^2 + 1 = 4x \Rightarrow x^2 - 4x = -1$$

$$(x - 2)^2 = 4 - 1 = 3$$

Sol.64.(b) We know the roots of an

$$\text{equation} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

here, $a = 1$, $b = -4$ and $c = a$

According to question

$$\frac{-(-4) + \sqrt{(-4)^2 - 4a(1)}}{2(1)} \dots\dots\dots (1)$$

$$= \frac{-(-4) - \sqrt{(-4)^2 - 4a(1)}}{2(1)} \dots\dots\dots (2)$$

Equating eq. (1) and (2), we get

$$2\sqrt{16 - 4a} = 0$$

$$16 = 4a \text{ so, } a = 4 \text{ ans.}$$

Short trick :

For real and equal roots,

$$D = b^2 - 4ac = 0$$

$$a = 1, b = -4 \text{ and } c = a$$

$$\Rightarrow (-4)^2 - 4a = 0 \Rightarrow 4a = 16 \Rightarrow a = 4$$

Sol.65.(c)

$$(a - 2b) = 8, 2ab = 9$$

$$\therefore (a - 2b)^2 = (a + 2b)^2 - 8ab$$

$$(a + 2b)^2 = 8^2 + 8 \times \frac{9}{2}$$

$$(a + 2b)^2 = 64 + 36 = 100$$

$$(a + 2b) = 10$$

Sol.66.(d) According to question,

$$A + 10 = (B - 10) \times 2$$

$$A - 2B = -30 \dots\dots\dots (i)$$

$$A - 10 = B + 10$$

$$A - B = 20 \dots\dots\dots (ii)$$

From equation (ii) $\times 2$ - (i), we get, $A = 70$

$$\text{Sol.67.(a)} \text{ Given, } x + y = 2 \text{ and } \frac{x}{y} + \frac{y}{x} = 1$$

$$\Rightarrow x^2 + y^2 = xy$$

Formula :

$$x^3 + y^3 = (x + y)(x^2 + y^2 - xy)$$

$$\Rightarrow x^3 + y^3 = 2 \times 0 = 0$$

Sol.68.(a) According to the question,

$$\frac{A}{L} = 1 - \frac{M}{B} = \frac{B - M}{B}$$

$$\frac{L}{A} = \frac{B}{B - M} \text{ and } \frac{N}{C} = 1 - \frac{B}{M} = \frac{M - B}{M}$$

$$= \frac{C}{N} = \frac{M}{M - B}$$

$$= \frac{L}{A} + \frac{C}{N} = \frac{B}{B - M} - \frac{M}{B - M} = \frac{B - M}{B - M} = 1$$

$$\text{Sol.69.(c)} \text{ Given, } a \propto b \propto \frac{1}{c} \propto \frac{1}{d}$$

$$\text{So, } a \propto \frac{1}{d}$$

$$\text{Sol.70.(d)} A + \frac{1}{1 + \frac{1}{2 + \frac{1}{3}}} = \frac{9}{10}$$

$$\Rightarrow A + \frac{1}{1 + \frac{1}{\frac{7}{3}}} = \frac{9}{10}$$

$$\Rightarrow A + \frac{1}{1 + \frac{3}{7}} = \frac{9}{10} \Rightarrow A + \frac{7}{10} = \frac{9}{10}$$

$$\Rightarrow A = \frac{2}{10} \text{ or } \frac{1}{5}$$

$$\text{Sol.71.(d)} x(5 - \frac{2}{x}) = \frac{5}{x}$$

$$\Rightarrow x(\frac{5x - 2}{x}) = \frac{5}{x} \Rightarrow 5x^2 - 5 = 2x$$

Dividing both sides by $5x$

$$\Rightarrow x - \frac{1}{x} = \frac{2}{5}$$

$$\text{then } x^2 + \frac{1}{x^2} = \frac{4}{25} + 2 = \frac{54}{25}$$

Sol.72.(d) Formula used :-

$$a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$$

$$\frac{(157 \times 157) + (157 \times 133) + (133 \times 133)}{(157 \times 157 \times 157) - (133 \times 133 \times 133)}$$

Let $157 = a$ and $133 = b$ then,

$$\frac{a^2 + ab + b^2}{(a - b)(a^2 + b^2 + ab)} = \frac{1}{(a - b)}$$

$$= \frac{1}{157 - 133} = \frac{1}{24}$$

Sol.73.(a) Formula used : - $a^3 - b^3$

$$= (a - b)(a^2 + b^2 + ab)$$

Let the $a = 3.254$ and $b = 0.746$

$$\frac{(a - b)(a^2 + b^2 + ab)}{(a^2 + b^2 + ab)} = (a - b)$$

$$= 3.254 - 0.746 = 2.508$$

$$\text{Sol.74.(c)} \frac{a^2 - (b - c)^2}{(a + c)^2 - b^2} + \frac{b^2 - (a - c)^2}{(a + b)^2 - c^2}$$

$$+ \frac{c^2 - (a - b)^2}{(b + c)^2 - a^2}$$

$$= \frac{\{(a + (b - c))(a - (b - c))\}}{\{(a + c - b)(a + c + b)\}} +$$

$$\frac{\{(b + (a - c))(b - (a - c))\}}{\{(a + b - c)(a + b + c)\}} +$$

$$\frac{\{(c + (a - b))(c - (a - b))\}}{\{(b + c - a)(b + c + a)\}}$$

$$= \frac{\{(a + (b - c))\}}{\{(a + c + b)\}} + \frac{\{(b - (a - c))\}}{\{(a + b + c)\}} +$$

$$\frac{\{(c + (a - b))\}}{\{(b + c + a)\}}$$

$$= \frac{\{(a + (b - c)) + (b - (a - c)) + (c + (a - b))\}}{\{(a + c + b)\}}$$

$$= \frac{(a + c + b)}{(a + c + b)} = 1$$

Sol.75.(c)

$$\Rightarrow (1 + x)^3 + (1 - x)^3 + (-2)^3$$

$$\Rightarrow (1^3 + x^3 + 3x(1 + x) +$$

$$(1^3 - x^3 - 3x(1 - x)) + (-2)^3$$

$$\Rightarrow 6x^2 + 2 \cdot 6x^2 - 6 = -6(1 - x^2)$$

Sol.76.(a) Formula :

$$(a + b)^2 - (a - b)^2 = 4ab$$

$$(1 - 2x)^2 - (1 + 2x)^2$$

$$= -4 \times 1 \times 2x = -8x$$

Sol.77.(a) $a + b + c = 5 \dots\dots\dots (1)$

and $ab + bc + ca = 7 \dots\dots\dots (2)$

Squaring both side in equation (1)

$$(a + b + c)^2 =$$

$$a^2 + b^2 + c^2 + 2(ab + bc + ca)]$$

$$\Rightarrow a^2 + b^2 + c^2 = 25 - 14 = 11$$

$$\text{Then, } a^3 + b^3 + c^3 - 3abc =$$

$$(a + b + c)[a^2 + b^2 + c^2 - (ab + bc + ca)]$$

$$\Rightarrow a^3 + b^3 + c^3 - 3abc = 5(11 - 7) = 20$$

$$\text{Sol.78.(d)} \Rightarrow (x - 2y)(y - 3x) + (x + y)(x - y) + (x - 3y)(2x + y)$$

$$\Rightarrow xy - 3x^2 - 2y^2 + 6xy + x^2 - y^2 + 2x^2 +$$

$$xy - 6xy - 3y^2$$

$$\Rightarrow 2xy - 6y^2 = 2y(x - 3y)$$

Sol.79.(b) Put $x = 2$ in equation

$$2x^2 + 12kx - 25k = 0$$

$$2 \times (2)^2 + 12k \times 2 - 25k = 0 \Rightarrow k = 8$$

Sol.80.(d) $2x + \frac{2}{x} = 5 \Rightarrow x + \frac{1}{x} = \frac{5}{2}$

$$x^3 + \frac{1}{x^3} = \left(\frac{5}{2}\right)^3 - 3 \times \left(\frac{5}{2}\right) = \frac{65}{8}$$

$$x^3 + \frac{1}{x^3} + 2 = \frac{65}{8} + 2 = \frac{81}{8}$$

Sol.81.(d)

$$(a^3 - b^3) = (a - b)\{(a - b)^2 + 3ab\}$$

$$4401 = 9\{81 + 3ab\}$$

$$4401 = 9\{81 + 3ab\}$$

$$489 - 81 = 3ab \Rightarrow ab = \frac{408}{3} = 136$$

Sol.82.(c) According to question,

$$a + \frac{1}{a} = 6 \Rightarrow a^2 + \frac{1}{a^2} = (6)^2 - 2 = 34$$

$$\text{Now, } \frac{3}{4} \left(a^2 + \frac{1}{a^2}\right) = \frac{3}{4} \times 34 = \frac{51}{2} = 25.5$$

Sol.83.(b) $a = \frac{1}{a - \sqrt{6}} \Rightarrow a - \frac{1}{a} = \sqrt{6}$

$$a + \frac{1}{a} = \sqrt{(\sqrt{6})^2 + 4} = \sqrt{10}$$

Sol.84.(c)

$$(a + b) = 30 \Rightarrow a^2 + b^2 = 900 - 2ab$$

$$\sqrt{\frac{a}{b}} = \frac{8}{3} + \sqrt{\frac{b}{a}}$$

$$\frac{a-b}{\sqrt{ab}} = \frac{8}{3} \Rightarrow \frac{a^2 + b^2 - 2ab}{ab} = \frac{64}{9}$$

$$\frac{900 - 4ab}{ab} = \frac{64}{9} \Rightarrow ab = \frac{8100}{100} = 81$$

Sol.85.(c) $2x + 3y = 9$ and $xy = 3$

Then,

$$8x^3 + 27y^3 = (2x + 3y)$$

$$\{(2x + 3y)^2 - 3(2x \times 3y)\}$$

$$= (9)\{(9)^2 - 18xy\} = (9)\{81 - 18 \times 3\}$$

$$= 243$$

Sol.86.(b) $a^2 + \frac{1}{a^2} = \frac{7}{3}$

$$a - \frac{1}{a} = \sqrt{\left(\frac{7}{3} - 2\right)} = \frac{1}{\sqrt{3}}$$

$$\text{Now, } a^3 - \frac{1}{a^3} = \left(\frac{1}{\sqrt{3}}\right)^3 + 3 \times \frac{1}{\sqrt{3}}$$

$$= \frac{1}{3\sqrt{3}} + \sqrt{3} = \frac{10}{3\sqrt{3}}$$

Sol.87.(d) $a + b = 10$ and $a^2 + b^2 = 58$

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$100 = 58 + 2ab \Rightarrow ab = 21$$

Sol.88.(c) $a^3 + b^3 + c^3 - 3abc =$

$$(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

Sol.89.(c)

$$2a + 3b = 14 \text{ and } 2a - 3b = 10$$

On adding, we get

$$4a = 24 \Rightarrow a = 6 \text{ and } b = \frac{2}{3}$$

Now, $ab = 4$

Sol.90.(a) $w - \frac{1}{w} = \sqrt{(6)^2 - 4} = 4\sqrt{2}$

Sol.91.(a) $(a - b) = 10$

$$\Rightarrow a^2 + b^2 = 100 + 2ab$$

$$\sqrt{\frac{a}{b}} = \frac{8}{3} - \sqrt{\frac{b}{a}}$$

$$\frac{a+b}{\sqrt{ab}} = \frac{8}{3} \Rightarrow \frac{a^2 + b^2 + 2ab}{ab} = \frac{64}{9}$$

$$\frac{100 + 4ab}{ab} = \frac{64}{9}$$

$$ab = \frac{900}{28} = \frac{225}{7} = 32\frac{1}{7}$$

Sol.92.(b)

$$\text{Identity :- } (a + b)^2 = a^2 + b^2 + 2ab$$

Sol.93.(c) $\frac{x^2 + 7x + 2}{x^2 + 13x + 2} = \frac{x + \frac{2}{x} + 7}{x + \frac{2}{x} + 13}$

$$= \frac{1 + 7}{1 + 13} = \frac{4}{7}$$

Sol.94.(d) $x + \frac{1}{x} = 7$

$$x - \frac{1}{x} = \sqrt{(7)^2 - 4} = 3\sqrt{5}$$

$$\text{Now, } x^3 - \frac{1}{x^3} = (3\sqrt{5})^3 + 3 \times (3\sqrt{5})$$

$$= 144\sqrt{5}$$

Sol.95.(b) $y - \frac{1}{y} = 9$

$$y^3 - \frac{1}{y^3} = (9)^3 + 3 \times 9 = 756$$

Sol.96.(c) $(2x + 3y)^2 = 4x^2 + 9y^2 + 12xy$

$$16 = 64 + 12xy \Rightarrow xy = (-4)$$

$$\text{Now, } (2x + 3y)^3 = 8x^3 + 27y^3 + 3 \times 2x \times 3y(2x + 3y)$$

$$64 = 8x^3 + 27y^3 - 72(4)$$

$$8x^3 + 27y^3 = 288 + 64 = 352$$

Sol.97.(d) $(a + b + c)^2 = a^2 + b^2 + c^2 + 2$

$$(ab + bc + ca)$$

$$a^2 + b^2 + c^2 = 225 - 2 \times 22 = 181$$

Sol.98.(b)

$$[(x - 5)(x - 1)] - [(9x - 5)(9x - 1)] \div 16x$$

$$\Rightarrow \{[x^2 - 6x + 5] - [81x^2 - 54x + 5]\} \div 16x$$

$$\Rightarrow \{[-80x^2 + 48x]\} \div 16x$$

$$\Rightarrow [16x\{-5x + 3\}] \div 16x$$

$$\Rightarrow -\{5x - 3\}$$

Sol.99.(a) $x + \frac{1}{x} = 2 \Rightarrow x = 1$

$$x^{1823} + \frac{1}{x^{1929}} = 1 + 1 = 2$$

Sol.100.(d) $2x^2 - 32 = 2(x^2 - 16)$

$$= 2(x - 4)(x + 4)$$

Sol.101.(d) $x - \frac{1}{x} = 0.4$

$$x^2 + \frac{1}{x^2} = (0.4)^2 + 2 = 2.16$$

Sol.102.(d) $x + \frac{1}{x} = 5 \Rightarrow x^2 + \frac{1}{x^2} = 23$

$$\Rightarrow x^3 + \frac{1}{x^3} = 110$$

Now,

$$x^5 + \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 + \frac{1}{x^3}) - (x + \frac{1}{x})$$

$$\Rightarrow 23 \times 110 - 5 = 2530 - 5 = 2525$$

Sol.103.(c)

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2$$

$$(xy + yz + zx)$$

$$(x + y + z) = \sqrt{628 + 2 \times 24}$$

$$= \sqrt{676} = 26$$

Sol.104.(b)

$$x + \frac{1}{x} = -1 \Rightarrow x^3 = 1 \text{ or } x^{15} = 1$$

Now,

$$x^{15} + \frac{1}{x^{15}} = 1 + 1 = 2$$

Sol.105.(d) $\frac{x^3 + y^3 + z^3 - 3xyz}{(x - y)^2 + (y - z)^2 + (z - x)^2}$

$$\frac{\frac{(x + y + z)}{2} \{(x - y)^2 + (y - z)^2 + (z - x)^2\}}{(x - y)^2 + (y - z)^2 + (z - x)^2}$$

$$= \frac{(x + y + z)}{2}$$

Sol.106.(c) $(x - 2) - \frac{1}{(x - 2)} = 5$

$$(x - 2)^3 - \frac{1}{(x - 2)^3} = 5^3 + 3 \times 5 = 140$$

Sol.107.(a) $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

$$256 = a^2 + b^2 + c^2 + 2 \times 81$$

$$a^2 + b^2 + c^2 = 94$$

Sol.108.(c)

$$(a + b + c)^2 - (a - b + c)^2 + 4ac$$

$$\Rightarrow 2ab + 2bc + 2ac + 2ab + 2bc - 2ac + 4ac$$

$$\Rightarrow 4(ab + bc + ca)$$

Sol.109.(b) $x + \frac{1}{x} = 5\sqrt{5}$

$$\Rightarrow x - \frac{1}{x} = \sqrt{(5\sqrt{5})^2 - 4} = 11$$

$$\Rightarrow x^3 - \frac{1}{x^3} = (11)^3 + 3 \times 11 = 1364$$

Sol.110.(c) $x^4 + \frac{1}{x^4} = 194$

$$x^2 + \frac{1}{x^2} = \sqrt{194 + 2} = 14$$

$$x + \frac{1}{x} = \sqrt{14 + 2} = 4$$

$$x^3 + \frac{1}{x^3} = 4^3 - 3 \times 4 = 52$$

$$\text{Now, } x^3 + \frac{1}{x^3} + x + \frac{1}{x} = 52 + 4 = 56$$

Sol.111.(a)

$$a^3 + b^3 + c^3 - 3abc = \frac{a+b+c}{2}$$

$$\left\{ 3(a^2 + b^2 + c^2) - (a+b+c)^2 \right\}$$

$$= \frac{5}{2} \{ 3(15) - (5)^2 \} = \frac{5}{2} \{ (20) \} = 50$$

$$\text{Now, } a^3 + b^3 + c^3 - 3abc - 27 = 50 - 27 = 23$$

$$\text{Sol.112.(c)} \quad x + y = \sqrt{427 + 404} = \sqrt{831}$$

$$x - y = \sqrt{427 - 404} = \sqrt{23}$$

$$\text{Now, } \frac{x+y}{x-y} = \sqrt{\frac{831}{23}}$$

Sol.113.(c)

$$(y - \frac{1}{y}) = -9 \Rightarrow (y^2 + \frac{1}{y^2}) = 83$$

$$\text{and } (y^3 - \frac{1}{y^3}) = (-9)^3 + 3 \times (-9)$$

$$= -756$$

$$y^5 - \frac{1}{y^5} = \{(y^2 + \frac{1}{y^2})(y^3 - \frac{1}{y^3})\} - (y - \frac{1}{y})$$

$$= \{83 \times (-756)\} - (-9)$$

$$= \{(-62748)\} - (-9) = (-62739)$$

$$\text{Sol.114.(a)} \quad 4x^2 - y^2 + 6y - 9$$

$$= 4x^2 - (y^2 - 6y + 9) \Rightarrow (2x)^2 - (y - 3)^2$$

$$= (2x + y - 3)(2x - y + 3)$$

Sol.115.(b)

$$\text{If } x - \frac{1}{x} = k \text{ then, } x^3 - \frac{1}{x^3} = k^3 + 3k$$

$$x^3 = \sqrt{6} - 1 \Rightarrow \frac{1}{x^3} = \frac{1}{\sqrt{6} - 1}$$

$$x^3 - \frac{1}{x^3} = (\sqrt{6} - 1) - \frac{1}{\sqrt{6} - 1}$$

$$= \frac{7 - 2\sqrt{6} - 1}{\sqrt{6} - 1} = \frac{6 - 2\sqrt{6}}{\sqrt{6} - 1} \times \frac{\sqrt{6} + 1}{\sqrt{6} + 1}$$

$$= \frac{6\sqrt{6} + 6 - 12 - 2\sqrt{6}}{5} = \frac{4\sqrt{6} - 6}{5}$$

$$\text{Sol.116.(b)} \quad ab + bc + ca$$

$$= \frac{(a+b+c)^2 - (a^2 + b^2 + c^2)}{2}$$

$$= \frac{(17)^2 - (101)}{2} = 94$$

Now,

$$(a-b)^2 + (b-c)^2 + (c-a)^2 =$$

$$2(a^2 + b^2 + c^2) - 2(ab + bc + ca)$$

$$= 2 \times 101 - 2 \times 94 = 14$$

$$\text{Sol.117.(a)} \quad x + \frac{1}{x} = 5$$

$$x^2 + \frac{1}{x^2} = 5^2 - 2 = 23$$

$$x^4 + \frac{1}{x^4} = 23^2 - 2 = 527$$

$$\text{Now, } x^4 + \frac{1}{x^4} - 36 = 527 - 36 = 491$$

$$\text{Sol.118.(c)} \quad a^2 + b^2 + c^2 = ab + bc + ca$$

only possible when $a = b = c$

$$\text{So, } \frac{11a^4 + 13b^4 + 15c^4}{16a^2b^2 + 19b^2c^2 + 17c^2a^2} = \frac{39a^4}{52a^4}$$

$$= \frac{3}{4}$$

$$\text{Sol.119.(a)} \quad (c+d)^2 - (c-d)^2 = 4cd$$

Sol.120.(b)

$$\frac{x^2 - 3x + 2}{x^2 - 4} = \frac{x^2 + 1 - 2x - x + 1}{(x-2)(x+2)}$$

$$= \frac{(x-1)^2 - (x-1)}{(x-2)(x+2)} = \frac{(x-1)(x-2)}{(x-2)(x+2)}$$

$$= \frac{(x-1)}{(x+2)}$$

$$\text{Sol.121.(a)} \quad \text{If } (l+m+n) = 0 \text{ then,}$$

$$l^3 + m^3 + n^3 = 3lmn$$

$$\left(\frac{l^2}{mn} + \frac{m^2}{nl} + \frac{n^2}{lm} \right)^2$$

$$= \left(\frac{l^3 + m^3 + n^3}{lmn} \right)^2 = \left(\frac{3lmn}{lmn} \right)^2 = 9$$

$$\text{Sol.122.(a)} \quad 2x + 5y = 15, \quad xy = 6$$

$$(2x + 5y)^2 = 4x^2 + 25y^2 + 20xy$$

$$4x^2 + 25y^2 = 225 - 120 = 105$$

$$\text{Sol.123.(c)} \quad x + \frac{1}{x} = \sqrt{9.76}$$

$$x - \frac{1}{x} = \sqrt{(\sqrt{9.76})^2 - 4}$$

$$= \sqrt{9.76 - 4} = 2.4$$

$$x^3 - \frac{1}{x^3} = (2.4)^3 + 3 \times 2.4$$

$$= 13.824 + 7.2 = 21.024$$

$$\text{Sol.124.(c)} \quad 5a + \frac{4}{a} = 15$$

$$25a^2 + \frac{16}{a^2} = (15)^2 - 2 \times 5 \times 4$$

$$= 225 - 40 = 185$$

Sol.125.(b)

$$(x^3 - y^3) = (x - y)(x^2 + y^2 + xy)$$

Here,

$$\frac{(x^3 - y^3)(x + y)}{x^2 + y^2 + xy} = (x - y)(x + y)$$

$$= x^2 - y^2$$

Sol.126.(c)

$$\frac{36a^2 - 49b^2}{6a + 7b} = \frac{(6a + 7b)(6a - 7b)}{6a + 7b}$$

$$= (6a - 7b)$$

$$\text{Sol.127.(d)} \quad (a+b)^2 + (b+c)^2 +$$

$$(c+a)^2 = a^2 + b^2 + c^2 + (a+b+c)^2$$

$$= 115 + (17)^2 = 115 + 289 = 404$$

$$\text{Sol.128.(d)} \quad (2a - 3b - c)^2 - (a + 2b + c)^2$$

$$= 4a^2 + 9b^2 + c^2 - 12ab + 6bc - 4ac -$$

$$a^2 - 4b^2 - c^2 - 4ab - 4bc - 2ac$$

$$= 3a^2 + 5b^2 - 16ab + 2bc - 6ac$$

$$\text{Sol.129.(b)} \quad x + \frac{1}{x} = 3.2$$

$$x - \frac{1}{x} = \sqrt{(3.2)^2 - 4} = \sqrt{6.24}$$

$$\text{Now, } x^2 - \frac{1}{x^2} = 3.2 \times \sqrt{6.24} = 12.8$$

$$\sqrt{0.39}$$

$$\text{Sol.130.(c)} \quad x^2 + y^2 + z^2 = 2(x + z - 1)$$

$$x^2 - 2x + 1 + z^2 - 2z + 1 + y^2 = 0$$

$$(x-1)^2 + (z-1)^2 + (y)^2 = 0$$

$$\Rightarrow x = 1, z = 1 \text{ and } y = 0$$

$$\text{Now, } x^3 + y^3 + z^3 = 1 + 0 + 1 = 2$$

Sol.131.(c) Formula:-

$$(a+b)^3 = a^3 + b^3 + 3ab(a+b)$$

$$(x+3y)^3 = x^3 + 27y^3 + 9xy(x+3y)$$

$$216 = x^3 + 27y^3 + 54xy$$

$$\text{Sol.132.(c)} \quad \text{Given, } p+q=12, pq=14$$

$$\text{Then, } p^2 - pq + q^2 = (p+q)^2 - 3pq$$

$$= 144 - 3 \times 14 = 102$$

$$\text{Sol.133.(d)} \quad \frac{(x+3)^3 + (x-3)^3}{x^2 + 27}$$

$$= \frac{(x^3 + 3^3 + 9x(x+3) + (x^3 - 3^3 - 9x(x-3)))}{x^2 + 27}$$

$$= \frac{2x^3 + 27x + 27x}{x^2 + 27} \Rightarrow \frac{2x(x^2 + 27)}{x^2 + 27} = 2x$$

Sol.134.(a)

$$\frac{1}{8} \left[\frac{1}{b-1} - \frac{1}{b+1} - \frac{2}{b^2+1} - \frac{4}{b^4+1} \right]$$

$$= \frac{1}{8} \left[\frac{b+1-b+1}{b^2-1} - \frac{2}{b^2+1} - \frac{4}{b^4+1} \right]$$

$$= \frac{1}{8} \left[\frac{2}{b^2-1} - \frac{2}{b^2+1} - \frac{4}{b^4+1} \right]$$

$$= \frac{1}{8} \left[\frac{2(b^2+1-b^2+1)}{b^4-1} - \frac{4}{b^4+1} \right]$$

$$= \frac{1}{8} \left[\frac{2(2)}{b^4-1} - \frac{4}{b^4+1} \right]$$

$$= \frac{1}{8} \left[\frac{4(b^4+1-b^4+1)}{b^8-1} \right] = \frac{1}{8} \left[\frac{8}{b^8-1} \right]$$

$$= \frac{1}{b^8-1}$$

Sol.135.(c)

$$(d+e+f) = 14, \text{ squaring both side}$$

$$\{d^2 + e^2 + f^2 + 2(de+ef+fd)\} = 196$$

$$2(de+ef+fd) = 196 - 96 = 100$$

$$de+ef+fd = 50$$

Sol.136.(a)

$$\text{Given:- } g = (2 - \sqrt{3}), \frac{1}{g} = (2 + \sqrt{3})$$

$$\text{Then, } g - \frac{1}{g} = -2\sqrt{3}$$

$$\text{Therefore, } (g^3 - \frac{1}{g^3})$$

$$= (-2\sqrt{3})^3 - 3 \times 2\sqrt{3} = -24\sqrt{3} - 6\sqrt{3} = -30\sqrt{3}$$

Sol.137.(c)

$$\begin{aligned} \text{Given:- } & (2x + 13 - y)(2x - 13 - y) \\ & \{(2x - y) + 13\} \{(2x - y) - 13\} \\ & (2x - y)^2 - (13)^2 \\ & 4x^2 + y^2 - 4xy - 169 \end{aligned}$$

Sol.138.(d)

$$\begin{aligned} & (3x - 2y)(3x + 2y)(9x^2 + 4y^2) \\ & = (9x^2 - 4y^2)(9x^2 + 4y^2) \Rightarrow (81x^4 - 16y^4) \end{aligned}$$

Sol.139.(b)

$$\begin{aligned} \text{Formula : } & \{(a + b)(a - b) = a^2 - b^2\} \\ & \{(5p + 3q)(5p - 3q) = 25p^2 - 9q^2\} \end{aligned}$$

$$\text{Sol.140.(d) Given, } p^2 - 4p - 1 = 0 \text{ -----(1)}$$

Dividing both side by p ,

$$p - \frac{1}{p} = 4$$

Squaring both sides of equation (1)

$$p^2 + \frac{1}{p^2} = (4)^2 + 2 = 18$$

$$\text{So, } p^2 + 3p + \frac{1}{p^2} - \frac{3}{p} = p^2 + \frac{1}{p^2} + 3$$

$$(p - \frac{1}{p}) = 18 + 3(4) = 30$$

$$\text{Sol.141.(d) } \frac{a+b}{a-b} \div \frac{(a+b)^2}{(a^2-b^2)}$$

$$\begin{aligned} & \frac{a+b}{a-b} \times \frac{(a^2-b^2)}{(a+b)^2} \\ & = \frac{(a+b)(a+b)(a-b)}{(a-b)(a+b)(a+b)} = 1 \end{aligned}$$

$$\text{Sol.142.(d) Given } 3x - 2y = 3 \text{ and } xy = 9$$

$$\begin{aligned} \text{So, } & 3x + 2y = \sqrt{(3x - 2y)^2 + 4ab} \\ & = \sqrt{9 + 4 \times 9 \times 3 \times 2} = \sqrt{225} = 15 \\ & \{\text{where } a = 3x \text{ and } b = 2y\} \end{aligned}$$

$$\text{Sol.143.(d) } px + qy = 1$$

$$\text{Put } p = 1, q = 0$$

$$\text{Then, } x = 1$$

$$\text{and } qx + py = \frac{2pq}{p^2 + q^2} \text{ -----(1)}$$

Put $x = 1$ in the equation (1)

$$y = \frac{2 \times 1 \times 0}{1^2 + 0^2} = 0$$

$$\text{So, } (x^2 + y^2)(p^2 + q^2) = (1 + 0)(1 + 0) = 1$$

$$\text{Sol.144.(c) } 7b - \frac{1}{4b} = 7$$

Multiply by $\frac{4}{7}$, we get

$$4b - \frac{1}{7b} = 4$$

$$16b^2 + \frac{1}{49b^2} = (4)^2 + 2 \times 4 \times \frac{1}{7}$$

$$= 16 + \frac{8}{7} = \frac{120}{7}$$

$$\begin{aligned} \text{Sol.145.(d) } & 3x^3 + 5x^2y + 12xy^2 + 7y^3 \\ & = 3(-64) + 5(16)(-1) + 12(-4)(1) + 7(-1) \\ & = -192 - 80 - 48 - 7 = -327 \end{aligned}$$

Sol.146.(b)

$$\begin{aligned} (a + b + c)^2 &= \{a^2 + b^2 + c^2 + 2(ab + bc + ca)\} \\ 256 &= \{90 + 2(ab + bc + ca)\} \end{aligned}$$

$$(ab + bc + ca) = \frac{256 - 90}{2} = 83$$

$$\text{Sol.147.(c) Formula : } a^3 + b^3 + c^3 - 3abc$$

$$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\text{And we know, If } (a + b + c) = 0$$

Then,

$$a^3 + b^3 + c^3 = 3abc$$

Sol.148.(b)

$$\text{If, } a + \frac{1}{a} = k, \text{ then } a^2 + \frac{1}{a^2} = k^2 - 2$$

$$\text{and } a^3 + \frac{1}{a^3} = k^3 - 3k$$

$$\text{Here } x + \frac{1}{x} = 7,$$

$$x^3 + \frac{1}{x^3} = 7^3 - 3 \times 7 = 322$$

$$x^6 + \frac{1}{x^6} = 322^2 - 2 = 103684 - 2 = 103682$$

$$\text{Sol.149.(d) } (a^3 - b^3 - 3a^2b + 3ab^2)$$

$$= (a - b)^3 = (17 - 13)^3 = 64$$

$$\text{Sol.150.(c) } \frac{x^4 - 2x^2 + 1}{x^2 - 2x + 1} = \frac{(x^2 - 1)^2}{(x - 1)(x - 1)}$$

$$\begin{aligned} & = \frac{(x + 1)(x - 1)(x + 1)(x - 1)}{(x - 1)(x - 1)} \\ & = (x + 1)(x + 1) = x^2 + 2x + 1 \end{aligned}$$

$$\text{Sol.151.(b) } x - \frac{1}{x} = \sqrt{6} \Rightarrow (x - \frac{1}{x})^2 = 6$$

$$\Rightarrow x^2 + \frac{1}{x^2} = (\sqrt{6})^2 + 2 = 8$$

$$\text{Similarly, } x^4 + \frac{1}{x^4} = 62 \text{ and}$$

$$x + \frac{1}{x} = \sqrt{(\sqrt{6})^2 + 4} = \sqrt{10}$$

now

$$x^8 - \frac{1}{x^8} = [(x^4) + \frac{1}{(x^4)}] [(x^2 + \frac{1}{x^2})$$

$$[(x^2 - \frac{1}{x^2})]$$

$$= 62 \times 8 \times (x + \frac{1}{x})(x - \frac{1}{x})$$

$$= 496 \times \sqrt{10} \times \sqrt{6}$$

$$= 496 \times \sqrt{60} = 496 \times 2\sqrt{15} = 992\sqrt{15}$$

$$\text{So, } x^8 - \frac{1}{x^8} = 992\sqrt{15}$$

Sol.152.(b)

$$(a + b + c)^2 - a^2 - b^2 - c^2$$

$$\begin{aligned} & = a^2 + b^2 + c^2 + 2(ab + bc + ac) - \\ & a^2 - b^2 - c^2 = 2(ab + bc + ac) \end{aligned}$$

$$\text{Sol.153.(c) According to the question,}$$

$$\frac{X-5Y}{X+5Y} = \frac{7}{13}$$

$$13X - 65Y = 7X + 35Y$$

$$6X = 100Y$$

$$\frac{X}{Y} = \frac{100}{6} = \frac{50}{3}$$

Sol.154.(b)

$$x + \frac{1}{x} = 10 \text{ (on squaring both sides)}$$

$$x^2 + \frac{1}{x^2} = 98 \text{ (again squaring both sides)}$$

$$x^4 + \frac{1}{x^4} = 9604 - 2 = 9602$$

$$\text{Sol.155.(b) Let, the numbers be a and b.}$$

According to question,

$$(a - b)^3 = 1728 \Rightarrow (a - b) = 12$$

$$a = 12 + b \text{ ---- (1)}$$

$$ab = 108 \text{ ---- (2)}$$

From the above equations, we get

$$a = 18 \text{ and } b = 6$$

$$\text{Therefore, } 18^3 - 6^3 = 5616$$

Sol.156.(a)

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$(19)^2 = 155 + 2(ab + bc + ca)$$

$$361 = 155 + 2(ab + bc + ca)$$

$$2(ab + bc + ca) = 361 - 155 = 206$$

$$(ab + bc + ca) = \frac{206}{2} = 103$$

Now,

$$\begin{aligned} & (a - b)^2 + (b - c)^2 + (c - a)^2 \\ & = a^2 + b^2 - 2ab + b^2 + c^2 - 2bc + c^2 + a^2 \\ & \quad - 2ca \end{aligned}$$

$$\begin{aligned} & = 2(a^2 + b^2 + c^2) - 2(ab + bc + ca) \\ & = 2(155) - 2(103) = 310 - 206 = 104 \end{aligned}$$

Sol.157.(a)

$$a^2 + b^2 + c^2 = ab + bc + ac \text{(Given)}$$

only possible when, $a = b = c$

$$\begin{aligned} \text{So, } & \frac{11a^4 + 13b^4 + 17c^4}{17a^2b^2 + 9b^2c^2 + 15c^2a^2} \\ & = \frac{11a^4 + 13a^4 + 17a^4}{17a^2a^2 + 9a^2a^2 + 15a^2a^2} = \frac{41a^4}{41a^4} = 1 \end{aligned}$$

$$\text{Sol.158.(b) } (x - y)^2 = 324 \Rightarrow x - y = 18$$

$$(x + y)^2 = (x - y)^2 + 4xy$$

$$(x + y)^2 = 324 + 4 \times 144 = 900$$

$$x + y = 30$$

Now,

$$x^2 - y^2 = (x - y)(x + y) = 18 \times 30 = 540$$

$$\text{Sol.159.(c) Given,}$$

$$p = \frac{\sqrt{2} + 1}{\sqrt{2} - 1} \text{ and } q = \frac{\sqrt{2} - 1}{\sqrt{2} + 1}$$

$$p \times q = \frac{\sqrt{2}+1}{\sqrt{2}-1} \times \frac{\sqrt{2}-1}{\sqrt{2}+1} = 1 \text{ and } p+q$$

$$= \frac{\sqrt{2}+1}{\sqrt{2}-1} + \frac{\sqrt{2}-1}{\sqrt{2}+1} = 6$$

According to question,

$$\begin{aligned} \frac{p^2}{q} + \frac{q^2}{p} &= \frac{p^3 + q^3}{pq} \\ &= \frac{(p+q)^3 - 3pq(p+q)}{pq} \\ &= \frac{(6)^3 - 3(1)(6)}{1} = 198 \end{aligned}$$

$$\text{Sol.160.(a)} \quad x + \frac{1}{x} = 2\sqrt{2}$$

$$x - \frac{1}{x} = \sqrt{(2\sqrt{2})^2 - 4} = 2$$

$$x^3 - \frac{1}{x^3} = (x - \frac{1}{x})^3 + 3(x - \frac{1}{x})$$

$$= (2)^3 + 3 \times 2 = 14$$

$$x^3 + \frac{1}{x^3} = (x + \frac{1}{x})^3 - 3(x + \frac{1}{x})$$

$$= (2\sqrt{2})^3 - 3 \times 2\sqrt{2} = 16\sqrt{2} - 6\sqrt{2} = 10\sqrt{2}$$

Now,

$$x^6 - \frac{1}{x^6} = (x^3 - \frac{1}{x^3})(x^3 + \frac{1}{x^3})$$

$$x^6 - \frac{1}{x^6} = 14 \times 10\sqrt{2} = 140\sqrt{2}$$

$$\text{Sol.161.(a)}$$

$$a^3 + b^3 + c^3 - 3abc = \frac{(a+b+c)}{2}$$

$$\{3(a^2 + b^2 + c^2) - (a + b + c)^2\} = 6\{3 \times 50 - 144\} = 36$$

$$\text{Sol.162.(d)} \quad (3y - \frac{3}{y}) = 5$$

$$\text{Then, } y - \frac{1}{y} = \frac{5}{3}$$

$$\text{Now, } y^2 + \frac{1}{y^2} = (\frac{5}{3})^2 + 2 = \frac{43}{9}$$

$$\text{Sol.163.(b)} \quad x^2 - 7x + 1 = 0$$

$$x + \frac{1}{x} = 7$$

$$x - \frac{1}{x} = \sqrt{(7)^2 - 4} = \pm\sqrt{45} = \pm 3\sqrt{5}$$

Now,

$$x^2 - \frac{1}{x^2} = (x + \frac{1}{x})(x - \frac{1}{x})$$

$$= 7 \times \pm 3\sqrt{5} = \pm 21\sqrt{5}$$

Since, $0 < x < 1$

$$\text{So, } x^2 - \frac{1}{x^2} = -21\sqrt{5}$$

$$\text{Sol.164.(b)} \quad x + \frac{1}{x} = 2 \Rightarrow x = 1$$

$$\text{Then, } x^7 + \frac{1}{x^{117}} = 1 + 1 = 2$$

$$\text{Sol.165.(a)}$$

$$a^3 + b^3 + c^3 - 3abc = \frac{(a+b+c)}{2}$$

$$\begin{aligned} &\{3(a^2 + b^2 + c^2) - (a + b + c)^2\} \\ &= 10 \times \{3 \times 152 - 20^2\} \\ &= 10 \times (456 - 400) = 560 \end{aligned}$$

$$\text{Sol.166.(b)}$$

$$\frac{l^2 - m^2}{(l+m)^2} = \frac{(l+m)(l-m)}{(l+m)^2} = \frac{(l-m)}{(l+m)}$$

$$\text{Sol.167.(a)} \quad x^4 + \frac{1}{x^4} = 142$$

$$x^2 + \frac{1}{x^2} = \sqrt{142 + 2} = 12$$

$$x + \frac{1}{x} = \sqrt{14}$$

$$x^3 + \frac{1}{x^3} = (\sqrt{14})^3 - (3 \times \sqrt{14}) = 11\sqrt{14}$$

$$x^7 + \frac{1}{x^7} = (x^3 + \frac{1}{x^3})(x^4 + \frac{1}{x^4}) - (x + \frac{1}{x})$$

$$= 11\sqrt{14} \times 142 - \sqrt{14} = 1561\sqrt{14}$$

$$\text{Sol.168.(c)}$$

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)$$

$$\{(a+b+c)^2 - 3(ab+bc+ca)\}$$

$$\Rightarrow a^3 + b^3 + c^3 - 3(-1)$$

$$= 7 \times \{49 - 3 \times 11\}$$

$$\Rightarrow a^3 + b^3 + c^3 + 3 = 7 \times \{16\}$$

$$\Rightarrow a^3 + b^3 + c^3 = 109$$

$$\text{Sol.169.(a)} \quad (x - \frac{1}{x}) = 6$$

$$(x + \frac{1}{x}) = \sqrt{6^2 + 4} = \sqrt{40} = 2\sqrt{10}$$

$$\text{Now, } x^2 - \frac{1}{x^2} = (x + \frac{1}{x})(x - \frac{1}{x})$$

$$\Rightarrow x^2 - \frac{1}{x^2} = 2\sqrt{10} \times 6 = 12\sqrt{10}$$

$$\text{Sol.170.(c)} \quad (x + \frac{1}{x}) = 3\sqrt{2}$$

$$\Rightarrow (x - \frac{1}{x}) = \sqrt{(3\sqrt{2})^2 - 4} = \sqrt{14}$$

$$\Rightarrow (x^2 - \frac{1}{x^2}) = 3\sqrt{2} \times \sqrt{14} = 6\sqrt{7}$$

$$\text{Again, } (x^2 + \frac{1}{x^2}) = (3\sqrt{2})^2 - 2 = 16$$

$$(x^4 + \frac{1}{x^4}) = (16)^2 - 2 = 254$$

$$\Rightarrow (x^4 - \frac{1}{x^4}) = 16 \times 6\sqrt{7} = 96\sqrt{7}$$

$$\text{Now, } (x^8 - \frac{1}{x^8}) = (x^4 + \frac{1}{x^4})(x^4 - \frac{1}{x^4})$$

$$= 254 \times 96\sqrt{7} = 24384\sqrt{7}$$

$$\text{Sol.171.(a)}$$

$$\frac{2a-5}{a} - \frac{4b-5}{b} + \frac{6c+5}{c} = 0$$

$$\Rightarrow bc(2a-5) - ac(4b-5) + ab(6c+5) = 0$$

$$\Rightarrow 2abc - 5bc - 4abc + 5ac + 6abc + 5ab = 0$$

$$\Rightarrow 4abc - 5(bc - ac - ab) = 0$$

$$\Rightarrow 5(bc - ac - ab) = 4abc$$

$$\Rightarrow \frac{(bc - ac - ab)}{abc} = \frac{4}{5}$$

$$\Rightarrow (\frac{1}{a} - \frac{1}{b} - \frac{1}{c}) = \frac{4}{5}$$

$$\text{Sol.172.(a)}$$

$$[(1+P)(1+P^2)(1+P^4)(1+P^8)(1+P^{16})(1-P)-1]$$

$$\Rightarrow [(1-P^2)(1+P^2)(1+P^4)(1+P^8)(1+P^{16})-1]$$

$$\Rightarrow [(1-P^4)(1+P^4)(1+P^8)(1+P^{16})-1]$$

$$\Rightarrow [(1-P^8)(1+P^8)(1+P^{16})-1]$$

$$\Rightarrow [(1-P^{16})(1+P^{16})-1]$$

$$\Rightarrow [(1-P^{32})-1] = -P^{32}$$

$$\text{Sol.173.(b)} \quad (x-y)^2 = x^2 + y^2 - 2xy$$

$$x^2 + y^2 = 25 + 2 \times 150 = 325$$

Now,

$$(x+y)^2 = x^2 + y^2 + 2xy$$

$$(x+y)^2 = 325 + 300 = 625$$

$$(x+y) = 25$$

Short tricks :-

$$(x-y) = 5 \text{ and } xy = 150$$

Clearly, value of $x = 15$ and $y = 10$

$$\text{So, } (x+y) = 15 + 10 = 25$$

$$\text{Sol.174.(d)} \quad (x^2 - \frac{1}{x^2}) = 4\sqrt{6}$$

Then,

$$(x^2 + \frac{1}{x^2}) = \sqrt{(4\sqrt{6})^2 + 4} = 10$$

$$x - \frac{1}{x} = \sqrt{8}$$

Now,

$$(x^3 - \frac{1}{x^3}) = (x - \frac{1}{x})(x^2 + \frac{1}{x^2} + 1)$$

$$= \sqrt{8}(10+1) = 11\sqrt{8} = 22\sqrt{2}$$

$$\text{Sol.175.(a)}$$

$$a^3 - b^3 = (a-b)(a^2 + b^2 + ab)$$

According to the question,

$$a-b=1$$

$$\text{Hence, } a^3 - b^3 = (a^2 + b^2 + ab)$$

$$\text{Sol.176.(d)} \quad (x^2 + \frac{1}{x^2}) = 18$$

$$x + \frac{1}{x} = \sqrt{18 + 2} = 2\sqrt{5}$$

$$x^3 + \frac{1}{x^3} = (2\sqrt{5})^3 - 3(2\sqrt{5})$$

$$= 40\sqrt{5} - 6\sqrt{5} = 34\sqrt{5}$$

$$\text{Sol.177.(a) Identity used :-}$$

$$a^2 - b^2 = (a+b)(a-b)$$

$$\frac{p^2 - (q-r)^2}{(p+r)^2 - q^2} + \frac{q^2 - (p-r)^2}{(p+q)^2 - r^2} + \frac{r^2 - (p-q)^2}{(q+r)^2 - p^2} =$$

$$\frac{[p+(q-r)][p-(q-r)]}{[(p+r)+q][(p+r)-q]} + \frac{[q+(p-r)][q-(p-r)]}{[(p+q)+r][(p+q)-r]}$$

$$+ \frac{[r+(p-q)][r-(p-q)]}{[(q+r)+p][(q+r)-p]}$$

$$= \frac{(p+q-r)[p-q+r]}{(p+r+q)[p-q+r]} + \frac{(q+p-r)[q-p+r]}{(p+q+r)[p-q-r]}$$

$$+ \frac{(p-q+r)[r-p+q]}{(q+r+p)[q-r-p]}$$

$$= \frac{(p+q-r)}{(p+q+r)} + \frac{(q-p+r)}{(p+q+r)} + \frac{(p-q+r)}{(q+r+p)}$$

$$= \frac{(p+q-r)+(q-p+r)+(p-q+r)}{(p+q+r)}$$

$$= \frac{p+q-r+q-p+r+p-q+r}{(p+q+r)} = 1$$

Short trick :-

Put $p, q = 0$ then,

$$\frac{p^2-(q-r)^2}{(p+r)^2-q^2} + \frac{q^2-(p-r)^2}{(p+q)^2-r^2} + \frac{r^2-(p-q)^2}{(q+r)^2-p^2}$$

$$= \frac{-r^2}{r^2} + \frac{(-r^2)}{(-r^2)} + \frac{r^2}{r^2} = 1$$

Sol.178.(b) $x^3 - y^3 = 6272$ and $x - y = 8$

$$x^3 - y^3 = (x - y)\{(x - y)^2 + 3xy\}$$

$$\{(8)^2 + 3xy\} = \frac{6272}{8}$$

$$3xy = 784 - 64 = 720 \Rightarrow xy = 240$$

Sol.179.(c) As per question, $\frac{a}{b} = \frac{7}{6}$

$$\text{So, } \frac{6a + 13b}{6a - 13b} = \frac{6 \times 7 + 13 \times 6}{6 \times 7 - 13 \times 6}$$

$$= \frac{42 + 78}{42 - 78} = -\frac{120}{36} = -\frac{10}{3}$$

Sol.180.(a) $\frac{1}{2+2p} + \frac{1}{2+2q} + \frac{1}{2+2r}$

According to question,

$$\frac{1}{2+2\frac{x}{y+z}} + \frac{1}{2+2\frac{y}{z+x}} + \frac{1}{2+2\frac{z}{x+y}}$$

$$= \frac{y+z}{2(y+z+x)} + \frac{z+x}{2(z+x+y)} + \frac{x+y}{2(x+y+z)} = \frac{2(x+y+z)}{2(x+y+z)} = 1$$

Sol.181.(d) $y - \frac{1}{y} = 4$

$$y + \frac{1}{y} = \sqrt{4^2 + 4} = 2\sqrt{5}$$

$$y^3 + \frac{1}{y^3} = (2\sqrt{5})^3 - 3 \times 2\sqrt{5} = 34\sqrt{5}$$

$$y^6 + \frac{1}{y^6} = (34\sqrt{5})^2 - 2 = 5778$$

Sol.182.(b) $x^2 + 1 = 15x$

Squaring both side

$$x^4 + 1 + 2x^2 = 225x^2$$

$$x^4 - 223x^2 + 1 = 0$$

$$x^4 - 223x^2 + 6 = 5$$

Sol.183.(a) $p^3 + q^3 + r^3 - 3pqr$

$$= (p + q + r)\{(p + q + r)^2 - 3(pq + qr + pr)\} \text{ ----(1)}$$

According to the question ,

$$p + q + r = pqr = \frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1$$

$$\Rightarrow p + q + r = pqr = \frac{pq + qr + rp}{pqr} = 1 \text{ ---- (2)}$$

By Putting the value of equation(2) into equation (1) we get :-

$$p^3 + q^3 + r^3 - 3 = (1)((1)^2 - 3(1))$$

$$p^3 + q^3 + r^3 = 1 - 3 + 3 = 1$$

Sol.184.(c)

$$\frac{256x^4 - 16y^4}{(80x^2 - 20y^2)(16x^2 + 4y^2)}$$

$$= \frac{16(16x^4 - y^4)}{20(4x^2 - y^2)4(4x^2 + y^2)}$$

$$= \frac{16(16x^4 - y^4)}{80((4x^2)^2 - (y^2)^2)}$$

$$= \frac{16(16x^4 - y^4)}{80(16x^4 - y^4)} = \frac{1}{5}$$

Sol.185.(a)

$$7a - \frac{7}{a} + 4 = 0 \Rightarrow a - \frac{1}{a} = -\frac{4}{7}$$

$$\text{Now, } a^3 - \frac{1}{a^3} = \left(-\frac{4}{7}\right)^3 + 3 \times \left(-\frac{4}{7}\right)$$

$$= \frac{-64 - 49 \times 12}{343} = -\frac{652}{343}$$

$$\text{So, } a^3 - \frac{1}{a^3} - 1 = -\frac{652}{343} - 1 = \frac{-995}{343}$$

Sol.186.(d)

$$\frac{x^2 - 2x - 63}{x^2 + 14x + 49} = \frac{x^2 - 9x + 7x - 63}{x^2 + 7x + 7x + 49}$$

$$= \frac{x(x-9) + 7(x-9)}{x(x+7) + 7(x+7)} = \frac{(x-9)(x+7)}{(x+7)(x+7)}$$

$$= \frac{(x-9)}{(x+7)}$$

Sol.187.(a) $\frac{s^2+t^2+2st-u^2}{s^2-t^2-2tu-u^2} = \frac{(s+t)^2-u^2}{s^2-(t+u)^2}$

$$\frac{(s+t+u)(s+t-u)}{(s+t+u)(s-t-u)} = \frac{(s+t-u)}{(s-t-u)}$$

Sol.188.(d) $x^2 - 8x - 1 = 0 \Rightarrow x - \frac{1}{x} = 8$

$$\text{So, } x^2 + \frac{1}{x^2} = (8)^2 + 2 = 64 + 2 = 66$$

Sol.189.(b) $x^2 + \frac{1}{x^2} = 6$

Given that, $0 < x < 1$, so the value of

$$\left(x^2 - \frac{1}{x^2}\right) \text{ will always be -ve.}$$

$$x^2 - \frac{1}{x^2} = -\left(\sqrt{(6)^2 - 4}\right) = -4\sqrt{2}$$

$$x^4 - \frac{1}{x^4} = \left(x^2 + \frac{1}{x^2}\right)\left(x^2 - \frac{1}{x^2}\right)$$

$$= 6 \times (-4\sqrt{2})$$

$$x^4 - \frac{1}{x^4} = -24\sqrt{2}$$

Sol.190.(d) $x = \frac{1}{x-3} \Rightarrow x^2 - 3x = 1$

Dividing both sides by x we get;

$$\Rightarrow x - \frac{1}{x} = 3$$

$$\text{So, } x + \frac{1}{x} = \sqrt{(3)^2 + 4} = \sqrt{13}$$

Sol.191.(b) $x + \frac{1}{x} = 1, \Rightarrow x^2 + 1 = x$

Now,

$$\frac{x^2 + 7x + 1}{x^2 + 11x + 1} = \frac{x + 7x}{x + 11x} = \frac{8x}{12x} = \frac{2}{3}$$

Sol.192.(c) Given, $x + \frac{1}{x} = \sqrt{6}$

$$\left(x^2 + \frac{1}{x^2}\right) = (\sqrt{6})^2 - 2 = 4$$

$$\left(x^2 - \frac{1}{x^2}\right) = \sqrt{4^2 - 4} = \sqrt{12} = 2\sqrt{3}$$

Then,

$$\left(x^4 - \frac{1}{x^4}\right) = \left(x^2 - \frac{1}{x^2}\right)\left(x^2 + \frac{1}{x^2}\right)$$

$$= 2\sqrt{3} \times 4 = 8\sqrt{3}$$

Since, $\left(x^2 + \frac{1}{x^2}\right) = 4$ then,

$$\left(x^4 + \frac{1}{x^4}\right) = (4)^2 - 2 = 14$$

Now,

$$\left(x^8 - \frac{1}{x^8}\right) = \left(x^4 + \frac{1}{x^4}\right)\left(x^4 - \frac{1}{x^4}\right)$$

$$= 14 \times 8\sqrt{3} = 112\sqrt{3}$$

Sol.193.(b) $\left(\frac{5}{b} + 5b\right)\left(\frac{25}{b^2} - 25 + 25b^2\right)$

$$= \left(\frac{5}{5} + 5 \times 5\right)\left(\frac{25}{5 \times 5} - 25 + 25 \times 5 \times 5\right)$$

$$= (26)(1 - 25 + 625) = 15626$$

Sol.194.(b) Given, $x - \frac{1}{x} = 10$

$$x^2 + \frac{1}{x^2} = (10)^2 + 2 = 102$$

$$\text{Now, } x^4 + \frac{1}{x^4} = 102 \times 102 - 2 = 10402$$

Sol.195.(b) $(x - 5)^2 + (x + 3)^2 + 4x$

$$= x^2 + 25 - 10x + x^2 + 9 + 6x + 4x$$

$$= 2x^2 + 34 = 2(x^2 + 17)$$

Sol.196.(c) formula used :-

$$a^3 + b^3 = (a + b)[(a + b)^2 - 3ab]$$

$$8x^3 + 27y^3 = 16[16^2 - 3 \times 2 \times 3 \times 9]$$

$$= 16[256 - 162] = 16 \times 94 = 1504$$

Sol.197.(c)

$$x^3 + y^3 = (x + y)[(x + y)^2 - 3xy]$$

$$= 25[25^2 - 3 \times 20] = 25[625 - 60]$$

$$= 25 \times 565 = 14125$$

Sol.198.(c) $x^2 - \frac{1}{x^2} = 4\sqrt{2}$

$$x^2 + \frac{1}{x^2} = \sqrt{(4\sqrt{2})^2 + 4} = 6$$

Now,

$$x^4 - \frac{1}{x^4} = \left(x^2 + \frac{1}{x^2}\right)\left(x^2 - \frac{1}{x^2}\right)$$

$$= 6 \times 4\sqrt{2} = 24\sqrt{2}$$

Sol.199.(b) $\left(x^2 + \frac{1}{x^2}\right) = 18$

$$\left(x^2 + \frac{1}{x^2} - 2\right) = 18 - 2 \Rightarrow \left(x - \frac{1}{x}\right) = 4$$

$$\Rightarrow (x^3 - \frac{1}{x^3}) = (4)^3 + 3 \times 4 = 76$$

Sol.200.(b) $3x - z = 0 \Rightarrow z = 3x$

$$3x - 9y + 4z = 5$$

$$\text{or } 15x - 9y = 5 \dots \text{e.q. (1)}$$

$$2x + 7y + z = 12 \text{ or } 5x + 7y = 12 \dots \text{e.q. (2)}$$

on solving e.q. (1) and e.q. (2), we get

$$y = \frac{31}{30} \text{ and } x = \frac{143}{150}, z = 3x = \frac{143}{50}$$

Sol.201.(a) $(2a + b)^2 = 10^2$

$$4a^2 + b^2 + 4ab = 100$$

$$4a^2 + b^2 = 100 - 18 = 82$$

Now,

$$(2a - b) = \sqrt{4a^2 + b^2 - 4ab} \\ = \sqrt{82 - 18} = \sqrt{64} = 8$$

Sol.202.(c) $x^2 + \frac{1}{x^2} = 2\sqrt{5}$

$$x^4 + \frac{1}{x^4} = (2\sqrt{5})^2 - 2 = 18$$

Now,

$$x^4 - \frac{1}{x^4} = \sqrt{(18^2 - 4)} = \sqrt{320} = 8\sqrt{5}$$

Sol.203.(b) $x^4 + \frac{1}{x^4} = 79$

$$x^4 + \frac{1}{x^4} + 2 = 81 \Rightarrow x^2 + \frac{1}{x^2} = 9$$

$$x^2 + \frac{1}{x^2} - 2 = 7 \Rightarrow x - \frac{1}{x} = \sqrt{7}$$

Sol.204.(b)

$$x + y = \sqrt{13} \Rightarrow x^2 + y^2 + 2xy = 13$$

$$x - y = \sqrt{11} \Rightarrow x^2 + y^2 - 2xy = 11$$

On adding both, we get

$$2(x^2 + y^2) = 24 \Rightarrow (x^2 + y^2) = 12$$

$$\text{Again, } x^2 + y^2 + 2xy = 13$$

$$xy = \frac{13 - 12}{2} = \frac{1}{2}$$

$$\text{Now, } xy(x^2 + y^2) = \frac{1}{2} \times 12 = 6$$

Sol.205.(a)

$$ab + bc + ca = 0 \Rightarrow ab + ca = -bc$$

$$\frac{1}{a^2 - bc} = \frac{1}{a^2 + (ab + ca)} = \frac{1}{a(a + b + c)}$$

$$\text{Similarly, } \frac{1}{b^2 - ca} = \frac{1}{b(a + b + c)}$$

$$\text{and } \frac{1}{c^2 - ab} = \frac{1}{c(a + b + c)}$$

Now,

$$\frac{1}{a^2 - bc} + \frac{1}{b^2 - ca} + \frac{1}{c^2 - ab}$$

$$= \frac{1}{a(a + b + c)} + \frac{1}{b(a + b + c)} +$$

$$\frac{1}{c(a + b + c)}$$

$$= \frac{bc + ca + ab}{abc(a + b + c)} = 0$$

Sol.206.(c)

Let the required value of A be x

According to question,

$$\frac{52}{91} = \frac{x}{126} \Rightarrow x = 72$$

Sol.207.(c) $ax + by + c_1 = 0$

$$px + qy + c_2 = 0$$

For infinite no. of solutions :-

$$\frac{a}{p} = \frac{b}{q} = \frac{c_1}{c_2}$$

$$\text{So, } \frac{38}{46} = \frac{q}{414} \Rightarrow q = \frac{414 \times 38}{46}$$

$$q = 9 \times 38 = 342$$

Sol.208.(b)

$$x^3 + y^3 + z^3 - 3xyz = \frac{(x + y + z)}{2}$$

$$\{(x - y)^2 + (y - z)^2 + (z - x)^2\} \\ = \frac{14 + 15 + 17}{2}$$

$$\{(14 - 15)^2 + (15 - 17)^2 + (17 - 14)^2\} \\ = 23 \{1 + 4 + 9\} = 322$$

Sol.209.(b) Given,

$$18x + 36y + 45 = 0 \text{ -----(1)}$$

$$px - 54y + 67 = 0 \text{ -----(2)}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \text{ (no solution)}$$

$$\frac{18}{p} = \frac{36}{(-54)} \neq \frac{45}{67}$$

$$p = \frac{18 \times (-54)}{36} = -27$$

Sol.210.(a) $(a + b)^2 - (a - b)^2 = 4ab$

$$\Rightarrow (\frac{9}{8} + \frac{8}{9})^2 - (\frac{9}{8} - \frac{8}{9})^2$$

$$\Rightarrow 4 \times \frac{9}{8} \times \frac{8}{9} \Rightarrow 4$$

Sol.211.(c)

$$\frac{1 + x^2}{1 - x} \div \frac{1 - x^4}{x - 1} \times \frac{x(1 - x)}{(1 + x)}$$

$$\Rightarrow \frac{1 + x^2}{1 - x} \times \frac{x - 1}{1 - x^4} \times \frac{x(1 - x)}{(1 + x)}$$

$$\Rightarrow \frac{1 + x^2}{1 - x} \times \frac{-(1 - x)}{(1 - x^2)(1 + x^2)} \times \frac{x(1 - x)}{(1 + x)}$$

$$\Rightarrow \frac{1 + x^2}{1 - x} \times \frac{-(1 - x)}{(1 - x)(1 + x)(1 + x^2)}$$

$$\times \frac{x(1 - x)}{(1 + x)}$$

$$\Rightarrow \frac{-x}{(1 + x)^2}$$

Sol.212.(a) $K + \frac{1}{K} - 2 = 0 \Rightarrow K + \frac{1}{K} = 2$

Here the value of k = 1

$$\text{Therefore, } K^{17} + \frac{1}{K^{12}} = 1^{17} + \frac{1}{1^{12}} = 2$$

Sol.213.(d)

Formula :- $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

Therefore, $(a + b + 2c)(a^2 + b^2 + 4c^2 - ab - 2bc - 2ca)$

$$= a^3 + b^3 + (2c)^3 - 3ab(2c)$$

$$\Rightarrow a^3 + b^3 + 8c^3 - 6abc$$

Sol.214.(c) According to the question,

$$a^4 + \frac{1}{a^4} = 194$$

$$\Rightarrow a^2 + \frac{1}{a^2} = \sqrt{194 + 2} = 14$$

$$\Rightarrow a + \frac{1}{a} = \sqrt{14 + 2} = 4$$

$$\Rightarrow a^3 + \frac{1}{a^3} = 64 - 3 \times 4 = 52$$

Now, we can write

$$\frac{a^6 + 1}{a^3} \text{ as } \frac{\frac{a^6}{a^3} + \frac{1}{a^3}}{\frac{a^3}{a^3}} = a^3 + \frac{1}{a^3} = 52$$

Sol.215.(c)

$$(a - b)(b + a)(a^2 + b^2)(2b^4 + 2a^4)$$

$$= (a^2 - b^2)(a^2 + b^2)(2b^4 + 2a^4)$$

$$\Rightarrow (a^4 - b^4)2(b^4 + a^4) = 2(a^8 - b^8)$$

Sol.216.(c)

$$\frac{(15)^3 + (4)^3}{(15^2 + 4^2 - k)} = 19$$

$$\Rightarrow \frac{(15 + 4)(15)^2 + (4)^2 - (60)}{(15^2 + 4^2 - k)} = 19$$

$$\Rightarrow (15)^2 + (4)^2 - 60 = (15)^2 + (4)^2 - k \\ \Rightarrow k = 60$$

Sol.217.(a) $x^2 + \frac{1}{x^2} = 38$

$$\Rightarrow x - \frac{1}{x} = \sqrt{38 - 2} = 6$$

Sol.218.(d)

Formula:- $p^3 - r^3 = (p - r)[p^2 + r^2 + pr]$

$$(21)^2 = p^2 + r^2 + 2 \times 38$$

$$\Rightarrow p^2 + r^2 = 365$$

$$(p - r)^2 = (p + r)^2 - 4pr$$

$$\Rightarrow (p - r)^2 = 441 - 4 \times 38$$

$$(p - r) = \sqrt{289} = 17$$

$$p^3 - r^3 = 17[365 + 38]$$

$$\Rightarrow p^3 - r^3 = 17[403] = 6851$$

Sol.219.(b) Formula:- $a^3 + b^3 + c^3 - 3abc$

$$= \frac{1}{2} (a + b + c)\{(a - b)^2 + (b - c)^2 + (c - a)^2\}$$

Therefore, $(a + b + c)$

$$\{(a - b)^2 + (b - c)^2 + (c - a)^2\}$$

$$= 2a^3 + 2b^3 + 2c^3 - 6abc$$

Sol.220.(d)

Formula:- $a^3 + b^3 + c^3 - 3abc =$

$$(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

By Comparing

$$(a^2 + b^2 + 4c^2 - ab - 2bc - 2ca)(a+b+2c)$$

with the given formula

We get

$$\Rightarrow (a^2 + b^2 + 4c^2 - ab - 2bc - 2ca)$$

$$(a + b + 2c) = a^3 + b^3 + 8c^3 - 6abc$$

Sol.221.(d)

$$x + \frac{1}{x} = -2 \text{ (clearly visible that } x = -1 \text{)}$$

$$\text{Therefore, } x^7 + \frac{1}{x^7} + x^2 + \frac{1}{x^2}$$

$$= (-1)^7 + \frac{1}{(-1)^7} + (-1)^2 + \frac{1}{(-1)^2}$$

$$= -2 + 2 = 0$$

$$\text{Sol.222.(a)} \frac{4x^2 + 9y^2 + 12xy}{144}$$

$$= \frac{(2x + 3y)^2}{(12)^2} = \left(\frac{x}{6} + \frac{y}{4}\right)^2$$

Sol.223.(c)

$$(a - 4)^3 = a^3 - 64 - 3 \times a^2 \times 4 + 3 \times a \times 16$$

$$\Rightarrow a^3 - 12a^2 + 48a - 64$$

Sol.224.(c)

$$x - y = 8 \text{ and } 8xy = 8 \Rightarrow xy = 1$$

$$(8)^2 = x^2 + y^2 - 2 \Rightarrow x^2 + y^2 = 66$$

$$\text{then, } 2(x^3 - y^3) = 2 \times 8(66 + 1)$$

$$= 16 \times 67 = 1072$$

Sol.225.(b) $a^3 = b^3 + 988$

$$\Rightarrow a^3 - b^3 = 988 \text{ and } a - b = 4$$

$$2a^2 + 2ab + 2b^2 = 2(a^2 + ab + b^2)$$

$$= 2\left(\frac{a^3 - b^3}{a - b}\right) = 2\left(\frac{988}{4}\right) = 494$$

Sol.226.(c) $a^3 + 3a^2 + 3a = 63$

$$\Rightarrow a^3 + 3a^2 + 3a - 63 = 0$$

$$\Rightarrow (a - 3)(a^2 + 6a + 21) = 0$$

$$\Rightarrow a = 3 \Rightarrow a^2 + 2a = 9 + 6 = 15$$

Sol.227.(b) $(a + b + c)^2$

$$= a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$(6)^2 = 40 + 2(ab + bc + ca)$$

$$(ab + bc + ca) = -2 \Rightarrow a^3 + b^3 + c^3 - 3abc$$

$$= (a + b + c)(a^2 + b^2 + c^2 - (ab + bc + ca))$$

$$\Rightarrow a^3 + b^3 + c^3 - 3abc = (6)[(40 - (-2))]$$

$$= 252$$

Sol.228.(d) $4ab + 4bc + 4ca = 160$

$$\Rightarrow ab + bc + ca = 40$$

$$9a + 9b + 9c = 81 \Rightarrow (a + b + c) = 9$$

$$(a + b + c)^2$$

$$= (a^2 + b^2 + c^2 + 2ab + 2bc + 2ca)$$

$$81 = a^2 + b^2 + c^2 + 80$$

$$a^2 + b^2 + c^2 = 1 \Rightarrow 6a^2 + 6b^2 + 6c^2 = 6$$

Sol.229.(b)

$$N + \frac{1}{N} = \sqrt{3} \text{ (squaring both side)}$$

$$N^2 + \frac{1}{N^2} = 3 - 2 = 1 \text{ (cubing both side)}$$

$$N^6 + \frac{1}{N^6} = 1 - 3 = -2$$

So the value of,

$$N^6 + \frac{1}{N^6} + 11 \Rightarrow -2 + 11 = 9$$

$$\text{Sol.230.(b)} \frac{(a+b)^2 - (a-b)^2}{ab}$$

$$= \frac{a^2 + b^2 + 2ab - a^2 - b^2 + 2ab}{ab} = 4$$

$$\text{Sol.231.(d)} (a - b)^3 - a^3 + b^3$$

$$= (a)^3 - (b)^3 - 3ab(a - b) - a^3 + b^3$$

$$= -3ab(a - b)$$

Sol.232.(d)

$$(300 - 1)^2 = 90000 + 1 - 600 = 89401$$

Sol.233.(d)

$$\frac{(k + \frac{1}{k})}{(k + \frac{1}{k})} \times (k - \frac{1}{k})(k^2 + \frac{1}{k^2})(k^4 + \frac{1}{k^4})$$

$$(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}}) \text{ Multiply by}$$

$$\frac{(k + \frac{1}{k})}{(k + \frac{1}{k})}$$

$$\frac{(k^2 - \frac{1}{k^2})(k^2 + \frac{1}{k^2})(k^4 + \frac{1}{k^4})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})}{(k + \frac{1}{k})}$$

$$= \frac{(k^4 - \frac{1}{k^4})(k^4 + \frac{1}{k^4})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})}{(k + \frac{1}{k})}$$

$$= \frac{(k^8 - \frac{1}{k^8})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})}{(k + \frac{1}{k})}$$

$$= \frac{(k^{16} - \frac{1}{k^{16}})(k^{16} + \frac{1}{k^{16}})}{(k + \frac{1}{k})} = \frac{(k^{32} - \frac{1}{k^{32}})}{(k + \frac{1}{k})}$$

Sol.234.(b) Let $a + \frac{1}{a} = x$

$$a^4 + \frac{1}{a^4} = 194 = [x^2 - 2]^2 - 2$$

$$\Rightarrow [x^2 - 2]^2 = 196 \Rightarrow x = 4$$

$$a^3 + \frac{1}{a^3} = x^3 - 3x = 4^3 - 3 \times 4 = 52$$

Sol.235.(a) $x + \frac{1}{x} = \frac{K}{2}$

$$\frac{x^8 + 1}{x^4} \text{ [dividing by } x^4]$$

$$x^4 + \frac{1}{x^4} = \left[\left(\frac{K}{2}\right)^2 - 2\right]^2 - 2$$

$$= \left[\left(\frac{K^2 - 8}{4}\right)^2 - 2\right] = \frac{K^4 - 16K^2 + 32}{16}$$

Sol.236.(a) Formula :-

$$(a^3 + b^3 + c^3 - 3abc) = \frac{a + b + c}{2}$$

$$\{(a - b)^2 + (b - c)^2 + (c - a)^2\}$$

$$\Rightarrow \frac{(p^3 + q^3 + r^3 - 3pqr)}{(p + q + r)}$$

$$= \frac{\{(6)^2 + (-5)^2 + (3)^2\}}{2} = 35$$

Sol.237.(d) If $A + \frac{1}{A} = -1$, then $A^3 = 1$

According to the question,

$$\frac{A^6 + A^3 - 1}{A^9 + A^3 - 1} = \frac{1 + 1 - 1}{1 + 1 - 1} = 1$$

Sol.238.(b) $(x + 1)^2 + (x + 2)^2 = 16$

$$x^2 + 1 + 2x + x^2 + 4 + 4x = 16$$

$$2x^2 + 6x + 5 = 16$$

$$\Rightarrow 4x^2 + 12x + 10 = 32$$

(multiply by 2 both side)

$$4x^2 + 12x + 10 + 30 = 32 + 30$$

(add 30 both side)

$$4x^2 + 12x + 40 = 62$$

Sol.239.(d)

$$a^2 + 4b^2 + 9c^2 - 4ab + 12bc - 6ca$$

$$= a^2 + (2b)^2 + (3c)^2 - 2 \times a \times 2b + 2 \times 2b \times 3c - 2 \times 3c \times a$$

$$= (a - 2b - 3c)^2$$

Sol.240.(d) Formula :-

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$(1)^2 = a^2 + b^2 + c^2 + 2(-1)$$

$$\Rightarrow a^2 + b^2 + c^2 = 3$$

Now,

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)$$

$$[a^2 + b^2 + c^2 - (ab + bc + ca)]$$

$$\Rightarrow a^3 + b^3 + c^3 - 3(-1)$$

$$= (1)[3 - (-1)] \Rightarrow a^3 + b^3 + c^3 = 1$$

Sol.241.(c) $(a + b)^3 - a^3 - b^3$

$$= a^3 + b^3 + 3ab(a + b) - a^3 - b^3$$

$$= 3ab(a + b)$$

Sol.242.(d) $\frac{1}{(x - 2)} + x = 8$

Subtracting 2 from both side; we get

$$\frac{1}{(x - 2)} + (x - 2) = 8 - 2 = 6$$

$$\frac{1}{(x - 2)^2} + (x - 2)^2 = 6^2 - 2 = 34$$

Sol.243.(b)

$$\frac{1 + x}{1 - x^2} \div \frac{1 + x}{1 - x^4} - \frac{1 - x^4}{1 - x} \times \frac{1 + x}{1 - x^2}$$

$$= \frac{1 + x}{1 - x^2} \times \frac{(1 - x^2)(1 + x^2)}{1 + x} - \frac{(1 + x^2)(1 + x)}{1 - x}$$

$$= (1 + x^2) - \frac{1 + x + x^2 + x^3}{1 - x}$$

$$= \frac{1-x+x^2-x^3-1-x-x^2-x^3}{1-x}$$

$$= \frac{-2x-2x^3}{1-x} = \frac{2x+2x^3}{x-1} = \frac{2x(1+x^2)}{x-1}$$

Sol.244.(b) If $x + y + z = 0$

Then, $x^3 + y^3 + z^3 = 3xyz$

$$\frac{x}{(yz)^2} + \frac{y}{(xz)^2} + \frac{z}{(xy)^2} = \frac{x^3+y^3+z^3}{x^2y^2z^2}$$

$$= \frac{3}{xyz}$$

Sol.245.(d) $\frac{K}{K^2-2K+1} = \frac{1}{6}$

$$\Rightarrow K^2 + 1 = 8K \Rightarrow K + \frac{1}{K} = 8$$

$$\Rightarrow \frac{1}{K^3} + K^3 = (8)^3 - 3 \times 8 = 488$$

Sol.246.(d) $x - \frac{1}{x} = 4$

$$\Rightarrow x^2 + \frac{1}{x^2} = 18 \Rightarrow x^4 + \frac{1}{x^4} = 322$$

$$(x^6 + \frac{1}{x^6}) = (x^2 + \frac{1}{x^2})(x^4 + \frac{1}{x^4}) -$$

$$(x^2 + \frac{1}{x^2})$$

$$\Rightarrow (x^6 + \frac{1}{x^6}) = 322 \times 18 - 18 = 5778$$

Sol.247.(a) Two lines have no solution, if these two lines are parallel to each other. The lines are parallel to each other means that the slopes of the lines are equal.

$$18x - 72y + 13 = 0 \Rightarrow y = \frac{18}{72}x + \frac{13}{72}$$

$$\text{Slope } (m) = \frac{1}{4}$$

$$\text{and } 7x - my - 17 = 0 \Rightarrow y = \frac{7}{m}x - \frac{17}{m}$$

$$\text{Value of } (m) \text{ for no solution } \Rightarrow \frac{7}{m} = \frac{1}{4}$$

$$m = 28$$

Sol.248.(c)

$$2^x + 3^y = 17 \text{ and } 2^{x+2} - 3^{y+1} = 5,$$

By option,

Putting the value of $x = 3$ and $y = 2$, we get

$$2^x + 3^y = 2^3 + 3^2 = 17 \text{satisfied}$$

$$\text{and } 2^{x+2} - 3^{y+1} = 2^5 - 3^3 = 32 - 27 = 5 \text{satisfied}$$

Sol.249.(c) $x^2 - 5x + 1 = 0$

$$x + \frac{1}{x} = 5 \text{ (eq .1)}$$

$$\{\text{When } x + \frac{1}{x} = k,$$

$$\text{then } x^3 + \frac{1}{x^3} = k^3 - 3k\}$$

$$x^3 + \frac{1}{x^3} = 110 \text{ (eq .2)}$$

Now, given eq . $= \frac{x^6 + x^4 + x^2 + 1}{5x^3}$

Divide numerator and denominator by x^3

$$\frac{x^3 + x + \frac{1}{x} + \frac{1}{x^3}}{5} = \frac{110 + 5}{5} = 23$$

Sol.250.(b) $k + k^{-1} = -2$

$$k + \frac{1}{k} = -2$$

Clearly, $K = -1$

Putting the value in given equation :-

$$\frac{k^2 + 4k - 2}{k^2 + k - 5} = \frac{1 - 4 - 2}{1 - 1 - 5} = \frac{-5}{-5} = 1$$

Sol.251.(d) Formula :- When, $x + \frac{1}{x}$

$$= k, \text{ Then } x^2 + \frac{1}{x^2} = k^2 - 2$$

$$\text{Given that, } x + \frac{1}{x} = 5\sqrt{2}$$

$$\text{Then, } x^2 + \frac{1}{x^2} = 48$$

$$\text{And } x^4 + \frac{1}{x^4} = 2302$$

Sol.252.(a) Formula :-

$$a^4 + b^4 + 2a^2b^2 = (a^2 + b^2)^2$$

$$a^2 + b^2 = 121 - 70 = 51$$

Now,

$$a^4 + b^4 = 51^2 - 2 \times (35)^2$$

$$= 2601 - 2450 = 151$$

Sol.253.(d)

Given, $x + y = 36$ then, $x = 36 - y$

$$= (x - 27)^3 + (y - 9)^3$$

$$= (36 - y - 27)^3 + (y - 9)^3$$

$$= (9 - y)^3 + (y - 9)^3$$

$$= (9 - y)^3 - (9 - y)^3 = 0$$

Sol.254.(a) Formula:-

$$(a + b - c)^2 = a^2 + b^2 + c^2 + 2ab - 2bc - 2ca$$

$$\text{Given, } \frac{a^2 + b^2 + c^2 - 1024}{ab - bc - ca} = -2$$

$$a^2 + b^2 + c^2 - 1024 = -2ab + 2bc + 2ca$$

$$a^2 + b^2 + c^2 + 2ab - 2bc - 2ca = 1024$$

$$(a + b - c)^2 = 1024$$

$$(5c - c)^2 = 1024$$

$$\text{.....(Given, } a + b = 5c) \Rightarrow c = 8$$

Sol.255.(d) $x = \frac{\sqrt{5} - 2}{\sqrt{5} + 2} = \frac{\sqrt{5} - \sqrt{4}}{\sqrt{5} + \sqrt{4}}$

Rationalizing,

$$x = \frac{\sqrt{5} - \sqrt{4}}{\sqrt{5} + \sqrt{4}} \times \frac{\sqrt{5} - \sqrt{4}}{\sqrt{5} - \sqrt{4}} = \frac{9 - 4\sqrt{5}}{1}$$

$$= 9 - 4\sqrt{5} \text{ eq (1)}$$

$$\text{Then, } \frac{1}{x} = 9 + 4\sqrt{5} \text{ eq (2)}$$

Adding eq . (1) and eq . (2) , we get

$$x + \frac{1}{x} = 18, \text{ Then, } x^2 + \frac{1}{x^2} = 322$$

Short Trick:- If $x = \frac{\sqrt{a} - \sqrt{b}}{\sqrt{a} + \sqrt{b}}$,

$$\text{Then } x + \frac{1}{x} = \frac{2(a+b)}{(a-b)}$$

According to question ,

$$x = \frac{\sqrt{5} - \sqrt{4}}{\sqrt{5} + \sqrt{4}}, \text{ Then}$$

$$x + \frac{1}{x} = \frac{2(5+4)}{(5-4)} = 18$$

$$\text{Then, } x^2 + \frac{1}{x^2} = 322$$

Sol.256.(b) $x^4 + \frac{1}{x^4} = 2207 \text{eq .(1)}$

Adding 2 in eq .(1), we get ,

$$x^2 + \frac{1}{x^2} = 47 \text{eq .(2)}$$

Adding 2 in eq .(2), we get ,

$$\Rightarrow x + \frac{1}{x} = 7$$

$$\text{Then, } x^3 + \frac{1}{x^3} = 7^3 - 3 \times 7 = 322$$

Now , using Identities :-

$$\Rightarrow x^7 + \frac{1}{x^7}$$

$$= (x^3 + \frac{1}{x^3})(x^4 + \frac{1}{x^4}) - (x + \frac{1}{x})$$

$$= 322 \times 2207 - 7 = 710647$$

Sol.257.(d) Given that ,

$$x + \frac{y^2}{x} = 5 \text{eq .(1)}$$

$$\frac{x^2 + y^2}{x} = 5 \Rightarrow x^2 + y^2 = 5x$$

$$\Rightarrow x^2 - 5x = -y^2 \text{eq .(2)}$$

$$\frac{x^2 + 2x + y^2}{x^3 - 5x^2} = \frac{x(x + 2 + \frac{y^2}{x})}{x(x^2 - 5x)} \text{eq .(3)}$$

Putting value of eq .(1) and eq .(2) in eq .(3), we get

$$\frac{x(x + 2 + \frac{y^2}{x})}{x(x^2 - 5x)} = \frac{(5+2)}{(-y^2)} = -\frac{7}{y^2}$$

Sol.258.(d) Given ,

$$x + \frac{1}{x} = 8 \Rightarrow x - 8 = \frac{-1}{x}$$

.... (eq .1)

$$\frac{5}{x^2 - 8x + 2} = \frac{5}{x(x - 8 + \frac{2}{x})} \text{ (eq .2)}$$

Putting the value of eq . (1) in eq . (2) , we get

$$\frac{5}{x(x - 8 + \frac{2}{x})} = \frac{5}{x(\frac{2}{x} - \frac{1}{x})} = 5$$

Sol.259.(c) Identities :- when $ax + \frac{y}{a} = k$

$$, \text{ then } (ax)^2 + \frac{y^2}{a^2} = k^2 - 2xy$$

Now,

$$4a + \frac{5}{a} + 5 = 14 \Rightarrow 4a + \frac{5}{a} = 9 \text{ ...e.q. (1)}$$

On squaring e.q. (1), we get

$$16a^2 + \frac{25}{a^2} = 9^2 - 2 \times 4 \times 5 = 41$$

Sol.260.(d) Identity : when $x - \frac{1}{x} = k$,

then,

$$x^2 + \frac{1}{x^2} = k^2 + 2 \Rightarrow x - \frac{1}{x} = 13,$$

$$\text{then, } x^2 + \frac{1}{x^2} = 13^2 + 2 = 171$$

$$\text{And } x^4 + \frac{1}{x^4} = (171)^2 - 2 = 29241 - 2 = 29239$$

Sol.261.(b)

$$\begin{aligned} (x^4 + x^2 + 25) &= (x^4 + x^2 + 5^2) \\ [(x^2 + 5)^2 - 2 \times 5 \times x^2] + x^2 & \\ = [(x^2 + 5)^2 - 10x^2 + x^2] & \\ = [(x^2 + 5)^2 - 9x^2] &= [(x^2 + 5)^2 - (3x)^2] \\ = (x^2 + 3x + 5)(x^2 - 3x + 5) & \end{aligned}$$

Sol.262.(c) $x = 3 + 2\sqrt{2}$

$$x = (\sqrt{2})^2 + (1)^2 + 2\sqrt{2} = (\sqrt{2} + 1)^2$$

$$\sqrt{x} = (\sqrt{2} + 1) \text{eq. (1)}$$

Taking reciprocal of eq. (1), we get

$$\frac{1}{\sqrt{x}} = \frac{1}{(\sqrt{2} + 1)} \times \frac{(\sqrt{2} - 1)}{(\sqrt{2} - 1)} = \frac{(\sqrt{2} - 1)}{1}$$

$$= (\sqrt{2} - 1) \text{eq. (2)}$$

Now, From eq. (1) and eq. (2), we get,

$$\sqrt{x} - \frac{1}{\sqrt{x}} = (\sqrt{2} + 1) - (\sqrt{2} - 1) = 2$$

Sol.263.(d) when, $x + \frac{1}{x} = -2$

then, $x = -1$

Now,

$$x^{17} + x^{-17} + x^{12} + x^{-12} = -1 - 1 + 1 + 1 = 0$$

Sol.264.(b)

A number is the factor of its square. .

Squaring the identity $(mx+n)$

$$(mx + n)^2 = (m^2 x^2 + 2mnx + n^2)$$

Sol.265.(d)

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$$

When $(x + y + z) = 0$,

$$\text{then } x^3 + y^3 + z^3 = 3xyz \text{eq. (1)}$$

$(x + y + z) = 0$ (given)

$$\frac{x^2}{(yz)} + \frac{y^2}{(xz)} + \frac{z^2}{(xy)} = \frac{x^3 + y^3 + z^3}{xyz} = \frac{3xyz}{xyz} = 3$$

Sol.266.(d) $(x + y + z)^2$

$$= x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$(a + b + c)^2 = 9 + 2 \times 8$$

$$a + b + c = 5$$

$$\text{Now, } (a + b + c) - 3 = 5 - 3 = 2$$

Sol.267.(a) Formula:-

$$(a - b)^3 = a^3 - b^3 - 3ab(a - b)$$

$$a - b = 2 \text{ and } a^3 - b^3 = 80 \text{(given)}$$

$$\text{Now, } (2)^3 = 80 - 3ab(2) \Rightarrow ab = 12$$

$$\text{Sol.268.(d)} \frac{1}{x^2 + a^2} = x^2 - a^2$$

$$x^4 - a^4 = 1 \Rightarrow x^4 = a^4 + 1$$

$$x = \sqrt[4]{a^4 + 1}$$

Sol.269.(c) Statement 1.

$$100^2 - 99^2 + 98^2 - 97^2 \text{} + 22^2 - 21^2$$

$$\Rightarrow (100 + 99)(100 - 99) + (98 + 97)(98 -$$

$$97) \text{} + (22 + 21)(22 - 21)$$

$$\Rightarrow 100 + 99 + 98 + 97 \text{} 22 + 21$$

$$\text{Sum} = \frac{n}{2} \times (a + l)$$

$$= \frac{80}{2}(100 + 21) = 40 \times 121 = 4840$$

Statement 2.

$$\Rightarrow (k^2 + \frac{1}{k^2})(k - \frac{1}{k})(k^4 + \frac{1}{k^4})(k + \frac{1}{k})$$

$$(k^4 - \frac{1}{k^4})$$

$$= (k^2 + \frac{1}{k^2})(k^2 - \frac{1}{k^2})(k^4 + \frac{1}{k^4})(k^4 - \frac{1}{k^4})$$

$$= (k^4 - \frac{1}{k^4})(k^8 - \frac{1}{k^8}) \text{eq. (1)}$$

$$\text{Now, } (k^{16} - \frac{1}{k^{16}}) = (k^8 - \frac{1}{k^8})(k^8 + \frac{1}{k^8})$$

.....eq. (2)

Where, e.q. (1) $\neq$ e.q. (2)

Sol.270.(a) Clearly, option 'a' is the correct algebraic expression/.

$$(a - 1)(b - 1) = ab - a - b + 1$$

Sol.271.(d) $x = 1$ (given)

$$x^2 + y^2 + z^2 = xy + yz + zx \text{(given)}$$

Put, $y = 1$ and $z = 1$. which satisfies the given condition.

$$\frac{10x^4 + 5y^4 + 7z^4}{13x^2y^2 + 6y^2z^2 + 3z^2x^2} = \frac{10 + 5 + 7}{13 + 6 + 3} = 1$$

$$\text{Sol.272.(c)} x + \frac{1}{2x} = 3 \text{e.q. (1)}$$

$$2x + \frac{1}{x} = 6$$

Now,

$$(2x + \frac{1}{x})^3 = 8x^3 + \frac{1}{x^3} + 6(2x + \frac{1}{x})$$

$$6^3 - 6 \times 6 = 8x^3 + \frac{1}{x^3}$$

$$\text{Hence, } 8x^3 + \frac{1}{x^3} = 216 - 36 = 180$$

Sol.273.(c) $x - y = -10$

$$x^2 - 2xy = 84$$

Adding y^2 both sides

$$x^2 - 2xy + y^2 = 84 + y^2$$

$$(x - y)^2 = 84 + y^2$$

$$y^2 = 100 - 84 \Rightarrow y = 4$$

Sol.274.(a) When, $x + \frac{1}{x} = 1$,

$$\text{then } x^3 = -1$$

$$x^3 + 1 = 0 \text{e.q. (1)}$$

$$\text{Now, } x^{12} + x^9 + x^6 + x^3 + 1$$

$$\Rightarrow x^9(x^3 + 1) + x^3(x^3 + 1) + 1 \Rightarrow 1$$

Sol.275.(a) $a + b = 2$ eq. (1)

$$\frac{a}{b} + \frac{b}{a} = 1$$

$$\frac{a^2 + b^2}{ab} = 1 \Rightarrow a^2 + b^2 = ab$$

$$a^2 + b^2 - ab = 0 \text{eq. (2)}$$

$$a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

From eq (1) and (2)

$$a^3 + b^3 = 0$$

Sol.276.(a) $4x^2 + y^2 = 40$

Adding $(4xy)$ both sides, we get

$$(2x)^2 + y^2 + 4xy = 40 + 4xy \quad xy = 6 \text{ (given)}$$

$$(2x + y)^2 = 64 \Rightarrow 2x + y = 8$$

Short trick :-

$$4x^2 + y^2 = 40 \text{ and } xy = 6 \text{(given)}$$

Put $x = 3$ and $y = 2$ which satisfy the both given equations.

$$\text{Now, the value of } 2x + y = 2 \times 3 + 2 = 8$$

Sol.277.(c) $ab = 3$ (given)

$$2a + 3b = 10$$

On squaring,

$$4a^2 + 9b^2 + 12ab = 100$$

$$4a^2 + 9b^2 = 100 - 12 \times 3 = 64$$

Sol.278.(c) $xy = 8$ (given)

$$(4x - 7y) = 11$$

On squaring, we get

$$16x^2 + 49y^2 - 56xy = 121$$

$$16x^2 + 49y^2 = 121 + 56 \times 8 = 569$$

Sol.279.(d) $x + \frac{1}{x} = 2$,

it is only possible when $x = 1$

$$x^{57} + \frac{1}{x^{57}} = 1 + 1 = 2$$

Sol.280.(d) $mx^m - nx^n = 0$

$$x^m = \frac{nx^n}{m} \text{ ----(i) and } x^n = \frac{mx^m}{n}$$

$$\Rightarrow x^{2n} = \frac{m^2 x^{2m}}{n^2} \text{ ----(ii)}$$

$$\text{Now, } \frac{1}{x^m + x^n} + \frac{1}{x^m - x^n}$$

$$= \frac{x^m + x^n + x^m - x^n}{x^{2m} - x^{2n}} = \frac{2x^m}{x^{2m} - x^{2n}}$$

$$= \frac{2x^m}{\frac{n^2 x^{2n}}{m^2} - x^{2n}} = \frac{2x^m}{\frac{x^{2n}(n^2 - m^2)}{m^2}}$$

{using eq. (i) and (ii)}

$$\Rightarrow \frac{2x^m m^2}{x^{2n}(n^2 - m^2)} = \frac{2 \times \frac{nx^n}{m} \times m^2}{x^{2n}(n^2 - m^2)}$$

$$= \frac{2mn}{x^n(n^2 - m^2)}$$

Sol.281.(d) $5x^2 + 7x + 5 = 0$

Dividing both sides by $5x$.

$$x + \frac{1}{x} = \frac{-7}{5}$$

$$x^3 + \frac{1}{x^3} = \left(\frac{-7}{5}\right)^3 - \left(\frac{-7}{5}\right) \times 3$$

$$= \frac{-343}{125} + \frac{21}{5} = \frac{182}{125}$$

Sol.282.(d) Concept :- If $(x)^2 + (y)^2 = 0$, only possible when $x = 0$ and $y = 0$.

$$x^2 + y^2 + 2y + 4x + 5 = 0$$

$$\Rightarrow (x^2 + 4x + 4) + (y^2 + 2y + 1) = 0$$

$$\Rightarrow (x + 2)^2 + (y + 1)^2 = 0$$

$$\text{So, } x + 2 = 0 \Rightarrow x = -2$$

$$\text{And, } y + 1 = 0 \Rightarrow y = -1$$

$$\text{Now, } \frac{x+y}{x-y} = \frac{-3}{-1} = 3$$

Sol.283.(b) Concept :

When $a + b + c = 0$,

$$\text{then } a^3 + b^3 + c^3 = 3abc$$

$$\frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{6(x-y)(y-z)(z-x)}$$

$$= \frac{3(x-y)(y-z)(z-x)}{6(x-y)(y-z)(z-x)} = \frac{1}{2}$$

Sol.284.(b) $k + \frac{1}{k} + 2 = 0$

$$\Rightarrow k + \frac{1}{k} = -2 \quad (\text{Only possible when } k = -1)$$

$$\text{Then, } k^{10} + \frac{1}{k^{11}} = 1 + (-1) = 0$$

Sol.285.(b) We know that,

$$(a-b)^3 + 3ab(a-b) = a^3 - b^3$$

So,

$$3\left[a - \frac{1}{a}\right] + \left[a - \frac{1}{a}\right]^3 = a^3 - \frac{1}{a^3}$$

Sol.286.(a) If, $x^4 + \frac{1}{x^4} = 194$ then,

$$x^2 + \frac{1}{x^2} = \sqrt{194 + 2} = 14$$

$$\text{When, } x^2 + \frac{1}{x^2} = 14 \text{ then}$$

$$x + \frac{1}{x} = \sqrt{14 + 2} = 4$$

$$\text{Now, } x^3 + \frac{1}{x^3} = 4^3 - 4 \times 3 = 64 - 12 = 52$$

Sol.287.(b) $(a+1)^2 + (a+2)^2 = 16$

$$\Rightarrow a^2 + 2a + 1 + a^2 + 4a + 4 = 16$$

$$\Rightarrow 2a^2 + 6a + 5 = 16$$

$$\Rightarrow 2(2a^2 + 6a + 5) = 16 \times 2$$

$$\Rightarrow 4a^2 + 12a + 10 = 32$$

$$\text{So, } 4a^2 + 12a + 10 + 30 = 32 + 30$$

$$\Rightarrow 4a^2 + 12a + 40 = 62$$

Sol.288.(a) I. $K + \frac{1}{K} = 12$

$$\text{then } K^2 + \frac{1}{K^2} = 144 - 2 = 142 \text{ (True)}$$

II. The value of

$$(K^2 + \frac{1}{K^2})(K - \frac{1}{K})(K^4 + \frac{1}{K^4})(K + \frac{1}{K})$$

$$= (K^2 + \frac{1}{K^2})(K^2 - \frac{1}{K^2})(K^4 + \frac{1}{K^4})$$

$$= (K^4 - \frac{1}{K^4})(K^4 + \frac{1}{K^4})$$

$$= (K^8 - \frac{1}{K^8}) \neq K^{16} - \frac{1}{K^{16}} \text{ (not true)}$$

So, only 1 statement is correct.

Sol.289.(c) $2x - 1 = 0 \Rightarrow x = \frac{1}{2}$,

$$\text{Put } x = \frac{1}{2} \text{ in the given equation.}$$

$$3x^4 - 2x^2 + 4x - 1$$

$$= 3\left(\frac{1}{2}\right)^4 - 2\left(\frac{1}{2}\right)^2 + 4\left(\frac{1}{2}\right) - 1$$

$$= \frac{3}{16} - \frac{1}{2} + 2 - 1 = \frac{11}{16}$$

$$\text{So, the remainder} = \frac{11}{16}$$

Sol.290.(d) $(a+b)^2 - (a-b)^2$

$$= (a^2 + b^2 + 2ab) - (a^2 + b^2 - 2ab)$$

$$= a^2 + b^2 + 2ab - a^2 - b^2 + 2ab = 4ab$$

Sol.291.(d) $9m^2 + n^2 = 40$ and $mn = 4$

$$\Rightarrow 9m^2 + n^2 + 6mn = 40 + 6mn$$

$$\Rightarrow (3m + n)^2 = 40 + 24 = 64$$

$$\Rightarrow 3m + n = 8$$

Sol.292.(a) $x^2 + 4y^2 + 4y - 4xy - 2x - 8$

$$= (x^2 - 4xy + 4y^2) + 4y - 2x - 8$$

$$= (x - 2y)^2 - 2(x - 2y) - 8$$

$$(x - 2y)^2 - 4(x - 2y) + 2(x - 2y) - 8$$

$$= (x - 2y)\{(x - 2y) - 4\}$$

$$+ 2\{(x - 2y) - 4\}$$

$$= \{x - 2y - 4\}\{x - 2y + 2\}$$

Sol.293.(a) $y = 1 + \sqrt{3} + \sqrt{4}$

$$y - 1 = \sqrt{3} + \sqrt{4}$$

Squaring both side, we have :

$$(y-1)^2 = (\sqrt{3} + \sqrt{4})^2$$

$$y^2 + 1 - 2y = 3 + 4 + 4\sqrt{3}$$

$$y^2 + 1 - 2y = 7 + 4\sqrt{3} \Rightarrow y^2 - 2y - 6 = 4\sqrt{3}$$

Again, squaring both side, we have :

$$(y^2 - 2y - 6)^2 = (4\sqrt{3})^2$$

$$y^4 + 4y^2 + 36 - 4y^3 + 24y - 12y^2 = 48$$

$$y^4 - 8y^2 - 12 - 4y^3 + 24y = 0$$

Multiplying both side by 2,

$$2y^4 - 16y^2 - 24 - 8y^3 + 48y = 0$$

$$2y^4 - 8y^3 - 6y^2 + 28y - 84 - 10y^2 + 60 + 20y = 0$$

$$2y^4 - 8y^3 - 6y^2 + 28y - 84 - 10(y^2 - 6 - 2y) = 0$$

$$2y^4 - 8y^3 - 6y^2 + 28y - 84 = 10(y^2 - 6 - 2y)$$

$$= 10 \times 4\sqrt{3} = 40\sqrt{3}$$

Sol.294.(d) $x = \frac{\sqrt{5} - \sqrt{4}}{\sqrt{5} + \sqrt{4}}$

$$= \frac{(\sqrt{5} - \sqrt{4})(\sqrt{5} - \sqrt{4})}{(\sqrt{5} + \sqrt{4})(\sqrt{5} - \sqrt{4})} = (\sqrt{5} - \sqrt{4})^2$$

$$y = \frac{\sqrt{5} + \sqrt{4}}{\sqrt{5} - \sqrt{4}} = \frac{(\sqrt{5} + \sqrt{4})(\sqrt{5} + \sqrt{4})}{(\sqrt{5} - \sqrt{4})(\sqrt{5} + \sqrt{4})}$$

$$= (\sqrt{5} + \sqrt{4})^2$$

$$x^2 + y^2 = [\sqrt{5} - \sqrt{4}]^4 + [\sqrt{5} + \sqrt{4}]^4$$

$$= \{(\sqrt{5} - \sqrt{4})^2 + (\sqrt{5} + \sqrt{4})^2\}^2 - 2(\sqrt{5} - \sqrt{4})^2(\sqrt{5} + \sqrt{4})^2$$

$$= 18^2 - 2 = 324 - 2 = 322$$

$$\Rightarrow \frac{x^2 - xy + y^2}{x^2 + xy + y^2} = \frac{322 - 1}{322 + 1} = \frac{321}{323}$$

Sol.295.(b) $\frac{2p}{p^2 - 5p + 1} = \frac{1}{10}$

$$\Rightarrow p^2 - 5p + 1 = 20p$$

$$\Rightarrow p^2 - 25p + 1 = 0$$

Dividing both side by p.

$$\Rightarrow p + \frac{1}{p} = 25$$

Sol.296.(c) On checking all the options for the values a and b one by one, option (c) gets satisfied.

$$a^2 = b^2 + (ab)^2$$

$$\cot^2 x = \cos^2 x + \cos^2 x \cot^2 x$$

$$\cot^2 x = \cos^2 x(1 + \cot^2 x) = \cos^2 x \cdot \operatorname{cosec}^2 x$$

$$\cot^2 x = \cot^2 x$$

LHS = RHS (satisfied)

Sol.297.(b)

$$\frac{1}{1+a+\frac{1}{b}} + \frac{1}{1+b+\frac{5}{c}} + \frac{1}{1+\frac{c}{5}+\frac{1}{a}}$$

putting $a = b = 1$ and $c = 5$ in the given expression

$$\frac{1}{1+1+1} + \frac{1}{1+1+\frac{5}{5}} + \frac{1}{1+\frac{5}{5}+1}$$

$$= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{3}{3} = 1$$

Sol.298.(b)

$$4(z+7)(2z-1) = (4z+28)(2z-1)$$

$$= 8z^2 - 4z + 56z - 28 = 8z^2 + 52z - 28$$

$$\text{Now, } 8z^2 + 52z - 28 = Az^2 + Bz + C$$

On comparing both side, we have :

$$A = 8, B = 52, C = -28$$

$$\text{So, } A + B + C = 8 + 52 - 28 = 32$$

Sol.299.(c) $p^3 + 27 = p^3 + 3^3$

$$= (p+3)(p^2 + 3^2 - 3p) = (p+3)(p^2 + 9 - 3p)$$

Sol.300.(d) Formula used :

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$a^2 + b^2 + c^2 = 9^2 - 2 \times 23 = 81 - 46 = 35$$

$$\text{Sol.301.(b)} \quad x - \frac{1}{x} = \sqrt{(2\sqrt{5})^2 - 4}$$

$$= \sqrt{20 - 4} = \sqrt{16} = 4$$

$$x^3 - \frac{1}{x^3} = (4)^3 + 3 \times 4 = 64 + 12 = 76$$

Sol.302.(a) Formula : $x^3 + y^3 + z^3 = 3xyz$
(when $x + y + z = 0$)

$$\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy} = \frac{x^3 + y^3 + z^3}{xyz}$$

$$= \frac{3xyz}{xyz} = 3$$

Sol.303.(c) Formula : $a^3 + b^3 + c^3 = 3abc$,
if $a + b + c = 0$

$$(p - q) + (q - r) + (r - p) = 0$$

$$\text{So, } \frac{(p - q)^3 + (q - r)^3 + (r - p)^3}{12(p - q)(q - r)(r - p)} = \frac{3(p - q)(q - r)(r - p)}{12(p - q)(q - r)(r - p)} = \frac{1}{4}$$

$$\text{Sol.304.(b)} \quad (a + b)^3 - a^3 - b^3$$

$$= a^3 + b^3 + 3ab(a + b) - a^3 - b^3$$

$$= 2b^3 + 3ab(a + b)$$

$$\text{Sol.305.(b)} \quad \text{If } x^2 + \frac{1}{x^2} = 66$$

$$(x - \frac{1}{x})^2 = 66 - 2 = 64 \Rightarrow x - \frac{1}{x} = \sqrt{64} = 8$$

$$\text{Sol.306.(c)} \quad P - \frac{1}{P} = 6$$

$$P^2 + \frac{1}{P^2} = 6^2 + 2 = 38$$

$$P^4 + \frac{1}{P^4} = 38^2 - 2 = 1444 - 2 = 1442$$

Sol.307.(d)

$$x^4 - 15x^3 + 15x^2 - 15x + 40; \text{ (given } x = 14)$$

$$x^4 - 14x^3 - x^3 + 14x^2 + x^2 - 14x - x + 14 + 26$$

$$x^3(x - 14) - x^2(x - 14) + x(x - 14) - 1(x - 14) + 26$$

Putting $x = 14$, we get,

$$x^4 - 15x^3 + 15x^2 - 15x + 40 = 26$$

$$\text{Sol.308.(b)} \quad y - \frac{1}{y} = \sqrt{11^2 - 4}$$

$$= \sqrt{121 - 4} = \sqrt{117} = 3\sqrt{13}$$

$$y^3 - \frac{1}{y^3} = (3\sqrt{13})^3 + 3 \times 3\sqrt{13}$$

$$= 351\sqrt{13} + 9\sqrt{13} = 360\sqrt{13}$$

$$\text{Sol.309.(b)} \quad (x - \frac{1}{x})^2 = 12$$

$$x - \frac{1}{x} = \sqrt{12} = 2\sqrt{3}$$

$$\text{Now, } x + \frac{1}{x} = \sqrt{(2\sqrt{3})^2 + 4} = \sqrt{16} = 4$$

$$\text{So, } x^2 - \frac{1}{x^2} = 4 \times 2\sqrt{3} = 8\sqrt{3}$$

$$\text{Sol.310.(b)} \quad a^3 + b^3 + c^3 = 3abc,$$

either $a + b + c = 0$ or $a = b = c$

$$\text{Sol.311.(d)} \quad m + \frac{1}{m - 2} = 4$$

Subtracting both side by 2. We get :

$$m - 2 + \frac{1}{m - 2} = 2$$

$$\text{So, } (m - 2)^2 + \frac{1}{(m - 2)^2} = 2^2 - 2 = 2$$

$$\text{Sol.312.(a)} \quad 5x - \frac{5}{x} + 6 = 0$$

$$5x - \frac{5}{x} = -6 \Rightarrow 5(x - \frac{1}{x}) = -6$$

$$(x - \frac{1}{x}) = \frac{-6}{5}$$

$$x^2 + \frac{1}{x^2} = (\frac{-6}{5})^2 + 2 = \frac{86}{25}$$

$$\text{Sol.313.(c)} \quad \frac{a^3 - b^3}{a^2 - b^2} - \frac{3ab}{a + b}$$

$$= \frac{(a - b)(a^2 + b^2 + ab)}{(a - b)(a + b)} - \frac{3ab}{a + b}$$

$$= \frac{(a^2 + b^2 + ab)}{a + b} - \frac{3ab}{a + b} = \frac{a^2 + b^2 - 2ab}{a + b}$$

$$= \frac{(a - b)^2}{a + b} = \frac{(26 - 22)^2}{26 + 22} = \frac{16}{48} = \frac{1}{3}$$

$$\text{Sol.314.(b)} \quad (a - b + c)^2$$

$$= a^2 + b^2 + c^2 - 2ab - 2bc + 2ca$$

$$\text{Sol.315.(b)} \quad \frac{x}{8} + \frac{1}{\frac{x}{8}} = 1$$

By applying formula, $[a^3 + b^3 = (a + b)(a^2 - ab + b^2)]$ we get,

$$(\frac{x}{8})^3 + \frac{1}{(\frac{x}{8})^3} = 1 - 3 = -2 \Rightarrow \frac{x}{8} = -1$$

$$x = -8 \text{ So, } x^3 = (-8)^3 = -512$$

Sol.316.(d) Formula used :

$$p^3 + q^3 + r^3 = 3pqr \text{ (when } p + q + r = 0)$$

$$P + q + r = 8.15 + 9.06 - 17.21 = 0$$

$$\text{So, } p^3 + q^3 + r^3 - 3pqr = 3pqr - 3pqr = 0$$

$$\text{Sol.317.(b)} \quad x^4 + (\frac{2}{x})^4 = 15617$$

$$(x^2 + \frac{4}{x^2})^2 = 15617 + 8 = 15625$$

$$x^2 + (\frac{2}{x})^2 = \sqrt{15625} = 125$$

$$(x + \frac{2}{x})^2 = 125 + 4 = 129$$

$$x + \frac{2}{x} = \sqrt{129}$$

Sol.318.(a) Formula used ; $(x + y + z)^2$

$$= x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$0 = 40 + 2(xy + yz + zx)$$

$$-40 = 2(xy + yz + zx)$$

$$xy + yz + zx = \frac{-40}{2} = -20$$

Sol.319.(a)

$$\text{I. } x^3 + y^3 + z^3 = 12^3 - 2^3 - 10^3$$

$$= 1728 - 8 - 1000 = 720$$

$$\text{II. } 4(x^2 + y^2) = 4[(x + y)^2 - 2xy]$$

$$= 4[48^2 - 64] = 4[2304 - 64]$$

$$= 4 \times 2240 = 8960$$

Clearly, we can see that neither statement I nor II is correct..

$$\text{Sol.320.(a)} \quad K + \frac{1}{K} + 2 = 0$$

$$K + \frac{1}{K} = -2 \Rightarrow k = -1$$

$$\text{So, } K^{11} + \frac{1}{K^4} = (-1)^{11} + (-1)^4 = (-1) + 1 = 0$$

Sol.321.(b) It is given that $x + \frac{1}{x} = 6$

$$\text{Now, } \frac{3x}{2x^2 - 5x + 2}$$

On dividing numerator and denominator by x , we get :

$$\frac{3}{2x - 5 + \frac{2}{x}} = \frac{3}{2(x + \frac{1}{x}) - 5}$$

$$= \frac{3}{12 - 5} = \frac{3}{7}$$

Sol.322.(d)

$$8p^3 + q^3 = (2p + q)[(2p + q)^2 - 6pq]$$

$$361 = 19[361 - 6pq]$$

$$19 = 361 - 6pq \Rightarrow 342 = 6pq$$

$$pq = \frac{342}{6} = 57$$

$$\text{Sol.323.(a)} \quad \frac{a}{b} + \frac{b}{a} = -1$$

$$\frac{a^2 + b^2}{ab} = -1 \Rightarrow a^2 + b^2 = -ab$$

$$\text{Now, } (a - b)^2 = a^2 + b^2 - 2ab$$

$$4 = a^2 + b^2 - 2ab = -ab - 2ab = -3ab$$

$$ab = -\frac{4}{3}$$

$$a^3 - b^3 = (a - b)[(a - b)^2 + 3ab]$$

$$= 2[4 + 3 \times (\frac{-4}{3})] = 2 \times 0 = 0$$

$$\text{Sol.324.(d)} \quad x = 2 - 2^{\frac{1}{3}} + 2^{\frac{2}{3}}$$

$$x - 2 = 2^{\frac{2}{3}} - 2^{\frac{1}{3}}$$

Taking cube on both side,

$$(x - 2)^3 = (2^{\frac{2}{3}} - 2^{\frac{1}{3}})^3$$

$$x^3 - 8 - 6x^2 + 12x$$

$$= 4 - 2 - 3 \times 2^{\frac{2}{3}} \times 2^{\frac{1}{3}} (2^{\frac{2}{3}} - 2^{\frac{1}{3}})$$

$$x^3 - 10 - 6x^2 + 12x = -6(x - 2)$$

$$x^3 - 10 - 6x^2 + 12x + 6x - 12 = 0$$

$$x^3 - 6x^2 + 18x = 22$$

Sol.325.(b)

$$\frac{2abc}{9} = \frac{2 \times (-12)(-6)18}{9} = 288$$

Sol.326.(a) $x^4 + \frac{1}{x^4} = 47$

Adding both side by 2,

$$\left(x^2 + \frac{1}{x^2}\right)^2 = 49 \Rightarrow x^2 + \frac{1}{x^2} = \sqrt{49} = 7$$

Again, Adding both side by 2

$$\left(x + \frac{1}{x}\right)^2 = 9 \Rightarrow x + \frac{1}{x} = 3$$

$$\text{So, } x + \frac{1}{x} - 2 = 3 - 2 = 1$$

Sol.327.(c) Formula used :

$$a^3 + b^3 + c^3 = 3abc \text{ (when } a + b + c = 0)$$

$$a^2 - b^2 = (a - b)(a + b)$$

$$\text{Now, } (x - y) + (y - z) + (z - x) = 0$$

$$\Rightarrow (x - y)^3 + (y - z)^3 + (z - x)^3 = 3(x - y)(y - z)(z - x)$$

$$(x^2 - y^2) + (y^2 - z^2) + (z^2 - x^2) = 0$$

$$\Rightarrow (x^2 - y^2)^3 + (y^2 - z^2)^3 + (z^2 - x^2)^3 = 3(x^2 - y^2)(y^2 - z^2)(z^2 - x^2)$$

$$\begin{aligned} \text{So, } & \frac{(x - y)^3 + (y - z)^3 + (z - x)^3}{(x^2 - y^2)^3 + (y^2 - z^2)^3 + (z^2 - x^2)^3} \\ &= \frac{3(x - y)(y - z)(z - x)}{3(x^2 - y^2)(y^2 - z^2)(z^2 - x^2)} \\ &= \frac{1}{(x + y)(y + z)(z + x)} \end{aligned}$$

Sol.328.(c) As we know that,

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$10^2 = 38 + 2(ab + bc + ca)$$

$$62 = 2(ab + bc + ca) \Rightarrow 31 = ab + bc + ca$$

$$\text{Now, } (a - b)^2 + (b - c)^2 + (c - a)^2$$

$$= 2[a^2 + b^2 + c^2 - (ab + bc + ca)]$$

$$= 2[38 - 31] = 2 \times 7 = 14$$

Sol.329.(a) $16y^2 - k = \left(4y + \frac{3}{2}\right)\left(4y - \frac{3}{2}\right)$

$$\Rightarrow 16y^2 - k = \left(4y\right)^2 - \left(\frac{3}{2}\right)^2$$

$$[(a + b)(a - b) = (a^2 - b^2)]$$

$$\Rightarrow 16y^2 - k = 16y^2 - \frac{9}{4} \Rightarrow k = \frac{9}{4}$$

Sol.330.(c) Formula used :

$$(m - n)^3 = m^3 - n^3 - 3mn(m - n)$$

$$\Rightarrow \frac{\{(m^2 + n^2)(m - n) - (m - n)^3\}}{(m^2 n - mn^2)} =$$

$$\frac{\{m^3 - m^2 n + n^2 m - n^3 - (m^3 - n^3 - 3m^2 n + 3mn^2)\}}{(m^2 n - mn^2)}$$

$$= \frac{2(m^2 n - mn^2)}{m^2 n - mn^2} = 2$$

Sol.331.(d) Formula used :

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)$$

$$[x^2 + y^2 + z^2 - (xy + yz + zx)]$$

$$x + y + z = 8, x^2 + y^2 + z^2 = 20$$

$$\text{Now, } xy + yz + zx = \frac{8^2 - 20}{2} = 22$$

$$\begin{aligned} \text{So, } x^3 + y^3 + z^3 - 3xyz &= 8(20 - 22) \\ &= 8 \times (-2) = -16 \end{aligned}$$

Sol.332.(d)

$$\{(1 \div x) \times (x^3 - 2x^2) + (4^3 \div 2^5) \times x\} \div 25 = 1$$

$$\left\{\frac{1}{x} \times x(x^2 - 2x) + (64 \div 32) \times x\right\} = 25$$

$$\{(x^2 - 2x) + 2x\} = 25 \Rightarrow x^2 = 25 \Rightarrow x = \pm 5$$

So, the possible value of x is 5 and -5.

Sol.333.(a)

$$(7x + 4y)^2 + (7x - 4y)^2 = 2[(7x)^2 + (4y)^2]$$

$$= 2[49x^2 + 16y^2] = 98x^2 + 32y^2$$

Sol.334.(c) $6x + 7y = 5xy$ ----- (1)

$$10y - 4x = 4xy$$

$$5y - 2x = 2xy$$
 ----- (2)

$$\text{Then, [eq. (2)]} \times 3$$

$$15y - 6x = 6xy$$
 ----- (3)

Adding eqn. (1) and (3) we get:

$$22y = 11xy \Rightarrow x = 2$$

Putting x = 2 in eqn(2) we get : y = 4

Sol.335.(b) The best approach to solve his question is the value putting method.

By putting x = 6 and y = 4, we get :

$$x + y = 6 + 4 = 10 \text{ and } 2xy = 2 \times 6 \times 4$$

$$= 48(\text{satisfied})$$

$$\text{So, } 2x - y = 2 \times 6 - 4 = 8$$

Sol.336.(a) $a^2 + b^2 + 1 = 2a$

$$\Rightarrow (a - 1)^2 + b^2 = 0$$

$$\Rightarrow a = 1 \text{ and } b = 0$$

$$\text{So, } a^4 + b^7 = 1^4 + 0 = 1$$

Sol.337. (a) Formula used :

$$x^3 + y^3 + z^3 = 3abc \text{ (when } a + b + c = 0)$$

$$\text{Now, } \frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab} = \frac{a^3 + b^3 + c^3}{abc}$$

$$= \frac{3abc}{abc} = 3$$

Sol.338.(a) Formula used :

$$(x - y)^3 = x^3 - y^3 - 3xy(x - y)$$

$$(x + y)^3 - (x - y)^3 - 6y(x^2 - y^2)$$

$$= (x + y)^3 - (x - y)^3 - 3(x + y)(x - y)2y$$

$$= (x + y)^3 - (x - y)^3 - 3(x + y)(x - y)$$

$$[x + y - (x - y)]$$

$$= [(x + y) - (x - y)]^3 = [2y]^3 = 8y^3$$

Sol.339.(d) $x^2 + 6x + 1 = 0$

Dividing both side by x,

$$x + 6 + \frac{1}{x} = 0 \Rightarrow x + \frac{1}{x} = -6$$

$$x^3 + \frac{1}{x^3} = (-6)^3 - 3 \times (-6)$$

$$= -216 + 18 = -198$$

$$\text{Also, } x + 6 = -\frac{1}{x}$$

$$\begin{aligned} \text{Now, } (x + 6)^3 + \frac{1}{(x + 6)^3} &= -\left(\frac{1}{x^3} + x^3\right) \\ &= 198 \end{aligned}$$

Sol.340.(d) Putting b = 1, we have ;

$$a^2 = 82 - 1 = 81 \Rightarrow a = 9$$

$$\text{and } c^2 = 65 - 1 = 64 \Rightarrow c = 8$$

Now ,

$$\begin{aligned} 2a + 7b - 3c &= 2 \times 9 + 7 \times 1 - 3 \times 8 \\ &= 18 + 7 - 24 = 1 \end{aligned}$$

Sol.341.(b) $x(x - 5) = -1$

$$x - 5 = \frac{-1}{x} \Rightarrow x + \frac{1}{x} = 5 \text{..... (1)}$$

$$\text{Now, } x^3 + \frac{1}{x^3} = 5^3 - 3 \times 5 = 110$$

$$\Rightarrow x^6 + 1 = 110x^3$$

$$\Rightarrow x^6 - 110x^3 = -1$$

$$\Rightarrow x^3(x^3 - 110) = -1$$

Sol.342.(c) Given, $a + \frac{1}{a} = p^2$,

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$\text{Then, } a^2 + \frac{1}{a^2} = p^4 - 2$$

Sol.343.(a) a = -9.6, b = 4.44

and c = 5.16,

If a + b + c = 0

$$\Rightarrow -9.6 + 4.44 + 5.16 = 0$$

$$\text{then } a^3 + b^3 + c^3 - 3abc = 0$$

Sol.344.(b) $\frac{(187 \div 17 \times a - 3 \times 3)}{(8^2 - 9 \times 7 + a^2)} = 1$

$$\Rightarrow (11a - 9) = (64 - 63 + a^2)$$

$$\Rightarrow a^2 - 11a + 10 = 0$$

$$\Rightarrow a^2 - 10a - a + 10 = 0$$

$$\Rightarrow (a - 10)(a - 1) = 0$$

$$\Rightarrow a = 1, 10$$

Since, a is less than 10, a = 1

Sol.345.(c) $\sqrt{1 + \frac{x}{529}} = \frac{24}{23}$

$$1 + \frac{x}{529} = \frac{576}{529}$$

$$\frac{x}{529} = \frac{47}{529} \Rightarrow x = 47$$

Sol.346.(a)

$$k(21x^2 + 24) + rx + (14x^2 - 9) = 0 \text{ ----(i)}$$

$$k(7x^2 + 8) + px + (2x^2 - 3) = 0 \text{ ----(ii)}$$

Multiply equation (ii) by 3 and subtract from equation (i),

$$k(21x^2 + 24) + rx + (14x^2 - 9) - 3 \times [k(7x^2 + 8) + px + (2x^2 - 3)] = 0$$

$$8x^2 - (3p - r)x + 0 = 0 \text{ ----(iii)}$$

So let roots of equation (iii) are α and β

$$\alpha + \beta = \frac{3p - r}{8} \Rightarrow \alpha \beta = 0$$

Equations have both roots common.

$$\text{so, } \frac{3p - r}{8} = 0 \Rightarrow \frac{p}{r} = \frac{1}{3}$$

Sol.347.(c)

$$\frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}} \div \frac{\sqrt{14} + \sqrt{10}}{\sqrt{14} - \sqrt{10}} + \frac{\sqrt{10}}{\sqrt{5}}$$

$$= \frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}} \times \frac{\sqrt{14} - \sqrt{10}}{\sqrt{14} + \sqrt{10}} + \frac{\sqrt{2} \times \sqrt{5}}{\sqrt{5}}$$

$$= \frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}} \times \frac{\sqrt{2}(\sqrt{7} - \sqrt{5})}{\sqrt{2}(\sqrt{7} + \sqrt{5})} + \sqrt{2} = 1 + \sqrt{2}$$

Sol.348.(d) $(x + \frac{1}{x})^2 = 3$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 3 \Rightarrow x^2 + \frac{1}{x^2} = 1$$

On cubing above equation, we get

$$(x^2 + \frac{1}{x^2})^3 = 1^3 \Rightarrow x^6 + \frac{1}{x^6} + 3(x^2)(\frac{1}{x^2}) = 1$$

$$x^6 + \frac{1}{x^6} + 3 = 1 \Rightarrow x^6 + \frac{1}{x^6} = -2$$

So, $x^6 + x^{-6} = -2$

Sol.349.(a)

$$2X + 3Y = 12 \dots\dots(1)$$

$$3X - 2Y = 5 \dots\dots(2)$$

Multiplying eqn (1) by 3 and eqn (2) by 2, we get

$$6X + 9Y = 36 \dots\dots(3)$$

$$6X - 4Y = 10 \dots\dots(4)$$

Solving equation (3) and (4), we get

$$X = 3 \text{ and } Y = 2$$

So, option (a) is the correct answer.

Sol.350.(c) Let $x = 3$ and $y = -2$

Putting above values of x and y , we get

$$xy = (3)(-2) = -6 \dots\dots(\text{satisfied})$$

$$x^3 + y^3 = 3^3 + (-2)^3$$

$$= 27 - 8 = 19 \dots\dots(\text{satisfied})$$

Hence, $x = 3$ and $y = -2$

So,

$$\frac{1}{x^{-1}} + \frac{1}{y^{-1}} = x + y = 3 + (-2) = 3 - 2 = 1$$

Sol.351.(c) $2^x = 3^y = 6^z = K$

$$\Rightarrow 2 = K^{\frac{1}{x}}, 3 = K^{\frac{1}{y}}, 6 = K^{\frac{1}{z}}$$

Now, $2 \times 3 = 6$

$$\Rightarrow K^{\frac{1}{x}} \times K^{\frac{1}{y}} = K^{\frac{1}{z}} \Rightarrow K^{\frac{1}{x} + \frac{1}{y}} = K^{\frac{1}{z}}$$

$$\Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{z} \Rightarrow \frac{1}{x} + \frac{1}{y} - \frac{1}{z} = 0$$

Sol.352.(d) $a + b = \sqrt{11}$

$$\Rightarrow a^2 + b^2 = 11 - 2ab \dots\dots(1)$$

$$\Rightarrow a - b = \sqrt{7} \Rightarrow a^2 + b^2 = 7 + 2ab \dots\dots(2)$$

From equation (1) and (2) we have:

$$\Rightarrow 11 - 2ab = 7 + 2ab \Rightarrow 4 = 4ab \Rightarrow ab = 1$$

Putting $ab = 1$ in eq.(1) we get:

$$\Rightarrow a^2 + b^2 = 11 - 2 = 9$$

Now, $a^2 b^2 (a^2 + b^2) = 1^2 \times 9 = 9$

Sol.353.(a)

Arithmetic mean of x and $y = \frac{x+y}{2} = 8$

$$\Rightarrow x + y = 16$$

Geometric mean of x and $y = \sqrt{xy} = 3\sqrt{7}$

$$\Rightarrow xy = (3\sqrt{7})^2 = 63$$

$$x^3 + y^3 = (x+y)[(x+y)^2 - 3xy]$$

$$= 16[16^2 - 3 \times 63] = 16[256 - 189]$$

$$= 16 \times 67 = 1072$$

Sol.354.(c)

$$25x^2 + 16y^2 + 9z^2 - 40xy + 24yz - 30zx$$

$$= (-5x)^2 + (4y)^2 + (3z)^2 - (2.5x.4y)$$

$$+ (2.4y.3z) - (2.5x.3z)$$

$$= (-5x + 4y + 3z)^2$$

Sol.355.(a) $xy = 6$ and

$$x^2y + xy^2 + x + y = 63$$

$$\Rightarrow xy(x+y) + (x+y) = 63$$

$$\Rightarrow 6(x+y) + (x+y) = 63$$

$$\Rightarrow 7(x+y) = 63 \Rightarrow x+y = 9$$

Squaring both sides,

$$\Rightarrow x^2 + y^2 + 2xy = 81$$

$$\Rightarrow x^2 + y^2 = 81 - 12 \Rightarrow x^2 + y^2 = 69$$

Sol.356.(b) Formula: $a^3 + b^3 + c^3 - 3abc$

$$= (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$x^3 + 27y^3 + 64z^3 - 36xyz$$

$$= (x)^3 + (3y)^3 + (4z)^3 - 3.x.3y.4z$$

$$= (x+3y+4z)(x^2 + 9y^2 + 16z^2 - 3xy - 12yz - 4xz)$$

Sol.357.(b)

$$(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$$

$$= (5x+4y)^2 + (5x-4y)^2$$

$$= 2(25x^2 + 16y^2) = 50x^2 + 32y^2$$

Sol.358.(a) If $\frac{p}{q} + \frac{q}{p} = 2$

$$\Rightarrow p^2 + q^2 = 2pq$$

It is only possible when $p = q$.

$$\text{Then, } (\frac{p}{q})^{10} + (\frac{q}{p})^{10} = 1 + 1 = 2$$

Sol.359.(a) Let the required no. be A

ATQ,

$$(x^3 - 3x^2y + 3xy^2 - y^3) - A = x^3 - y^3$$

$$A = (x^3 - 3x^2y + 3xy^2 - y^3) - x^3 + y^3$$

$$A = -3x^2y + 3xy^2$$

Sol.360.(b) $(a+b)^2 = (a-b)^2 + 4ab$

$$= 10^2 + 4 \times 4 = 100 + 16 = 116$$

$$a^3 - b^3 = 10(10^2 + 3 \times 4) = 10(100 + 12)$$

$$= 10 \times 112 = 1120$$

$$\text{So, } a^3 - b^3 + 4(a+b)^2 = 1120 + 4 \times 116$$

$$= 1120 + 464 = 1584$$

Sol.361.(c) Putting $x = y = 1$, we get $a = 2$,

$$b = 0 \text{ and } c = 1$$

$$\text{So, } a^2 + b^2 + c^2 - 2ab + 2ac - 2bc = 2^2 + 0$$

$$+ 1^2 - 0 + 2 \times 2 \times 1 - 0 = 4 + 1 + 4 = 9$$

Now, on putting $x = y = 1$ in all the options given, option (c) gets satisfied.

$$(2x + 2y - 1)^2 = (2 + 2 - 1)^2$$

$$= 3^2 = 9 \text{ (satisfied)}$$

Sol.362.(a)

$$y^6 + \frac{1}{y^6} = (y^3 + \frac{1}{y^3})^2 - 2$$

$$\Rightarrow y^3 + \frac{1}{y^3} = (\sqrt{5})^3 - 3 \times \sqrt{5} = 5\sqrt{5} - 3\sqrt{5}$$

$$= 2\sqrt{5}$$

$$\text{So, } y^6 + \frac{1}{y^6} = (2\sqrt{5})^2 - 2 = 20 - 2 = 18$$

Sol.363.(b)

$$P = \frac{x^2 - 100}{x^2 - 16} = \frac{(x-10)(x+10)}{(x-4)(x+4)}$$

$$Q = \frac{x+10}{x-4}$$

$$\text{So, } \frac{P}{Q} = \frac{(x-10)(x+10)}{(x-4)(x+4)} \times \frac{x-4}{(x+10)}$$

$$= \frac{x-10}{x+4}$$

Sol.364.(a)

$$\frac{a}{2} = \frac{b}{3} = \frac{c}{4} = \frac{2a-3b+5c}{k} = n(\text{let})$$

$$a = 2n, b = 3n, c = 4n \text{ and}$$

$$2a - 3b + 5c = kn \dots\dots(i)$$

Put value of a, b and c in equation (i),

$$2 \times 2n - 3 \times 3n + 5 \times 4n = kn$$

$$\Rightarrow 4n - 9n + 20n = kn \Rightarrow 15n = kn \Rightarrow k = 15$$

Sol.365.(c) $\frac{x-1}{x} = 5 \Rightarrow x-1 = 5x$

$$\Rightarrow 4x = -1 \Rightarrow \frac{1}{x} = -4$$

$$\text{So, } 1 + \frac{1}{x^2} = 1 + 16 = 17$$

Sol.366.(c) $2x + \frac{1}{3x} = 12$

Multiplying both side by $\frac{3}{2}$,

$$3x + \frac{1}{2x} = 18$$

$$\text{So, } 9x^2 + \frac{1}{4x^2} = 18^2 - (2 \times 3 \times \frac{1}{2})$$

$$= 324 - 3 = 321$$

Sol.367.(c) $\frac{4}{x} - \frac{3}{2x} = 20$

$$\Rightarrow \frac{8-3}{2x} = 20 \Rightarrow 5 = 40x \Rightarrow x = \frac{5}{40} = \frac{1}{8}$$

Sol.368.(a) Formula used:

$$x^3 - y^3 = (x-y)[(x-y)^2 + 3xy]$$

$$\Rightarrow 520 = 5[5^2 + 3xy] \Rightarrow 104 = 25 + 3xy$$

$$\Rightarrow 79 = 3xy \Rightarrow xy = \frac{79}{3}$$

Sol.369.(d) Formula used:

$$x^3 + y^3 + z^3 = 3xyz,$$

$$\text{when } (x+y+z) = 0$$

$$\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy} = \frac{x^3 + y^3 + z^3}{xyz}$$

$$= \frac{3xyz}{xyz} = 3$$

Sol.370.(c)

$$(2a - b)^2 = (2a + b)^2 - 4(2a)b$$

$$(2a - b)^2 = 10^2 - 4 \times 9$$

$$(2a - b)^2 = 100 - 36 = 64 \Rightarrow 2a - b = 8$$

$$\text{Sol.371.(b)} \quad a^2 + b^2 = (a + b)^2 - 2ab$$

$$= 12^2 - 2 \times 35 = 144 - 70 = 74$$

$$a^4 + b^4 = (a^2 + b^2)^2 - 2a^2b^2 =$$

$$74^2 - 2 \times 35^2 = 5476 - 2450 = 3026$$

$$a^4 + b^4 - 10^3 = 3026 - 1000 = 2026$$

$$\text{Sol.372.(a)} \quad (a^3 + 8b^3) = (a+2b)$$

$$\{(a + 2b)^2 - 3 \times 2ab\}$$

$$5427 = 27(27^2 - 3 \times 2ab)$$

$$201 = 729 - 3 \times 2ab \Rightarrow 2ab = \frac{528}{3} = 176$$

$$\text{Sol.373.(c)} \quad x^2 + ax = b$$

By cubing both side, we get:

$$x^6 + x^3a^3 + 3ax^3 \times b = b^3$$

$$x^3(x^3 + a^3 + 3ab) = b^3$$

Now, comparing this equation with the

given equation i.e. $x^3(x^3 + c) = d$

$$c = a^3 + 3ab \text{ and } d = b^3$$

Sol.374.(c) As we know that,

$$x^3 + y^3 + z^3 - 3xyz$$

$$= (x + y + z)(x^2 + y^2 + z^2 - xy - yz - xz)$$

So, when $x + y + z = 0$,

$$\text{we have, } x^3 + y^3 + z^3 - 3xyz = 0$$

$$\text{Sol.375.(a)} \quad r + \frac{64}{r} = 16$$

On putting value, $r = 8$

$$8 + \frac{64}{8} = 16$$

$$16 = 16 \text{ (satisfy)}$$

$$r^4 + \frac{1}{r^3} = 8^4 + \frac{1}{8^3} = 4096 \frac{1}{512}$$

Sol.376.(c) On putting value, $p = 11$

$$p(p^2 + 3p + 3) \Rightarrow 11(11^2 + 3 \times 11 + 3)$$

$$\Rightarrow 11(121 + 33 + 3) \Rightarrow 11 \times 157 = 1727$$

$$\text{Sol.377.(a)} \quad \frac{X^2}{y^2} + \frac{y^2}{X^2} = 7$$

$$\text{Let } \frac{X}{y} = k \text{ then } \frac{y}{X} = \frac{1}{k}$$

$$k^2 + \frac{1}{k^2} = 7 \Rightarrow k^2 + \frac{1}{k^2} + 2 = 7 + 2$$

$$k + \frac{1}{k} = 3 \Rightarrow \frac{X^3}{y^3} + \frac{y^3}{X^3} = k^3 + \frac{1}{k^3}$$

$$k^3 + \frac{1}{k^3} = 3^3 - 3 \times 3$$

$$k^3 + \frac{1}{k^3} = 18 \text{ So, } \frac{X^3}{y^3} + \frac{y^3}{X^3} = 18$$

$$\text{Sol.378.(c)} \quad a^2 - 4a + 1 = 0$$

Dividing both side by a

$$a + \frac{1}{a} = 4; a^2 + \frac{1}{a^2} = 4^2 - 2 = 14$$

$$a^2 + \frac{1}{a^2} + a + \frac{1}{a} = 14 + 4 = 18$$

Sol.379.(b) Formula used :

$$x^3 - \frac{1}{x^3} = (x - \frac{1}{x})^3 + 3(x - \frac{1}{x})$$

$$x^3 - \frac{1}{x^3} = (\frac{7}{3})^3 + 3 \times \frac{7}{3} = \frac{343}{27} + 7$$

$$= \frac{532}{27} = 19\frac{19}{27}$$

Sol.380.(a)

$$x^3 + \frac{1}{x^3} = (\frac{11}{5})^3 - 3 \times \frac{11}{5}$$

$$= \frac{1331}{125} - \frac{33}{5} = \frac{506}{125} = 4\frac{6}{125}$$

Sol.381.(d) Formula used :

$$(1 + x + x^2)(1 - x + x^2) = 1 + x^2 + x^4$$

$$\text{So, } 1 + x + x^2 = \frac{585}{65} = 9$$

Sol.382.(c)

$$\text{Formula used : } a^2 + b^2 + c^2 - ab - bc - ca$$

$$= \frac{1}{2}[(a - b)^2 + (b - c)^2 + (c - a)^2]$$

$$\Rightarrow a^2 + b^2 + c^2 - ab - bc - ca$$

$$= \frac{1}{2}[(2022 - 2021)^2 +$$

$$(2021 - 2020)^2 + (2020 - 2022)^2]$$

$$\Rightarrow \frac{1}{2} \times [1 + 1 + 4] = \frac{1}{2} \times 6 = 3$$

$$\text{Sol.383.(d)} \quad (x - 2y)^2 = x^2 + 4y^2 - 4xy$$

$$= 40 - 4 \times 6 = 40 - 24 = 16 \Rightarrow x - 2y = 4$$

$$\text{Sol.384.(b)} \quad x^2 - 8x + 1 = 0$$

Dividing both sides by x. We get

$$x + \frac{1}{x} = 8 \Rightarrow x^2 + \frac{1}{x^2} = 64 - 2 = 62$$

$$x^4 + \frac{1}{x^4} = 3844 - 2 = 3842$$

Multiplying both sides by x^4

$$x^8 + 1 = 3842x^4 \Rightarrow x^8 - 3842x^4 + 1 = 0$$

$$\text{Sol.385.(c)} \quad 144x^2 - 36x + \frac{9}{4}$$

$$= (12x)^2 - 2 \times 12 \times \frac{3}{2} + (\frac{3}{2})^2 = (12x - \frac{3}{2})^2$$

$$\text{So, square of } 12x - \frac{3}{2} = 144x^2 - 36x + \frac{9}{4}$$

Sol.386.(b) It is given that,

$$a + b = 17, ab = 34$$

$$a^2 + b^2 = (a + b)^2 - 2ab$$

$$\Rightarrow 17^2 - 2 \times 34 = 289 - 68 = 221$$

Sol.387.(a) Formula used :

$$x^3 + y^3 + z^3 = 3xyz, \text{ when } x + y + z = 0$$

$$8x^3 + 27y^3 + 64z^3 = 72xyz$$

$$(2x)^3 + (3y)^3 + (4z)^3 = 3 \times 2x \times 3y \times 4z$$

$$\text{So, } 2x + 3y + 4z = 0 \Rightarrow 2x + 3y = -4z$$

Sol.388.(a) Formula used :

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$= \frac{[5.4^3 - 0.4^3]}{[5.4^2 + 5.4 \times 0.4 + 0.4^2]}$$

$$= \frac{(5.4 - 0.4)[5.4^2 + 5.4 \times 0.4 + 0.4^2]}{[5.4^2 + 5.4 \times 0.4 + 0.4^2]} = 5$$

Sol.389.(b)

$$x^5 - 10x^4 + 10x^3 - 10x^2 + 10x - 1$$

$$x^5 - 9x^4 - x^4 + 9x^3 + x^3 - 9x^2 - x^2 + 9x + x - 1$$

$$x^4(x - 9) - x^3(x - 9) + x^2(x - 9) - x(x - 9) + x - 1$$

Putting $x = 9$, we get

$$0 + 9 - 1 = 8$$

Sol.390.(a) Formula used :

$$(x + y)^3 = x^3 + y^3 + 3xy(x + y)$$

$$(p + 1)^3 - 1 = p^3 + 3p^2 + 3p = 26$$

$$(p + 1)^3 = 3^3 \Rightarrow p + 1 = 3 \Rightarrow p = 2$$

$$\text{Now, } p^2 + 2p = 2^2 + 2 \times 2 = 4 + 4 = 8$$

$$\text{Sol.391.(c)} \quad \frac{a}{b} + \frac{b}{a} = -1$$

$$\Rightarrow \frac{a^2 + b^2}{ab} = -1$$

$$\Rightarrow a^2 + b^2 = -ab \Rightarrow a^2 + b^2 + ab = 0 \dots (1)$$

$$\Rightarrow 6(a^3 - b^3) = 6(a - b)(a^2 + b^2 + ab)$$

$$\Rightarrow 6(a^3 - b^3) = 6(a - b) \times 0 \text{ (From eqn. 1)}$$

$$\Rightarrow 6(a^3 - b^3) = 0$$

$$\text{Sol.392.(a)} \quad p^4 + \frac{1}{p^4} + 2 = 4354 + 2 = 4356$$

$$p^2 + \frac{1}{p^2} = \sqrt{4356} = 66$$

$$p^2 + \frac{1}{p^2} - 2 = 66 - 2$$

$$p - \frac{1}{p} = \sqrt{64} = 8$$

$$p^3 - \frac{1}{p^3} = 8^3 + 3 \times 8 = 512 + 24 = 536$$

Sol.393.(a)

$$x^3 + y^3 = (x + y)[(x + y)^2 - 3xy]$$

$$= 14[14^2 - 3xy] = 2744 - 42xy$$

$$x^3 + y^3 + 42xy = 2744 - 42xy + 42xy = 2744$$

Short trick :

Putting $x = 14, y = 0$

$$x^3 + y^3 + 42xy = 14^3 + 0 + 0 = 2744$$

$$\text{Sol.394.(c)} \quad y = \frac{1}{3}(t + 1)$$

$$3y = t + 1 \Rightarrow x = 5t = t + 1$$

$$4t = 1 \Rightarrow t = \frac{1}{4}$$

$$\text{Sol.395.(a)} \quad x^4 + \frac{1}{x^4} = 2207$$

$$x^2 + \frac{1}{x^2} = \sqrt{2207 + 2} = 47$$

$$x + \frac{1}{x} = \sqrt{47 + 2} = 7$$

$$x^3 + \frac{1}{x^3} = 7^3 - 3 \times 7 = 343 - 21 = 322$$

$$x^5 + \frac{1}{x^5} = (x^3 + \frac{1}{x^3})(x^2 + \frac{1}{x^2}) - (x + \frac{1}{x})$$

$$= (47 \times 322) - 7 = 15134 - 7 = 15127$$

Sol.396.(b) $(2x + 3y + 4z) \times$
 $(4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8xz)$
 $= (2x)^3 + (3y)^3 + (4z)^3 - 3 \times 2x \times 3y \times 4z$
 $= 8x^3 + 27y^3 + 64z^3 - 72xyz$

Sol.397.(d)
 $\Rightarrow 3ab \times (a+b)^{-1} \times (a^{-1} + b^{-1})$
 $\Rightarrow 3ab \times \frac{1}{a+b} \times (\frac{1}{a} + \frac{1}{b})$
 $\Rightarrow 3ab \times \frac{1}{a+b} \times \frac{a+b}{ab} = 3$

Sol.398.(d) $x^4 + \frac{1}{x^4} = 14159$

$$x^2 + \frac{1}{x^2} = \sqrt{14159 + 2} = 119$$

$$x + \frac{1}{x} = \sqrt{119 + 2} = 11$$

Sol.399.(c) Formula used:
 $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$
 $8x^3 + 9x + 8 = 0$
 Dividing both sides by $8x$. We get
 $x + \frac{1}{x} = -\frac{9}{8}$
 $x^3 + \frac{1}{x^3} = (-\frac{9}{8})^3 - 3 \times x \times \frac{1}{x} \times (-\frac{9}{8})$
 $= -\frac{729}{512} + \frac{27}{8} = \frac{999}{512}$

Sol.400.(b) x can be written as $x + 2 - 2$
 So, $x^2 + 2x + 3$
 $= \{(x+2) - 2\}^2 + 2\{(x+2) - 2\} + 3$
 $\Rightarrow (x+2)^2 + 2^2 - 2 \times 2(x+2) + 2(x+2) - 4 + 3$
 $\Rightarrow (x+2)^2 + 4 - 4(x+2) + 2(x+2) - 4 + 3$
 $\Rightarrow (x+2)^2 - 2(x+2) + 3$

Sol.401.(b) $a + b = 2c$ So, $a - c = c - b$
 According to the question,
 $\frac{a}{a-c} + \frac{c}{b-c} = \frac{a}{c-b} + \frac{c}{b-c}$
 $= \frac{a}{-(b-c)} + \frac{c}{b-c} = \frac{c-a}{b-c} = \frac{a-c}{c-b} = 1$

Sol.402.(b) Formula used :
 $x^3 + y^3 + z^3 - 3xyz$
 $= (x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx)$
 $85 - 3 \times 20 = 25(x^2 + y^2 + z^2 - xy - yz - zx)$
 $85 - 60 = 25(x^2 + y^2 + z^2 - xy - yz - zx)$
 $(x^2 + y^2 + z^2 - xy - yz - zx) = \frac{25}{25} = 1$

Sol.403.(c) $(x^2 - 14x + 49)(x^2 + 6x + 9)$
 $(x^2 - 7x - 7x + 49)(x^2 + 3x + 3x + 9)$
 $\{x(x-7) - 7(x-7)\} \{x(x+3) + 3(x+3)\}$

$$(x-7)(x-7)(x+3)(x+3)$$

So,

$$\sqrt{(x-7)(x-7)(x+3)(x+3)}$$

$$= (x-7)(x+3)$$

Sol.404.(a) $(x^3 + y^3)^{\frac{2}{3}} = (x^2 + y^2)$

On cubing both sides, we get,

$$(x^3 + y^3)^2 = (x^2 + y^2)^3$$

$$\Rightarrow x^6 + y^6 + 2x^3y^3 = x^6 + y^6 + 3x^2y^2(x^2 + y^2)$$

$$\Rightarrow \frac{2x^3y^3}{3x^2y^2} = (x^2 + y^2) \Rightarrow (x^2 + y^2) = \frac{2}{3}xy$$

$$\Rightarrow \frac{(x^2 + y^2)}{xy} = \frac{2}{3}$$

Sol.405.(b) $a^2 - (x^2 + 2x + 1)$
 $\Rightarrow a^2 - (x+1)^2 \Rightarrow (a+x+1)(a-x-1)$
 So, the factors of $a^2 - x^2 - 2x - 1$
 are $(a+x+1)(a-x-1)$

Sol.406.(a) $x - 2y = 3$
 $(x+2y)^2 = (x-2y)^2 + 4 \times 2xy$
 $\Rightarrow (x+2y)^2 = 3^2 + 4 \times 5 = 9 + 20 = 29$

Sol.407.(d) $a^2 + b^2 + \frac{1}{a^2} + \frac{1}{b^2} = 4$
 $\Rightarrow a^2 + \frac{1}{a^2} - 2 + b^2 + \frac{1}{b^2} - 2 = 0$
 $\Rightarrow (a - \frac{1}{a})^2 + (b - \frac{1}{b})^2 = 0$
 $\Rightarrow (\frac{a^2-1}{a})^2 + (\frac{b^2-1}{b})^2 = 0$
 $\Rightarrow a^2 = 1, b^2 = 1$
 So, $a^4 + b^4 = 1 + 1 = 2$

Sol.408.(c) $a + b + c = 0, b + c = -a$
 $(a+b+c)^2 =$
 $a^2 + b^2 + c^2 + 2(ab+bc+ca)$
 $a^2 + b^2 + c^2 = -2\{a(b+c) + bc\}$
 $= 2(a^2 - bc) \Rightarrow \frac{a^2 + b^2 + c^2}{a^2 - bc} = 2$

Sol.409.(d)
 $a^3 - b^3 = (a-b)[(a-b)^2 + 3ab]$
 $999 = 3[3^2 + 3ab] \Rightarrow 333 = 9 + 3ab$
 $3ab = 333 - 9 = 324 \Rightarrow ab = \frac{324}{3} = 108$
 $(a+b)^2 = (3)^2 + 4 \times 108 = 9 + 432 = 441$
 $\Rightarrow a+b = 21$
 Now, $a^2 - b^2 = (a+b)(a-b) = 21 \times 3 = 63$

Sol.410.(a)
 $(a-1)^2 + (b+2)^2 + (c-1)^2 = 0$
 $a-1 = 0, b+2 = 0, c-1 = 0$
 $a = 1, b = -2, c = 1$
 $a^2 + b^2 + c^2 = 1^2 + (-2)^2 + 1^2 = 6$

Sol.411.(b)
 $16a^2 + 4b^2 + 9c^2 - 16ab + 12bc - 24ca$

$$= (-4a)^2 + (2b)^2 + (3c)^2 + 2 \times (-4a) \times 2b$$

$$+ 2 \times 2b \times 3c + 2 \times (-4a) \times 3c$$

$$= (-4a + 2b + 3c)^2 = (4a - 2b - 3c)^2$$

Sol.412.(d)
 $p(p^2 + 3p + 3) = p^3 + 3p^2 + 3p + 1 - 1$
 $= (p+1)^3 - 1 = 39^3 - 1$
 $= 59319 - 1 = 59318$

Sol.413.(a) $a + b = \sqrt{7}, a - b = \sqrt{5}$
 $(a+b)^2 = (a-b)^2 + 4ab$
 $(\sqrt{7})^2 = (\sqrt{5})^2 + 4ab \Rightarrow 7 - 5 = 4ab$
 $\Rightarrow ab = \frac{2}{4} = \frac{1}{2}$
 Now, $a^2 + b^2 = (\sqrt{7})^2 - 2 \times \frac{1}{2} = 7 - 1 = 6$
 $\Rightarrow 8ab(a^2 + b^2) - (a-b)^2 = 8 \times \frac{1}{2} \times 6 - (\sqrt{5})^2$
 $= 24 - 5 = 19$

Sol.414.(b) $P(x) = (x^3 - 8)(x+1)$
 $= (x-2)(x^2 + 2x + 4)(x+1)$
 $Q(x) = (x^3 + 1)(x-2)$
 $= (x+1)(x^2 - x + 1)(x-2)$
 LCM of $P(x)$ and $Q(x) =$
 $(x+1)(x-2)(x^2 + 2x + 4)(x^2 - x + 1)$

Sol.415.(a) Formula used : $(x+y+z)^2$
 $= x^2 + y^2 + z^2 + 2(xy + yz + zx)$
 Since $x + y + z = 0$,
 So, $x^2 + y^2 + z^2 = -2(xy + yz + zx)$
 $\Rightarrow x^2 + y^2 + z^2 = -2\{y(x+z) + zx\}$
 $= -2(-y^2 + zx) = 2(y^2 - zx) \quad (\because x + z = -y)$
 Similarly,
 $x^2 + y^2 + z^2 = 2(x^2 - yz)$ and $2(z^2 - xy)$
 Now, $\frac{x^2}{x^2 - yz} + \frac{y^2}{y^2 - zx} + \frac{z^2}{z^2 - xy}$
 $= \frac{2x^2}{x^2 + y^2 + z^2} + \frac{2y^2}{x^2 + y^2 + z^2} +$
 $\frac{2z^2}{x^2 + y^2 + z^2} = \frac{2(x^2 + y^2 + z^2)}{x^2 + y^2 + z^2} = 2$

Sol.416.(a) Given
 $x = \sqrt{10} + \sqrt{11}, y = \sqrt{10} - \sqrt{11}$
 then
 $= 7x^2 - 50xy + 7y^2$
 $= 7(21 + 2\sqrt{110}) - 50(-1) + 7(21 - 2\sqrt{110})$
 $= 147 + 14\sqrt{110} + 50 + 147 - 14\sqrt{110} = 344$

Sol.417.(a)
 $2^2(x)^{(a-b)(a+b)} \times 3^2(x)^{(b-c)(b+c)} \times 6^{-2}$
 $(x)^{(c-a)(c+a)}$
 $= 36 \times (x)^{a^2 - b^2 + b^2 - c^2} \times \frac{1}{36} \times (x)^{c^2 - a^2}$
 $= 1$

Sol.418.(b) $x - \frac{1}{x} = 11$
 On squaring both sides,

$$x^2 + \frac{1}{x^2} = 121 + 2 = 123$$

$$x^4 + \frac{1}{x^4} + 2 = 15129 \Rightarrow x^4 + \frac{1}{x^4} = 15127$$

On subtracting 2 both sides,

$$x^4 + \frac{1}{x^4} - 2 = 15127 - 2$$

$$\Rightarrow (x^2 - \frac{1}{x^2})^2 = 15125 \Rightarrow x^2 - \frac{1}{x^2} = 55\sqrt{5}$$

Sol.419.(d) $px^3 + x^2 + 3x + q$ is divisible by $(x + 2)$ and $(x - 2)$

When $x = -2$

$$p(-2)^3 + (-2)^2 + 3(-2) + q = 0$$

$$\Rightarrow -8p + 4 - 6 + q = 0 \Rightarrow 8p - q = -2 \dots(i)$$

$$\text{When } x = 2 \Rightarrow p(2)^3 + (2)^2 + 3(2) + q = 0$$

$$8p + 4 + 6 + q = 0 \Rightarrow 8p + q = -10 \dots(ii)$$

On solving eq (i) and (ii)

$$p = \frac{-3}{4}, q = -4$$

Sol.420.(d) Formula used :-

$$x^3 + y^3 = (x + y) \left[(x + y)^2 - 3xy \right]$$

$$\Rightarrow 416 = 8 \left[8^2 - 3xy \right]$$

$$52 = 64 - 3xy \Rightarrow 12 = 3xy \Rightarrow xy = \frac{12}{3} = 4$$

$$x^2 + y^2 = (x + y)^2 - 2xy$$

$$= 8^2 - 2 \times 4 = 64 - 8 = 56$$

$$x^4 + y^4 = 56^2 - 2 \times 4^2 = 3136 - 32 = 3104$$

Sol.421.(c)

$$x + y = 2z \Rightarrow x + y - z = 2z - z$$

$$x - z = z - y$$

Again,

$$x + y - 2y = 2z - 2y \Rightarrow x - y = 2(z - y)$$

$$\frac{x}{x - z} + \frac{y}{y - z} = \frac{x}{z - y} - \frac{y}{z - y}$$

$$= \frac{x - y}{z - y} = \frac{2(z - y)}{z - y} = 2$$

Short trick :

Putting $z = 0$, we get $x = -y$

$$\frac{x}{x - z} + \frac{y}{y - z} = \frac{x}{x - 0} + \frac{y}{y - 0}$$

$$= \frac{x}{x} + \frac{y}{y} = 1 + 1 = 2$$

Sol.422.(b)

$$ab + bc + ca = 0 \Rightarrow a(b + c) = -bc$$

$$\text{Similarly, } b(c + a) = -ca, c(a + b) = -ab$$

$$= \frac{1}{a^2 - bc} + \frac{1}{b^2 - ca} + \frac{1}{c^2 - ab}$$

$$= \frac{1}{a^2 + a(b + c)} + \frac{1}{b^2 + b(c + a)} + \frac{1}{c^2 + c(a + b)}$$

$$= \frac{1}{a(a + b + c)} + \frac{1}{b(a + b + c)} + \frac{1}{c(a + b + c)} = \frac{bc + ca + ab}{abc(a + b + c)} = 0$$

Sol.423.(b) $(x + 2)$ and $(x - 3)$ are the

factors of $x^2 + k_1x + k_2$.

$$(x + 2)(x - 3) = x^2 - x - 6$$

$$\text{Now, equate } x^2 - x - 6 \text{ and } x^2 + k_1x + k_2$$

We get, $k_1 = -1$ and $k_2 = -6$

Sol.424.(a) Putting $k = 1$, we get LHS = RHS i.e. $2 = 2$

$$\text{So, } 8 \times k \times k \times k = 8 \times 1^3 = 8$$

$$\text{Sol.425.(d)} p + \frac{1}{9} \sqrt{p} + k^2$$

$$= (\sqrt{p})^2 + 2 \times \frac{1}{18} \times \sqrt{p} + \left(\frac{1}{18}\right)^2$$

On comparing both side we get,

$$k^2 = \left(\frac{1}{18}\right)^2 \Rightarrow k = \pm \frac{1}{18}$$

Sol.426.(a) $x + y + 3 = 0$

$$x + y = -3 \dots(1)$$

On cubing equation (1), we get

$$x^3 + y^3 + 3xy(x + y) = -27$$

$$x^3 + y^3 + 3xy(-3) = -27$$

$$x^3 + y^3 - 9xy = -27 \dots(2)$$

Add 9 in equation (2) both side, we get

$$\Rightarrow x^3 + y^3 - 9xy + 9 = -27 + 9 = -18$$

$$\text{Sol.427.(c)} (4x + 2y)^3 + (4x - 2y)^3$$

$$= 16(Ax^3 + Bxy^2)$$

$$\Rightarrow [64x^3 + 8y^3 + 24xy(4x + 2y)] +$$

$$[64x^3 - 8y^3 - 24xy(4x - 2y)]$$

$$= 16(Ax^3 + Bxy^2)$$

$$\Rightarrow [64x^3 + 8y^3 + 96x^2y + 48xy^2] +$$

$$[64x^3 - 8y^3 - 96x^2y + 48xy^2]$$

$$= 16(Ax^3 + Bxy^2)$$

$$\Rightarrow 128x^3 + 96xy^2 = 16(Ax^3 + Bxy^2)$$

$$\Rightarrow 16(8x^3 + 6xy^2) = 16(Ax^3 + Bxy^2)$$

$$\Rightarrow 8x^3 + 6xy^2 = Ax^3 + Bxy^2$$

On comparing, we get $A = 8$ and $B = 6$

$$\frac{1}{2}(\sqrt{A^2 + B^2}) = \frac{1}{2}(\sqrt{8^2 + 6^2})$$

$$= \frac{1}{2}(\sqrt{10^2}) = \frac{10}{2} = 5$$

$$\text{Sol.428.(d)} x = 4 + \sqrt{15}$$

$$\text{So, } \frac{1}{x} = \frac{1}{4 + \sqrt{15}} \times \frac{4 - \sqrt{15}}{4 - \sqrt{15}}$$

$$= 4 - \sqrt{15}$$

$$\Rightarrow \left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$$

$$= (4 + \sqrt{15} + 4 - \sqrt{15})^2 - 2$$

$$= (8)^2 - 2 = 64 - 2 = 62$$

$$\text{Sol.429.(a)} x^3 + y^3 + z^3 - 3xyz$$

$$= (x + y + z) [x^2 + y^2 + z^2 - xy - yz - zx]$$

$$\Rightarrow x^3 + y^3 + z^3 - 3xyz$$

$$= (x + y + z) [(x + y + z)^2 - 3(xy + yz + zx)]$$

$$(xy + yz + zx)]$$

$$\Rightarrow 1604 - 3xyz = (11)[(11)^2 - 3(-6)]$$

$$\Rightarrow 1604 - 3xyz = (11)[121 + 18]$$

$$\Rightarrow 1604 - 3xyz = (11)[139]$$

$$\Rightarrow 3xyz = 1604 - 1529 \Rightarrow 3xyz = 75$$

$$\Rightarrow xyz = 25$$

$$\text{Sol.430.(a)} a^2 + b^2 + 49c^2 + 18$$

$$= 2(b - 28c - a)$$

$$a^2 + b^2 + 49c^2 + 18$$

$$= 2b - 56c - 2a$$

$$a^2 + 2a + 1 + b^2 - 2b$$

$$+ 1 + 49c^2 + 56c + 16 = 0$$

$$(a + 1)^2 + (b - 1)^2 + (7c + 4)^2 = 0$$

$$\text{So, } a = -1, b = 1 \text{ and } c = -\frac{4}{7}$$

$$a + b - 7c = -1 + 1 - 7\left(-\frac{4}{7}\right) = 0 + 4 = 4$$

Sol.431.(a) Let $k^3 = x$,

Now, $8k^6 + 15k^3 - 2 = 0$ becomes

$$8x^2 + 15x - 2 = 0.$$

$$8x^2 + 16x - x - 2 = 0$$

$$(8x - 1)(x + 2) = 0 \Rightarrow x = \frac{1}{8} \text{ or } -2$$

$$\text{For positive value we take } x = \frac{1}{8}$$

$$x = k^3 = \frac{1}{8}, k = \frac{1}{2}$$

$$\left(k + \frac{1}{k}\right) = \frac{1}{2} + 2 = 2\frac{1}{2}$$

Sol.432.(a) If $x - y + z = 0$ then

$$\Rightarrow x^3 - y^3 + z^3 = -3xyz$$

$$\Rightarrow -(-x^3 + y^3 - z^3) = -3xyz$$

$$\Rightarrow y^3 - x^3 - z^3 = 3xyz \dots\dots(1)$$

Now,

$$\frac{y^2}{2xz} - \frac{x^2}{2yz} - \frac{z^2}{2xy} = \frac{y^3 - x^3 - z^3}{2xyz}$$

$$= \frac{3xyz}{2xyz} = \frac{3}{2}$$

Sol.433.(c)

$$2\sqrt{2}x^3 - 3\sqrt{3}y^3 = (\sqrt{2}x - \sqrt{3}y)$$

$$(Ax^2 - Bxy + Cy^2) \dots\dots(1)$$

$$2\sqrt{2}x^3 - 3\sqrt{3}y^3 = (\sqrt{2}x - \sqrt{3}y)$$

$$(2x^2 + \sqrt{6}xy + 3y^2) \dots\dots(2)$$

On comparing we get,

$$A = 2, B = -\sqrt{6} \text{ and } C = 3$$

$$A^2 + B^2 + C^2 = 2^2 + (-\sqrt{6})^2 + 3^2$$

$$= 4 + 6 + 9 = 19$$

Sol.434.(a)

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)$$

$$[(x + y + z)^2 - 3(xy + yz + zx)]$$

$$\Rightarrow x^3 + y^3 + z^3 - 3xyz$$

$$= 18 \times [(18)^2 - 3(90)] = 972$$

$$\Rightarrow x^3 + y^3 + z^3 - 3xyz + 4xyz = 972 + 4xyz$$

$$\Rightarrow x^3 + y^3 + z^3 + xyz = 972 + 4 \times 81$$

$$\Rightarrow x^3 + y^3 + z^3 + xyz = 972 + 324 = 1296$$

$$\sqrt[4]{x^3 + y^3 + z^3 + xyz} = \sqrt[4]{1296} = 6$$

Sol.435.(b) $(a + b - c)^2 =$
 $a^2 + b^2 + c^2 + 2(ab - bc - ac)$
 $(5)^2 = a^2 + b^2 + c^2 + 2(10)$
 $a^2 + b^2 + c^2 = 25 - 20 = 5$

Sol.436.(d) $\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{3}$

On squaring, we get

$$x + \frac{1}{x} - 2 = 3 \Rightarrow x + \frac{1}{x} = 5$$

On squaring, we get

$$x^2 + \frac{1}{x^2} + 2 = 25 \Rightarrow x^2 + \frac{1}{x^2} = 23$$

On squaring, we get

$$x^4 + \frac{1}{x^4} + 2 = 529 \Rightarrow x^4 + \frac{1}{x^4} = 527$$

Sol.437.(d) Given,

$$a^3 + b^3 = 218 \text{ and } a + b = 2,$$

$$(a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

$$2^3 = 218 + 3ab \times 2$$

$$\Rightarrow -210 = 6ab \Rightarrow ab = -35$$

$$\sqrt{(1 - ab)} = \sqrt{1 - (-35)} = \sqrt{36} = 6$$

Short trick :

by hit and trial method $a = 7$ and $b = -5$

$$\sqrt{(1 - ab)} = \sqrt{1 - (-35)} = \sqrt{36} = 6$$

Sol.438.(d) $x^2 - 5x - 1 = 0$

$$\Rightarrow x^2 - 1 = 5x \text{ dividing both sides by } x$$

$$\Rightarrow x - \frac{1}{x} = 5$$

then

$$\Rightarrow \frac{x^6 - x^4 + x^2 - 1}{x^3} = x^3 - x + \frac{1}{x} - \frac{1}{x^3}$$

$$\Rightarrow (x^3 - \frac{1}{x^3}) - (x - \frac{1}{x})$$

Used identity

$$(x^3 - \frac{1}{x^3}) = (x - \frac{1}{x})^3 + 3 \times (x - \frac{1}{x})$$

Now,

$$\Rightarrow (5^3 + 3 \times 5) - 5 = 140 - 5 = 135$$

Sol.439.(c)

$$(x + y)^3 - (x - y)^3 - 3y(2x^2 - 3y^2) = ky^3$$

$$\Rightarrow x^3 + y^3 + 3x^2y + 3xy^2 - (x^3 - 3x^2y +$$

$$3xy^2 - y^3) - 6x^2y + 9y^3 = ky^3$$

$$\Rightarrow x^3 + y^3 + 3x^2y + 3xy^2 - x^3 + 3x^2y - 3xy^2$$

$$+ y^3 - 6x^2y + 9y^3 = ky^3$$

$$\Rightarrow 11y^3 = ky^3 \Rightarrow k = 11$$

Sol.440.(d) $\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{5}$ squaring

both sides

$$x + \frac{1}{x} = (\sqrt{5})^2 + 2 = 7$$

$$\frac{(x^4 + \frac{1}{x^2})}{(x^2 + 1)}$$

dividing numerator and

denominator by x

$$\Rightarrow \frac{(x^3 + \frac{1}{x^3})}{(x + \frac{1}{x})} = \frac{(7)^3 - 3 \times 7}{7} = 46$$

Sol.441.(a) Given that

$$a + b + c = 6, \quad a^2 + b^2 + c^2 = 32$$

$$\text{and } a^3 + b^3 + c^3 = 189$$

now,

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$\Rightarrow 6^2 = 32 + 2(ab + bc + ca)$$

$$\Rightarrow ab + bc + ca = 2$$

now,

$$a^3 + b^3 + c^3 - 3abc$$

$$= (a + b + c)[a^2 + b^2 + c^2 - (ab + bc + ca)]$$

$$\Rightarrow 189 - 3abc = 6 \times [32 - 2]$$

$$\Rightarrow 3abc = 9 \Rightarrow abc = 3 \Rightarrow 4abc = 12$$

Sol.442.(a) $x^4 - 79x^2 + 1 = 0$

$$\Rightarrow x^4 + 1 = 79x^2 \text{ divide eq by } x^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 79 \text{ add 2 both sides}$$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 81 \text{ ... take square root}$$

$$\text{both sides } \Rightarrow x + \frac{1}{x} = 9$$

Sol.443.(a) $4x - 3y = 12$

Squaring both sides

$$16x^2 + 9y^2 - 24xy = 144$$

$$16x^2 + 9y^2 = 144 + 24x(5)$$

$$16x^2 + 9y^2 = 144 + 120 = 264$$

$$\frac{16x^2 + 9y^2}{8} = \frac{264}{8} = 33$$

Sol.444.(b) $(a + \frac{1}{a} + 3)^2 = 16$

$$a + \frac{1}{a} + 3 = \pm 4 \Rightarrow a + \frac{1}{a} = 1 \text{ or } -7$$

$$\text{For } a + \frac{1}{a} = 1, a^2 + \frac{1}{a^2} = 1^2 - 2 = -1$$

(No option matches)

$$\text{For } a + \frac{1}{a} = -7, a^2 + \frac{1}{a^2}$$

$$= (-7)^2 - 2 = 49 - 2 = 47$$

Sol.445.(d) $a^3 + b^3 + c^3 - 3abc$

$$= (a + b + c)[(a + b + c)^2 - 3(ab + bc + ca)]$$

$$\Rightarrow a^3 + b^3 + c^3 - 3 \times (-40)$$

$$= 1 \times [1^2 - 3 \times (-22)]$$

$$a^3 + b^3 + c^3 + 120 = 67$$

$$a^3 + b^3 + c^3 = 67 - 120 = -53$$

Sol.446.(b) $2x^2 + 5x + 1 = 0$

On dividing by 2x,

$$x + \frac{5}{2} + \frac{1}{2x} = 0 \Rightarrow x + \frac{1}{2x} = -\frac{5}{2}$$

$$x - \frac{1}{2x} = \sqrt{\left(x + \frac{1}{2x}\right)^2 - 4 \times \frac{1}{2}}$$

$$x - \frac{1}{2x} = \sqrt{\left(-\frac{5}{2}\right)^2 - 2}$$

$$x - \frac{1}{2x} = \sqrt{\frac{25}{4} - 2} = \frac{\sqrt{17}}{2}$$

Sol.447.(a) $\frac{22\sqrt{2}}{4\sqrt{2} - \sqrt{3} + \sqrt{5}},$

dividing numerator and denominator by

$\sqrt{2}$, we get

$$\frac{22}{4 - \sqrt{\frac{3+\sqrt{5}}{2}}} = \frac{22}{4 - \sqrt{\frac{6+2\sqrt{5}}{4}}}$$

$$\frac{22}{4 - \sqrt{\left(\frac{\sqrt{5}+1}{2}\right)^2}} = \frac{22}{4 - \left(\frac{\sqrt{5}+1}{2}\right)}$$

$$= \frac{44}{7 - \sqrt{5}} = \frac{44(7 + \sqrt{5})}{(7 - \sqrt{5})(7 + \sqrt{5})}$$

$$= \frac{44(7 + \sqrt{5})}{44}$$

$$= 7 + \sqrt{5} \Rightarrow 7 + \sqrt{5} = a + \sqrt{5}b,$$

On comparing, $a = 7$ and $b = 1$

$$(ab) : (a + b) = 7 \times 1 : 7 + 1 = 7 : 8$$

Sol.448.(a) $2x - y = 2$

Both side divided by 2, then

$$x - \frac{y}{2} = 1 \Rightarrow (x - \frac{y}{2})^3 = 1$$

$$x^3 - \frac{y^3}{8} - 3x \frac{y}{2} (x - \frac{y}{2}) = 1$$

$$x^3 - \frac{y^3}{8} - \frac{3}{2} \times \frac{3}{2} (1) = 1$$

$$x^3 - \frac{y^3}{8} = 1 + \frac{9}{4} = \frac{13}{4}$$

Sol.449.(c)

$$x = \sqrt{1 + \frac{\sqrt{3}}{2}} - \sqrt{1 - \frac{\sqrt{3}}{2}}$$

$$= \sqrt{\frac{4 + 2\sqrt{3}}{4}} - \sqrt{\frac{4 - 2\sqrt{3}}{4}}$$

$$= \sqrt{\left(\frac{\sqrt{3}+1}{2}\right)^2} - \sqrt{\left(\frac{\sqrt{3}-1}{2}\right)^2}$$

$$= \frac{\sqrt{3}+1 - \sqrt{3}+1}{2} = 1$$

$$\text{so, } \frac{\sqrt{3}-x}{\sqrt{3}+x} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2 - \sqrt{3} = 2 - 1.73 = 0.27$$

Sol.450.(d) $x^2 - \sqrt{7}x + 1 = 0$

$$x + \frac{1}{x} = \sqrt{7} \Rightarrow x^2 + \frac{1}{x^2} = (\sqrt{7})^2 - 2 = 5$$

$$x^3 + \frac{1}{x^3} = (\sqrt{7})^3 - 3 \times \sqrt{7} = 4\sqrt{7}$$

$$x^5 + \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 + \frac{1}{x^3}) - (x + \frac{1}{x})$$

$$x^5 + \frac{1}{x^5} = 5 \times 4\sqrt{7} - \sqrt{7} = 19\sqrt{7}$$

Sol.451.(a)

$$x^3 + y^3 + z^3 - 3xyz = 0.98k$$

$$\Rightarrow \frac{1}{2} (x + y + z)[(x - y)^2$$

$$+ (y - z)^2 + (z - x)^2] = 0.98k$$

$$\Rightarrow \frac{1}{2}(32.5 + 34.6 + 30.9)[(32.5 - 34.6)^2 + (34.6 - 30.9)^2 + (30.9 - 32.5)^2]$$

$$= 0.98k$$

$$\Rightarrow \frac{1}{2}(98)[(-2.1)^2 + (3.7)^2 + (-1.6)^2]$$

$$= 0.98k$$

$$\Rightarrow (49)[(4.41) + (13.69) + (2.56)]$$

$$= 0.98k$$

$$\Rightarrow (49)[20.66] = 0.98k$$

$$\Rightarrow 1012.34 = 0.98k \Rightarrow k = 1033$$

Sol.452.(b) $x + \frac{1}{x} = 3 \dots (1)$

On squaring eqn (1), we get

$$x^2 + \frac{1}{x^2} = 7 \dots (2)$$

On squaring eqn (2), we get

$$x^4 + \frac{1}{x^4} = 47 \dots (3)$$

On cubing eqn (1), we get

$$x^3 + \frac{1}{x^3} = (3)^3 - (3)(3) = 27 - 9 = 18$$

So, $x^3 + \frac{1}{x^3} = 18 \dots (4)$

On multiplying (3) and (4), we get

$$(x^4 + \frac{1}{x^4})(x^3 + \frac{1}{x^3}) = (47)(18)$$

$$(x^7 + \frac{1}{x^7}) + (x + \frac{1}{x}) = 846$$

$$x^7 + \frac{1}{x^7} + 3 = 846 \text{ So, } x^7 + \frac{1}{x^7} = 843$$

Sol.453.(a) $x^4 + y^4 + x^2y^2$

$$= (x^2 - xy + y^2)(x^2 + xy + y^2)$$

$$\Rightarrow \frac{273}{16} = \frac{21}{4} \times (x^2 + xy + y^2)$$

$$(x^2 + xy + y^2) = \frac{273}{16} \times \frac{4}{21} = \frac{13}{4} \dots (i)$$

$$x^2 - xy + y^2 = \frac{21}{4} \dots (ii) \text{ (Given that)}$$

From equation (i) and (ii)

$$x^2 + y^2 = \frac{\frac{13}{4} + \frac{21}{4}}{2} = \frac{17}{4}$$

$$xy = \frac{\frac{13}{4} - \frac{21}{4}}{2} = -1$$

$$(x - y)^2 = x^2 + y^2 - 2xy = \frac{17}{4} + 2$$

$$(x - y)^2 = \frac{25}{4} \Rightarrow (x - y) = \frac{5}{2}$$

Sol.454.(c) According to the question, let x be negative and y be positive.

So, $x : y = -4 : 9$

$$\frac{3x + 4y}{2x - 3y} \dots (1)$$

In equation (1) dividing above by y in both numerator and denominator, we get

$$= \frac{3(\frac{x}{y}) + 4}{2(\frac{x}{y}) - 3} = \frac{3(\frac{-4}{9}) + 4}{2(\frac{-4}{9}) - 3}$$

$$= \frac{-12 + 36}{-8 - 27} = \frac{24}{-35} = -\frac{24}{35}$$

Sol.455.(c) Given, $\frac{11a^3 - 15b^3}{10a^3 + 4b^3} = \frac{5}{7}$

$$\Rightarrow 77a^3 - 105b^3 = 50a^3 + 20b^3$$

$$\Rightarrow 27a^3 = 125b^3 \Rightarrow \frac{a}{b} = \frac{5}{3}$$

$$(9a + 5b) : (5a - 3b)$$

$$\Rightarrow (9 \times 5 + 5 \times 3) : (5 \times 5 - 3 \times 3)$$

$$\Rightarrow (45 + 15) : (25 - 9) \Rightarrow 60 : 16 = 15 : 4.$$

Sol.456.(b)

$$\frac{1}{x} + \frac{1}{y} = \frac{16}{15} = \frac{x + y}{xy} = \frac{16}{15}$$

$$\frac{4}{xy} = \frac{16}{15}, xy = \frac{15}{4}$$

$$x^3 + y^3 = (x + y)(x^2 + y^2 - xy)$$

$$= (x + y)[(x + y)^2 - 3xy]$$

$$\Rightarrow x^3 + y^3 = 4(4^2 - 3 \times \frac{15}{4}) = 19$$

Sol.457.(b) Given that, $\frac{1}{x} - \frac{1}{y} = \frac{11}{24}$

$$\frac{y - x}{xy} = \frac{11}{24} \text{ So, } xy = -24$$

$$x^3 - y^3 + x^2y^2 = (x - y)\{(x - y)^2 + 3xy\} + x^2y^2$$

$$x^3 - y^3 + x^2y^2 = 11(121 - 72) + (-24)^2$$

$$x^3 - y^3 + x^2y^2 = 11 \times 49 + 576$$

$$= 539 + 576 = 1115$$

Short-trick:

Put $x = 8$ and $y = -3$

$$x - y = 11 \Rightarrow 8 - (-3) = 11 \Rightarrow xy = -24$$

$$x^3 - y^3 + x^2y^2 = 512 - (-27) + 64 \times 9$$

$$= 512 + 27 + 576 = 1115$$

Sol.458.(a) $16x^2 + y^2 = 48$

adding $8xy$ on both the sides

$$16x^2 + y^2 + 8xy = 48 + 16 = 64$$

$$(4x + y)^2 = 8^2$$

$$4x + y = 8 \text{ taking cube on both the side}$$

$$\Rightarrow 64x^3 + y^3 + 3 \times 4x \times y(4x + y) = 512$$

$$\Rightarrow 64x^3 + y^3 = 512 - 192 = 320$$

Sol.459.(c) If $x - \frac{1}{x} = 5$ then

$$x^3 - \frac{1}{x^3} = 125 + 15 = 140$$

$$\frac{x^6 - 5x^3 - 1}{x^6 + 7x^3 - 1} \text{ dividing the numerator and denominator by } x^3$$

$$\Rightarrow \frac{x^3 - 5 - \frac{1}{x^3}}{x^3 + 7 - \frac{1}{x^3}} = \frac{x^3 - \frac{1}{x^3} - 5}{x^3 - \frac{1}{x^3} + 7}$$

$$= \frac{140 - 5}{140 + 7} = \frac{45}{49}$$

Sol.460.(d) $x^4 + \frac{1}{x^4} = 727$ adding 2 and

taking root on both side

$$\Rightarrow x^2 + \frac{1}{x^2} = 27 \text{ subtracting 2 from both}$$

side and taking root

$$\Rightarrow x - \frac{1}{x} = 5 \text{ and } -5, \text{ But } x > 1 \text{ So, } x = 5$$

Sol.461.(c) $x - \frac{1}{x} = 1$

(On squaring both sides)

$$\Rightarrow x^2 + \frac{1}{x^2} = 1^2 + 2 = 3$$

Similarly,

$$\Rightarrow x^4 + \frac{1}{x^4} = 3^2 - 2 = 7$$

$$\Rightarrow x^8 + \frac{1}{x^8} = 7^2 - 2 = 47$$

Sol.462.(c)

$$x^3 + y^3 = (x + y)(x^2 + y^2 - xy)$$

$$= (x + y)[(x + y)^2 - 3xy]$$

$$405 = 9(81 - 3ab) \Rightarrow 45 = 81 - 3ab$$

$$3ab = 36 \Rightarrow ab = 12$$

Sol.463.(a) $(2x - \frac{3}{x}) = 2$

Squaring both side

$$4x^2 + \frac{9}{x^2} - 2 \times 2 \times 3 = 4$$

$$4x^2 + \frac{9}{x^2} = 16 \text{ (squaring both side)}$$

$$16x^4 + \frac{81}{x^4} + 2 \times 4 \times 9 = 256$$

$$16x^4 + \frac{81}{x^4} = 256 - 72 = 184$$

Sol.464.(b) $a - \frac{12}{a} = 1$

$$a^2 - 12 = a \Rightarrow a^2 - a - 12 = 0$$

$$a^2 - 4a + 3a - 12 = 0$$

$$a(a - 4) + 3(a - 4) = 0$$

$$(a + 3)(a - 4) = 0 \Rightarrow a = -3 \text{ and } 4$$

Put $a = 4$

$$a^2 + \frac{16}{a^2} = 16 + \frac{16}{16} = 16 + 1 = 17$$

Sol.465.(a)

$$4x^4 - 37x^2 + 9 = 0 \text{ dividing both side by } x^2$$

$$\text{Then } 4x^2 + \frac{9}{x^2} = 37 \text{ subtracting 12 and}$$

taking root from both side

$$2x - \frac{3}{x} = 5 \text{ taking cube on both side}$$

$$8x^3 - \frac{27}{x^3} = 5^3 + 3 \times 5 \times 2 \times 3 = 125 + 90 = 215$$

Sol.466.(a) $\frac{1}{x} + \frac{1}{y} = \frac{18}{5}$

$$\frac{x + y}{xy} = \frac{18}{5} \Rightarrow xy = \frac{5}{9}$$

$$x^3 + y^3 = (x + y)\{(x + y)^2 - 3xy\}$$

$$x^3 + y^3 = 2(4 - \frac{5}{3}) = \frac{14}{3} = 4\frac{2}{3}$$

Sol.467.(b) $(x - 2y)^2 = x^2 + 4y^2 - 4xy$

$$25 = 53 - 4xy \Rightarrow xy = 7$$

$$x^3 - 8y^3 = (x - 2y)(x^2 + 4y^2 + 2xy)$$

$$x^3 - 8y^3 = 5(53 + 14)$$

$$x^3 - 8y^3 = 5 \times 67 = 335$$

Sol.468.(b) $a^2 + c^2 + 17 = 2(a - 8b - 2b^2)$

$$a^2 + c^2 + 17 - 2a + 16b + 4b^2 = 0$$

$$(a - 1)^2 + (2b + 4)^2 + c^2 = 0$$

$$a = 1, b = -2, c = 0$$

$$a^3 + b^3 + c^3 = (1)^3 + (-2)^3 + 0 = -7$$

Sol.469.(a) Short-Trick: Put $x = 1$

$$4x^4 = 5x^2 - 1 \Rightarrow 4(1) = 5(1) - 1$$

$$\Rightarrow 4 = 4 \text{ (satisfy)}$$

$$(2x^2 - x - 1) = 2(1) - 1 - 1 = 0$$

Sol.470.(b)

$$x^2 + 9y^2 + 4z^2 = 12(x - 2y + 2z) - 88$$

$$x^2 - 12x + 9y^2 + 24y + 4z^2 - 24z + 88 = 0$$

$$x^2 - 12x + 36 + 9y^2 + 24y + 16 + 4z^2 - 24z + 36 = 0$$

$$(x - 6)^2 + (3y + 4)^2 + (2z - 6)^2 = 0$$

$$x = 6, y = -\frac{4}{3}, z = 3$$

$$(x - 3y + z) = (6 + 4 + 3) = 13$$

Sol.471.(c) Short-Trick:

$$(a + b)^2 - (a - b)^2 = 4ab$$

$$\frac{(375 + 125)^2 - (125 - 375)^2}{375 \times 375 - 125 \times 125}$$

$$= \frac{4 \times 375 \times 125}{500 \times 250} = \frac{3}{2}$$

Sol.472.(b)

$$x^2 + 1 = 5\sqrt{2}x \text{ (dividing both side by } x)$$

$$x + \frac{1}{x} = 5\sqrt{2} \Rightarrow x^2 + \frac{1}{x^2} = 48$$

$$\frac{(x^3 + \frac{1}{x})}{x^2 + 1} = \frac{(x^2 + \frac{1}{x^2})}{x + \frac{1}{x}} = \frac{48}{5\sqrt{2}} = \frac{24\sqrt{2}}{5}$$

Sol.473.(c) $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

Constant term would be $3ab^2$

$$= 3 \times 5x^2 \times \frac{1}{x^2} = 15$$

Sol.474.(a) $a^3 - b^3 = 2349$

$$(a - b)(a^2 + b^2 + ab) = 2349$$

$$(a^2 + b^2 + ab) = 261$$

$$(a + b)^2 - ab = a^2 + b^2 + ab = 261$$

Sol.475.(d) Short-Trick:

$$\text{Put } a = 2, b = -1, c = -1$$

$$\frac{(b+c)^2}{bc} + \frac{(c+a)^2}{ca} + \frac{(a+b)^2}{ab}$$

$$\frac{(-1-1)^2}{(-1) \times (-1)} + \frac{(-1+2)^2}{(-1) \times 2} + \frac{(2-1)^2}{2 \times (-1)}$$

$$= 4 - \frac{1}{2} - \frac{1}{2} = 3$$

Sol.476.(a)

$$a^2 + b^2 + c^2 + 216 = 12(a + b - 2c)$$

$$(a - 6)^2 + (b - 6)^2 + (c + 12)^2 = 0$$

$$a = 6, b = 6, c = -12$$

$$\sqrt{ab - bc - ca} =$$

$$\sqrt{6 \times 6 - \{6 \times (-12)\} - \{(-12) \times 6\}}$$

$$= \sqrt{180} = 6\sqrt{5}$$

Sol.477.(c)

$$(56\sqrt{7}x^3 - 2\sqrt{2}y^3) \div (2\sqrt{7}x - \sqrt{2}y)$$

$$= Ax^2 + By^2 - Cxy,$$

$$[(2\sqrt{7}x)^3 - (\sqrt{2}y)^3] \div (2\sqrt{7}x - \sqrt{2}y)$$

$$= Ax^2 + By^2 - Cxy$$

$$28x^2 + 2y^2 + 2\sqrt{14}xy = Ax^2 + By^2 - Cxy$$

Comparing the above equation we get,

$$A = 28, B = 2, C = -2\sqrt{14}$$

$$A + B - \sqrt{14}C = 28 + 2 + 28 = 58$$

Sol.478.(b) $x = 7, y = 3$ satisfies given equation

$$x^4 - y^4 = 2401 - 81 = 2320$$

Sol.479.(a)

$$8 + 2px^2 - 36x - 27x^3 = (2 - 3x)^3$$

$$8 + 2px^2 - 36x - 27x^3 = 8 + 54x^2 - 36x - 27x^3$$

Comparing coefficient of x^2 term, $p = 27$

Sol.480.(b) $2x^2 - 7x + 5 = 0$

$$\Rightarrow x + \frac{5}{2x} = \frac{7}{2}$$

Cubing both side,

$$\Rightarrow x^3 + \frac{125}{8x^3} = \frac{343}{8} - (3 \times \frac{7}{2} \times \frac{5}{2})$$

$$\Rightarrow x^3 + \frac{125}{8x^3} = \frac{133}{8} \Rightarrow x^3 + \frac{125}{8x^3} = 16\frac{5}{8}$$

$$\text{Sol.481.(d)} \quad x + \frac{1}{x} = \frac{17}{4}$$

Squaring both side,

$$x^2 + \frac{1}{x^2} = \frac{289}{16} - 2$$

$$x^2 + \frac{1}{x^2} - 2 = \frac{225}{16} \Rightarrow x - \frac{1}{x} = \frac{15}{4}$$

$$\text{Short-Trick: } x + \frac{1}{x} = \frac{17}{4}$$

$$x = 4, \text{ satisfy}$$

$$x - \frac{1}{x} = 4 - \frac{1}{4} = \frac{15}{4}$$

Sol.482.(b)

$$9(a^2 + b^2) + c^2 + 20 = 12(a + 2b)$$

$$(3a - 2)^2 + (3b - 4)^2 + c^2 = 0$$

$$a = \frac{2}{3}, b = \frac{4}{3}, c = 0$$

$$\Rightarrow \sqrt{6a + 9b + 2c}$$

$$= \sqrt{6 \times \frac{2}{3} + 9 \times \frac{4}{3} + 2 \times 0}$$

$$= \sqrt{4 + 12} = 4$$

Sol.483.(a) $2x + 3y + 1 = 0$

$$2x + 3y = -1$$

On cubing both sides,

$$(2x + 3y)^3 = (-1)^3$$

$$8x^3 + 27y^3 + 3 \times 2x \times 3y \times (-1) = -1$$

$$8x^3 + 27y^3 - 18xy = -1$$

Adding 8 both sides,

$$8x^3 + 27y^3 - 18xy + 8 = 7$$

Sol.484.(a)

$$\frac{5[(a^6 - b^6)^3 + (b^6 - c^6)^3 + (c^6 - a^6)^3]}{2[(a^3 - b^3)^3 + (b^3 - c^3)^3 + (c^3 - a^3)^3]}$$

As we know that when $a + b + c = 0$ then

$$a^3 + b^3 + c^3 = 3abc$$

$$\text{So, } \frac{5[3(a^6 - b^6)(b^6 - c^6)(c^6 - a^6)]}{2[3(a^3 - b^3)(b^3 - c^3)(c^3 - a^3)]} =$$

$$\frac{5[3(a^3 + b^3)(b^3 + c^3)(c^3 + a^3)(a^3 - b^3)(b^3 - c^3)(c^3 - a^3)]}{2[3(a^3 - b^3)(b^3 - c^3)(c^3 - a^3)]}$$

$$= \frac{5}{2}(a^3 + b^3)(b^3 + c^3)(c^3 + a^3)$$

Sol.485.(c) $y = 2x + 1,$

$$y - 2x = 1$$

Cubing both sides,

$$y^3 - 8x^3 - 3 \times y \times 2x \times (y - 2x) = 1$$

$$y^3 - 8x^3 - 6xy = 1$$

$$8x^3 - y^3 + 6xy = -1$$

Sol.486.(b)

$$x^8 - 34x^4 + 1 = 0 \text{ dividing both side by } x^4$$

$$x^4 + \frac{1}{x^4} = 34 \text{ adding 2 and taking root}$$

on both side

$$x^2 + \frac{1}{x^2} = 6 \text{ subtracting 2 and taking root}$$

on both side

$$x - \frac{1}{x} = 2 \text{ taking cube on both side}$$

$$x^3 - \frac{1}{x^3} = 2^3 + 2 \times 3 = 8 + 6 = 14$$

Sol.487.(c) $x^4 + y^4 + x^2y^2 = 117$

$$x^2 + y^2 - xy = 3(4 + \sqrt{3})$$

$$= 12 + 3\sqrt{3} \dots (i)$$

$$x^4 + y^4 + x^2y^2 =$$

$$(x^2 + y^2 - xy)(x^2 + y^2 + xy)$$

$$117 = 3(4 + \sqrt{3})(x^2 + y^2 + xy)$$

$$(x^2 + y^2 + xy) = \frac{39}{4 + \sqrt{3}} = 3(4 - \sqrt{3})$$

$$= 12 - 3\sqrt{3} \dots (ii)$$

On adding (i) and (ii)

$$x^2 + y^2 = 12$$

Sol.488.(b) $x^2 - 3x + 1 = 0$

$$x + \frac{1}{x} = 3 \Rightarrow x^2 + \frac{1}{x^2} = 7$$

$$x^4 + \frac{1}{x^4} = 47 \Rightarrow x^8 + \frac{1}{x^8} = 2207$$

$$2(x^8 + \frac{1}{x^8}) - 5(x^2 + \frac{1}{x^2})$$

$$= 2 \times 2207 - 5 \times 7 = 4379$$

Sol.489.(b) $x - y - z = 0$

$$x = 2, y = 1, z = 1,$$

satisfies given equation

$$(x^2 + y^2 + z^2) \div (y^2 + zx) = 2$$

Sol.490.(d) Using the identity

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$\text{Let } a = 2x + y, b = x + 2y$$

$$a^2 + b^2 = (2x + y)^2 + (x + 2y)^2$$

$$= 5x^2 + 5y^2 + 8xy$$

$$ab = (2x + y)(x + 2y) = 2x^2 + 2y^2 + 5xy$$

$$(2x + y)^3 - (x + 2y)^3$$

$$= (x - y)(5x^2 + 5y^2 + 8xy + 2x^2 + 2y^2 + 5xy)$$

$$(2x + y)^3 - (x + 2y)^3$$

$$= (x - y)(7x^2 + 7y^2 + 13xy)$$

On comparing

$$(2x + y)^3 - (x + 2y)^3 = (x - y) \times$$

$$[A(x^2 + y^2) + Bxy]$$

$$A = 7 \text{ and } B = 13$$

$$2A - B = 14 - 13 = 1$$

Sol.491.(d) $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$

$$(12)^3 = 468 + 3xy \times 12$$

$$1728 - 468 = 36xy$$

$$\frac{1260}{36} = xy, xy = 35$$

$$(x + y)^2 = x^2 + y^2 + 2xy$$

$$x^2 + y^2 = 144 - 70 = 74$$

$$(x^2 + y^2)^2 = x^4 + y^4 + 2x^2y^2$$

$$x^4 + y^4 = (74)^2 - 2(35)^2$$

$$= 5476 - 2450 = 3026$$

Sol.492.(c)

$$a^2 + b^2 + c^2 + 170 = 2(8a + 5b - 9c)$$

$$a^2 + b^2 + c^2 + 64 + 25 + 81 - 16a - 10b + 18c = 0$$

$$(a - 8)^2 + (b - 5)^2 + (c + 9)^2 = 0$$

$$a = 8, b = 5 \text{ and } c = -9$$

$$4a + 8b - c = 32 + 40 + 9 = 81$$

$$\Rightarrow \sqrt{4a + 8b - c} = 9$$

Sol.493.(b) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$

$$\Rightarrow (xy + yz + zx) = 0$$

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)[(x^2 + y^2 + z^2) - (xy + yz + zx)]$$

$$\Rightarrow 151 - 36 = 5 \times [(x^2 + y^2 + z^2) - 0]$$

$$x^2 + y^2 + z^2 = 23$$

Sol.494.(d) $\frac{1}{x} + \frac{1}{y} = \frac{20}{9}$

$$\Rightarrow xy = \frac{9}{20} \times (x + y) \Rightarrow xy = \frac{9}{4}$$

$$x^3 + y^3 = (x + y)^3 - 3xy(x + y)$$

$$= 125 - 3 \times \frac{9}{4} \times 5$$

$$= 125 - \frac{135}{4} = \frac{365}{4}$$

Sol.495.(a)

$$(7x + 3)^3 + (x - 2)^3 + 27(2x - 5)^3 - 9(7x + 3)$$

$$(x - 2)(2x - 5) = 0$$

$$\Rightarrow 7x + 3 + x - 2 + 6x - 15 = 0$$

$$14x = 14 \Rightarrow x = 1$$

$$\text{So, } (5x + 3) = 5 + 3 = 8$$

Sol.496.(d) $(x + \frac{2}{x}) = 7$

Squaring both sides

$$x^2 + \frac{4}{x^2} + 2 \times x \times \frac{2}{x} = 49$$

$$x^2 + \frac{4}{x^2} + 4 = 49 \Rightarrow x^2 + \frac{4}{x^2} = 45$$

Multiply by 2 both sides

$$(2x^2 + \frac{8}{x^2}) = 90$$

Sol.497.(d) $(\frac{x}{y} + 1) = 4 \Rightarrow \frac{x}{y} = 3$

Squaring both side, then we get $\frac{x^2}{y^2} = 9$

Add 1 both side, then we get

$$\Rightarrow \frac{x^2}{y^2} + 1 = 9 + 1 \Rightarrow \frac{x^2 + y^2}{y^2} = 10$$

Short-trick:

Put, $x = 3$ and $y = 1$

$$\frac{x^2 + y^2}{y^2} = \frac{9 + 1}{1} = 10$$

Sol.498.(b) $a^2 + 4b^2 + 25c^2 + 18$

$$= 2(a - 2b + 20c)$$

$$\Rightarrow (a^2 - 2a + 1) + (4b^2 + 4b + 1)$$

$$+ (25c^2 - 40c + 16) = 0$$

$$(a - 1)^2 + (2b + 1)^2 + (5c - 4)^2 = 0$$

$$a = 1, b = -0.5, c = 0.8$$

$$\text{Now, } a + 2b + 5c = 1 - 1 + 4 = 4$$

Sol.499.(c)

$$\frac{0.8 \times 0.8 \times 0.8 + 0.6 \times 0.6 \times 0.6}{0.08 \times 0.08 + 0.06 \times 0.06 - 0.08 \times 0.06}$$

$$= \frac{10000 \times (8^3 + 6^3)}{1000 \times (8^2 + 6^2 - 8 \times 6)}$$

$$= \frac{10 \times (8 + 6) \times (8^2 + 6^2 - 8 \times 6)}{(8^2 + 6^2 - 8 \times 6)} = 140$$

Sol.500.(d) $(x - y)^2 = x^2 + y^2 - 2xy$

$$25 = 45 - 2xy \Rightarrow xy = 10$$

$$\text{Now, } (x - y)^3 = x^3 - y^3 - 3xy(x - y)$$

$$125 = x^3 - y^3 - 3 \times 10 \times 5$$

$$\Rightarrow x^3 - y^3 = 275$$

Sol.501.(a)

$$a^3 + b^3 + c^3 - 3abc =$$

$$(a + b + c)[(a + b + c)^2 - 3(ab + bc + ca)]$$

$$a^3 + b^3 + c^3 + 54 = 4[16 + 42]$$

$$\Rightarrow a^3 + b^3 + c^3 = 178$$

$$\sqrt{4a^3 + 4b^3 + 4c^3 - 36} =$$

$$\sqrt{4(178 - 9)} = \sqrt{4 \times 169} = 26$$

Sol.502.(d) Short-Trick: Put $x = 1$

$$\frac{(x^4 + x^2 + 1)}{(x^2 + x + 1)} = 1$$

Check option (d)

$$(x^2 - x + 1) = 1$$

Sol.503.(d) $a + b = p$

Squaring both side

$$a^2 + b^2 + 2ab = p^2 \Rightarrow a^2 + b^2 = p^2 - 2q$$

Squaring both side

$$a^4 + b^4 + 2a^2b^2 = p^4 + 4q^2 - 4p^2q$$

$$a^4 + b^4 = p^4 - 4p^2q + 2q^2$$

Short trick :

$$\text{Let } a = 1, b = 1 \Rightarrow p = 2, q = 1$$

$$\Rightarrow (a^4 + b^4) = 2$$

Check option (d)

$$p^4 - 4p^2q + 2q^2 = 2$$

Sol.504.(c) $\sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{6}$

Squaring both side

$$x + \frac{1}{x} = 4$$

Squaring both side

$$x^2 + \frac{1}{x^2} = 14 \text{ ----- (1)}$$

Taking cube both side

$$x^6 + \frac{1}{x^6} = 14^3 - 14 \times 3$$

$$x^6 + \frac{1}{x^6} = 2702$$

Sol.505.(c) $x^2 + 1 - 2x = 0$

$$x^2 + 1 = 2x \text{ (On squaring both side)}$$

$$\Rightarrow x^4 + 1 + 2x^2 = 4x^2$$

$$\Rightarrow x^4 + 1 = 2x^2 \Rightarrow x^2(x^2 - 2) = -1$$

Sol.506.(c)

$$x^2 + 4y^2 + 3z^2 + \frac{19}{4} = 2\sqrt{3}(x + y + z)$$

$$\Rightarrow x^2 - 2\sqrt{3}x + 4y^2 - 2\sqrt{3}y + 3z^2 - 2\sqrt{3}z + \frac{19}{4} = 0$$

$$(x - \sqrt{3})^2 + (2y - \frac{\sqrt{3}}{2})^2 + (\sqrt{3}z - 1)^2 = 0$$

$$(x - \sqrt{3}) = 0, (2y - \frac{\sqrt{3}}{2}) = 0, (\sqrt{3}z - 1) = 0$$

$$x = \sqrt{3}, y = \frac{\sqrt{3}}{4}, z = \frac{1}{\sqrt{3}}$$

$$(x - 4y + 3z) = \sqrt{3} - 4 \times \frac{\sqrt{3}}{4} + 3 \times \frac{1}{\sqrt{3}} = \sqrt{3}$$

Sol.507.(c) $9x^2 - 3x - 3x + 1 = 0$

$$3x(3x - 1) - 1(3x - 1) = 0$$

$$(3x - 1)(3x - 1) = 0 \Rightarrow x = \frac{1}{3}$$

$$27x^3 + (27x^3)^{-1} = 27(\frac{1}{3})^3 + \{27(\frac{1}{3})^3\}^{-1}$$

$$= 1 + 1 = 2$$

Sol.508.(c) $(4x - 5)^3 + (x - 2)^3 + 27(2x - 5)^3$

$$= 9(4x - 5)(x - 2)(2x - 5)$$

$$(4x - 5)^3 + (x - 2)^3 + (6x - 15)^3 - 3(4x - 5)(x - 2)(6x - 15) = 0$$

The given equation is the form of $(a^3 + b^3 + c^3 - 3abc = 0)$

If $a^3 + b^3 + c^3 - 3abc = 0$, then $a = b = c$ or $a + b + c = 0$

$$(4x - 5) + (x - 2) + (6x - 15) = 0$$

$$11x = 22 \Rightarrow x = 2$$

$$x + \frac{3}{2} = 2 + \frac{3}{2} = \frac{7}{2}$$

Sol.509.(d) Given that, $a + b = 24$

$$\text{and } a^2 + b^2 = 306$$

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$\Rightarrow 576 = 306 + 2ab$$

$$\Rightarrow ab = 135 = 15 \times 9$$

$$\text{and } a + b = 15 + 9 = 24$$

$$a = 15 \text{ and } b = 9$$

$$\text{So, } 4a - 5b = 4(15) - 5(9) = 15$$

Short-Trick:

You can directly put the value of $a = 15$ and $b = 9$ So, $4a - 5b = 4(15) - 5(9) = 15$

Sol.510.(b) If $a + b - c = 0$ then

$$b - c = -a, c - a = b, a + b = c$$

$$\Rightarrow \frac{(b-c)^2}{4bc} + \frac{(c-a)^2}{4ca} - \frac{(a+b)^2}{4ab}$$

$$= \frac{(-a)^2}{4bc} + \frac{(b)^2}{4ca} - \frac{(c)^2}{4ab}$$

$$= \frac{a^3 + b^3 - c^3}{4abc} = \frac{-3abc}{4abc} = \frac{-3}{4}$$

Sol.511.(c) $(2a - b - 3c)(4a^2 + b^2 + 9c^2 + 2ab + 6ac - 3bc)$

$$\Rightarrow (2a - b - 3c)\{(2a)^2 + (b)^2 + (3c)^2 + 2ab + 6ac - 3bc\}$$

We know, $a^3 - b^3 - c^3 - 3abc =$

$$(a - b - c)(a^2 + b^2 + c^2 + ab + ac - bc)$$

On comparing, we get,

$$(2a)^3 - (b)^3 - (3c)^3 - 3(2a)(b)(3c)$$

$$= 8a^3 - b^3 - 27c^3 - 18abc$$

Sol.512.(c) $(x^2 + \frac{1}{49x^2}) = 15\frac{5}{7}$

$$(x + \frac{1}{7x})^2 = x^2 + \frac{1}{49x^2} + 2 \times x \times \frac{1}{7x}$$

$$(x + \frac{1}{7x})^2 = \frac{110}{7} + \frac{2}{7} = 16$$

$$x + \frac{1}{7x} = \pm 4$$

Sol.513.(d) $(3x + 2y)^3 + (3x - 2y)^3$

$$= (3x)^3 + (2y)^3 + 3(3x)^2(2y) + 3(3x)(2y)^2 +$$

$$(3x)^3 - (2y)^3 - 3(3x)^2(2y) + 3(3x)(2y)^2$$

$$= (3x + 2y)^3 + (3x - 2y)^3 = 54x^3 + 72xy^2$$

$$= (3x + 2y)^3 + (3x - 2y)^3 = 18x(3x^2 + 4y^2)$$

$$3kx(3x^2 + 4y^2) = 18x(3x^2 + 4y^2)$$

On comparing,

$$3k = 18 \Rightarrow k = 6$$

Sol.514.(c) $x + y = 27$

Squaring both side

$$\Rightarrow x^2 + y^2 + 2xy = 729$$

$$\Rightarrow 2xy = 729 - 425 = 304$$

$$\Rightarrow (x - y)^2 = x^2 + y^2 - 2xy$$

$$= 425 - 304 = 121$$

Sol.515.(a) $x + \frac{1}{15x} = 3$

Multiplying both sides by '3'

$$3x + \frac{1}{5x} = 9$$

$$27x^3 + \frac{1}{125x^3} = 729 - 16.2$$

$$27x^3 + \frac{1}{125x^3} = 712.8$$

Divide both side by 3

$$9x^3 + \frac{1}{375x^3} = 237.6$$

Sol.516.(a) We know, $a^3 + b^3 + c^3 - 3abc$

$$= (a + b + c) \times (a^2 + b^2 + c^2 - ab - bc - ca)$$

Here, $a = 2x, b = 3y, c = 4z$

$$\Rightarrow 8x^3 + 27y^3 + 64z^3 - 72xyz$$

$$= (2x + 3y + 4z) (4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8xz)$$

$$\Rightarrow 105 - 72 = 11 (4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8xz)$$

$$4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8xz = 3$$

Sol.517.(b)

$$a^3 + b^3 + c^3 = 3abc \Rightarrow a + b + c = 0$$

$$x - 1.5 + x - 4 + x - 3.5 = 0$$

$$3x = 9 \Rightarrow x = 3$$

Sol.518.(c) $x^4 + x^{-4} = 47$

$$x^4 + \frac{1}{x^4} + 2 = 49 \Rightarrow x^2 + \frac{1}{x^2} = 7$$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 9 \Rightarrow x + \frac{1}{x} = 3$$

$$\Rightarrow x^2 - 3x = -1 \Rightarrow 4x^2 - 12x = -4$$

$$\Rightarrow (2x - 3)^2 = 4x^2 - 12x + 9 = -4 + 9 = 5$$

Sol.519.(b)

$$a^2 + b^2 + c^2 + 48 = 8(a + b + c)$$

$$\Rightarrow a^2 - 8a + 16 + b^2 - 8b + 16 + c^2 - 8c + 16 = 0$$

$$\Rightarrow (a - 4)^2 + (b - 4)^2 + (c - 4)^2 = 0$$

$$a = 4, b = 4, c = 4$$

$$\sqrt[3]{a^3 - b^3 + c^3} = \sqrt[3]{4^3 - 4^3 + 4^3} = 4$$

Sol.520.(d) $a - \frac{24}{a} = 5$

$$\Rightarrow a^2 - 5a - 24 = 0 \Rightarrow a^2 - 8a + 3a - 24 = 0$$

$$\Rightarrow (a - 8)(a + 3) = 0$$

$$a = 8, -3$$

$$a^2 + \frac{64}{a^2} = 8^2 + \frac{64}{8^2} = 65$$

Sol.521.(d)

$$3x - 2y + 3 = 0, 3x - 2y = -3$$

After cubing both sides,

$$27x^3 - 8y^3 - 18xy(3x - 2y) = -27$$

$$27x^3 - 8y^3 + 54xy = -27$$

$$27x^3 - 8y^3 + 54xy + 30 = 3$$

Sol.522.(d) On rationalizing,

$$a = \frac{\sqrt{5} + 2}{\sqrt{5} - 2} \times \frac{\sqrt{5} + 2}{\sqrt{5} + 2}$$

$$= (\sqrt{5} + 2)^2 = 9 + 4\sqrt{5}$$

$$\text{Similarly } b = 9 - 4\sqrt{5} \text{ and } ab = 1$$

$$2a^2 + 2b^2 - 5ab = 2(9 + 4\sqrt{5})^2 + 2(9 - 4\sqrt{5})^2 - 5$$

$$\Rightarrow 2(81 + 80 + 72\sqrt{5}) + 2(81 + 80 - 72\sqrt{5}) - 5$$

$$\Rightarrow 2 \times 161 + 2 \times 161 - 5 = 644 - 5 = 639$$

Sol.523.(c) $x^2 + (4 - \sqrt{3})x - 1 = 0,$

$$x - \frac{1}{x} = -(4 - \sqrt{3})$$

On squaring both sides,

$$x^2 + \frac{1}{x^2} = 21 - 8\sqrt{3}$$

Sol.524.(a)

$$a^2 + b^2 + c^2 + 216 = 12(a + b - 2c)$$

Taking all to LHS and rearranging all we get

$$\Rightarrow a^2 - 12a + 36 + b^2 - 12b + 36 + c^2 + 24c + 144 = 0$$

$$\Rightarrow (a - 6)^2 + (b - 6)^2 + (c + 12)^2 = 0$$

Now we can easily say that $a = 6, b = 6,$ and $c = -12,$

Now we have to find out $\sqrt{ab - bc + ca}$

Putting the values of a, b and c we get

$$\sqrt{36 + 72 - 72} = \sqrt{36} = 6$$

Sol.525.(b)

$$\frac{0.325 \times 0.325 + 0.175 \times 0.175 + 25 \times 0.00455}{5 \times 0.0065 \times 3.25 - 7 \times 0.175 \times 0.025} - \frac{0.5}{1.5}$$

$$= \frac{(0.325)^2 + (0.175)^2 + 2 \times 0.175 \times 0.325}{(0.325)^2 - (0.175)^2} - \frac{0.5}{1.5}$$

$$= \frac{(0.325 + 0.175)^2}{(0.325 + 0.175)(0.325 - 0.175)} - \frac{0.5}{1.5}$$

$$= \frac{0.325 + 0.175}{0.325 - 0.175} - \frac{0.5}{1.5}$$

$$= \frac{0.5}{0.15} - \frac{0.5}{1.5} = \frac{5 - 0.5}{1.5} = \frac{4.5}{1.5} = 3$$

Sol.526.(c)

$$x^2 + 8y^2 - 12y - 4xy + 9 = 0,$$

$$\Rightarrow (x - 2y)^2 + (2y - 3)^2 = 0$$

$$x = 2y \text{ and } y = \frac{3}{2} \text{ then, } x = 3 \text{ Now,}$$

$$\Rightarrow 7x - 8y = 21 - 12 = 9$$

Sol.527.(d) $x + y + z = 19,$

$$x^2 + y^2 + z^2 = 133 \text{ and } xz = y^2$$

Using $(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz$

+ $2xz$ And putting the values we get

$$361 = 133 + 2(xy + yz + xz)$$

$$228 = 2(xy + yz + xz)$$

$$\Rightarrow (xy + yz + xz) = 114$$

Now putting $xz = y^2$ and taking out y common simultaneously

$$\text{We get } y(x + y + z) = 114$$

$$\text{Putting } x + y + z = 19 \text{ we get } y = 6$$

$$\text{Now } x + z = 13 \text{ and } x^2 + z^2 = 97$$

Solving these equations we get

$$x = 9 \text{ and } z = 4 \text{ As } x > z > 0 (\text{given } 0)$$

$$\text{So } x - z = 9 - 4 = 5$$

Sol.528.(a)

$$x^8 + 1 = 34x^4 \text{ dividing both side by } x^4$$

$$x^4 + \frac{1}{x^4} = 34 \text{ adding 2 to both side}$$

$$x^4 + \frac{1}{x^4} + 2 = 36 \text{ taking root on both side}$$

$$x^2 + \frac{1}{x^2} = 6 \text{ again applying above process}$$

$$x + \frac{1}{x} = 2\sqrt{2} \text{ taking cube on both side}$$

$$x^3 + \frac{1}{x^3} = 8\sqrt{8} - 3\sqrt{8} \Rightarrow x^3 + \frac{1}{x^3} = 5\sqrt{8}$$

Sol.529.(a)

Taking the square of $(3x + y)^2$

$$(3x + y)^2 = 9x^2 + y^2 + 6xy$$

$$(3x + y)^2 = 37 + 6 \times 2$$

$$(3x + y)^2 = 49 \Rightarrow 3x + y = 7$$

Taking cube on both side

$$27x^3 + y^3 + 9xy(3x + y) = 343$$

$$27x^3 + y^3 = 343 - 9 \times 2 \times 7$$

$$27x^3 + y^3 = 217$$

$$\text{Sol.530.(a)} \quad \frac{0.0203 \times 2.92}{0.7 \times 0.0365 \times 2.9} \div$$

$$\frac{(12.12)^2 - (8.12)^2}{(0.25)^2 + (0.25)(19.99)}$$

$$= \frac{203 \times 292 \times 10000 \times 100}{7 \times 365 \times 29 \times 10 \times 10000 \times 10} \div$$

$$\frac{(12.12 + 8.12)(12.12 - 8.12)}{0.25 \times 0.25 + 0.25 \times 19.99}$$

$$= \frac{203 \times 292}{7 \times 365 \times 29} \div \frac{(20.24)(4)}{0.25(0.25 + 19.99)}$$

$$= \frac{4}{5} \times \frac{20.24 \times 0.25}{20.24 \times 4} = 0.05$$

$$\text{Sol.531.(a)} \quad \frac{x^6 + 3x^3 - 1}{x^6 - 8x^3 - 1}$$

dividing numerator and denominator by x^3

$$\Rightarrow \frac{x^3 + 3 - \frac{1}{x^3}}{x^3 - 8 - \frac{1}{x^3}}$$

$$\Rightarrow x^3 - \frac{1}{x^3} = 125 + 15 = 140$$

$$\Rightarrow \frac{x^3 + 3 - \frac{1}{x^3}}{x^3 - 8 - \frac{1}{x^3}} = \frac{140 + 3}{140 - 8} = \frac{143}{132} = \frac{13}{12}$$

$$\text{Sol.532.(a)} \quad \frac{27 \times (0.25)^3 + 125(0.05)^3}{(0.75)^2 - 0.25 \times 0.5}$$

$$= \frac{3^3 \times (0.25)^3 + 5^3(0.05)^3}{(0.75)^2 - 0.25 \times 0.5}$$

$$= \frac{(0.75)^3 + (0.25)^3}{(0.75)^2 - 0.25 \times (0.75 - 0.25)}$$

$$= \frac{(0.75)^3 + (0.25)^3}{(0.75)^2 - 0.25 \times (0.75) + (0.25)^2}$$

$$= 0.75 + 0.25 = 1$$

$$\text{Sol.533.(a)} \quad \frac{8 + 2\sqrt{3}}{3\sqrt{3} + 5} \text{ on rationalizing}$$

$$\Rightarrow \frac{8 + 2\sqrt{3}}{3\sqrt{3} + 5} \times \frac{3\sqrt{3} - 5}{3\sqrt{3} - 5}$$

$$= \frac{2(4 + \sqrt{3}) \times (3\sqrt{3} - 5)}{2}$$

$$= (4 + \sqrt{3}) \times (3\sqrt{3} - 5)$$

$$= 12\sqrt{3} - 20 + 9 - 5\sqrt{3} = 7\sqrt{3} - 11$$

$$a = 7 \text{ and } b = 11 \Rightarrow a + b = 18$$

$$\text{Sol.534.(d)}$$

$$x(3 - \frac{2}{x}) = \frac{3}{x}$$

$$\Rightarrow 3x - \frac{3}{x} = 2 \Rightarrow x - \frac{1}{x} = \frac{2}{3}$$

$$\Rightarrow x^3 - \frac{1}{x^3} = \frac{8}{27} + 3 \times \frac{2}{3}$$

$$\Rightarrow x^3 - \frac{1}{x^3} = \frac{62}{27}$$

$$\text{Sol.535.(b)} \quad 5 - \frac{8 + 2\sqrt{15}}{4} - \frac{1}{8 + 2\sqrt{15}}$$

on rationalization,

$$= 5 - \frac{8 + 2\sqrt{15}}{4} - \frac{1}{8 + 2\sqrt{15}} \times \frac{8 - 2\sqrt{15}}{8 - 2\sqrt{15}}$$

$$= 5 - \frac{8 + 2\sqrt{15}}{4} - \frac{8 - 2\sqrt{15}}{4}$$

$$= (20 - 16) \div 4 = 1$$

$$\text{Sol.536.(b)}$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0 \Rightarrow (ab + bc + ca) = 0$$

$$ac = \frac{4}{b} \Rightarrow abc = 4$$

$$a^3 + b^3 + c^3 = 28$$

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)$$

$$(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\Rightarrow 28 - 12 = 2(a^2 + b^2 + c^2)$$

$$\Rightarrow 16 = 2(a^2 + b^2 + c^2)$$

$$\Rightarrow (a^2 + b^2 + c^2) = 8$$

$$\text{Sol.537.(b)}$$

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)$$

$$(x^2 + y^2 + z^2 - xy - yz - zx)$$

$$\Rightarrow 75 - 3 \times 15 = 10(x^2 + y^2 + z^2 - xy - yz - zx)$$

$$\Rightarrow x^2 + y^2 + z^2 - xy - yz - zx = 3$$

$$\text{Sol.538.(d)}$$

$$a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$$

$$a^2 + b^2 + ab = 22 \dots\dots\dots \text{eq(1)}$$

$$(a - b)^2 = a^2 + b^2 - 2ab$$

$$a^2 + b^2 - 2ab = 16 \dots\dots\dots \text{eq(2)}$$

Sub eq(2) from eq (1), we get

$$ab = 2, \text{ Now, } (a - b)^2 = a^2 + b^2 - 2ab$$

$$a^2 + b^2 = 20$$

$$\text{And } (a + b)^2 = a^2 + b^2 + 2ab$$

$$(a + b) = 2\sqrt{6}$$

therefore,

$$a^2 - b^2 = (a + b)(a - b) = (2\sqrt{6})4 = 8\sqrt{6}$$

$$\text{Sol.539.(c)} \text{ Let two numbers are a and b.}$$

sum of two numbers, $a + b = 11$,

sum of their squares, $a^2 + b^2 = 65$,

Now,

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$(11)^2 = 65 + 2ab \Rightarrow ab = 28$$

$$a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

$$= (11)(65 - 28) = 407$$

Short-trick: -

Put $a = 7$ and $b = 4$

According to the question,

$$a + b = 11 \text{ and } a^2 + b^2 = 65$$

Now the value of a and b satisfied the above equation

So,

$$\Rightarrow a^3 + b^3 = 7^3 + 4^3 = 343 + 64 = 407$$

$$\text{Sol.540.(b)} \quad 2^{x+y-2z} = 8^{8z-5-y}$$

$$\Rightarrow 2^{x+y-2z} = (2^3)^{8z-5-y}$$

$= x + y - 2z = 3(8z - 5 - y)$ { base is same then power is also same }

$$= x + 4y - 26z = -15 \dots\dots\dots \text{eq(1)}$$

$$5^{4y-6z} = 25^{y+z} \Rightarrow 5^{4y-6z} = (5^2)^{y+z}$$

{base is same then power is also same }

$$\Rightarrow 4y - 6z = 2(y + z)$$

$$= 2y - 8z = 0 \dots\dots\dots \text{eq(2)}$$

$$3^{4x-3z} = 9^{x+z} \Rightarrow 3^{4x-3z} = (3^2)^{x+z}$$

$= 4x - 3z = 2(x + z)$ { base is same then power is also same }

$$= 2x - 5z = 0 \dots\dots\dots \text{eq(3)}$$

Add eq(2) and eq(3), we get

$$2x + 2y - 13z = 0 \dots\dots\dots \text{eq(4)}$$

Multiply eq(4) by 2 and subtract eq(1)

from it, we get $3x = 15$ and $x = 5$,

Put value of x in eq(3), we get $z = 2$

Put value of z in eq(2), we get $y = 8$

Now, the value of $2x + 3y + 5z$

$$= 2 \times 5 + 3 \times 8 + 5 \times 2 = 44$$

$$\text{Sol.541.(a)}$$

$$\frac{1}{(9 - 4\sqrt{5})^2} + \frac{1}{(9 + 4\sqrt{5})^2}$$

$$= \frac{1}{(a - b)^2} + \frac{1}{(a + b)^2}$$

$$\{a = 9, b = 4\sqrt{5}\}$$

$$= \frac{(a + b)^2 + (a - b)^2}{(a - b)^2(a + b)^2} = \frac{2(a^2 + b^2)}{(a^2 - b^2)^2}$$

$$= \frac{2[9^2 + (4\sqrt{5})^2]}{[9^2 - (4\sqrt{5})^2]^2} = 322$$

$$\text{Sol.542.(a)} \quad x^4 + \frac{1}{x^4} = \frac{257}{16}$$

$$\Rightarrow x^4 + \frac{1}{x^4} + 2 = \frac{257}{16} + 2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = \frac{17}{4}$$

$$\Rightarrow x + \frac{1}{x} + 2 = \frac{17}{4} + 2 = \frac{25}{4}$$

$$\Rightarrow x + \frac{1}{x} = \frac{5}{2}$$

Put $x = 2$

$$\frac{8}{13}(2^3 + \frac{1}{2^3}) = \frac{8}{13} \times \frac{65}{8} = 5$$

$$\text{Sol.543.(d)} \quad a + a^2b + b + ab^2 = 128$$

$$\Rightarrow a + b + ab(a + b) = 128$$

$$\Rightarrow (a + b)(1 + ab) = 128$$

$$\Rightarrow 8(1 + ab) = 128 \Rightarrow ab = 15$$

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$\Rightarrow 64 = a^2 + b^2 + 30$$

$$\Rightarrow a^2 + b^2 = 34$$

$$a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

$$a^3 + b^3 = 8(34 - 15) \Rightarrow a^3 + b^3 = 152$$

Short-trick:

$$\text{Put } a = 5, b = 3$$

$$a^3 + b^3 = 5^3 + 3^3 = 152$$

$$\text{Sol.544.(b)} \quad 3x^2 - 5x - 8$$

$$\Rightarrow 3x^2 - 8x + 3x - 8 = 0$$

$$\Rightarrow x(3x - 8) + 1(3x - 8) = 0$$

$$\Rightarrow (x + 1)(3x - 8)$$

$$\text{Sol.545.(c)} \quad (a + 2b)^2 - (a - 2b)^2$$

$$= a^2 + 4b^2 + 4ab - a^2 - 4b^2 + 4ab$$

$$= 8ab$$

$$\text{Sol.546.(c)}$$

$$5\sqrt{3} + 7\sqrt{2} - \sqrt{6} - \frac{23}{\sqrt{2} + \sqrt{3} + \sqrt{6}}$$

$$(\text{put the value of } \sqrt{3} = 1.73 \text{ and } \sqrt{2} = 1.41$$

$$\text{and } \sqrt{6} = 2.45$$

$$= 5(1.73) + 7(1.41) - 2.45 -$$

$$\frac{23}{1.41 + 1.73 + 2.45}$$

$$= 8.65 + 9.87 - 2.45 - 4.11 = 11.96 \approx 12$$

$$\text{Sol.547.(c)} \quad a + b = 27, a^3 + b^3 = 5427$$

$$a^3 + b^3 = (a + b)\{(a + b)^2 - 3ab\}$$

$$\Rightarrow 5427 = 27(729 - 3ab)$$

$$\Rightarrow 201 = 729 - 3ab \Rightarrow 3ab = 528$$

$$\Rightarrow ab = 176$$

$$\text{Sol.548.(c)} \quad \sqrt[3]{15625} - \sqrt{x} = 4$$

$$\Rightarrow 25 - \sqrt{x} = 4 \Rightarrow 21 = \sqrt{x} \Rightarrow x = 441$$

$$\text{Sol.549.(a)}$$

Let the two numbers are P and Q

$$P + Q = 47, PQ = 550$$

$$(P + Q)^2 = P^2 + Q^2 + 2PQ$$

$$\Rightarrow (47)^2 = P^2 + Q^2 + 1100$$

$$\Rightarrow 1109 = P^2 + Q^2$$

$$\text{Sol.550.(b)} \quad x = 2 + \sqrt{3}$$

$$x - 2 = \sqrt{3}$$

After squaring both sides

$$\Rightarrow x^2 - 4x + 4 = 3$$

$$\Rightarrow x^2 - 4x = -1$$

Again after squaring both sides

$$\Rightarrow x^4 - 8x^3 + 16x^2 = 1$$

$$\text{Sol.551.(a)} \quad \text{Go through option (a)}$$

$$\Rightarrow 4x^2 + 2x - 3y + 9y^2 - 12xy$$

$$\Rightarrow (2x - 3y)^2 + (2x - 3y)$$

$$\text{So, } (2x - 3y) \text{ is a factor of } 4x^2 + 2x - 3y + 9y^2 - 12xy$$

$$\text{Sol.552.(d)} \quad a : b = 3 : \sqrt{5}$$

$$(2a + b) : (3a - 2b) = \{2(3) + \sqrt{5}\} : \{3(3) - 2(\sqrt{5})\}$$

$$= \frac{6 + \sqrt{5}}{9 - 2\sqrt{5}} \text{ (after rationalization)}$$

$$= \frac{6 + \sqrt{5}}{9 - 2\sqrt{5}} \times \frac{9 + 2\sqrt{5}}{9 + 2\sqrt{5}}$$

$$= \frac{54 + 12\sqrt{5} + 9\sqrt{5} + 10}{81 - 20}$$

$$= \frac{64 + 21\sqrt{5}}{61} = \frac{1}{61}(64 + 21\sqrt{5})$$

$$\text{Sol.553.(a)}$$

$$x^3 + y^3 = (x + y)\{(x + y)^2 - 3xy\}$$

$$\Rightarrow 12 = 4(16 - 3xy)$$

$$\Rightarrow 3 = 16 - 3xy$$

$$\Rightarrow xy = \frac{13}{3} \Rightarrow x^2 y^2 = \frac{169}{9}$$

$$x^3 + y^3 = (x + y)\{(x^2 + y^2 - xy)\}$$

$$\Rightarrow 12 = 4(x^2 + y^2 - \frac{13}{3}) \Rightarrow x^2 + y^2 = \frac{22}{3}$$

$$x^4 + y^4 = (x^2)^2 + (y^2)^2$$

$$\Rightarrow (x^2)^2 + (y^2)^2 = (x^2 + y^2)^2 - 2x^2 y^2$$

$$\Rightarrow (x^2)^2 + (y^2)^2 = \frac{484}{9} - \frac{338}{9}$$

$$\Rightarrow (x^2)^2 + (y^2)^2 = \frac{146}{9} \Rightarrow x^4 + y^4 = \frac{146}{9}$$

$$\text{Sol.554.(d)} \quad x = 3 + 2\sqrt{2}$$

$$x^2 = 17 + 12\sqrt{2} \Rightarrow \frac{1}{x^2} = 17 - 12\sqrt{2}$$

$$x^2 + \frac{1}{x^2} = 17 + 12\sqrt{2} + 17 - 12\sqrt{2} = 34$$

Alternate method:

$$x = 3 + 2\sqrt{2} \text{ then, } \frac{1}{x} = 3 - 2\sqrt{2}$$

$$x + \frac{1}{x} = 6 \text{ then, } x^2 + \frac{1}{x^2} = 6^2 - 2 = 34$$

$$\text{Sol.555.(c)} \quad x + \frac{4}{x} - 4 = 0$$

$$\Rightarrow x^2 + 4 - 4x = 0$$

$$\Rightarrow (x - 2)^2 = 0 \Rightarrow x = 2$$

$$\text{So, } x^2 - 4 = 2^2 - 4 = 4 - 4 = 0$$

$$\text{Sol.556.(d)}$$

$$\text{When, } a^3 + b^3 + c^3 = 3abc$$

$$\text{Then, } a = b = c \text{ or } a + b + c = 0$$

$$x^3 + (3y)^3 + (4z)^3 = 3(x)(3y)(4z)$$

$$\text{Therefore, } x + 3y + 4z = 0$$

$$\text{Sol.557.(a)} \quad x(5 - \frac{2}{x}) = \frac{5}{x}$$

$$5(x - \frac{1}{x}) = 2 \Rightarrow (x - \frac{1}{x}) = \frac{2}{5}$$

Square both sides, we get

$$x^2 + \frac{1}{x^2} = 2\frac{4}{25}$$

$$\text{Sol.558.(c)} \quad \sqrt{x} = \sqrt{3} - \sqrt{5}$$

Square both sides, we get

$$x = 8 - 2\sqrt{15} \Rightarrow x - 8 = -2\sqrt{15}$$

Again square both sides, we get

$$x^2 - 16x + 64 = 60 \Rightarrow x^2 - 16x + 4 = 0$$

$$\Rightarrow x^2 - 16x + 6 = 2$$

$$\text{Sol.559.(b)} \quad x = \frac{\sqrt{3}}{2}$$

$$1 + x = \frac{(2 + \sqrt{3})}{2} = \frac{(4 + 2\sqrt{3})}{4}$$

$$\Rightarrow \sqrt{1 + x} = \frac{(\sqrt{3} + 1)}{2}$$

$$1 - x = \frac{(2 - \sqrt{3})}{2} = \frac{(4 - 2\sqrt{3})}{4}$$

$$\Rightarrow \sqrt{1 - x} = \frac{(\sqrt{3} - 1)}{2}$$

Now, put both values

$$\frac{\sqrt{1 + x} + \sqrt{1 - x}}{\sqrt{1 + x} - \sqrt{1 - x}} = \frac{\sqrt{3}}{1} = \sqrt{3}$$

$$\text{Sol.560.(a)}$$

$$\text{if } x + \frac{1}{x} = \sqrt{3} \text{ then, } x^6 = -1$$

$$\text{Put this value in } x^{18} + x^{12} + x^6 + 1, \\ \text{we get, } -1 + 1 - 1 + 1 = 0$$

$$\text{Sol.561.(a)} \quad a^2 + b^2 + 2b + 4a + 5 = 0$$

$$\Rightarrow (a + 2)^2 + (b + 1)^2 = 0$$

$$\Rightarrow a = -2, b = -1$$

Put the value of a and b, we get

$$\frac{2a - 3b}{2a + 3b} = \frac{(-4 + 3)}{(-4 - 3)} = \frac{1}{7}$$

$$\text{Sol.562.(a)} \quad 2a + \frac{1}{a} = 4,$$

Divide both sides by 2, we get

$$a + \frac{1}{2a} = 2,$$

Squaring both sides, we get

$$a^2 + \frac{1}{4a^2} = 4 - 1 = 3$$

$$\text{Sol.563.(a)}$$

$$\text{Given:- } A = \frac{1 + 2x}{1 - 2x} \text{ and } B = \frac{1 - 2x}{1 + 2x}$$

$$A + B = \frac{1 + 2x}{1 - 2x} + \frac{1 - 2x}{1 + 2x}$$

$$= \frac{(1 + 2x)^2 + (1 - 2x)^2}{1 - 4x^2}$$

$$= \frac{1 + 4x^2 + 4x + 1 + 4x^2 - 4x}{1 - 4x^2} = \frac{2 + 8x^2}{1 - 4x^2}$$

$$A - B = \frac{1 + 2x}{1 - 2x} - \frac{1 - 2x}{1 + 2x}$$

$$= \frac{(1 + 2x)^2 - (1 - 2x)^2}{1 - 4x^2}$$

$$= \frac{1 + 4x^2 + 4x - 1 - 4x^2 + 4x}{1 - 4x^2} = \frac{8x}{1 - 4x^2}$$

$$\frac{A + B}{A - B} = \frac{2 + 8x^2}{1 - 4x^2} \times \frac{1 - 4x^2}{8x}$$

$$= \frac{2 + 8x^2}{8x} = \frac{4x^2 + 1}{4x} = x + \frac{1}{4x}$$

$$\text{Sol.564.(a)} \quad x = 3 + 2\sqrt{2} = (\sqrt{2} + 1)^2$$

$$\Rightarrow \sqrt{x} = \sqrt{2} + 1$$

$$\Rightarrow \frac{1}{\sqrt{x}} = \frac{1}{\sqrt{2} + 1} = \sqrt{2} - 1$$

$$\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{2} + 1 - (\sqrt{2} - 1)$$

$$= \sqrt{2} + 1 - \sqrt{2} + 1 = 2$$

Sol.565.(d) Given:- $a + b + c = 0$

Let $a = 2$, $b = -1$ and $c = -1$

Then,

$$\frac{2a^2}{3bc} + \frac{2b^2}{3ca} + \frac{2c^2}{3ab}$$

$$\frac{2(2)^2}{3(-1)(-1)} + \frac{2(-1)^2}{3(-1)(2)} + \frac{2(-1)^2}{3(2)(-1)}$$

$$= \frac{8}{3} - \frac{1}{3} - \frac{1}{3} = \frac{8-1-1}{3}$$

$$= \frac{6}{3} = 2$$

Sol.566.(d)

$$\frac{3x\sqrt{y} + 2y\sqrt{x}}{3x\sqrt{y} - 2y\sqrt{x}} - \frac{3x\sqrt{y} - 2y\sqrt{x}}{3x\sqrt{y} + 2y\sqrt{x}} = \sqrt{x}\sqrt{y}$$

$$\Rightarrow \frac{(3x\sqrt{y} + 2y\sqrt{x})^2 - (3x\sqrt{y} - 2y\sqrt{x})^2}{(3x\sqrt{y} - 2y\sqrt{x})(3x\sqrt{y} + 2y\sqrt{x})}$$

$$= \sqrt{x}\sqrt{y}$$

$$\Rightarrow \frac{4(3x\sqrt{y})(2y\sqrt{x})}{9x^2y - 4y^2x} = \sqrt{x}\sqrt{y} \Rightarrow 9x - 4y = 24$$

Sol.567.(c) We know, The sum of odd numbers = (no. of terms)²

In $1 + 3 + 5 + 7 + \dots (2n - 1)$; let the no. of terms = x

$$2n - 1 = 1 + (x - 1)2$$

$$2n - 2 = (x - 1)2$$

$$n - 1 = x - 1 \Rightarrow x = n$$

Therefore, no. of terms, $x = n$

$$1 + 3 + 5 + 7 + \dots (2n - 1) = n^2$$

$$= n \times n$$

Sol.568.(c) $2(a^2 + b^2) = (a + b)^2$

$$\Rightarrow 2a^2 + 2b^2 = a^2 + b^2 + 2ab$$

$$\Rightarrow 2a^2 + 2b^2 - a^2 - b^2 - 2ab = 0$$

$$\Rightarrow a^2 + b^2 - 2ab = 0 \Rightarrow (a - b)^2 = 0 \Rightarrow a = b$$

Sol.569.(a) $21^{\sqrt{x}} + 20^{\sqrt{x}} = 29^{\sqrt{x}}$

We know triplet :- $29^2 = 21^2 + 20^2$

Therefore, $\sqrt{x} = 2$, $x = 4$

Sol.570.(a)

Put $a = 1$, $b = -1$ and $c = 0$ in

$$\Rightarrow [(a^2 - b^2)^3 + (b^2 - c^2)^3 + (c^2 - a^2)^3] \div$$

$$[(a - b)^3 + (b - c)^3 + (c - a)^3]$$

$$\Rightarrow [(1^2 - (-1)^2)^3 + ((-1)^2 - 0^2)^3 +$$

$$(0^2 - 1^2)^3] \div [(1 - (-1))^3 +$$

$$((-1) - 0)^3 + (0 - 1)^3]$$

$$\Rightarrow [(0)^3 + (1)^3 + (-1)^3] \div [(2)^3 +$$

$$(-1)^3 + (-1)^3]$$

$$\Rightarrow [(1)^3 + (-1)^3] \div [8 + (-1 - 1)]$$

$$\Rightarrow [0] \div [8 + (-2)] = 0$$

Now check options, $(a + b)(b + c)(c + a)$

$$= (1 - 1)(-1 + 0)(0 + 1) = 0$$

Sol.571.(b) $A = \frac{x-1}{x+1}$

$$\frac{1}{A} = \frac{x+1}{x-1}$$

$$A - \frac{1}{A} = \frac{x-1}{x+1} - \frac{x+1}{x-1}$$

$$A - \frac{1}{A} = \frac{(x-1)^2 - (x+1)^2}{(x+1)(x-1)}$$

$$A - \frac{1}{A} = \frac{x^2 + 1 - 2x - x^2 - 1 - 2x}{x^2 - 1}$$

$$A - \frac{1}{A} = \frac{-4x}{x^2 - 1}$$

Sol.572.(c)

$$x + y = 8, y + z = 13 \text{ and } z + x = 17$$

On adding above equations,

$$2(x + y + z) = 38 \Rightarrow x + y + z = 19$$

$$x = 6, y = 2, z = 11$$

$$\frac{x^2}{yz} = \frac{6 \times 6}{2 \times 11} = \frac{18}{11}$$

Sol.573.(a) $(a + b - c)^3 + (a - b + c)^3 - 8a^3$

$$\Rightarrow (a + b - c)^3 + (a - b + c)^3 + (-2a)^3$$

We know that if $x + y + z = 0$

$$\text{Then } x^3 + y^3 + z^3 = 3xyz$$

$$\text{Here, } x = a + b - c, y = a - b + c, z = -2a$$

$$\text{So, } (a + b - c)^3 + (a - b + c)^3 - 8a^3$$

$$= 6a(a - b + c)(c - a - b)$$

Short trick :

Put $a = 1$ and $b = c = 0$

$$(a + b - c)^3 + (a - b + c)^3 - 8a^3$$

$$\Rightarrow (1 + 0 - 0)^3 + (1 - 0 + 0)^3 - 8(1)^3$$

$$= 1 + 1 - 8 = -6$$

Now on checking the options, option (a) is satisfied.

$$6a(a - b + c)(c - a - b)$$

$$\Rightarrow 6(1)(1 - 0 + 0)(0 - 1 - 0) = -6$$

condition satisfied hence option (a) is the right answer.

Sol.574.(a) In the equation:

$$(a + b - c + d)^2 - (a - b + c - d)^2$$

$$\text{Use formula : } x^2 - y^2 = (x - y)(x + y)$$

$$(a + b - c + d)^2 - (a - b + c - d)^2 =$$

$$(a + b - c + d + a - b + c - d)(a + b - c + d - a + b + c - d)$$

$$= (2a)[2(b - c + d)]$$

$$= 4a(b - c + d)$$

Sol.575.(c) We know,

$$x^3 - y^3 = (x - y)(x^2 + y^2 + xy)$$

$$27a^3 - 2\sqrt{2}b^3 = (3a)^3 - (\sqrt{2}b)^3$$

$$= (3a - \sqrt{2}b)(9a^2 + 2b^2 + 3\sqrt{2}ab)$$

Sol.576.(c) $5x + \frac{1}{3x} = 4$

Multiply by $\frac{3}{5}$ on both sides

We get:

$$3x + \frac{1}{5x} = \frac{12}{5}$$

On square both sides:

$$9x^2 + \frac{1}{25x^2} + \frac{6}{5} = \frac{144}{25}$$

$$9x^2 + \frac{1}{25x^2} = \frac{114}{25}$$

Sol.577.(a)

$$\frac{x^3 - y^3}{x[(x+y)^2 - 3xy]} \div \frac{y[(x-y)^2 + 3xy]}{x^3 + y^3}$$

$$\times \frac{(x+y)^2 - (x-y)^2}{x^2 - y^2}$$

$$\Rightarrow \frac{x^3 - y^3}{x[(x+y)^2 - 3xy]} \times \frac{x^3 + y^3}{y[(x-y)^2 + 3xy]}$$

$$\times \frac{(x+y)^2 - (x-y)^2}{x^2 - y^2}$$

$$\Rightarrow \frac{x-y}{x} \times \frac{x+y}{y} \times \frac{4xy}{(x-y)(x+y)} = 4$$

Sol.578.(c) $12x^2 - 21x + 1 = 0$

$$\Rightarrow 12x^2 + 1 = 21x \Rightarrow 4x + \frac{1}{3x} = 7$$

On multiplying both sides by $(\frac{3}{4x})$

$$9x + \frac{3}{4x} = \frac{63}{4}$$

$$3x + \frac{1}{4x} = \frac{21}{4} \text{ -----(i)}$$

squaring equation (i),

$$\Rightarrow 9x^2 + (16x^2)^{-1} + 2 \times 3x \times \frac{1}{4x} = \frac{441}{16}$$

$$\Rightarrow 9x^2 + (16x^2)^{-1} = \frac{441}{16} - \frac{3}{2}$$

$$\Rightarrow 9x^2 + \frac{1}{16x^2} = \frac{417}{16}$$

Sol.579.(a) $P = \frac{x^3 + y^3}{(x-y)^2 + 3xy}$,

$$Q = \frac{(x+y)^2 - 3xy}{x^3 - y^3} \text{ and}$$

$$R = \frac{(x+y)^2 + (x-y)^2}{x^2 - y^2}$$

$$(P \div Q) \times R = \left(\frac{x^3 + y^3}{(x-y)^2 + 3xy} \div \right.$$

$$\left. \frac{(x+y)^2 - 3xy}{x^3 - y^3} \right) \times \frac{(x+y)^2 + (x-y)^2}{x^2 - y^2}$$

$$= \frac{x^3 + y^3}{x^2 + y^2 + xy} \times \frac{x^3 - y^3}{x^2 + y^2 - xy} \times \frac{2(x^2 + y^2)}{x^2 - y^2}$$

$$= 2(x^2 + y^2)$$

Sol.580.(b)

$$\frac{x^2(x-4)^2}{(x+4)^2 - 4x} \div \frac{(x^2 - 4x)^3}{(x+4)^2} \times \frac{64 - x^3}{16 - x^2}$$

$$= \frac{x^2(x-4)^2}{(x+4)^2 - 4x} \times \frac{(x+4)^2}{(x^2 - 4x)^3} \times \frac{64 - x^3}{16 - x^2}$$

$$= \frac{(x^2 - 4x)^2}{x^2 + 16 + 4x} \times \frac{(x+4)^2}{(x^2 - 4x)^3} \times \frac{4^3 - x^3}{4^2 - x^2}$$

$$= \frac{x+4}{x(x-4)}$$

Sol.581.(d) $3^a = 3^{3b} = 3^{4c}$

$$\Rightarrow a = 3b = 4c$$

$$\Rightarrow a : b : c = 12 : 4 : 3$$

Given: $abc = 144$

Thus, $a = 12$; $b = 4$; $c = 3$

Put values in $12\left(\frac{1}{a} + \frac{1}{2b} + \frac{1}{5c}\right)$

$$= 12\left(\frac{1}{12} + \frac{1}{8} + \frac{1}{15}\right) = \frac{33}{10}$$

Sol.582.(c) $(a + b - 2)^2 + (b + c - 5)^2 + (c + a - 5)^2 = 0$

Square of any number is always positive and their sum can be 0 only when they are themselves zero.

Thus, $a + b = 2$, $b + c = 5$ and $c + a = 5$

$a = 1$, $b = 1$, $c = 4$

$$\sqrt{(b + c)^a + (c + a)^b - 1}$$

$$= \sqrt{(5)^1 + (5)^1 - 1} = 3$$

Sol.583.(c) $a^3 + \frac{1}{a^3} = 52$

$$\Rightarrow a + \frac{1}{a} = 4 \Rightarrow 2\left(a + \frac{1}{a}\right) = 8$$

Sol.584.(c)

$b + c = ax$, $a + c = by$, $a + b = cz$

$\Rightarrow a + b + c = a + ax$, $b + a + c = b + by$,

$c + a + b = c + cz$

$$\Rightarrow \frac{a + b + c}{a} = 1 + x, \frac{a + b + c}{b} = 1 + y,$$

$$\frac{a + b + c}{c} = 1 + z$$

$$\frac{1}{9}\left[\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1}\right] = \frac{1}{9}$$

Sol.585.(a) $a^4 + \frac{1}{a^4} = 50$

$$a^2 + \frac{1}{a^2} = \sqrt{52}$$

$$a + \frac{1}{a} = \sqrt{2\sqrt{13} + 2}$$

$$\Rightarrow a^3 + \frac{1}{a^3} + 3(\sqrt{2\sqrt{13} + 2})$$

$$= (2\sqrt{13} + 2) \times (\sqrt{2\sqrt{13} + 2})$$

$$= 2\sqrt{13}(\sqrt{2\sqrt{13} + 2}) + 2(\sqrt{2\sqrt{13} + 2})$$

$$\Rightarrow a^3 + \frac{1}{a^3} = 2\sqrt{13}(\sqrt{2\sqrt{13} + 2})$$

$$- (\sqrt{2\sqrt{13} + 2})$$

$$a^3 + \frac{1}{a^3} = \sqrt{2(1 + \sqrt{13})}(-1 + 2\sqrt{13})$$

Sol.586.(a) $a^3 + b^3 + c^3 - 3abc$

$$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

Similarly,

$$(a + b + 2c)(a^2 + b^2 + 4c^2 - ab - 2bc - 2ca)$$

$$= a^3 + b^3 + 8c^3 - 6abc$$

Sol.587.(a) if $x + y + z = 0$, then $x^3 + y^3 + z^3 = 3xyz$

Similarly, $a^2 + b^2 - c^2 = 0$

Therefore, $a^6 + b^6 - c^6 = 3a^2b^2c^2$

$$\frac{2(a^6 + b^6 - c^6)}{3a^2b^2c^2} = 2$$

Sol.588.(a) $a = 2b = 8c$

$$a : b : c = 8 : 4 : 1$$

$$a + b + c = 13$$

Thus, $a = 8$; $b = 4$ and $c = 1$

$$\frac{\sqrt{a^2 + b^2 + c^2}}{2c} = \frac{\sqrt{64 + 16 + 1}}{2} = \frac{\sqrt{81}}{2} = \frac{9}{2}$$

Sol.589.(a) $x + y = 13$, $y + z = 15$, $z + x = 16$

$$\Rightarrow 2(x + y + z) = 13 + 15 + 16 = 44$$

$$x + y + z = 22$$

$$x = 7, y = 6, z = 9$$

$$\frac{xy + xz}{xyz} = \frac{y + z}{yz} = \frac{15}{54} = \frac{5}{18}$$

Sol.590.(c) $x + 3y + 2 = 0$

$$x + 3y = -2$$

$$\Rightarrow x^3 + 27y^3 - 18xy = -8$$

$$\Rightarrow x^3 + 27y^3 - 18xy + 8 = 0$$

Sol.591.(a)

$$\frac{1}{8}\left\{(x + \frac{1}{y})^2 - (x - \frac{1}{y})^2\right\}$$

$$= \frac{1}{8}\left\{(x^2 + \frac{1}{y^2} + \frac{2x}{y}) - (x^2 + \frac{1}{y^2} - \frac{2x}{y})\right\}$$

$$= \frac{1}{8}\left\{\frac{2x}{y} + \frac{2x}{y}\right\} = \frac{1}{8}\left\{\frac{4x}{y}\right\} = \frac{x}{2y}$$

Sol.592.(a)

$$\frac{0.325 \times 0.325 + 0.175 \times 0.175 - 25 \times 0.00455}{5 \times 0.0065 \times 3.25 - 7 \times 0.175 \times 0.025}$$

$$= \frac{(0.325 - 0.175)^2}{0.105625 - 0.030625} = \frac{(0.150)^2}{0.075} = 0.3$$

Sol.593.(a)

$$\frac{0.74 \times 1.23 \times 0.13}{(0.37)^3 + (0.41)^3 - 8(0.39)^3}$$

We can write $0.74 = 2 \times 0.37$

$1.23 = 3 \times 0.41$ and 0.13 as it is

On substituting these values in the given equation we get,

$$= \frac{(2 \times 0.37) \times (3 \times 0.41) \times 0.13}{(0.37)^3 + (0.41)^3 - [2(0.39)]^3}$$

$$= \frac{(0.37) \times (0.41) \times 0.78}{(0.37)^3 + (0.41)^3 - [(0.78)]^3}$$

As per the formula $a + b + c = 0$

$$= 0.37 + 0.41 - 0.78 = 0$$

We know that, $a^3 + b^3 + c^3 = 3abc$

$$= \frac{(0.37) \times (0.41) \times 0.78}{3 \times 0.37 \times 0.41 \times (-0.78)} = \frac{-1}{3}$$

Sol.594.(d) In such cases, let's assume the values of $a = 5$, $b = 3$ and $c = -2$ as $a + b + c = 6$ in order to get the answer easily

On putting these values in the given equation we get

$$a(b^2 + c^2) + b(c^2 + a^2) + c(a^2 + b^2) + 3abc$$

$$= 5(9 + 4) + 3(4 + 25) - 2(25 + 9) + (3 \times 5 \times 3 \times -2)$$

on solving

$$= 84 - 90 = -6$$

Sol.595.(d) $\sqrt{x} + \frac{1}{\sqrt{x}} = 3$

Now squaring both sides of the equation we get,

$$x + \frac{1}{x} = 9 - 2 \Rightarrow x + \frac{1}{x} = 7$$

On squaring again

$$x^2 + \frac{1}{x^2} = 47 \Rightarrow (x^2 - 47) = -\frac{1}{x^2}$$

Now,

$$x^2(x^2 - 47) = -1$$

Sol.596.(b)

$$4(x - 2)^2 + (y - 3)^2 - 2(x - 3)^2$$

.....(1)

In order to get least value of $(y - 3)^2$, we need to put $y = 3$

$$\text{Now, } 4(x - 2)^2 - 2(x - 3)^2$$

After differentiation we get,

$$8(x - 2) - 4(x - 3) = 0$$

$$8x - 16 - 4x + 12 = 0 \Rightarrow x = 1$$

Now putting $x = 1$ and $y = 3$ in eq (1)

$$= 4(1) + 0 - 2 \times 4 = -4$$

Sol.597.(a)

$$\frac{4.669 \times 4.669 - 9 \times (0.777)^2}{(4.669)^2 + (2.331)^2 + 14(0.667)(2.331)} = 1 - K$$

$$\frac{(4.669)^2 - (2.331)^2}{(4.669)^2 + (2.331)^2 + 2(4.669)(2.331)} = 1 - K$$

$$\frac{4.669 - 2.331}{4.669 + 2.331} = 1 - K$$

$$\Rightarrow \frac{2.338}{7} = 1 - K \Rightarrow 0.334 = 1 - K$$

$$\Rightarrow K = 0.666$$

Sol.598.(b) $\sqrt{11 + 2\sqrt{18}}$

$$= \sqrt{9 + 2 + 2 \times 3 \times \sqrt{2}}$$

$$= \sqrt{(3 + \sqrt{2})^2}$$

$$= 3 + \sqrt{2} = 3 + 1.41 = 4.41$$

Sol.599.(b) As per question

$$\sqrt{6 - \sqrt{17 - 2\sqrt{72}}} \text{eq(i)}$$

Firstly, We can simplify this part:

$$\sqrt{17 - 2\sqrt{72}}$$

$$= \sqrt{9 + 8 - 2\sqrt{8} \cdot \sqrt{9}}$$

$$= \sqrt{(3 - 2\sqrt{2})^2}$$

$$= 3 - 2\sqrt{2} \text{eq(ii)}$$

Putting the value from eq(ii) into eq(i)

$$= \sqrt{6 - 3 + 2\sqrt{2}} = \sqrt{3 + 2\sqrt{2}}$$

$$= \sqrt{(\sqrt{2} + 1)^2} = \sqrt{2} + 1$$

$$= 1.41 + 1 = 2.41$$

Sol.600.(c) $\frac{8x}{2x^2 + 7x - 2} = 1$

Dividing both numerator and denominator by 'x'

$$\frac{8}{2x + 7 - \frac{2}{x}} = 1 \Rightarrow \frac{8}{7 + 2[x - \frac{1}{x}]} = 1$$

$$\frac{1}{2} = [x - \frac{1}{x}] \text{eq(1)}$$

Now calculating $x + \frac{1}{x}$

Squaring both sides of eq(1)

$$x^2 + \frac{1}{x^2} = \frac{1}{4} + 2 = \frac{9}{4}$$

On adding '2' to both sides of the above equation

We get

$$(x + \frac{1}{x})^2 = \frac{17}{4} \Rightarrow x + \frac{1}{x} = \frac{\sqrt{17}}{2}$$

we know that $(x^3 + \frac{1}{x^3}) = k^3 - 3k$

$$\Rightarrow x^3 + \frac{1}{x^3} = \frac{17\sqrt{17}}{8} - \frac{3\sqrt{17}}{2}$$

$$\Rightarrow x^3 + \frac{1}{x^3} = \frac{17\sqrt{17} - 12\sqrt{17}}{8} = \frac{5\sqrt{17}}{8}$$

Sol.601.(d)

$$(x^3 + \frac{1}{x^3} - k)^2 + (x + \frac{1}{x} - p)^2 = 0$$

$$x^3 + \frac{1}{x^3} - k = 0 \dots\dots\text{eq(i)}$$

$$x + \frac{1}{x} - p = 0, p = x + \frac{1}{x}$$

$$x^3 + \frac{1}{x^3} = p^3 - 3p$$

Putting the above value in eq(i)

$$\Rightarrow p^3 - 3p - k = 0$$

$$\Rightarrow p[p^2 - 3] = k \Rightarrow \frac{k}{p} = p^2 - 3$$

Sol.602.(c) $x^6 - 512y^6$

$$= (x^2 + Ay^2)(x^4 - Bx^2y^2 + Cy^4) \dots\dots(1)$$

$$x^6 - 512y^6$$

$$= (x^2 - 8y^2)(x^4 + 64y^4 + 8x^2y^2) \dots\dots(2)$$

comparing (1) and (2), we get

$$A = -8, B = -8, C = 64$$

$$A + B - C = -16 - 64 = -80$$

Sol.603.(d) Short-trick: Put $a = b = c = 1$

$$(a + b + c)(ab + bc + ca) - abc$$

$$= (1 + 1 + 1)(1 + 1 + 1) - 1 = 9 - 1 = 8$$

Now, Put $a = b = c = 1$ in all options and check which option gives 8.

Option(d)

$$(a + b)(b + c)(c + a) = 2 \times 2 \times 2 = 8$$

$$\text{Sol.604.(d)} \frac{1}{\sqrt{17 + 12\sqrt{2}}}$$

$$= \frac{1}{\sqrt{9 + 8 + 2 \times 3 \times 2\sqrt{2}}}$$

$$= \frac{1}{\sqrt{(3 + 2\sqrt{2})^2}} = \frac{1}{(3 + 2\sqrt{2})}$$

$$= 3 - 2\sqrt{2} = 3 - 2 \times 1.41 = 0.18$$

Sol.605.(d)

$$(a + b)^2 - ab = \frac{a^3 - b^3}{a - b} = \frac{216}{6} = 36$$

Sol.606.(c)

$$(a + b)^2 - 3ab = a^2 + b^2 + 2ab - 3ab$$

We know that

$$a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

$$432 = 12(a^2 + b^2 - ab)$$

$$36 = (a^2 + b^2 - ab)$$

Sol.607.(c) Given, $a^2 + b^2 + c^2 +$

$$96 = 8(a + b - 2c)$$

$$\Rightarrow a^2 + b^2 + c^2 + 96 - 8(a + b - 2c) = 0$$

$$\Rightarrow a^2 + 16 - 8a + b^2 + 16 - 8b + c^2 +$$

$$64 + 16b = 0$$

$$\Rightarrow (a - 4)^2 + (b - 4)^2 + (c + 8)^2 = 0$$

Sum of the square of the terms is equal to zero only if the terms themselves are zero.

$$(a - 4) = 0, (b - 4) = 0 \text{ and } (c + 8) = 0$$

$$\Rightarrow a = 4 \text{ and } b = 4 \text{ and } c = -8$$

$$\sqrt{ab - bc + ca} =$$

$$\sqrt{(4 \times 4) - 4 \times (-8) + (-8) \times 4} = 4$$

Sol.608.(a) Given,

$$x = \sqrt{1 + \frac{\sqrt{3}}{2}} - \sqrt{1 - \frac{\sqrt{3}}{2}}$$

$$\Rightarrow x = \sqrt{\frac{2 + \sqrt{3}}{2}} - \sqrt{\frac{2 - \sqrt{3}}{2}}$$

Multiply and divide by 2

$$\begin{aligned} \Rightarrow x &= \sqrt{\frac{4 + 2\sqrt{3}}{4}} - \sqrt{\frac{4 - 2\sqrt{3}}{4}} \\ &= \sqrt{\left(\frac{\sqrt{3} + 1}{2}\right)^2} - \sqrt{\left(\frac{\sqrt{3} - 1}{2}\right)^2} \\ &= \frac{\sqrt{3} + 1}{2} - \frac{\sqrt{3} - 1}{2} = 1 \end{aligned}$$

$$\text{Now, } \frac{\sqrt{2} - x}{\sqrt{2} + x} = \frac{\sqrt{2} - 1}{\sqrt{2} + 1}$$

$$= \frac{\sqrt{2} - 1}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1} = 3 - 2\sqrt{2} \approx 0.17$$

$$\text{Sol.609.(a)} \text{ Given, } x + \frac{1}{16x} = 3$$

Multiplying both sides by 4

$$\Rightarrow 4x + \frac{1}{4x} = 12$$

$$64x^3 + \frac{1}{64x^3} = (12^3 - 3 \times 12) = 1692$$

$$(\text{If } a + \frac{1}{a} = k, \text{ then } a^3 + \frac{1}{a^3} = k^3 - 3k)$$

Exploring question term

$$16x^3 + \frac{1}{256x^3} = \frac{1}{4} \times 4x$$

$$(16x^3 + \frac{1}{256x^3}) = \frac{1}{4}(64x^3 + \frac{1}{64x^3})$$

$$\Rightarrow \frac{1}{4} \times 1692 = 423$$

$$\text{Sol.610.(a)} x = \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} + \sqrt{3}}$$

$$\Rightarrow x = \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} + \sqrt{3}} \times \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} - \sqrt{3}}$$

$$\Rightarrow \frac{(\sqrt{5} - \sqrt{3})^2}{2} = 4 - \sqrt{15}$$

$$\text{Now, } y = \frac{1}{x} = \frac{1}{4 - \sqrt{15}}$$

Applying rationalization

$$= \frac{1}{4 - \sqrt{15}} \times \frac{4 + \sqrt{15}}{4 + \sqrt{15}} = 4 + \sqrt{15}$$

Now,

$$x + y = x + \frac{1}{x} = 4 - \sqrt{15} + 4 + \sqrt{15} = 8$$

$$(x^3 + y^3) = x^3 + \frac{1}{x^3} = 8^3 - 3(8) = 488$$

$$(\text{if } a + \frac{1}{a} = k, \text{ then } a^3 + \frac{1}{a^3} = k^3 - 3k)$$

Sol.611.(b)

$$\text{Given, } \sqrt{10 - 2\sqrt{21}} + \sqrt{8 + 2\sqrt{15}}$$

$$= \sqrt{a} + \sqrt{b} \dots\dots(1)$$

Now,

$$\sqrt{10 - 2\sqrt{21}} =$$

$$\sqrt{7 + 3 - 2 \times \sqrt{7} \times \sqrt{3}}$$

$$= \sqrt{(\sqrt{7} - \sqrt{3})^2} = (\sqrt{7} - \sqrt{3})$$

And

$$\sqrt{8 + 2\sqrt{15}} = \sqrt{5 + 3 + 2 \times \sqrt{5} \times \sqrt{3}}$$

$$= \sqrt{(\sqrt{5} + \sqrt{3})^2} = (\sqrt{5} + \sqrt{3})$$

$$\sqrt{10 - 2\sqrt{21}} + \sqrt{8 + 2\sqrt{15}}$$

$$= (\sqrt{7} - \sqrt{3}) + (\sqrt{5} + \sqrt{3})$$

$$= (\sqrt{7} + \sqrt{5})$$

Comparing this with the equation (1)

$$\Rightarrow a = 7 \text{ and } b = 5$$

$$\sqrt{ab} = \sqrt{7 \times 5} \approx 5.9$$

Sol.612.(d) Short-trick:

Put $a = 2, b = 1$ and $c = 0$

$$ab(a - b) + bc(b - c) + ca(c - a) =$$

$$2 \times 1(2 - 1) + 1 \times 0(1 - 0) + 0 \times 2(0 - 2) = 2$$

Now, Go throw options Choose option (d)

$$\Rightarrow (b - a)(b - c)(c - a)$$

$$= (1 - 2)(1 - 0)(0 - 2) = 2$$

Condition satisfied.

Similarly you can put these values in other options but only option (d) will satisfy the condition

Sol.613.(b) Given,

$$(5x - 3)^3 + (2x + 5)^3 + 27(4 -$$

$$3x)^3 = 9(3 - 5x)(2x + 5)(3x - 4)$$

$$\Rightarrow (5x - 3)^3 + (2x + 5)^3 + 27(4 - 3x)^3 - 9(3 - 5x)(2x + 5)(3x - 4) = 0$$

$$\Rightarrow (5x - 3)^3 + (2x + 5)^3 + \{3(4 - 3x)\}^3 - 9(-1)(5x - 3)(2x + 5)(-1)(4 - 3x) = 0$$

$$\Rightarrow (5x - 3)^3 + (2x + 5)^3 + (12 - 9x)^3 - 3(5x - 3)(2x + 5)\{3(4 - 3x)\} = 0$$

$$\Rightarrow (5x - 3)^3 + (2x + 5)^3 + (12 - 9x)^3 - 3(5x - 3)(2x + 5)(12 - 9x) = 0$$

Now, $(5x - 3) \neq (2x + 5) \neq (12 - 9x)$

$$\text{Clearly, } 5x - 3 + 2x + 5 + 12 - 9x = 0$$

$$\Rightarrow -2x + 14 = 0 \Rightarrow x = 7$$

$$(2x + 1) = 2(7) + 1 = 15$$

Sol.614.(d) Given, $\frac{3(x^2+1)-7x}{3x} = 6$

$$\Rightarrow \frac{3x^2+3-7x}{3x} = 6 \Rightarrow x + \frac{1}{x} = \frac{25}{3}$$

$$\Rightarrow \sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{\frac{25}{3}} + 2 \dots (\text{If } a^2 + \frac{1}{a^2})$$

$$= k, \text{ then } a + \frac{1}{a} = \sqrt{k+2} = \sqrt{\frac{31}{3}}$$

Sol.615.(c)

$$(x+y+z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$\Rightarrow x^2 + y^2 + z^2 = 361 - 228 = 133$$

$$\therefore \sqrt{x^3 + y^3 + z^3 - 3xyz} =$$

$$\sqrt{(x+y+z)(x^2+y^2+z^2 - xy - yz - zx)}$$

$$= \sqrt{19(133 - 114)} = \sqrt{19 \times 19} = 19$$

Sol.616.(d) Short-Trick: Put $a = 1$

then, $x = 2$ and $y = 0$

$$\sqrt{x^4 + y^4 - 2x^2y^2} = \sqrt{2^4} = 4$$

Sol.617.(b) We know that

$$\text{If } x + \frac{1}{x} = k, \text{ then } x^2 + \frac{1}{x^2} = k^2 - 2$$

$$\text{Here, } a + \frac{1}{a} = 3 \Rightarrow a^2 + \frac{1}{a^2} = 3^2 - 2 = 7$$

$$\Rightarrow a^4 + \frac{1}{a^4} = 7^2 - 2 = 47$$

Sol.618.(b) Given,

$$x^4 - 6x^2 - 1 = 0, \quad x^2 - \frac{1}{x^2} = 6$$

Cubing both sides

$$(x^2 - \frac{1}{x^2})^3 = 6^3$$

$$\Rightarrow x^6 - \frac{1}{x^6} - 3 \times x^2 \times \frac{1}{x^2} (x^2 - \frac{1}{x^2}) = 216$$

$$\dots [(a-b)^3 = a^3 - b^3 - 3ab(a-b)]$$

$$\Rightarrow x^6 - \frac{1}{x^6} = 216 + 18 = 234$$

$$x^6 - 5x^2 + \frac{5}{x^2} - \frac{1}{x^6} + 5$$

$$\Rightarrow x^6 - \frac{1}{x^6} - 5(x^2 - \frac{1}{x^2}) + 5$$

$$\Rightarrow 234 - 5(6) + 5 = 209$$

Sol.619.(c) $\sqrt{x} - \frac{1}{\sqrt{x}} = 4$

$$\Rightarrow x + \frac{1}{x} = 4^2 + 2 = 18$$

$$\dots (\text{If } a - \frac{1}{a} = k, \text{ then } a^2 + \frac{1}{a^2} = k^2 + 2)$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 18^2 - 2 = 322$$

$$(\text{If } a + \frac{1}{a} = k, \text{ then } a^2 + \frac{1}{a^2} = k^2 - 2)$$

Sol.620.(d) Given, $\sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{7}$

$$x + \frac{1}{x} = (\sqrt{7})^2 - 2 = 5$$

$$(\text{If } x + \frac{1}{x} = k, \text{ then } x^2 + \frac{1}{x^2} = k^2 - 2)$$

$$x^3 + \frac{1}{x^3} = 5^3 - 3(5) = 110$$

$$(\text{If } x + \frac{1}{x} = k, \text{ then } x^3 + \frac{1}{x^3} = k^3 - 3k)$$